THREE ESSAYS ON THE ECONOMETRIC ANALYSIS OF EMERGING ISSUES IN ELECTRICITY MARKETS

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Abstract of the Dissertation

Three Essays on the Econometric Analysis of Emerging Issues in Electricity Markets

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This dissertation comprises three essays. Each essay examines an emerging issue in the electricity market and impacting the energy transition using econometric methods. The dissertation maps the challenges facing the market up and down the supply chain and provides counterfactual insights for incorporation by policy makers.

The first essay estimates how transmission expansion affects generator investment in electricity markets. Renewable resources are essential for decarbonization, but their unequal geographic distribution limits adoption. I estimate a dynamic model of generator behavior combining a short-run dispatch problem with transmission constraints and a long-run dynamic investment game, using an originally constructed dataset combining EIA, ISO, and proprietary data from 2018–2023. I find that upgrading inter-zonal transmission to modern HVDC standards would reduce average wholesale prices by 6%, generating approximately \$4 billion in annual welfare gains, with heterogeneous regional effects. Major transmission

projects such as the Grain Belt Express increase solar and wind adoption in the Eastern Interconnection by 11.5% by 2050. I use my estimates to evaluate the Big Wires Act, the Inflation Reduction Act, and the Bipartisan Infrastructure Law.

This second essay studies how learning-by-doing shapes cost dynamics, technological adoption, and the effectiveness of industrial policies in the solar panel industry. We develop and estimate a dynamic model of firm entry, exit, production, and technology choice using firm-level data on Chinese manufacturers and global data from 2000 to 2013. The model allows production experience to reduce future costs, generating a dynamic feedback loop between output and productivity. We find that learning-by-doing is a key driver of cost reductions, particularly for crystalline silicon (c-Si) technology: doubling cumulative production reduces marginal costs significantly. Learning also amplifies the effects of industrial policies, leading to larger long-run impacts on output and prices. It further shapes technological competition, contributing to the dominance of c-Si technology. Finally, earlier policy implementation yields substantially larger effects by accelerating movement along the experience curve.

The third essay measures the grid-level impacts of artificial intelligence. We develop a game-theoretic model identifying a timing wedge, the mismatch between data center consumption and credited renewable generation, as the central mechanism through which AI demand degrades reliability, raises prices, and increases emissions even under full REC coverage. AI demand significantly increases fossil generation, wholesale prices (up to 25% in treated PJM zones), and outage frequency (0.5–1 additional outages per year) near data centers, with impacts scaling in model size. Data centers with on-site generation exhibit a sign reversal in power-quality effects, consistent with the model's prediction that behind-the-meter capacity absorbs demand spikes.

To electricity grid and the people who make a better future possible by planning, inventing, building and maintaining it.

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Chapter 1

Investment and the Transfer of Power: Dynamic Effects of Transmission in Electricity Markets

Abstract. Renewable resources are essential for energy transition, but their unequal geographic distribution limits broader adoption. Long-distance transmission offers a potential solution by connecting renewables-rich areas to high-demand markets. I examine how expanded transmission capacity affects investment decisions by generators. I estimate a dynamic model of generator behavior consisting of a short-run optimal dispatch problem incorporating line losses and transmission constraints and a long-run dynamic game for capacity investment. Using an originally constructed dataset combining EIA, ISO, and proprietary data to estimate the model from 2018–2023, I find that upgrading all inter-zonal transmission to modern HVDC standards would reduce average wholesale prices by 6%, generating approximately $4 billion in annual welfare gains. Regional effects are heterogeneous: the Northeast experiences the largest price declines while the Midwest sees modest increases. In the long run, adding major transmission projects such as the Grain Belt Express increases solar and wind adoption in the Eastern Interconnection by 11.5% by 2050. I use these estimates to evaluate proposed and enacted policies including the Big Wires Act, the Inflation Reduction Act, and the Bipartisan Infrastructure Law.

Keywords: Electricity, Transmission, Capacity Investment, Renewable Energy, Dynamic Games

JEL Codes: Q40, L94, Q54, C73

1.1 Introduction

Climate change remains the most important environmental issue facing the world at present. In order to prevent catastrophic impacts of climate change, economies need to decrease greenhouse gas emissions. Electricity is associated with the highest share of carbon emissions in the United States, 31 percent in 2022 according to the Energy Information Administration, and has continuously provided the largest source of greenhouse gas emissions for decades (EIA, 2022). Renewable energy provides the clearest path to a reduction in electricity-related greenhouse gas emissions. Currently, a massive transition is underway throughout the developed world and the United States in particular to replace fossil fuel generators with renewable generation.

However, two persistent problems plague the renewable energy transition. First, the unequal dispersion of the potential for renewable resources makes renewable generation infeasible in vast swaths of the United States. The climate of states like New York and New Hampshire presents significantly less potential for renewable generation than that of states like Texas and Nevada. Texas and Nevada have abundant sun and wind, which have allowed renewable energy to flourish. The area required by solar and wind also means that the locations with the highest propensity for renewable generation are often further away from cities. Figure 1.10 illustrates this geographic heterogeneity, showing combined technical potential for wind and solar generation in US counties. Second, renewable resources tend to be intermittent. The sun shines only for so many hours a day, and wind patterns demonstrate marked variability. As electricity supply and demand must always balance, intermittency can cause instability in electric grids. If too much electricity enters the system at one time, it can cause an overload and lead to system outages. Although storage offers a potential solution to intermittency, it remains costly and physically limited at present. Storage also does not provide a solution in areas with limited ability to take advantage of any renewable energy.

Increasing transmission, the ability to send electricity between zones, provides another potential solution to the problem of intermittency and dispersion. Time-zone and geographic differences ensure that the peak hours for sunlight in Kansas coincide well with the peak demand hours in Indiana, and increasing long-distance transmission increases market interconnectedness, which decreases risks of instability and renewable energy curtailment (the forced shutdown of renewable resources due to over-supply). While long-distance transmission has previously proved largely infeasible both technically and economically, advancement in technologies such as High-Voltage Direct Current (HVDC) and High-Voltage Alternating Current (HVAC) provide a method through which to increase the range of transmission and potentially facilitate greater connection of renewables to the grid. Figure 1.11 shows the current state of HVDC infrastructure in the United States, with existing lines shown as solid and proposed expansions shown as dashed. Similarly, the transition from fossil fuel resources that could be easily transported to renewable resources increases economic viability of transmission expansion. Despite its potential merits as a technique for increasing adoption rates of renewable energy, econometric benefit analysis of transmission expansion has largely escaped the examination of economists.

How does long-run investment in transmission impact capacity investment in renewable resources? At face value, the direction of the welfare effects of transmission expansion may seem trivially positive, as is the case in most problems of friction reduction; however, because of the substantial cost of increasing transmission capacity combined with the dynamic effects of firm investment, the sign of welfare change remains ambiguous without further study.

Increasing transmission tends to lead to equilibration of prices across markets, and thus lower prices may drive firm exit in some markets, while higher prices spur entry in others. Exiting firms may cause blackout and decreased reliability, which can cause significant losses in consumer welfare due to the high inelasticity of the demand curve for electricity. Beyond first order welfare and reliability effects, emissions produced and distributional effects must be examined.

Three main sets of counterfactual experiments are performed to examine policy impacts. Broadly, these counterfactual experiments focus on transmission upgrades, new long-distance lines, and federal legislation. I examine transmission upgrades from current HVAC to more modern HVDC infrastructure to determine the impact of replacing current inter-zonal transmission with improved lines. From the perspective of line additions, I examine the limits of transmission with a counterfactual where all constraints are removed. I also consider the impact of further interconnection between ERCOT and the Eastern Interconnection and of the Grain Belt Express. I examine three main pieces of federal legislation, two passed, and one hypothetical. I examine the BIG WIRES Act, which provides a mandated amount of inter-regional transmission expansion. I further examine the impacts of generation investment subsidies from the Inflation Reduction Act and investments in transmission from the Bipartisan infrastructure Law. For all experiments, I consider overall welfare changes and distributional effects with the goal of understanding static and dynamic impacts but also providing insight into potential transfers that might make line additions Pareto improving.

Recent work within the industrial organization literature has taken a profound interest in the determinants of long-run investment in the electricity market as it relates to energy transition. Butters et al. (2021) examine the issue of battery investment to reduce intermittency as it relates to the California Independent System Operator (CAISO) market while Elliott (2022) uses an oligopolistic model to consider the potential impacts of carbon taxes and capacity payments on the resource mix in Western Australia. While both consider the dynamics of the electricity market, both also assume that prices are constant throughout the entire region and cannot be used to analyze locations with more than one settlement point. More recent work by Gowrisankaran et al. (2024) connects changes in the price differential between natural gas and coal to widespread changes in capacity investment away from coal to natural gas. Arkolakis and Walsh (2023) integrates the impacts of transmission into a macro model and considers how the implementation of proposed projects might impact further adoption of renewable energy. Though Arkolakis and Walsh's work considers a similar question, the question is approached from a macro perspective and so, the dynamics of individual firms are neglected, providing an important source of future research that the current work addresses.

Gonzales et al. (2022) considers the implications of interconnection between the Northern and Southern Chilean electricity markets for long-run entry and exit. My work extends their research by generalizing the model to deal with larger and more complex grids. My work also goes further by utilizing variation in spatial transmission capacity to examine transmission's impact without significant temporal variation in transmission infrastructure.

The inclusion of generator investment based on price differentials between locations marks a major contribution to the literature. Real-world limitations on the movement of electricity make ignoring transmission perilous, and the need for investment in renewables makes a valuation of transmission expansion, including its effects on renewables investment, increasingly pertinent. Consideration of how interconnected electricity markets function together even on short time scales also remains conspicuously absent from the industrial organization literature around electricity providing a gap for research.

I examine the behavior of generators providing electricity in the wholesale market by building a theoretical model of the short-run electricity market and connecting the model to a dynamic game in the long-run electricity market where generators enter and exit. The model builds on prior work in electricity by adding constrained trade between regions in electricity. I estimate the short-run model using maximum likelihood, and I estimate the long-run model using a fixed-point generalized method of moments methodology, applying the strategy for estimating capacity investment in dynamic games found in Gowrisankaran and Schmidt-Dengler (2024). I run the counterfactuals by adjusting the available set of long-range transmission lines in the United States and the fixed costs and entrance costs of the generators in line with the legislation considered.

The paper is organized as follows. Section 1.2 reviews the prior literature related to transmission as well as investment in the electricity market. Section 1.3 describes the structure of the analyzed market and presents descriptive statistics about the market. Section 1.4 presents a structural model, while Section 1.5 describes the estimation strategy using the structural model. Section 1.6 provides a description of the results and fit of the model. Section 1.7 details counterfactuals. Section 1.8 concludes the paper and provides further research directions.

1.2 Literature Review

The literature related to the question of how transmission impacts generator investment in electricity markets falls along three interrelated threads: long-run capacity investment decisions by firms, the role of transmission and congestion in shaping market outcomes, and the modeling of electricity markets undergoing a changing resource mix.

A central challenge in the literature on investment in the electricity market is modeling how firms choose to add or retire generation capacity over time. Butters et al. (2021) develops a dynamic competitive equilibrium model of battery storage adoption and operations using California data, linking short-run arbitrage in the day-ahead and real-time markets to the long-run market for storage capacity. The paper finds that California's storage mandate modestly increases social welfare even before accounting for reliability benefits, although equilibrium effects sharply diminish the marginal value of additional storage units. Elliott (2022) extends this type of framework to a setting with strategic interaction, building a dynamic oligopoly model of investment in the Western Australian electricity market. Elliott's model distinguishes between strategic firms, firms whose entry and exit decisions influence price, and fringe firms. Elliott (2022) uses this structure to evaluate the impacts of carbon taxes and capacity payments on the generation mix. Both papers demonstrate the importance of modeling the feedback between short-run market operations and long-run capacity decisions, a feature that my model shares.

A separate strand examines how the composition of the generation fleet responds to changing fuel economics and policy. Gowrisankaran et al. (2016) quantifies the social costs of solar intermittency in southeastern Arizona, finding that intermittency accounts for a substantial share of the costs associated with renewable generation and represents a significant barrier to a grid powered primarily by renewables. Gowrisankaran et al. (2024) examines why regulated utilities have retired coal more slowly than their counterparts in restructured markets despite the growing price advantage of natural gas. That paper uses an estimation approach developed by Gowrisankaran and Schmidt-Dengler (2024), which provides a computationally tractable method for solving dynamic oligopoly models with many ordered capacity levels. My estimation strategy draws on this computational approach but embeds it within a dynamic game that considers multiple generation companies simultaneously, accounts for the shift toward renewables, and integrates transmission constraints into the model.

The role of transmission in electricity markets has received growing attention, but remains underexplored in the investment context. Holland et al. (2022) provides one of the first models to consider multiple wholesale markets simultaneously, using EIA data on generation mix and load to evaluate how policies, including transmission expansion, affect renewables adoption and social welfare. However, the model treats regional markets as homogeneous and equates increased transmission with universal access by all generators to all load, a strong assumption that abstracts from the nodal structure of actual wholesale markets. The current work relaxes this assumption by modeling transmission and congestion with equilibria determined at the node level.

Arkolakis and Walsh (2023) builds a spatial macroeconomic model of clean energy adoption in which electricity transmission plays a role, incorporating learning-by-doing and regional comparative advantage in renewable resources. Although the model captures the aggregate and spatial implications of the energy transition, it does not endogenize the process by which individual generators make investment decisions in response to market signals. This is a mechanism that is central to my analysis.

The closest antecedent to my work is Gonzales et al. (2022), which estimates the investment effects of grid expansion connecting the northern and southern regions of Chile on the entry and exit of renewable and fossil fuel generators. That paper makes an important contribution by linking transmission infrastructure to investment incentives. However, its short-run model relates transmission to prices through a framework that does not allow for simultaneous settlement at multiple nodes, and it relies on generation data that are not available at comparable granularity for the United States. Moreover, Gonzales et al. (2022) does not model the strategic interdependence of generators in its long-run investment stage. My model addresses these limitations by using a dynamic game for the long-run stage, which captures the fact that within a transmission-constrained network, capacity choices are fundamentally interdependent: the decision to invest in solar in Nebraska alters the competitive landscape facing coal plant owners in Indiana.

My work synthesizes and extends several of these contributions. It combines the attention to transmission found in Gonzales et al. (2022) with a dynamic oligopoly investment model in the spirit of Elliott (2022), using the computational methods of Gowrisankaran and Schmidt-Dengler (2024) and building on the estimation framework of Pakes and Ericson (1998). By incorporating congestion data from multiple markets simultaneously and linking real-time price formation to long-run capacity dynamics, the model enables counterfactual analysis of how transmission expansion reshapes generator investment across the interconnected U.S. grid.

1.3 Setting and Data

1.3.1 Setting

In most of the United States, the electricity market is divided between the wholesale and retail markets. In wholesale markets, generators sell electricity to utilities, who then sell the electricity to end-users in the retail market. Wholesale market prices are generated through the locational marginal price (LMP) mechanism. Generators submit bids of price-quantity pairs that determine how much an individual generator is willing to supply at a series of set prices. The equilibrium price is thus the price at which the last economically dispatched generator bids for the final unit of electricity. This price is paid to all generators that are dispatched at a particular market location.

I focus on the wholesale market for electricity in the Eastern Interconnection and ERCOT (Electric Reliability Council of Texas). I choose to focus on regions with wholesale markets because wholesale markets have readily available price data and because vertically integrated retail markets function differently. The focus on the Eastern Interconnection comes from the fact that the majority of US ISOs are in the Eastern Interconnection. The addition of ERCOT provides the ability to perform counterfactuals examining expanded interconnection between ERCOT and the Eastern grid.

1.3.2 Dataset Creation

Figure 1.4 shows a full diagram of the creation of the dataset for analysis. Data for the short-term model came from EIA Form 923, EIA Form 860, EIA Form 930 data, EPA's Power Markets Data, the individual ISOs, S&P capital IQ Global, Homeland Infrastructure Foundation Level Database (HIFLD), and Meteostat. The model was estimated on data from 2018 to 2023 at the zonal level. The zonal level was chosen to increase the proportion of binding transmission constraints in the data and to increase the granularity of the dataset in terms of both generators and load sources.

From each ISO individually, I took price and load data at the zonal level. Then I took the hourly fuel mix data at the zonal level to estimate the generation by generator by hour. For fossil fuel generators covered by EPA's CAMPD (Clean Air Markets Program Data), the generation by hour was available. However, for other generators, the generation needed to be estimated. In order to estimate generation by hour by generator, EIA Forms 923 and

860 were combined with ISO data on fuel mix. Percentage of generation by the generator was found by taking the percentage of total generation for the zone by month by generation owner from Form 923. The value was then multiplied by the generation for the fuel type for the hour. If generation by the generator was greater than generation capacity, the generation was reduced to the generation capacity, and generation was redistributed to the remaining generators.

In order to estimate transmission capacity and line losses between zones, the HIFLD data on the electricity grid was utilized. The data allowed me to build a full dataset of transmission capacity and line losses in 2022. A line was considered to connect two zones if the line started at a substation in one zone and ended at a substation in another. Using this approach for connection estimation has the downside of neglecting flows that went between multiple substations and zones. However, this works well with the assumption of the grid as a bipartite network from the model. The HIFLD data contained information on type of line, voltage, length, and location. This was used to estimate flow limits and line losses based on engineering assumptions according to the technique discussed in Appendix 1.12 following Arkolakis and Walsh (2023). To get historical changes in transmission capacity and line losses, I solved for transmission capacity additions and retirements based on S&P Capital IQ's transmission projects dataset. Calculated additions and retirements were added to the transmission capacity and losses from the HIFLD data to get the transmission capacity from 2001 to 2024.

With the approximate generation data for generators that did not report to the EPA and the exact costs for generators that did report to the EPA, the exports from each zone to each other zone were estimated. To estimate exports, I first added net exports to zones outside of the area of interest. Net exports to outside areas by hour were taken from EIA Form 930, which provides data on flows between balancing authorities at an hourly level. Where a balancing authority outside of the dataset connected to multiple zones in the dataset, net exports are added to load proportionately to load in each of the connected regions.

Given data on load and approximate generation, an optimal dispatch problem was solved for each hour based on the generation by generator, transmission capacity and line losses. Generation was dispatched from zones with low LMP to zones with high LMP in order of marginal cost based on heat rate and fuel cost until all load was satisfied and all generation was distributed. Marginal cost was estimated using fuel cost data from EIA Form 860 and heat rate data from EPA CAMPD data. For resources without marginal cost data from EIA, an order of dispatch was provided by resource type with solar, hydro, geothermal, and wind dispatched first with ties distributed based on capacity size. Nuclear was then distributed followed by fossil fuel resources for which marginal cost data at the monthly level was available. Firms with lower marginal costs per unit of generation were dispatched to regions earlier. I use these export data to estimate hourly marginal cost curves in the short-term model.

The characteristics of the zones impacting marginal cost of the generators were taken from EIA Form 860, Meteostat, and S&P Capital IQ Global. EIA provided information about the generation companies in terms of ownership and characteristics of generators. Meteostat provided data on weather hourly in the zones. In order to get an average weather for the zones, a centroid was taken from each zone. Weather data thus represent the data at the weather station closest to the centroid of the zone.

Data for the long-term model came from EIA Form 860, EIA's Annual Energy Outlook (AEO), NREL's Standard Scenarios, and S&P Capital IQ Global. The long-term model was estimated on historical data going back to 2001 with forward-looking generators basing capacity decisions on revenues estimated until 2050 after which the fuel mix and load profile are assumed constant and an infinite sum is taken. The year 2050 was chosen because it was the last year of projections from AEO and because it is a date at which most pledges for energy transition are based.

EIA Form 860 provided data on capacity of generators between 2001 and 2024. These data were aggregated by owner, zone, and resource type to get capacity for long-term model estimation. For resources of types not modelled such as nuclear and hydroelectric, the capacity was taken from the aggregate capacity by zone. Meteostat and Prism provided historical long-term weather data. S&P Capital IQ Global and EIA provided data on historical natural gas prices and coal prices.

Most data from the projections for years in the future come from EIA's Annual Energy Outlook (AEO) for 2023 and NREL's Standard Scenarios with Cambium. The AEO has data on projected load by region, projected natural gas and coal price, and projected capacity and generation for nuclear and other generation types. In order to get projections for long-term changes in transmission capacity for the long-term model, NREL's Standard Scenarios were utilized. The Standard Scenarios also provided long-run data on transmission and distribution losses, load growth, and future capacity for fuels not in the model.

The current study will consider data only from Independent System Operators in the Eastern Interconnection and the Electric Reliability Council of Texas (ERCOT). For a map of the zones of interest, refer to Figure 1.5. The process of constructing the zone map is described in Appendix 1.11.2. This includes the Southwest Power Pool (SPP) which governs parts of the Midwest from Oklahoma to Wyoming. The Eastern Interconnection also includes the Midcontinent Independent System Operator (MISO) which governs the Midwest east of Mississippi to Nebraska as well as Pennsylvania-New Jersey-Maryland Interconnection (PJM), New York Independent System Operator (NYISO), and ISO New England, which governs all states on the Eastern Seaboard North of New York. Outside of the Eastern Interconnection, data are also provisioned from the ERCOT, which includes much of Texas.

1.3.3 Descriptive Statistics

Table 1.1 shows how prices and loads differ between the different Independent system operators. There is significant variation across ISOs with ISO-NE having the highest mean price and SPP having the lowest. Load also varies by ISO with PJM, the largest ISO also having the highest load. Figures 1.6 and 1.8 show how load varies across the sample by zone. Load differs both within a time dimension and across the spatial dimension with peaks in load profiles in differing by time zone and weather conditions and larger more densely populated zones having more load. Table 1.2 provides descriptive statistics on the estimated features of the hourly dispatch model including hourly exports by generator and hourly generation by generator type. Table 1.3 provides descriptive statistics for the short run characteristics and the variables that feed into the short-run model.

Figure 1.7 shows how estimated capacity factor differs by ISO. Capacity factor is the ratio between total generation and total capacity and indicates the efficiency of a particular generator or generators in a particular location.

1.4 Model

1.4.1 Preliminaries

The modeling framework borrows substantially from Butters et al. (2021), Elliott (2022), and Gonzales et al. (2022). The model consists of a static short-run model and a dynamic long-run model. The primary source of variation arises from the introduction of location-specific prices and spatial heterogeneity in generators, as well as the introduction of a dynamic game into a model with transmission constraints.

While previous models typically assumed either a single price for all locations or prices varying only by broad regions, this model allows all markets to clear interdependently. Allowing many settlement locations increases the size of the state space and makes a short-run dynamic model intractable for considering complex phenomena like curtailment; however, the tradeoff is necessary to capture granular transmission effects. Because decreasing curtailment is a benefit of transmission expansion, excluding it likely leads to an conservative estimate of the benefits of increasing transmission.

Within the short-run, a combined Independent System Operator (ISO) solves an optimal dispatch problem, choosing how much electricity to offer and at what nodes to minimize costs. In the long-run, a capacity game occurs where generator owners choose optimal capacity based on profit expectations derived from the short-run model's prices. While the short-run model functions primarily as an engineering problem solved by a social planner and estimated using econometrics, the long-run model is solved by competitive equilibrium.

1.4.2 Network Topology and Aggregation

The network is modeled as a bipartite graph consisting of demand nodes indexed by i and generation nodes indexed by j. Electricity flows from generation nodes j to demand nodes i. Figure 1.12 illustrates the mechanics of this network. The adjacency matrix A represents the transmission capacity, where A_{ij} is the maximum electricity that can flow between node j and node i. Line losses, governed by the particular technologies used at locations, are represented by the matrix B, where B_{ij} represents the proportion of electricity lost during transmission from j to i. At each generation node j, there are various fuel resources indexed by k (e.g., wind, solar, coal, natural gas) with individual generators indexed by g.

Fuel resources are divided into endogenous and exogenous categories in the long-run investment model. Wind, solar, coal, and natural gas are modeled endogenously. The "other major fuel sources" category, primarily hydroelectricity and nuclear, is taken as exogenous to reduce the state space. This is consistent with historical data, where year-to-year variation is much higher in fossil and renewable sources than in hydro or nuclear, which are more heavily influenced by government investment and regulatory constraints than by short-run market forces.

1.4.3 Short-run Operations Model

Under certain conditions, the generator's profit maximization problem is identical to the social planner's cost minimization problem. Following Gilbert et al. (2004) and Cramton (2017), the conditions for equivalency include: no market power among generators, no gamesmanship or non-competitive bidding, increasing and continuous marginal cost curves without jumps, and linear constraints.

The assumption of increasing marginal cost curves is consistent with both economic theory and bid offer data provided by generators. While gamesmanship is strictly policed by the Federal Energy Regulatory Commission, the assumption of no market power is the strongest. As Gilbert et al. (2004) notes, transmission constraints are the main source of market power; however, due to competitive pressures and bidding regulations, sustained market power in the electricity market proves elusive. Given the value of this equivalency for tractability, I proceed by solving the problem as a social planner's problem.

The ISO solves a static problem each hour t, choosing the quantity q_{ijkgt} to send to each individual demand node i from each resource k at supply node j for generator g. The ISO minimizes the total cost of generation subject to system constraints, which is equivalent to maximizing the negative of the cost function:

$$\pi_t = \max_{\{q_{ijkgt}\}_{i,j,k,g,t \in I,J,K,G,T}} \left[-\sum_{I=1}^{I} \sum_{j=1}^{J} \sum_{k=1}^{K} \sum_{g=1}^{G} c_{jkgt}(x_{jkgt}, q_{ijkgt}, \varepsilon_{ijkgt}) \right]. \tag{1.1}$$

The cost of dispatching a generator, c_{jkgt}, depends on resource characteristics x_{jkgt}, the quantity dispatched, and an error term ε_{ijkgt} representing incorrect specification of the cost function visible to the ISO but not the econometrician.

The optimization is subject to three primary linear constraints:

1. Resource Constraint (Kirchhoff's Law): The system must balance such that total load equals total generation minus line losses. Demand is assumed perfectly inelastic.

$$\sum_{j=1}^{J} \sum_{k=1}^{K} \sum_{g=1}^{G} (1 - B_{ij}) q_{ijkgt} = L_{it} \quad \forall i. \tag{1.2}$$

2. Transmission Constraint: The ISO cannot transmit more electricity between two locations than the line capacity permits.

$$\sum_{k=1}^{K}\sum_{g=1}^{G} q_{igjkt} \leq A_{ij} \quad \forall i,j,t. \tag{1.3}$$

3. Capacity Constraint: A resource cannot be dispatched beyond its maximum capacity O_{jt}^{MAX} at any time t. This capacity varies over time due to weather conditions (affecting solar/wind) and seasonal factors (affecting thermal efficiency).

$$0 \leq \sum_{i} q_{igjkt} \leq O_{jkgt}^{MAX} \quad \forall j,k,g,t. \tag{1.4}$$

Dispatch and Pricing

If no constraints bind, resources set marginal cost equal to price and produce where the price is highest. However, when constraints bind, prices diverge. If the lowest-cost resource faces a transmission constraint, it dispatches up to A_{ij} in the highest-price location, then moves to the second-highest, repeating until either the maximum generation O_{jgkt}^{MAX} is reached or marginal revenue equals marginal cost.

The Locational Marginal Price (LMP) at any node is equal to the marginal cost of the last economically dispatched resource serving that node. When transmission constraints bind, cheap power from low-cost zones cannot fully reach high-demand areas, causing LMPs to separate spatially.

I rely on the following assumptions for the solution:

1. Cost functions are continuous without jumps and quadratic with respect to inputs.

2. Blackouts are not allowed; there is sufficient electricity to satisfy all load at every node at any time.

3. No resource is ever transmission-constrained at *all* locations simultaneously.

4. Within each generation node j, at least one resource is not at maximum capacity at

any given time.

1.4.4 Econometric Specification

The solution method for the operations model (detailed in Appendix 1.9) is reminiscent of trade models. We can define a latent variable $\tilde{V}$ representing the conditions for the marginal resource. The error for a constrained generator equals the difference between its marginal cost and that of the marginal generator, adjusted for the shadow costs of transmission and capacity constraints.

The structural equation is given by:

$$\pi_t = \max_{\{q_{ijkgt}\}_{i,j,k,g,t \in I,J,K,G,T}} [-\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{g=1}^{G} c_{jkgt}(x_{jkgt}, q_{ijkgt}, \varepsilon_{ijkgt})] \quad s.t. \tag{1.5}$$

$$\underbrace{\sum_{j=1}^{J}\sum_{k=1}^{K}\sum_{g=1}^{G}(1-B_{ij})q_{ijkgt} = L_{it}}_{\text{Resource constraint}} \quad \forall i$$

$$\underbrace{\sum_{k=1}^{K}\sum_{g=1}^{G} q_{ijkgt} \leq A_{ij}}_{\text{Transmission constraint}} \quad \forall i,j,t$$

$$\underbrace{0 < \sum_{i} q_{ijkgt} \leq O^{MAX}_{jkgt}}_{\text{Capacity constraint}} \quad \forall j,k,g,t.$$

To identify the error terms, consider generator $jg'k$. For any nodes ijk' with transmission constraints but without capacity constraints:

$$\tilde{\varepsilon}_{igj'kt} = \varepsilon_{igjkt} - \varepsilon_{i'g'jkt}. \tag{1.6}$$

For nodes $i'jk'$ that are neither transmission nor capacity constrained:

$$\tilde{\varepsilon}_{i'jk'gt} = \varepsilon_{i'jk'g't}. \tag{1.7}$$

By assumption, each j node has one resource not capacity constrained, and each resource has at least one i node with which it is not transmission constrained. Plugging Equation 1.7 into 1.6 allows us to solve for each ε_{ijkgt}

Define P_t as a multivariate normal probability density function with mean zero and variance Ω_1 from which the errors ε_{ijkgt} are drawn. This yields the likelihood function for the set of resources excluding the marginal resource:

$$L_1 = \prod_{t=1}^{t=T} P_t(\tilde{V}_{111t}, \ldots, \tilde{V}_{IJKt}). \tag{1.8}$$

By maximizing this likelihood function, I estimate the parameters ν_k and β to solve the short-run model.

Resource Dynamics in the Short-Run

In my framework, renewable resources, such as wind and solar, have marginal costs averaging to zero, reflecting that their fuel inputs are free. There is an added mean-zero error to prevent stochastic degeneracy in estimation. Conversely, fossil fuel resources have consistently positive marginal costs and control the marginal generation. They choose when to burn fuel.

In the long run, these differences drive market entry and exit. Fossil fuel resources face increasing fixed costs due to regulation and maintenance, while renewables experience decreasing entry costs due to technological improvement. As transmission expands, prices equalize spatially: prices paid to fossil resources generally decrease, while those paid to renewables increase. As the market moves from autarky to trade, prices decrease on average due to dispatch of cheaper resources and an insurance effect. These short-run profit signals drive the long-run transition, leading to increased renewable capacity and fossil fuel exit.

1.4.5 Long-run Model

The short-run and long-run models interact primarily through prices and capacity. Figure 1.13 illustrates this relationship: the ISO solves the optimal dispatch problem hourly, which generates prices and costs that feed into generators' expected profits. These expected profits then drive long-run capacity investment decisions, which in turn affect future dispatch outcomes.

In the short run, expanding transmission capacity leads to price convergence between regions through spatial arbitrage. Figure 1.14 illustrates this mechanism: low-cost regions (net exporters) see prices rise as they gain access to distant markets, while high-cost regions (net importers) experience price declines as cheaper power flows in. The welfare effects include producer surplus gains in exporting regions and consumer surplus gains in importing regions.

These short-run price changes create dynamic investment incentives. Figure 1.15 shows how expanded market access shifts the marginal benefit curve for capacity investment outward in resource-rich regions, leading to increased equilibrium capacity. Higher prices in previously constrained exporting regions incentivize entry by renewable generators who can now sell to distant markets.

The long-run model characterizes the decision of firms to invest in or reduce capacity. Firms invest in capacity based on potential profits and retire capacity based on the expected losses. The value of a set amount of capacity given state variables in the current period $\upsilon_{gt}(\Theta)$ is equal to the max with respect to the current period's chosen investment ΔO_g of profits from all generators in the current period $\Pi_{jg}(\Theta)$ plus the unobservable error of the chosen amount of investment today plus the discounted expected value of the generator in the next period. Profits $\Pi_{jg}(\Theta)$ are the sum for a particular node of the aggregate demand times the average price minus the aggregate cost times the aggregate demand minus fixed costs times total capacity minus entrance cost times positive investment. The problem is shown in Equation 1.9.

The problem is not stationary as both entry costs and fixed costs are integrated AR(1) processes. Our primary parameters to estimate are ψ_{ik}, the per-period average increase or decrease in fixed and entry costs, ρ_{ik}, the persistence of fixed and entry costs, and $\sigma^2_{\zeta_i}$, the standard deviation of the error for fixed and entry costs. In theory, over time, fixed costs will increase for coal and natural gas leading to retirement of fossil fuel plants while entry costs will decrease for solar and wind plants leading to investments in renewable generation. Over time, the model simulates a renewable energy transition. The short-run transmission network impacts the long-run choices of investment and retirement through changes in revenue and costs, which impact the profits and change the desirability of owning and investing in capacity.

$$\upsilon_{gt}(\Theta) = \max_{\Delta O_g} \sum_j \Pi_{jg}(\Theta) + \varepsilon_{jg}(\Delta O_{jg}) + \beta \mathbb{E} \upsilon_{jgt+1}(\Theta') \quad s.t. \tag{1.9}$$

$$\Pi_{jg} = \textstyle\sum_k [\underbrace{D_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) P_{jkg}}_{\text{Revenue}}$$

$$-\underbrace{C_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) D_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg})}_{\text{Variable costs}} - \underbrace{F_k O_{jgk}}_{\text{Fixed cost}} - \underbrace{E_k \max(\Delta O_{jgk}, 0)}_{\text{Entry cost}}]$$

$$\underbrace{O'_{jgk} = O_{jgk} + \Delta O_{jgk}}_{\text{Evolution of capacity}}$$

$$\underbrace{F'_k = \psi_{1k} + \rho_{1k} F_k + \zeta'_{k1}, \zeta_1 \sim N(0, \sigma^2_{\zeta_1})}_{\text{Fixed cost process}}$$

$$\underbrace{E'_k = \psi_{2k} + \rho_{2k} E_k + \zeta'_{k2}, \zeta_2 \sim N(0, \sigma^2_{\zeta_2})}_{\text{Entry cost process}}$$

1.5 Estimation Strategy

1.5.1 Identification

The parameter set β is identified based on variations in choices to dispatch resources of the same fuel type with different characteristics. The parameter set ν_k is identified by variations chosen in quantities dispatched by resource type after controlling for within resource heterogeneity. The variance of any particular error is not identified separately from the parameters, but the covariance of the residuals identifies the correlation between errors at a particular timestep.

The ability to estimate the impact of changing transmission capacity through altering transmission capacity matrix A and line loss matrix B comes from the model's estimation of the parameters above, β and ν_k. Using these parameters, the system of resources for dispatch can be solved as a linear program giving the shadow prices related to transmission capacity. Using the fully solved model, counterfactuals can also be run by altering these matrices.

The eight primary parameters of interest in the long-run model are the per-period change in fixed cost by resource type, ψ_{1k}, the per-period change in entrance cost by resource type, ψ_{2k}, the auto-regressive coefficient in the law of motion for fixed costs ρ_{1k}, the auto-regressive coefficient in the law of motion for entry cost ρ_{2k}, the initial fixed cost, F_{0k}, the initial entrance cost, E_{0k}, and the variance of the error terms, $\sigma^2_{\zeta_{1k}}$ and $\sigma^2_{\zeta_{2k}}$. It should be noted that each of these parameters are estimated for coal, natural gas, solar, and wind, so in total 32 parameters are estimated or 8 parameters per resource. The total sample size for the long-run is equal to the total number of included generation companies times the total number of resources those companies hold times the number of zones the generators participate in times the total number of years in the long-run sample.

The initial fixed cost by resource type, F_{0k}, is identified by the capacity retirement choice in the starting period by generator type after accounting for expected demand, costs, and prices. This occurs when ΔO_{jk} is negative. The autoregressive coefficient in the entrance cost equation,ρ_{1k}, is identified by the level of correlation between negative capacity investment decisions in different periods after accounting for other parameters and factors. The per-period change in the fixed cost, ψ_{1k}, is identified by decisions to change capacity retirements over time after accounting for changes in prices, demand, and costs. The initial entrance cost by resource type, E_{0k}, is identified by the amount of increase in capacity in the starting period by resource type after accounting for fixed costs and changes in prices and expected demand. The autoregressive coefficient in the entrance cost equation, ρ_{2k}, is identified by the level of correlation between positive capacity investment decisions in different periods after accounting for other parameters and factors. The per-period change in entrance costs by resource type, ψ_{2k}, is identified by changes in entrance by resource type after accounting for other parameters and factors.

The variance of the error terms $\sigma^2_{\zeta_{1k}}$ and $\sigma^2_{\zeta_{2k}}$ are identified by variations in decisions by resources to decrease or increase capacity respectively for resources of the same type after accounting for differences in demand, prices, and costs.

1.5.2 Short-run Estimation

The data do not include information on where electricity generated by a particular resource in a particular location is transported. However, with the generation and transportation data available, the exact locations electricity is sent to can be determined by assuming that energy goes to the location of highest price, so all low-cost energy is dispatched to the region with highest returns given losses and transmission constraints before high-cost energy is dispatched. This reasonably approximates the unobserved quantity flows by resource type.

Where data on flows and generation are available at a more granular level than the ISO such as at the zonal level, the zonal level is used as the area of observation, and demand nodes i and supply nodes j are at the zonal level. This makes prices more consistent with the prices experienced by resources. Where these data are available only at the ISO level, these nodes are ISOs, and prices are aggregated to the ISO level by taking a weighted average based on load. Where available, these weighted averages are taken from official sources. When not available, I calculate them.

The data do not include information on losses for all transfers between all zones and

ISOs. As such, losses are calculated outside the model using the loss data that I have and data on line lengths, composition, and flows to estimate the losses. This is appropriate because losses are not endogenous to the model but are purely a result of technical constraints related to lines.

The operations model will be solved using the simulated maximum likelihood method. Errors will be drawn from a multivariate normal distribution for each time, which will allow me to solve for the logarithm of the likelihood function from Equation 1.8;

$$log(L_1) = \sum_{t=1}^{T} log(P_t(\tilde{V}_{1111t}, ..., \tilde{V}_{IJKGt})). \tag{1.10}$$

Errors will be simulated, and parameters will be changed based on gradient descent until the maximum of the log-likelihood function is found.

1.5.3 Long-run Estimation

The long-run model estimation utilizes the dynamic game investment estimation methodology from Pakes and Ericson (1998) and utilizes the algorithm from Gowrisankaran and Schmidt-Dengler (2024) to find feasible choices to speed up algorithmic performance. The estimation strategy is thus an iterative fixed point algorithm, which solves for a value function within the iterated step to find equilibrium choice probabilities. The estimation utilizes the generalized method of moments (GMM) framework to find parameters.

Following the methodology from Gowrisankaran and Schmidt-Dengler (2024), I compute the equilibrium through a nested fixed point procedure where I iterate start with a belief of each generation company of the probability of rival actions, solve the value function iteratively for the belief, and iterate until all beliefs are correct. Diverging from the framework, I use an oblivious equilibrium concept, so the beliefs must only be correct with regards to the beliefs of generation companies about rival competitive capacity.

The state space is discretized as follows. Fixed cost is divided into 20 bins for from zero to the maximum revenue per megawatt of a resource type. Entry cost is divided into 20 bins from zero to ten times the maximum revenue per megawatt of a resource type. Demand state space is divided into 5 bins from .75 to 1.5 times previously seen max load. The own capacity state space is divided into 15 bins from 0 to 1.5 times max generation of a generation owner. The alternative capacity state space is divided into ten bins from 0 to 1.5 times the maximum load. The average price state space is divided into 10 bins from 0 to 1.1 times the maximum average price in the data. The possible investments are restricted to between zero new investments and investment that would meet half zonal capacity for 10 possible investment choices. Investment is restricted, so negative capacity is not permitted.

In order to estimate the model, the parameters from the short-term model are estimated via maximum likelihood. These parameters are used within the marginal cost function to solve the linear program to get quantities and prices in the short-run with expected changes to the grid in the future. A subset of dates are solved using the parameters for all elements of the state space. The results are then fed into the long-run model, which uses the short-run results for profits for each year.

The long-term model has 32 parameters to estimate including the integrated and auto-regressive terms in the fixed cost and entry cost, the initial fixed cost and initial entry cost, and the standard deviation for both errors. Each parameter needed to be estimated for all four resource types. In order to estimate the 32 parameters, at least 32 moments must be used. I largely follow Gowrisankaran et al. (2024) in choice of moments to get 36 total moments for an overidentified model. For natural gas, solar, and wind, the moments include an indicator for positive investment interacted with quantity built, total quantity squared, and an indicator for investments in capacity of large size, which is specific to resource based on the data. I interact these moments with total capacity and also include investment variance as a moment. I add one additional moment, which is the amount of capacity retired to help identify fixed cost parameters. For coal, I do the same but focus on moments related to retirements instead of investment with one investment-related moment.

1.6 Results

1.6.1 Short-run Model Estimates

Table 1.4 presents estimates from the short-run dispatch model for export sensitivity and base cost parameters across resource types. The intercept terms represent base marginal costs in $/MWh, while the export coefficients capture how marginal costs vary with the quantity exported from each zone. Table 1.5 reports coefficients for natural gas generator characteristics. The natural gas spot price coefficient of 6.93 implies that a $1/MMBtu increase in the Henry Hub spot price raises the marginal cost of natural gas generation by approximately $6.93/MWh, consistent with average heat rates in the sample. The negative coefficient on heat content reflects that generators with higher fuel efficiency face lower marginal costs. Table 1.6 presents analogous results for coal resources, where the coal price coefficient of 8.68 reflects the higher heat rates typical of coal plants.

Figure 1.1 displays model fit for the short-run dispatch model. The model slightly underestimates average prices but accurately captures price dispersion across zones. The correlation between predicted and actual locational marginal prices is strong, indicating that the model captures the primary drivers of spatial price variation.

1.6.2 Long-run Model Estimates

GMM estimates of the evolution for both fixed and entry costs are derived based on the normalized empirical moments found in Table 1.7. Table 1.8 reports parameter estimates for the long-run capacity investment model. The estimates reveal substantial persistence in both fixed and entry cost processes across technologies, though with meaningful heterogeneity. Fixed cost persistence ρ_{fc} ranges from 0.806 for coal to 0.917 for wind, indicating highly persistent fixed cost environments across all four technologies. Entry cost persistence ρ_{ec} is similarly strong for coal (0.926) and natural gas (0.916), and for solar (0.869), but is notably lower for wind (0.390), suggesting that wind entry costs revert more quickly toward their long-run drift than do costs for the other technologies.

Figure 1.18 shows the evolution of cost structures through 2050 in the base case scenario, assuming costs follow their estimated paths. The drift in entry costs, governed by the entry cost increase parameter, highlights stark differences across technologies. Renewable technologies exhibit declining entry costs over time, with estimated annual decreases of approximately \$4.39 million for solar and \$4.17 million for wind, consistent with technological progress and ongoing cost reductions. In contrast, fossil technologies show increasing entry costs, particularly for coal at approximately \$16.69 million and to a lesser extent natural gas at about \$2.69 million, suggesting rising regulatory, input, or construction costs. This pattern is consistent with the observed rise in renewable generation and the decline in coal, alongside the more heterogeneous changes in natural gas investment.

Together, these estimates imply that while cost shocks are persistent across all technologies, and strongly so for most, the underlying trends strongly favor renewable investment over fossil generation in the long run. Transmission increases the value of investment in renewable technologies.

Figure 1.2 compares simulated capacity paths against historical data from 2001–2024. The model matches aggregate capacity levels well but overpredicts solar and wind investment relative to historical patterns while generating relatively flat coal and natural gas capacity trajectories. This discrepancy likely reflects factors outside the model including state-level renewable portfolio standards, federal tax credits with complex phase-out schedules, and permitting delays that constrained renewable development during portions of the sample period.

1.6.3 Robustness Checks

I perform a suite of robustness checks detailed in Appendix Section 1.15 and summarized in Table 1.12. I show the sensitivity of my estimates to a relaxation of assumptions. In particular, I show the robustness of my estimates to the allowance of blackouts and market

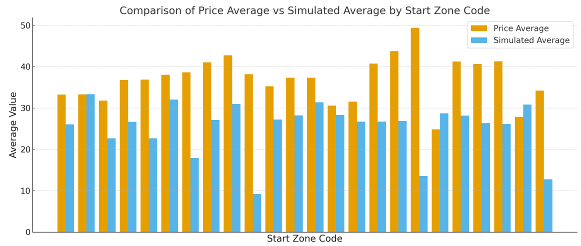


Figure 1.1: Model Fit for the Short-run.

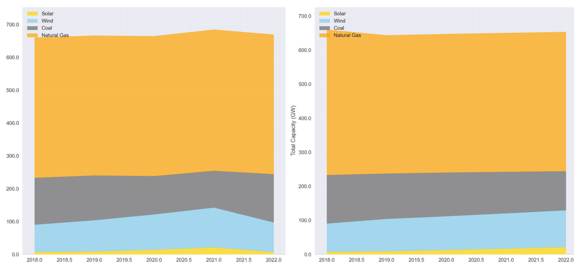


Figure 1.2: Model Fit for the Long-run.

power, varying assumptions about demand and its structure, changes to the specific dataset structure, and changes to other specific model features. The evidence presented is relatively compelling that the results are robust under slight relaxation. Some changes like market power have more impact on the estimated impacts as the level of deviation from the baseline assumption rises.

1.7 Counterfactuals

I use the estimated model to evaluate three sets of policy counterfactuals: upgrades to existing transmission infrastructure, addition of new long-distance transmission lines, and active federal legislation affecting both transmission and generation investment. Counterfactuals on the Grain Belt Express line addition and upgrade are completed and detailed here while further counterfactuals in progress are detailed in Appendix Section 1.13. In all counterfactual exercises, I hold estimated parameters fixed and modify only the primitives of the model—entry costs, line capacities, and per-MWh revenue—during simulation. This ensures that behavioral responses (investment, retirement, and dispatch) emerge endogenously from the estimated model rather than from re-fitting. The full set of counterfactuals run is described in Table 1.9 and Figure 1.17 with additional discussion of counterfactuals in progress in Appendix Section 1.13.

1.7.1 Transmission Infrastructure Upgrades

The first set of counterfactuals examines the effects of upgrading existing transmission lines to High-Voltage Direct Current (HVDC) technology, which reduces line losses and increases transfer capacity. I simulate outcomes under full replacement of all inter-zonal lines as well as partial replacement of randomly selected lines at varying intensities.

Full replacement of all lines with HVDC technology reduces average wholesale prices by 6% across the study region, generating welfare gains of approximately $4 billion per year. Shreve et al. (2025) estimate that congestion costs in the U.S. study region totaled approximately $12.4 billion in 2024; the $4 billion welfare gain reported here therefore represents roughly one-third (32%) of that estimate. Notably, most of these gains can be achieved with partial upgrades: replacing 50% of transmission lines still delivers price reductions exceeding 5%.

Figure 1.3 illustrates how price reductions scale with the percentage of lines upgraded. The relationship exhibits diminishing returns, with the marginal benefit of additional upgrades declining as congestion on the most constrained corridors is relieved.

The effects of transmission upgrades vary substantially by region. The Northeast, where congestion costs are highest and renewable resources are scarce, experiences the largest price declines. The Midwest, home to abundant wind resources that are often curtailed due to transmission constraints, sees modest price increases as expanded export capacity raises local prices toward the national average. This price convergence, while increasing costs for some Midwestern consumers, reflects more efficient allocation of generation resources and reduced curtailment of low-cost renewable energy.

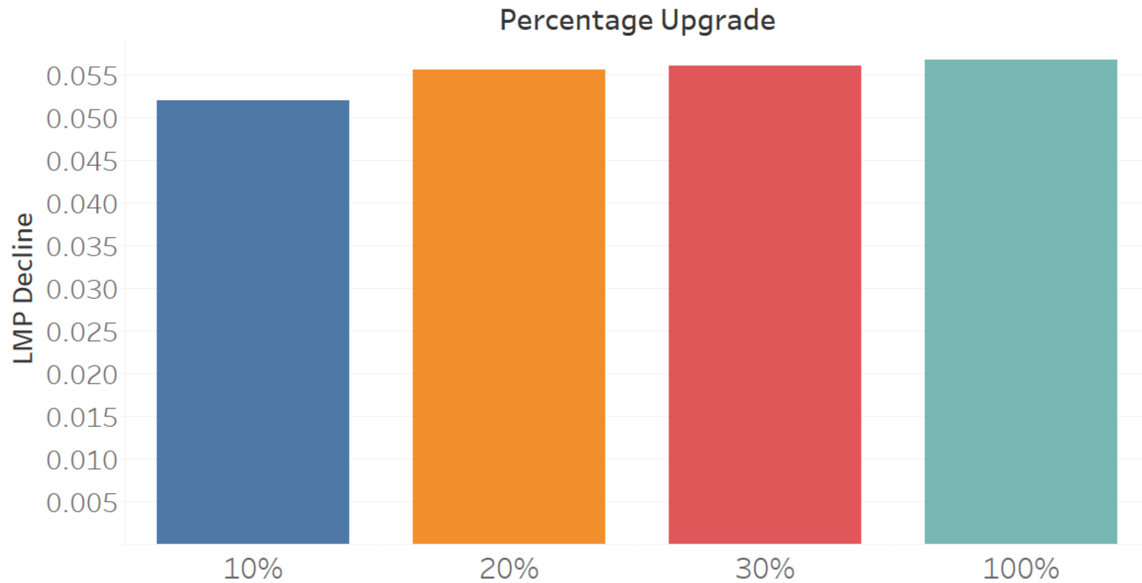


Figure 1.3: Price Decrease by Percentage Grid Upgrade

1.7.2 New Long-Distance Transmission

The second set of counterfactuals examines the addition of specific long-distance transmission projects. I focus on the case of the in-progress Grain Belt Express; further line-addition counterfactuals—including a major HVDC connection between ERCOT and the Eastern Interconnection—are still in development and are discussed as extensions in Appendix Section 1.13.

Grain Belt Express

The Grain Belt Express is a proposed 800-mile HVDC transmission line designed to carry 5,000 MW of wind power from western Kansas to Indiana and points east. As Figure 1.19 shows, adding this line increases solar and wind adoption in the Eastern Interconnection by 11.5% by 2050, with the largest increases concentrated in SPP and western MISO where the line originates.

ERCOT-Eastern Interconnection Connection

A major HVDC connection between ERCOT and the Eastern Interconnection would allow Texas's abundant wind and solar resources to serve load in the Southeast and beyond. The mechanism would mirror the Grain Belt Express counterfactual: a large new low-loss corridor reduces effective congestion between resource-rich and load-rich zones, raising expected revenues from renewable investment and pulling additional capacity into the equilibrium. Quantitative welfare and adoption magnitudes for this counterfactual are not yet available; the simulation specification and the qualitative direction of the expected effects are discussed alongside the other in-progress counterfactuals in Appendix Section 1.13.

1.7.3 Federal Legislation

The final set of counterfactuals evaluates three pieces of federal legislation affecting electricity markets. These counterfactuals and their relationships are described in Table 1.10. I implement these policies as *policy wedges*. *Policy wedges* are time-varying modifications to model primitives that activate at each policy's enactment date while holding all estimated parameters fixed. This approach isolates the causal effect of each policy from the underlying structural relationships estimated from pre-policy data. To ensure comparability across scenarios, I use identical random seeds and shock draws for all simulations, so that differences across scenarios reflect only the policy treatment and not sampling variation. I run four scenarios: (i) both policies as enacted, (ii) IRA only, (iii) BIL only, and (iv) neither policy (the baseline).

1.8 Conclusion

This paper develops and estimates a dynamic model of generator behavior in electricity markets with transmission constraints. I link short-run dispatch decisions by ISOs resulting from competitive equilibrium to long-run capacity investment by generators. The model captures how transmission infrastructure shapes price formation across interconnected markets and how these prices influence firms' incentives to invest in different generation technologies.

Three main findings emerge from the analysis. First, transmission constraints impose substantial costs on electricity consumers: upgrading existing infrastructure to modern HVDC standards would reduce wholesale prices by 6% and generate $4 billion in annual welfare gains. Second, the benefits of transmission investment are regionally heterogeneous, with the largest gains accruing to transmission-constrained regions in the Northeast. Third, in the long run, expanded transmission capacity accelerates the renewable energy transition by allowing generators in renewables-rich regions to access distant markets, increasing solar and wind adoption by 11.5% by 2050 under scenarios including major projects like the Grain Belt Express.

The policy counterfactuals indicate that transmission investment and renewable energy subsidies act as complements rather than substitutes. Policies like the Inflation Reduction Act increase the supply of renewable generation, while transmission investments under the Bipartisan Infrastructure Law and Big Wires Act ensure this generation can reach consumers. Implementing both types of policies together yields benefits exceeding the sum of their individual effects.

Several limitations suggest directions for future research. The model operates at the zonal rather than nodal level, preventing analysis of specific substation-to-substation connections. This limitation makes the model better suited to evaluating large-scale inter-regional transmission than individual line additions. Additionally, the model treats demand as exogenous; long-run shifts in the geographic distribution of load in response to electricity price changes represent an important extension. Finally, the model does not incorporate storage, which may interact with transmission as alternative solutions to renewable intermittency.

Despite acknowledged limitations, the results provide evidence that transmission investment offers a cost-effective complement to direct renewable subsidies in accelerating the energy transition. As policymakers evaluate proposals to expand inter-regional transmission capacity, models that capture both the short-run operational benefits and long-run investment effects can inform the efficient allocation of infrastructure spending.

Tables

ISO	Mean Price	Mean Load	Median Price	Median Load
ISO-NE	50	13,623	45	13,308
NYISO	43	17,686	39	17,297
ERCOT	40	44,328	23	41,901
SPP	37	30,420	30	29,246
MISO	38	73,945	34	72,179
PJM	41	146,112	37	88,236

Table 1.1: Average and Median Price and Load by ISO

Variable	Mean	Standard Deviation	Median
Load	5,075	7,055	2,296
Price	37.8	79.5	27.6
SO_2 Emissions	526	417	470
Solar Generation	2	13	0.03
Wind Generation	44	61	20
Natural Gas Generation	29	66	33
Coal Generation	162	217	64
Exports	18.9	76.3	0.53

Table 1.2: Descriptive Statistics: Short-run Generation

Variable	Mean	Standard Deviation	Median
Fuel Cost	603	2,093	324
Heat Content	3.6	6.0	1.03
Natural Gas Spot Price	3.8	2.3	2.7
Coal Spot Price	166	128	98
Temperature	13	12	15
Loss Percentage	0.05	0.06	0.05
Transmission Constraint	0.51	1.00	0.50
Capacity Constraint	0.06	66	33

Table 1.3: Descriptive Statistics: Short-run MC Inputs

Resource	Intercept	Signif.	Export Coeff.	Signif.
Electricity used for energy storage	30.97	***	0.02	***
Natural Gas	37.72	***	0.01	***
Nuclear (incl. Uranium, Plutonium, Thorium)	46.44	***	0.01	***
Coal	30.97	***	32.47	***
Distillate Fuel Oil (diesel, No. 1–4)	33.23	***	0.01	***
Landfill Gas	33.91	***	0.00	***
Water at hydro or hydrokinetic technologies	36.27	***	-0.02	***
Other Biomass Gas (digester, methane, etc.)	27.64	***	-0.11	***
Petroleum Coke	31.50	***	0.06	***
Wood/Wood Waste Solids	40.91	***	0.06	***
Blast Furnace Gas	30.34	***	0.00	***
Waste heat (no direct fuel source)	37.43	***	0.01	***
Black Liquor	30.97	***	0.01	***
Other Gas	41.99	***	-0.12	***
Other	30.97	***	0.10	***
Purchased Steam	30.97	***	-0.02	***
Agricultural By-Products	29.56	***	-0.06	***
Petroleum Products	34.94	***	0.00	***

Note: Export coefficients represent the marginal relationship between exports and output by energy resource. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The log-likelihood of the regression is 7.662301.

Table 1.4: Short-run Model Estimates by Resource Type

Variable	**Estimate**	**Significance**
Natural Gas Price	6.93	***
Dewpoint	2.46	***
Temperature	0.00	
Heat Content	-2.22	***
Generator Startup	-2.3	***
Generator Near Capacity	-2.3	***

Notes: Specification includes hourly and monthly dummies, zone-code dummies, and severe weather dummies.

Table 1.5: Estimates for Natural Gas Covariates

Variable	**Estimate**	**Significance**
Coal Price	8.68	***
Dewpoint	2.46	***
Temperature	4.00	***
Heat Rate	-1.30	***
Generator Startup	-2.3	***
Generator Near Capacity	-2.3	***

Notes: Specification includes hourly and monthly dummies, zone-code dummies, and severe weather dummies.

Table 1.6: Estimates for Coal Covariates

Table 1.7: Empirical Moments by Fuel Type

Moment	Solar	Wind	Coal	Natural Gas
Investment moments				
Mean investment	0.080	0.000	0.011	0.104
Mean large investment	1.900	0.000	0.367	3.100
Variance of investment	0.128	0.000	0.004	0.296
Variance of large inv.	0.000	0.000	0.000	0.000
Kurtosis of investment	19.705	−3.000	27.565	25.461
Return moments				
Mean Retirement	0.082	0.198	1.086	0.812
Variance of return	0.139	1.090	1.729	18.157
Kurtosis of return	20.446	25.618	15.543	25.377

Table 1.8: GMM Parameter Estimates: Dynamic Investment Model

	Solar	Wind	Coal	Natural Gas
Panel A: Fixed Cost AR(1) Process				
Persistence, ρ_{fc}	0.877	0.917	0.806	0.916
Innovation std. dev., σ_{fc} (\$M)	125.14	200.23	300.34	250.29
Trend, μ_{fc} (\$M)	−0.84	−1.66	6.26	1.89
Panel B: Entry Cost AR(1) Process				
Persistence, ρ_{ec}	0.869	0.390	0.926	0.916
Innovation std. dev., σ_{ec} (\$M)	876.00	1,251.43	1,626.86	2,002.29
Panel C: Initial Cost Levels and Trends				
Initial fixed cost, FC_0 (\$M)	0.56	1.50	150.02	2.82
Initial entry cost, EC_0 (\$M)	60.07	105.12	225.26	150.17
Fixed cost increment, ΔFC (\$M)	−0.84	−1.66	6.26	1.89
Entry cost increment, ΔEC (\$M)	−4.39	−4.17	16.69	2.69
Panel D: Auxiliary Weights				
GMM objective value		3,338.91		

Notes: Parameter estimates from GMM estimation of a dynamic electricity-market investment model in the spirit of Gowrisankaran, Schmidt, and Dengler. Monetary quantities are reported in millions of U.S. dollars; persistence parameters are dimensionless. Volatility parameters (σ_{fc}, σ_{ec}) and cost levels (FC_0, EC_0) are reported after the softplus transformation $\log(1+e^x)$, which enforces positivity during optimization. The trend parameter μ_{fc} and increment ΔFC coincide by construction in this specification: negative values for Solar and Wind are consistent with technological learning, while positive values for Coal and Natural Gas indicate rising fixed costs over the sample.

Table 1.9: Counterfactual Simulation Scenarios

		Modified Primitives		
Scenario	**Description**	Line Capacity A_{ij}	Entry Cost E_k	Revenue P_k^{eff}
Panel A: Transmission Infrastructure Upgrades				
Full HVDC	All inter-zonal lines replaced with HVDC	✓		
Partial HVDC	Random subset at intensity $\alpha \in (0,1)$	✓		
Panel B: New Long-Distance Transmission				
No constraints	All transmission limits removed	✓		
Grain Belt Express	5,000 MW HVDC, KS to IN	✓		
ERCOT–Eastern IC	HVDC link, ERCOT to Southeast	✓		
Panel C: Federal Legislation				
BIG WIRES Act	MITC $\geq$ 30% peak demand by 2035	✓		
IRA $\times$ BIL Factorial Design				
Baseline	Neither IRA nor BIL			
IRA only	IRA enacted, no BIL		✓	✓
BIL only	BIL enacted, no IRA	✓		
As enacted	Both IRA and BIL	✓	✓	✓
Panel D: Cross-Cutting Analysis				
Welfare decomposition	Producer/consumer surplus by zone for each scenario		*Applied to all scenarios*	

Notes: Each scenario holds estimated structural parameters fixed and modifies only the indicated model primitives. A_{ij} denotes inter-zonal transfer capacity, E_k denotes technology-specific entry cost, and P_k^{eff} denotes effective per-MWh revenue inclusive of production tax credits. All policy wedges phase in via a ramp function $\rho(t;t_0,R)$ to avoid discontinuities. Panel C scenarios use identical random seeds and shock draws to isolate policy effects from sampling variation.

Table 1.10: Federal Legislation: Factorial Design and Complementarity Test

	Bipartisan Infrastructure Law	
Inflation Reduction Act	Not enacted	Enacted
Not enacted	W^{baseline}	$W^{\text{BIL only}}$
Enacted	$W^{\text{IRA only}}$	$W^{\text{as enacted}}$

Comparison	**Definition**
IRA marginal effect	$\Delta W^{\text{IRA}} = W^{\text{IRA only}} - W^{\text{baseline}}$
BIL marginal effect	$\Delta W^{\text{BIL}} = W^{\text{BIL only}} - W^{\text{baseline}}$
Joint effect	$\Delta W^{\text{joint}} = W^{\text{as enacted}} - W^{\text{baseline}}$
Complementarity	$\Delta W^{\text{joint}} - \left(\Delta W^{\text{IRA}} + \Delta W^{\text{BIL}}\right) > 0$

Notes: W denotes total welfare (consumer plus producer surplus). The top panel displays the 2×2 factorial structure of the federal legislation counterfactuals. The bottom panel defines the comparisons used to test policy complementarity. Complementarity holds when the joint welfare gain exceeds the sum of individual gains, indicating that transmission investment (BIL) and renewable subsidies (IRA) are mutually reinforcing.

Figures

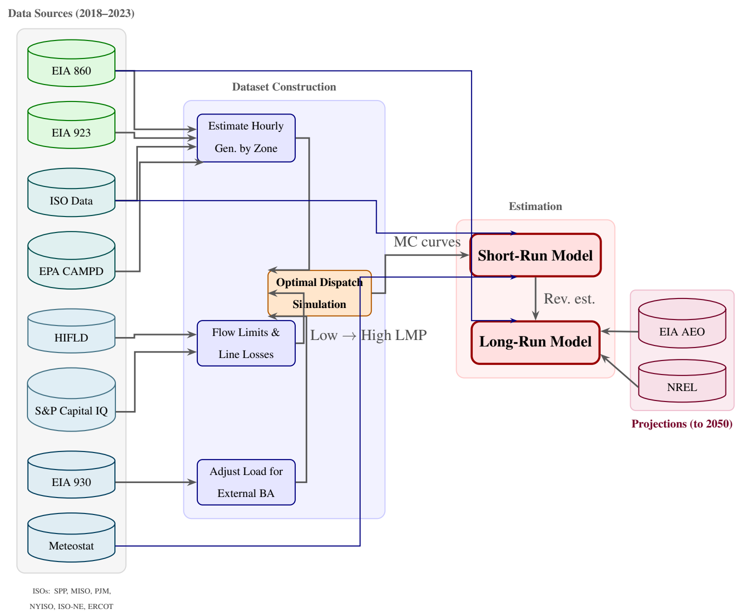


Figure 1.4: Dataset Construction Diagram.

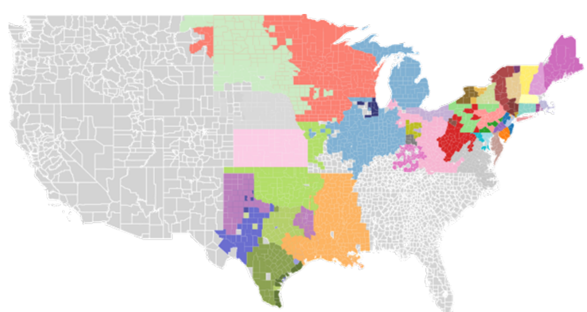


Figure 1.5: Mapped Zones from Market.

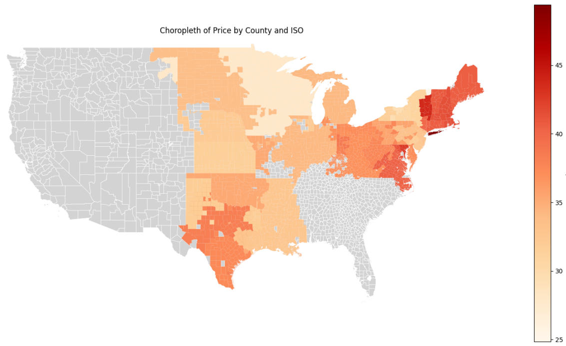


Figure 1.6: Locational Marginal Price by Zone in Data.

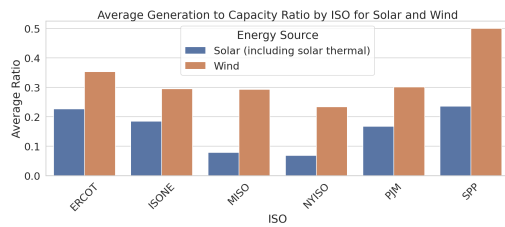


Figure 1.7: Estimated Capacity Factor for Solar and Wind by ISO.

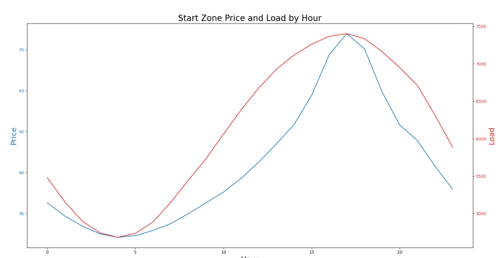


Figure 1.8: Price and Load by Hour for entire dataset.

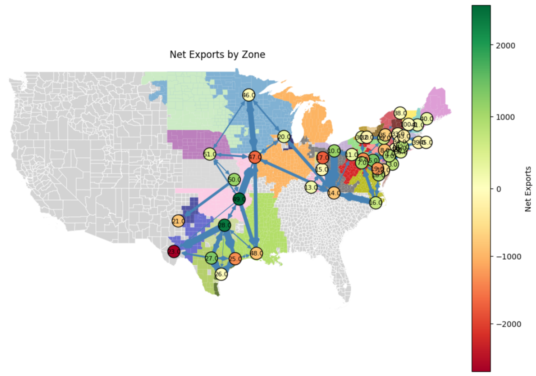


Figure 1.9: Map of Exports between Zones in the Model.

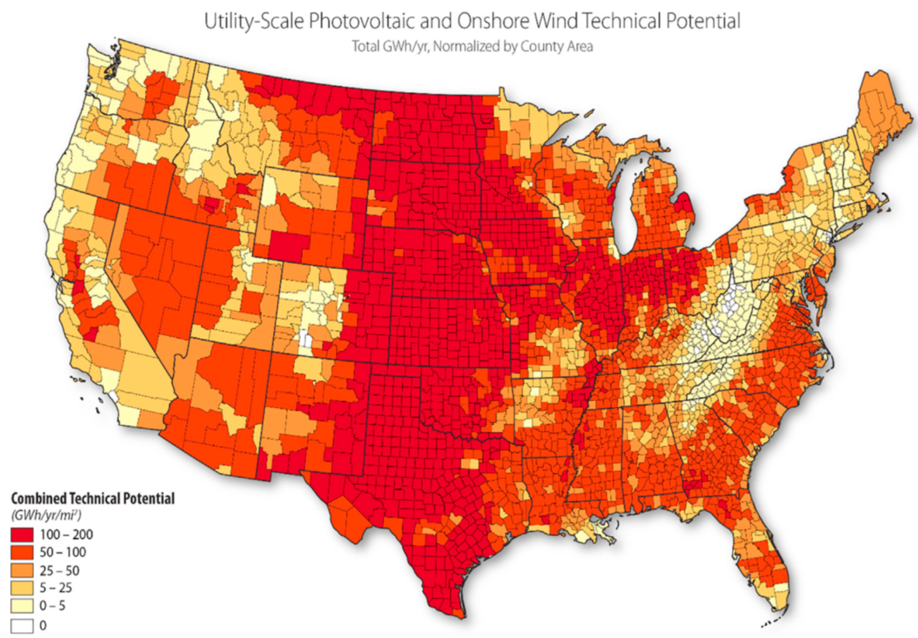


Figure 1.10: Combined Technical Potential for Wind and Solar Generation by County (GWh/yr/mi^2). Darker shading indicates higher combined renewable potential. Source: Department of Energy.

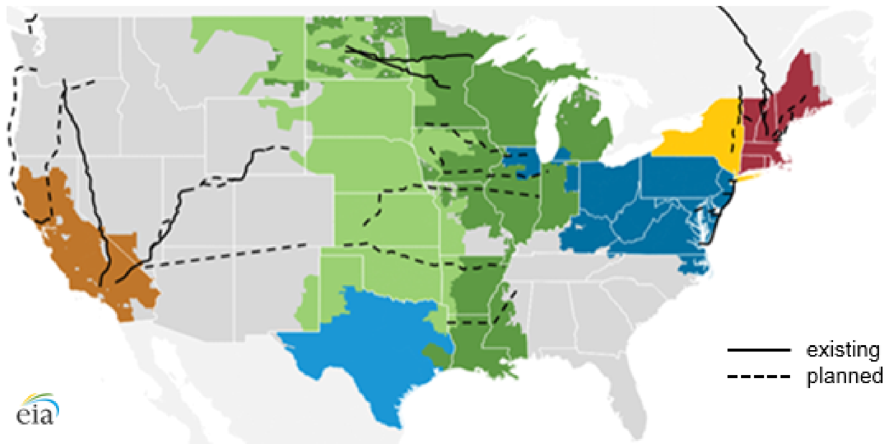


Figure 1.11: Existing and Planned High-Voltage Direct Current Transmission Lines. Solid lines indicate existing HVDC infrastructure; dashed lines indicate proposed expansions. Source: EIA.

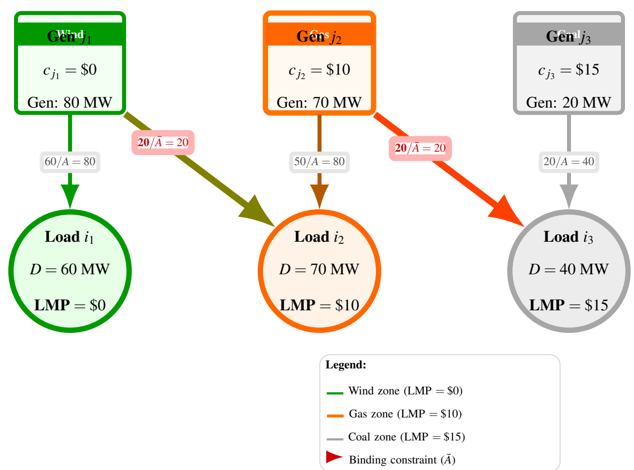


Figure 1.12: Network Topology Illustrating Congestion Pricing. Binding export constraints create price separation across zones. Each zone's LMP equals the local marginal cost when transmission constraints bind.

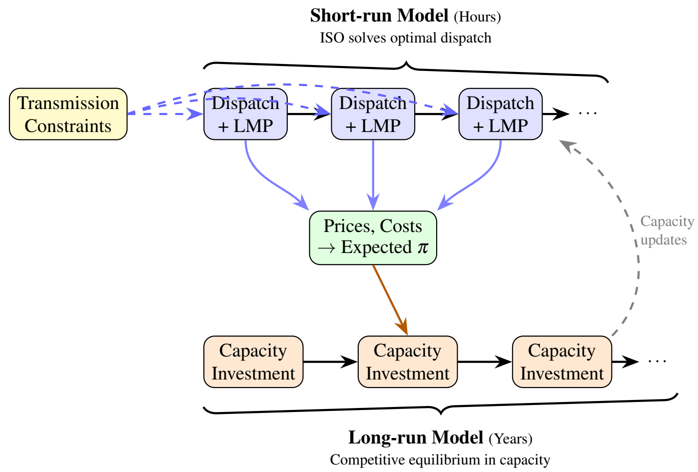


Figure 1.13: **Short-run**: A combined ISO minimizes dispatch costs subject to transmission constraints (engineering problem). **Long-run**: Generators choose capacity to maximize profits given expected prices (equilibrium).

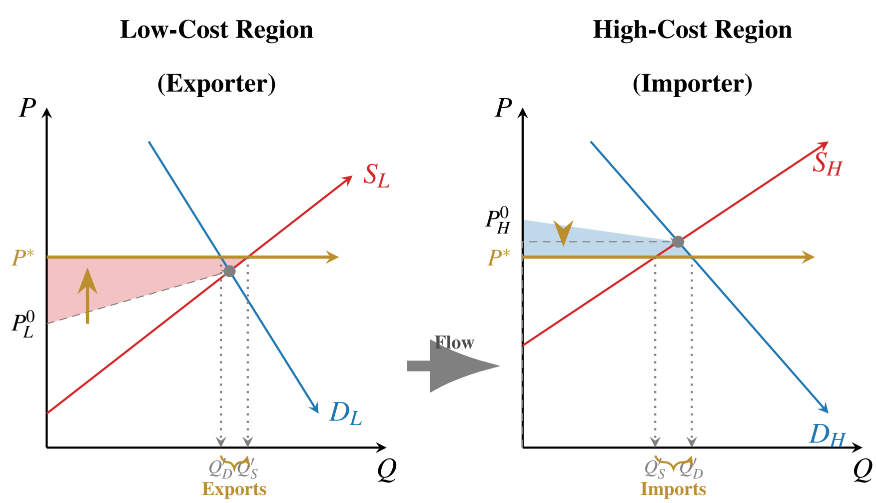


Figure 1.14: Static Effects of Transmission Expansion: Price Convergence and Spatial Arbitrage. In the low-cost (exporting) region, prices rise and producer surplus increases (red shaded area). In the high-cost (importing) region, prices fall and consumer surplus increases (blue shaded area).

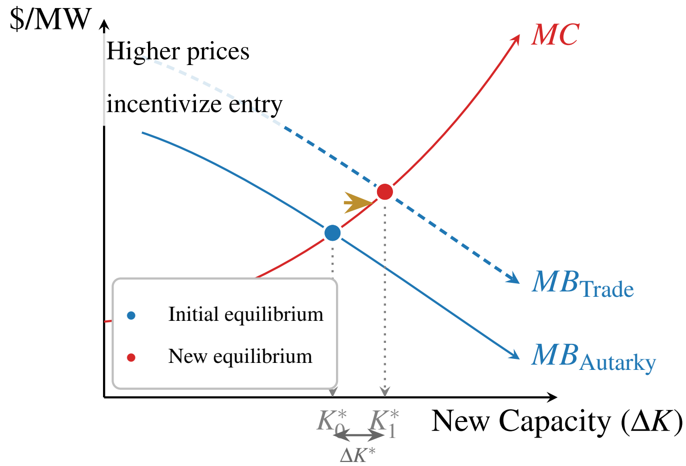


Figure 1.15: Dynamic Effects of Transmission Expansion: The Investment Signal. Expanded market access shifts the marginal benefit curve outward for generators in resource-rich regions, leading to increased equilibrium capacity investment.

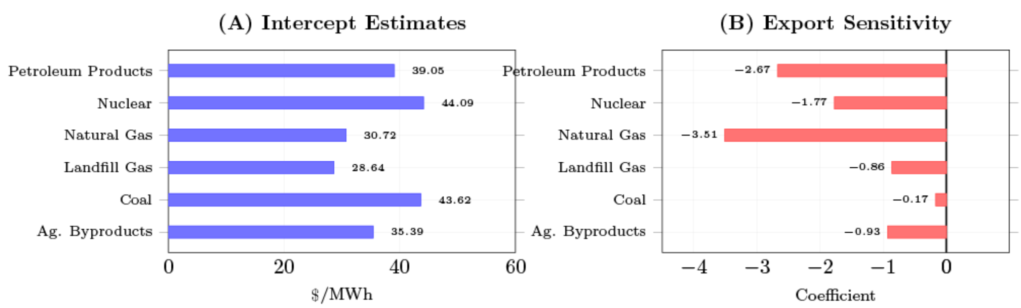


Figure 1.16: Intercept and Export Sensitivity Estimates by Resource.

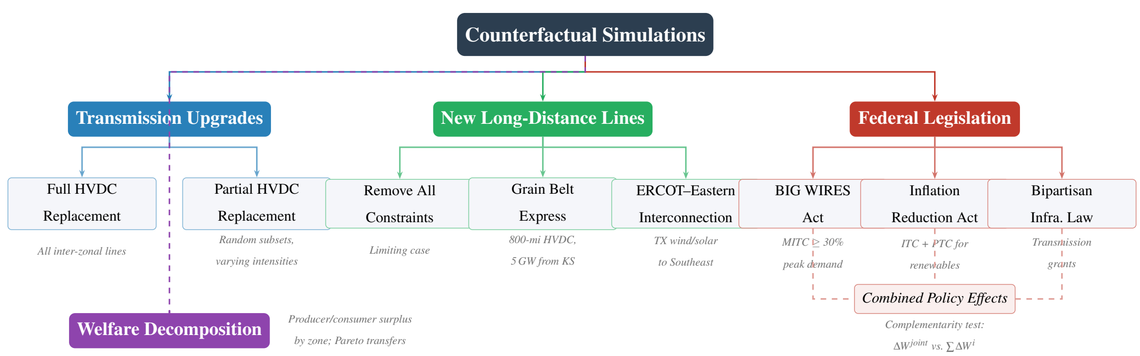
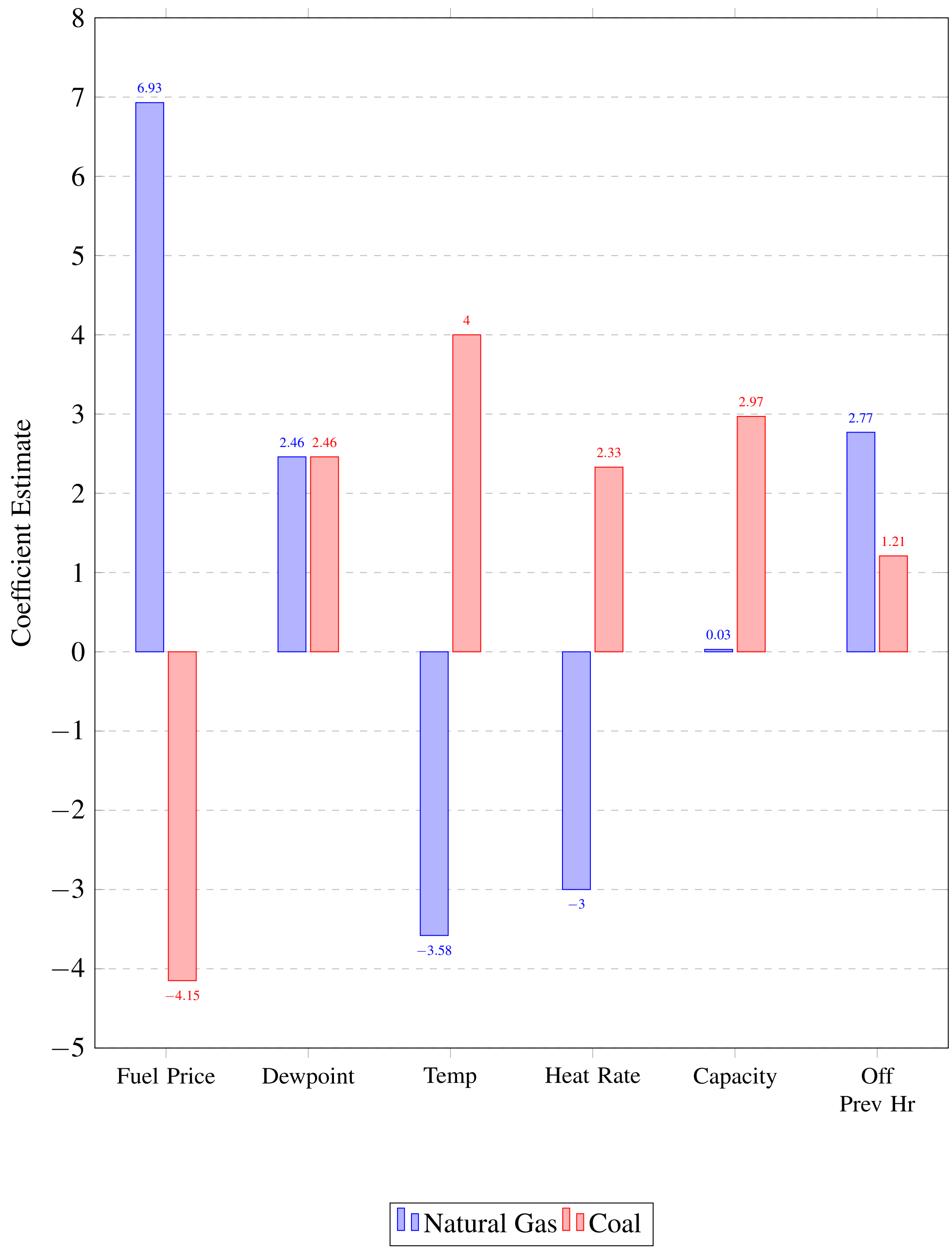


Figure 1.17: Visual description of counterfactuals run and how they relate.



8
7
6
5
4
3
2
1
0
−1
−2
−3
−4
−5
Coefficient Estimate
6.93
−4.15
2.46
2.46
−3.58
4
−3
2.33
0.03
2.97
2.77
1.21
Fuel Price
Dewpoint
Temp
Heat Rate
Capacity
Off
Prev Hr
Natural Gas
Coal

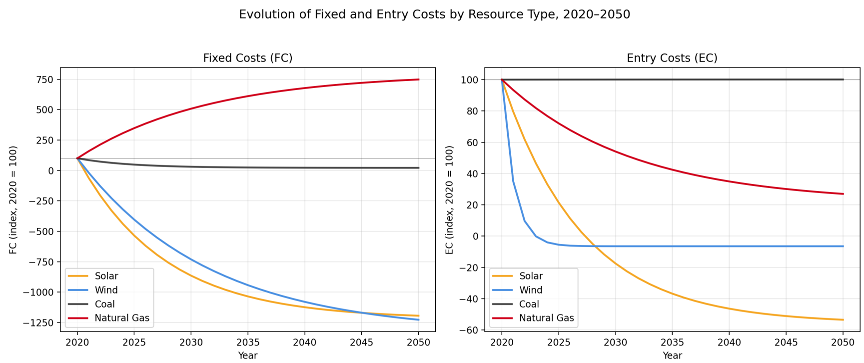


Figure 1.18: Evolution of fixed and entry cost over time until 2050 based on Model predictions.

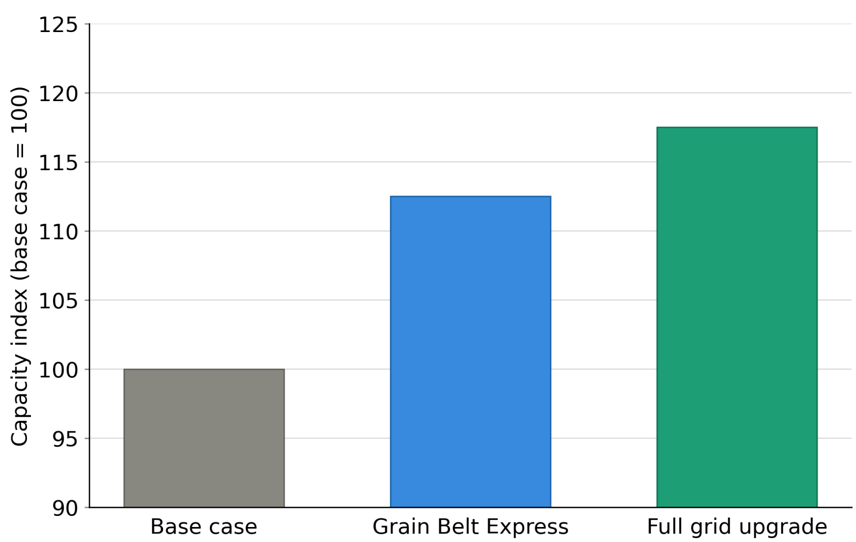


Figure 1.19: Counterfactual growth of renewable capacity under different transmission scenarios.

1.9 Solving the Operations Model

This appendix derives the likelihood function used to estimate the short-run operations model. Section 1.9.1 establishes the equivalence between the generator profit maximization problem and a social planner's cost minimization problem, then formulates the ISO's dispatch problem. Section 1.9.2 derives the first-order conditions from the Lagrangian. Section 1.9.3 specifies the cost function and derives the likelihood function for estimation.

1.9.1 The ISO Dispatch Problem

Under standard conditions, the generator's profit maximization problem is equivalent to a social planner's cost minimization problem. Following Gilbert et al. (2004) and Cramton (2017), the conditions for this equivalence include: (i) no market power among generators, (ii) no gamesmanship or non-competitive bidding, (iii) increasing marginal cost curves, and (iv) linear constraints. The assumption of increasing marginal costs is consistent with both bid offer data and economic theory. Gamesmanship and non-competitive bidding are prohibited and actively monitored by the Federal Energy Regulatory Commission. While transmission constraints can create local market power (Gilbert et al. (2004)), competitive pressures and regulatory limits on bidding behavior make sustained market power difficult to maintain.

I aggregate to the resource and ISO level due to data limitations: hourly generation data are available only at the resource level rather than the generator level. While this aggregation sacrifices granularity that would be valuable in a transmission model, I disaggregate where data permit.

Notation

Let $i \in \{1,\ldots,I\}$ index demand nodes, $j \in \{1,\ldots,J\}$ index supply (generation) nodes, $k \in \{1,\ldots,K\}$ index resources at each supply node, $g \in \{1,\ldots,G\}$ index generators, and

$t \in \{1, \ldots, T\}$ index time periods (hours). Define:

- q_{ijkgt}: Quantity of electricity sent from resource k at supply node j to demand node i by generator g at time t
- $c_{jkgt}(\cdot)$: Cost function for resource type k and generator g at supply node j at time t
- x_{jkgt}: Observable cost shifters for resource jk and generator g at time t
- ε_{ijkgt}: Unobserved cost shock
- A_{ij}: Transmission capacity between nodes i and j
- O^{MAX}_{jkgt}: Maximum generation capacity of resource k for generator g at node j at time t
- L_{it}: Load (demand) at node i at time t
- B_{ij}: Transmission loss factor between nodes i and j

The Optimization Problem

The ISO solves a static cost minimization problem each hour, choosing dispatch quantities q_{ijkgt} to minimize the total cost of meeting demand:

$$\pi_t = \max_{\{q_{ijkgt}\}_{i,j,k,g,t \in I,J,K,G,T}} \left[-\sum_{I=1}^{I} \sum_{j=1}^{J} \sum_{k=1}^{K} \sum_{g=1}^{G} c_{jkgt}(x_{jkgt}, q_{ijkgt}, \varepsilon_{ijkgt}) \right]. \tag{1.11}$$

subject to three constraints.

Resource Constraint (Kirchhoff's Law). Supply must equal demand at each node, accounting for transmission losses:

$$\sum_{j=1}^{J} \sum_{k=1}^{K} \sum_{g=1}^{G} (1 - B_{ij}) q_{ijkgt} = L_{it} \quad \forall i. \tag{1.12}$$

Transmission Constraint. Flows between any two nodes cannot exceed line capacity:

$$\sum_{k=1}^{K} \sum_{g=1}^{G} q_{igjkt} \leq A_{ij} \quad \forall i, j, t. \tag{1.13}$$

Capacity Constraint. Total dispatch from each generator cannot exceed its maximum capacity:

$$0 \leq \sum_{i} q_{igjkt} \leq O_{jkgt}^{MAX} \quad \forall j, k, g, t. \tag{1.14}$$

Three maintained assumptions ensure the problem is well-defined: (i) no blackouts, meaning sufficient generation capacity exists to serve all load at every node; (ii) no resource is transmission-constrained to all destinations; and (iii) at least one resource at each generation node has slack capacity at any given time.

When no constraints bind, resources set marginal cost equal to price and serve the highest-priced locations first. When the transmission constraint binds, a resource dispatches up to $A_{ij} - \sum_{k' \neq k} q_{ijk't}$ units to the highest-priced location, then up to A_{ij} units to the next highest-priced location, continuing until either maximum capacity O_{jkt}^{MAX} is reached or marginal revenue equals marginal cost. Prices equal the marginal cost of the last economically dispatched resource at each location.

1.9.2 First-Order Conditions

The Lagrangian for the ISO's problem at time t is:

$$
\begin{aligned}
\mathscr{L}_t = &- \sum_{k=1}^{K} \sum_{j=1}^{J} c_{jkt} \left(x_{jkt}, \sum_{i=1}^{I} q_{ijkt}, \sum_{i=1}^{I} \varepsilon_{ijkt} \right) \\
&+ \sum_{i,j} \lambda_{ijt} \left(A_{ij} - \sum_{k=1}^{K} q_{ijkt} \right) \\
&+ \sum_{i} \gamma_{it} \left(L_{it} - \sum_{k=1}^{K} \sum_{j=1}^{J} (1 - B_{ij}) q_{ijkt} \right) \\
&+ \sum_{j,k} \theta_{jkt} \left(O_{jkt}^{MAX} - \sum_{i=1}^{I} q_{ijkt} \right),
\end{aligned} \tag{1.15}
$$

where λ_{ijt} is the multiplier on the transmission constraint, γ_{it} is the multiplier on the load balance constraint, and θ_{jkt} is the multiplier on the capacity constraint.

Optimality Conditions

The first-order condition with respect to q_{ijkt} is:

$$
\frac{\partial \mathscr{L}_t}{\partial q_{ijkt}} = -\frac{\partial c_{jkt}}{\partial q_{ijkt}} - \lambda_{ijt} - \gamma_{it}(1 - B_{ij}) - \theta_{jkt} = 0. \tag{1.16}
$$

Complementary slackness requires:

$$
\lambda_{ijt} \left(A_{ij} - \sum_{k=1}^{K} q_{ijkt} \right) = 0 \quad \forall i, j, t, \tag{1.17}
$$

$$
\theta_{jkt} \left(O_{jkt}^{MAX} - \sum_{i=1}^{I} q_{ijkt} \right) = 0 \quad \forall j, k, t. \tag{1.18}
$$

Rearranging equation (1.16) yields three equivalent expressions:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \gamma_{it}(1 - B_{ij}) + \theta_{jkt} = \lambda_{ijt}, \tag{1.19}$$

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \gamma_{it}(1 - B_{ij}) + \lambda_{ijt} = \theta_{jkt}, \tag{1.20}$$

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \lambda_{ijt} + \theta_{jkt} = \gamma_{it}(1 - B_{ij}). \tag{1.21}$$

Intratemporal Euler Conditions

Three key relationships emerge from comparing first-order conditions across different resource-node combinations.

Condition 1: Same supply node, different resources. For two resources k and k' at the same supply node j, both sending to demand node i:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \theta_{jkt} = \frac{\partial c_{jk't}}{\partial q_{ijk't}} + \theta_{jk't}. \tag{1.22}$$

Condition 2: Same resource, different demand nodes. For resource k at supply node j sending to two demand nodes i and i':

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \gamma_{it}(1 - B_{ij}) + \lambda_{ijt} = \frac{\partial c_{jkt}}{\partial q_{i'jkt}} + \gamma_{i't}(1 - B_{i'j}) + \lambda_{i'jt}. \tag{1.23}$$

Condition 3: Different supply nodes, same demand node. For resources at different supply nodes jk and $j'k'$ both sending to demand node i:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \lambda_{ijt} + \theta_{jkt} = \frac{(1 - B_{ij})}{(1 - B_{ij'})} \left[\frac{\partial c_{j'k't}}{\partial q_{ij'k't}} + \lambda_{ij't} + \theta_{j'k't} \right]. \tag{1.24}$$

1.9.3 Cost Function Specification and Likelihood

Cost Function

For renewable resources (wind and solar), the cost function is linear in output:

$$c_{jkt}^{renew} = q_{ijkt}\varepsilon_{ijkt}. \tag{1.25}$$

For non-renewable resources, the cost function is quadratic:

$$c_{jkt} = \sum_{i=1}^{I} q_{ijkt}x_{jkt}\beta + \frac{1}{2}\left(\sum_{i=1}^{I} q_{ijkt}\right)^2 \nu_k + \sum_{i=1}^{I} q_{ijkt}\varepsilon_{ijkt}, \tag{1.26}$$

where β and ν_k are parameters to be estimated, with ν_k varying by resource type to capture differences in fuel costs. The marginal cost for non-renewable resources is:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} = x_{jkt}\beta + q_{ijkt}\nu_k + \varepsilon_{ijkt}. \tag{1.27}$$

Identifying the Central Resource

Define a *central resource* at each demand node as a resource for which neither the transmission constraint nor the capacity constraint binds: $\lambda_{ijt} = 0$ and $\theta_{jkt} = 0$. By the no-blackout assumption, every demand node has at least one central resource. For central resources, the observed locational marginal price (LMP) equals marginal cost:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} = LMP_{it}. \tag{1.28}$$

Using equation (1.24) with a central resource $j''k''$ at node i:

$$\frac{\partial c_{jkt}}{\partial q_{ijkt}} + \lambda_{ijt} + \theta_{jkt} = \frac{(1-B_{ij})}{(1-B_{ij''})}LMP_{it}. \tag{1.29}$$

Rearranging and substituting the marginal cost specification:

$$x_{jkt}\beta + q_{ijkt}\nu_k + \lambda_{ijt} + \theta_{jkt} - \frac{(1-B_{ij})}{(1-B_{ij''})}LMP_{it} = -\varepsilon_{ijkt}. \tag{1.30}$$

Solving for Shadow Prices

From equation (1.21) applied to the central resource at node i:

$$\gamma_{it} = \frac{LMP_{it}}{(1-B_{ij''})}. \tag{1.31}$$

Transmission Shadow Price. Using equation (1.23) and the assumption that each resource can send electricity to at least one unconstrained destination i' (so $\lambda_{i'jt} = 0$):

$$\lambda_{ijt} = q_{i'jkt}\nu_k - q_{ijkt}\nu_k + \frac{LMP_{i't}}{(1-B_{i'j''})}(1-B_{i'j}) - \frac{LMP_{it}}{(1-B_{ij'''})}(1-B_{ij}) + \varepsilon_{i'jkt} - \varepsilon_{ijkt}, \tag{1.32}$$

where j'' and j''' denote the central resources at nodes i' and i, respectively.

Capacity Shadow Price. Using equation (1.22) and the assumption that each generation node has at least one resource k' with slack capacity (so $\theta_{jk't} = 0$):

$$\theta_{jkt} = (x_{jk't}\beta + q_{ijk't}\nu_{k'}) - (x_{jkt}\beta + q_{ijkt}\nu_k) + \varepsilon_{ijk't} - \varepsilon_{ijkt}. \tag{1.33}$$

Constructing the Likelihood Function

Substituting the expressions for λ_{ijt} and θ_{jkt} into equation (1.30) yields the estimating equation. Define the composite error:

$$\tilde{\varepsilon}_{ijkt} = \varepsilon_{ijkt} - \varepsilon_{i'jkt} - \varepsilon_{ijk't}, \tag{1.34}$$

where i' is the unconstrained destination for resource jk and k' is the unconstrained resource at node j.

The deterministic component of the estimating equation is:

$$\begin{aligned}\tilde{V}_{ijkt} = & \underbrace{x_{jkt}\beta + q_{ijkt}\nu_k}_{\text{MC of resource } jk} - \underbrace{\frac{(1-B_{ij})}{(1-B_{ij''})}LMP_{it}}_{\text{MC of central resource}} \\ & \underbrace{+ q_{i'jkt}\nu_k - q_{ijkt}\nu_k + \frac{LMP_{i't}}{(1-B_{i'j''})}(1-B_{i'j}) - \frac{LMP_{it}}{(1-B_{ij'''})}(1-B_{ij})}_{\text{cost of transmission constraint}} \\ & + \underbrace{(x_{jk't}\beta + q_{ijk't}\nu_{k'}) - (x_{jkt}\beta + q_{ijkt}\nu_k)}_{\text{cost of capacity constraint}}. \end{aligned} \tag{1.35}$$

For resources with transmission constraints but no capacity constraints:

$$\tilde{\varepsilon}_{ijk't} = \varepsilon_{ijk't} - \varepsilon_{i'jk't}. \tag{1.36}$$

For resources with neither constraint binding:

$$\tilde{\varepsilon}_{i'jk't} = \varepsilon_{i'jk't}. \tag{1.37}$$

By substitution, each individual error ε_{ijkt} can be recovered.

Maximum Likelihood Estimation

Let $P_t(\cdot)$ denote a multivariate normal density with mean zero and variance-covariance matrix Ω. The likelihood function is:

$$\mathscr{L} = \prod_{t=1}^{T} P_t\left(\tilde{V}_{111t}, \ldots, \tilde{V}_{IJKt}\right). \tag{1.38}$$

Maximizing this likelihood function yields estimates of β (the effect of observable cost shifters) and ν_k (resource-type-specific marginal cost parameters), thereby recovering the short-run cost structure.

1.10 Long-run Model Computational Details

1.10.1 Algorithm Overview

This appendix details the computational methodology for estimating the long-run capacity investment model. The estimation combines a nested fixed point algorithm for computing equilibrium with generalized method of moments (GMM) estimation for parameter recovery. The approach follows the dynamic game estimation methodology developed by Pakes and Ericson (1998) and utilizes algorithmic improvements from Gowrisankaran and Schmidt-Dengler (2024) to identify feasible choices and accelerate convergence.

The computational procedure consists of three main components: (i) state space discretization, (ii) equilibrium computation via nested fixed point iteration, and (iii) GMM estimation using simulated moments. Each component is described in detail below.

1.10.2 State Space Discretization

The dynamic investment problem is solved over a discretized approximation to the continuous state space. Discretization renders the Bellman equation computationally tractable while preserving economically relevant variation in prices, quantities, and profits. The grid is constructed to capture empirically observed ranges of capacity, demand, and prices, with bounds chosen to exceed historical maxima to accommodate counterfactual investment paths.

State Variable Bins

The state space is discretized as shown in Table 1.11. Each state variable is discretized using evenly spaced bins over a bounded interval informed by historical data and engineering constraints.

Table 1.11: State Space Discretization

State Variable	Bins	Range
Fixed cost (F_k)	20	$[0, \bar{R}_k]$
Entry cost (E_k)	20	$[0, 10 \times \bar{R}_k]$
Demand state	5	$[0.75 \times \bar{L}, 1.5 \times \bar{L}]$
Own capacity (O_{jkg})	15	$[0, 1.5 \times \bar{G}_{jg}]$
Rival capacity (O^{-}_{jkg})	10	$[0, 1.5 \times \bar{L}]$
Average price (P_{jkg})	10	$[0, 1.1 \times \bar{P}]$

Notes: $\bar{R}_k$ denotes maximum observed revenue per MW for resource type k; $\bar{L}$ denotes the historical maximum system load; $\bar{G}_{jg}$ denotes the maximum observed generation by owner g in region j; and $\bar{P}$ denotes the historical maximum average price.

Demand uncertainty is modeled using a multiplicative demand scaling factor applied to historical load profiles. The five demand states span $\pm 25\%$ around baseline demand and are intended to capture persistent variation in system conditions rather than transitory hourly shocks.

Own capacity and rival capacity are discretized separately. Own capacity reflects firm-specific investment decisions, while rival capacity aggregates the competitive fringe and other strategic firms into a single state variable, consistent with an oblivious equilibrium approximation. Capacity bounds exceed historical maxima to allow for endogenous entry and expansion in counterfactual scenarios.

Short-Run Outcome Mapping

For each point in the discretized state space, the model computes short-run equilibrium outcomes by solving a linear or quadratic dispatch problem subject to generator capacity constraints, zonal load balance, and transmission limits (see Section 1.12). The solution yields locational marginal prices, dispatched quantities, production costs, and operating profits.

Outcomes are computed as weighted averages across representative operating conditions (e.g., seasonal or load-duration blocks), producing a smooth mapping from state variables to expected period profits. These expected profits are stored and treated as primitives in the dynamic investment problem.

Investment Choice Set

The firm's investment choice is restricted to a discrete set of ten capacity adjustment options. These include incremental additions for expanding technologies and retirement options for declining resources. Capacity is constrained to remain weakly non-negative, ruling out infeasible states. This restriction substantially reduces the dimensionality of the choice set while preserving the economically relevant margins of adjustment.

Approximation Accuracy and Computational Feasibility

The number of bins for each state variable reflects a tradeoff between approximation accuracy and computational feasibility. Finer grids are used for own capacity and cost states, which directly affect firm profits, while coarser grids are used for rival capacity and prices. Sensitivity checks using alternative grid densities yield similar qualitative investment patterns, indicating that results are not driven by discretization artifacts.

The full state space is evaluated using batched computation with intermediate checkpointing, enabling parallel execution and recovery from job interruption. This approach allows the model to evaluate tens of thousands of state points per resource type while maintaining feasible memory and runtime requirements. A schematic overview of the dataset construction is provided in Figure 1.4.

1.10.3 Equilibrium Computation

This section describes the numerical procedure used to compute equilibrium investment behavior in the dynamic oligopoly model. The equilibrium is solved using a nested fixed point algorithm, combining backward induction over time with an outer loop over firms' beliefs about rival behavior.

Nested Fixed Point Procedure

The equilibrium is computed through a nested fixed point procedure. The outer loop iterates on beliefs over rival investment behavior until beliefs are consistent with optimal policies, while the inner loop solves each firm's dynamic programming problem conditional on these beliefs.

This approach follows the standard structure of dynamic oligopoly models with forward-looking firms and strategic interactions, and is computationally necessary given the high-dimensional state space.

Algorithm (Outer Loop):

5. *Initialize beliefs:* Each generation company initializes beliefs over the probability distribution of rival investment actions. Beliefs are defined over aggregate rival capacity changes rather than individual firm actions, consistent with the equilibrium concept described below.

6. *Value function iteration:* Conditional on these beliefs, each firm solves its dynamic optimization problem via backward induction. This inner loop yields optimal investment policies and associated value functions for all states.

7. *Update beliefs:* Beliefs are updated using the implied distribution of investment actions generated by firms' optimal policies.

8. *Check convergence:* The procedure iterates until the maximum deviation between successive policy functions satisfies

$$\max|\sigma^{(n+1)} - \sigma^{(n)}| < \varepsilon.$$

If convergence is not achieved, the algorithm returns to Step 2.

Convergence is assessed jointly over value functions and policy functions to ensure stability of both expected payoffs and equilibrium behavior.

Equilibrium Concept

Rather than solving for a full Markov Perfect Equilibrium (MPE), I employ an oblivious equilibrium concept. Under this approach, firms condition their decisions on their own state variables and a low-dimensional summary of competitors' states, namely the distribution of aggregate rival competitive capacity.

Firms do not track the full joint distribution of rival capacities across individual competitors. Instead, beliefs must be correct with respect to the evolution of total rival capacity. This substantially reduces the dimensionality of the state space while preserving the key strategic channel through which rivals' investment decisions affect prices and profits.

This approximation is particularly well suited to electricity markets, where individual generators are small relative to the market but investment decisions influence prices through aggregate capacity.

Value Function Solution

In the final period T, firms solve a stationary problem with no further investment opportunities. The value function is given by:

$$\upsilon_{jgT}(\Omega) = \max_{\Delta O_{jkg}} \left\{ \Pi_{jg}(\Omega) + \sum_{k} \varepsilon_{jkg}(\Delta O_{jkg}) + \frac{1}{1-\beta}\Pi'_{jg}(\Omega') \right\} \tag{1.39}$$

subject to the law of motion for capacity:

$$O'_{jkg} = O_{jkg} + \Delta O_{jkg} \quad \forall k \tag{1.40}$$

The state vector Ω includes firm-specific capacity holdings, demand conditions, and cost states. The term ε_{jkg} represents an idiosyncratic choice shock that smooths the policy function and ensures well-defined choice probabilities in the numerical implementation. The final period is treated as stationary with no further capacity investments, so firms receive the discounted perpetuity of steady-state profits.

The final period is treated as stationary: after period T, firms receive the discounted perpetuity of steady-state profits conditional on post-investment capacity levels.

Period profits are defined as:

$$\Pi_{jg} = \sum_k \Big[D_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) P_{jkg} - C_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) D_{jkg}(d_{jkg}, O_{jkg}, O^-_{jkg}) \\ - F_k O_{jkg} - E_k \max(\Delta O_{jkg}, 0) \Big] \tag{1.41}$$

where D_{jkg} is dispatch, P_{jkg} is price, C_{jkg} is variable cost, F_k is per-period fixed cost, and E_k is entry cost for new capacity.

Prices, dispatch, and variable costs are outcomes of a short-run operations model that clears the electricity market given total installed capacity and demand conditions. Fixed costs are incurred on all installed capacity, while entry costs apply only to new investment.

First-Order Conditions

Ignoring the non-differentiable choice shock term, the first-order condition with respect to ΔO_{jkg} is:

$$\frac{\partial \upsilon_{jgT}}{\partial \Delta O_{jkg}} = \frac{\partial \Pi_{jkg}}{\partial \Delta O_{jkg}} + \frac{1}{1-\beta} \frac{\partial \Pi'_{jkg}}{\partial \Delta O_{jkg}} = 0. \tag{1.42}$$

For capacity reductions, $\max(\Delta O_{jkg}, 0) = 0$, yielding:

$$\frac{\partial \left[D'_{jkg} P'_{jkg} - C'_{jkg} D'_{jkg} \right]}{\partial \Delta O_{jkg}} = F_k. \tag{1.43}$$

For capacity expansions, $\max(\Delta O_{jkg}, 0) = \Delta O_{jkg}$:

$$E_k + \frac{\beta}{1-\beta} F_k = \frac{\beta}{1-\beta} \frac{\partial \left[D'_{jkg} P'_{jkg} - C'_{jkg} D'_{jkg} \right]}{\partial \Delta O_{jkg}}. \tag{1.44}$$

The derivatives of dispatch, prices, and costs with respect to capacity are well defined because the short-run operations problem is a linear program. Sensitivity of equilibrium outcomes to capacity constraints can therefore be computed using standard LP dual variables.

Handling Multiple Equilibria

The dynamic investment game may admit multiple equilibria. I impose a sequential move structure within each period, where firms move in descending order of installed capacity. The largest incumbent firm chooses its investment first, followed sequentially by smaller firms.

This equilibrium selection rule both reduces multiplicity and reflects empirical patterns in electricity markets, where large incumbents typically lead capacity expansion decisions.

Backward Induction

For periods $t < T$, the model is solved via backward induction. Relative to the terminal period, two features differ:

9. Firms account for expectations over future profit streams conditional on stochastic evolution of demand and cost states.

10. Capacity adjustment costs enter through the parameter ψ_{1k}, affecting average fixed costs and smoothing intertemporal investment behavior.

Expected continuation values are computed by integrating over discretized transition kernels for exogenous state variables and deterministic capacity evolution.

1.10.4 GMM Estimation

Parameter Space

The long-run model contains 32 parameters to estimate. For each of the four resource types $k \in \{\text{natural gas, coal, solar, wind}\}$, the parameters include:

- **Fixed cost process:** initial value (F_k^0), autoregressive coefficient (ρ_k^F), innovation standard deviation (σ_k^F), and integrated component coefficient
- **Entry cost process:** initial value (E_k^0), autoregressive coefficient (ρ_k^E), innovation standard deviation (σ_k^E), and integrated component coefficient

Moment Conditions

The estimation uses 36 moment conditions to identify the 32 parameters, yielding an overidentified model. Following Gowrisankaran et al. (2024), the moment conditions are constructed to capture key features of the investment distribution.

For natural gas, solar, and wind resources:

- $\mathbb{1}(\Delta O_{jkg} > 0) \times \Delta O_{jkg}$ — positive investment indicator interacted with quantity
- $(\Delta O_{jkg})^2$ — total quantity squared
- $\mathbb{1}(\Delta O_{jkg} > 500)$ — indicator for large investments (over 500 MW)
- Interactions of the above with total capacity $\sum_j O_{jkg}$
- $\text{Var}(\Delta O_{jkg})$ — investment variance
- $|\min(\Delta O_{jkg}, 0)|$ — quantity of capacity retired

For coal resources:

The moment conditions focus on retirement behavior rather than new investment, reflecting the empirical pattern of declining coal capacity. One investment-related moment is included to capture any residual new construction.

Estimation Procedure

The GMM estimator minimizes the weighted distance between simulated moments from the model and their empirical counterparts:

$$\hat{\theta} = \arg\min_{\theta} \left[m(\theta) - m^{\text{data}}\right]' W \left[m(\theta) - m^{\text{data}}\right] \tag{1.45}$$

where $m(\theta)$ is the vector of simulated moments, m^{data} is the vector of data moments, and W is the weighting matrix.

The estimation proceeds in two stages:

11. *First stage:* Estimate parameters using an identity weighting matrix $W = I$ to obtain consistent (but inefficient) estimates $\hat{\theta}^{(1)}$.

12. *Second stage:* Re-estimate using the optimal weighting matrix $W^* = \hat{\Sigma}^{-1}$, where $\hat{\Sigma}$ is the variance-covariance matrix of the moment conditions computed from first-stage residuals.

Standard errors are computed using the standard GMM sandwich formula:

$$\text{Var}(\hat{\theta}) = \frac{1}{n} \left(G'WG\right)^{-1} G'W\Sigma WG \left(G'WG\right)^{-1} \tag{1.46}$$

where $G = \partial m(\theta)/\partial \theta'$ is the Jacobian of moments with respect to parameters.

Integration with Short-Run Model

The long-run investment model is estimated conditional on parameters from the short-run operations model. The short-run model parameters including demand elasticities, cost function parameters, and market clearing conditions, are estimated first and held fixed during long-run estimation. This sequential approach is computationally necessary given the nested structure of the model, where long-run investment decisions depend on expectations of short-run market outcomes.

The derivatives of dispatch, prices, and costs with respect to capacity changes are computed from the short-run model. Since the short-run operations model consists of linear programs, these derivatives are well-defined almost everywhere and can be computed efficiently using sensitivity analysis of the LP solutions.

1.11 Construction of Short-Term Electricity Market Data

This section documents the construction of the short-term electricity market dataset used in the empirical analysis. The goal of this procedure is to produce an hourly, generator-level dataset that links production, fuel costs, prices, weather conditions, and market outcomes across U.S. wholesale electricity markets. The transmission grid and interzonal flow constraints are discussed separately in Appendix 1.12.

Figure 1.4 provides a schematic overview of the full dataset construction pipeline.

1.11.1 Overview

The short-term dataset is constructed in four stages. First, raw ISO-specific fuel mix data are combined with generator characteristics from EIA Forms 860 and 923 to recover hourly generation at the generator level. Second, these ISO-level datasets are standardized and merged into a unified panel. Third, generation is allocated to load-serving zones through an economic dispatch procedure that accounts for interzonal transfers (described in Appendix 1.12). Finally, the resulting data are augmented with weather conditions, fuel prices, and time indicators to produce the final short-term estimation sample.

The pipeline covers six major U.S. Independent System Operators (ISOs): ERCOT, ISO New England, MISO, PJM, SPP, and NYISO.

1.11.2 Construction of Zonal Geographic Maps

A central requirement of the analysis is the ability to match generators to geographic areas in a manner consistent with ISO market structure while preserving as much spatial detail as possible. Because publicly available GIS shapefiles for ISO-defined market zones are incomplete, inconsistent, or entirely unavailable, the construction of zonal maps required substantial manual effort. I view this process as a significant contribution of the paper.

The guiding principle throughout was to aggregate only where necessary to permit consistent geographic matching and econometric analysis, while keeping the data at the lowest feasible spatial resolution. In all cases, zonal boundaries were explicitly restricted to lie within ISO boundaries using GIS shapefiles defining Independent System Operator footprints. This prevents generators, loads, or weather observations from being incorrectly assigned across ISOs.

General Approach. The mapping procedure proceeds in three steps. First, ISO boundaries are imposed as a hard geographic constraint. Second, ISO-specific zones are mapped to counties or county aggregates using a combination of published documentation, utility maps, and GIS overlays. Third, generators are assigned to zones using EIA Form 860 identifiers where available, and county-level geographic matching otherwise. Hand-constructed crosswalk tables are used to reconcile inconsistent geographic schemas across datasets.

Where counties do not align cleanly with ISO zones, counties are split proportionally based on geographic overlap and engineering judgment, applied consistently across datasets.

ERCOT. For ERCOT, zones are based primarily on the system's published weather zones, which in most cases map cleanly to county boundaries. Where weather zones do not align perfectly with counties, I split counties across zones using GIS overlays to preserve contiguity and proportional area. Several ERCOT datasets rely on alternative geographic schemas (e.g., hub- or region-based reporting); in these cases, I construct hand-built mapping tables to reconcile these schemas with the zonal structure used in the analysis.

PJM. PJM required the most extensive manual mapping effort. PJM publishes a large number of zones, many of which do not correspond cleanly to county boundaries. I manually mapped each PJM zone to counties using a combination of PJM documentation, utility service maps, and GIS inspection. In cases where counties span multiple PJM zones, judgment was applied to split counties as cleanly as possible while preserving geographic continuity. Where appropriate for the analysis, zones were subsequently aggregated into regions to support the distribution of generation data.

SPP. SPP zones were mapped using a similar procedure, combining transmission zone definitions with county-level utility maps and GIS shapefiles. In several states, publicly available utility maps with county labels were used to guide assignments. As with PJM, counties were split only where necessary, and always within ISO boundaries.

NYISO. For NYISO, zones were mapped based on reliability assessment zones using a zonal county map. These zones align relatively well with county boundaries, allowing for a clean assignment in most cases.

MISO. MISO presents a comparatively clean case: all zonal boundaries correspond to state-level regions. As a result, counties were assigned directly to MISO zones based on state boundaries, eliminating the need for county splitting.

ISO New England. ISO New England zones are primarily defined at the state level, with the notable exception of Massachusetts, which is divided into three wholesale zones. For Massachusetts, I manually mapped counties to zones using GIS overlays and ISO documentation, splitting counties where necessary to maintain geographic consistency.

Generator Assignment. Generators are assigned to zones using ISO labels from EIA Form 860 wherever possible, which minimizes the risk of misassignment across ISOs. When ISO labels are unavailable or insufficiently granular, generators are matched using county-level geographic information derived from plant location data. This layered approach ensures consistency between generator assignment, zonal geography, and ISO market structure.

Overall, this mapping procedure enables the construction of consistent zonal maps that balance geographic precision with analytical tractability, forming the backbone of the spatial analysis in the paper.

1.11.3 Generator-Level Generation by ISO

Each ISO publishes hourly generation by fuel type rather than by individual generator. To recover generator-level output, I combine these fuel mix data with plant-level characteristics and monthly generation shares from EIA Forms 860 and 923.

For each ISO, hourly generation by fuel type is distributed across generators proportionally to their monthly generation shares within that fuel category. Let g index generators, k fuel types, t hours, and $m(t)$ the month corresponding to hour t. Generator-level hourly output is constructed as

$$q_{gt} = s_{gkm(t)} \cdot Q_{kt},$$

where Q_{kt} is total ISO-level generation for fuel type k at time t, and s_{gkm} is generator g's share of monthly generation within fuel type k.

This procedure is implemented separately for each ISO to accommodate differences in reporting conventions, fuel classifications, and renewable aggregation (e.g., hub-level wind and solar reporting in ERCOT and PJM). Special handling is applied where renewables are reported at regional or zonal levels rather than by unit.

1.11.4 Capacity Constraints

To ensure physical feasibility, generator output is capped at nameplate capacity. In earlier versions of the dataset, excess generation was redistributed across generators of the same fuel type within the same ISO and hour. In the current implementation, generator output is capped at nameplate capacity without redistribution, and a capacity-constrained indicator is recorded. This approach avoids introducing artificial correlations across generators while preserving information on binding capacity constraints.

1.11.5 Combining ISO-Level Datasets

After constructing generator-level output for each ISO, the datasets are combined into a unified panel. Datetime formats, fuel categories, and variable definitions are standardized across ISOs. Fuel types are consolidated into consistent categories (e.g., multiple coal types mapped to a single coal category).

The combined dataset is merged with EPA emissions data where available. When observed EPA generation data exist, these values override the constructed generation estimates to improve measurement accuracy.

1.11.6 Allocation to Load and Short-Term Market Outcomes

Generator-level production is next allocated to load-serving zones using an economic dispatch algorithm that determines how generation serves demand across zones and ISOs. This step produces measures of exports, marginal resources, congestion indicators, and delivered prices. The details of this procedure, including transmission constraints and losses, are described in Appendix 1.12.

1.11.7 Augmentation with Weather, Fuel Prices, and Controls

In the final stage, the dataset is augmented with hourly weather variables matched to generation zones, including temperature and wind conditions. Daily fuel prices for natural gas and coal are merged and interpolated to the hourly frequency.

Time indicators (hour, month, year), zone fixed effects, and interactions between fuel indicators and prices are constructed to support econometric analysis. The resulting dataset contains generator-level observations at the hourly frequency, with detailed information on costs, constraints, market conditions, and environmental factors.

1.11.8 Final Output

The final output is an hourly generator-level panel suitable for short-term market analysis. It forms the basis for the reduced-form and structural estimations in the main text.

1.12 Transmission Capacity and Line Loss Estimation

1.12.1 Overview

This appendix describes the construction of the transmission network dataset used in the empirical analysis, as well as the design of the counterfactual transmission scenarios. The goal is to construct a consistent panel of interregional transmission capacity and losses that can be incorporated into market equilibrium and dispatch models. The methodology combines geographic transmission line data, engineering-based approximations of electrical properties, and historical transmission project information to reconstruct the evolution of the U.S. transmission network over time.

1.12.2 Baseline Transmission Network Construction

Data Sources

The baseline transmission network integrates four primary data sources. First, a national shapefile of high-voltage electric power transmission lines provides geographic geometry, voltage class, and current type (AC or DC). Second, U.S. county boundary shapefiles are used to geolocate transmission line endpoints. Third, a commercial transmission projects database provides information on historical and planned line additions and retirements, including in-service dates, voltage levels, and construction type. Finally, Independent System Operator (ISO) county-to-zone mapping tables are used to associate transmission line endpoints with market pricing zones for ISONE, NYISO, PJM, MISO, SPP, and ERCOT.

Geographic Processing

Transmission line lengths are computed directly from shapefile geometries and converted into physical distance units. For each line, start and end points are extracted from the geometry using the first and last coordinates for `LineString` and `MultiLineString` objects. These endpoints are spatially joined to county boundaries to identify the corresponding county, state, latitude–longitude coordinates, and time zone.

Each endpoint is then mapped to an ISO pricing zone using a hierarchical merge procedure. Counties not directly covered by predefined ISO mapping tables are assigned to the ISO with the largest geographic overlap based on spatial intersection with ISO boundary shapefiles.

Temporal Network Construction (2000–2024)

The transmission network is constructed as a balanced annual panel spanning the years 2000–2024. The reference year for the network is 2022, corresponding to the vintage of the underlying shapefile. For years after 2022, new transmission lines are added based on project commissioning dates. For earlier years, the historical network is reconstructed by removing post-year additions and reinstating lines that were retired after the relevant year.

This procedure assumes that the reference shapefile is comprehensive and that the transmission project database captures the universe of major additions and retirements. While imperfect, this approach allows for a consistent reconstruction of interregional transmission capacity over time.

1.12.3 Electrical Characterization of Transmission Lines

Each transmission line is assigned electrical parameters using standard power-systems engineering approximations.

Resistance

Line resistance is computed as

$$R = \frac{\rho L}{A}, \tag{1.47}$$

where ρ denotes conductor resistivity, L is line length, and A is the conductor cross-sectional area. A uniform conductor type is assumed across lines.

Reactance and Impedance (AC Lines)

For overhead AC transmission lines, inductance per unit length is approximated as

$$\mathscr{L} = 2 \times 10^{-7} \ln\left(\frac{D}{r}\right), \tag{1.48}$$

where D is the distance between phase conductors and r is the conductor radius. Conductor spacing is interpolated based on voltage class using standard utility design practices. Inductive reactance is then computed as

$$X = 2\pi f \mathscr{L}, \tag{1.49}$$

with frequency $f = 60$ Hz. The magnitude of total impedance is given by

$$Z = \sqrt{R^2 + X^2}. \tag{1.50}$$

Flow Limits

For AC lines, maximum transferable power is approximated using a simplified surge-impedance-loading expression:

$$P_{AC}^{\max} = \frac{V^2}{X}, \tag{1.51}$$

where V denotes line voltage. For DC lines, thermal limits are used:

$$P_{DC}^{\max} = 2VI^{\max}, \tag{1.52}$$

assuming a bipole configuration.

Line Losses

Losses are calculated at maximum flow. For AC lines,

$$P_{AC}^{loss} = 3I^2R, \qquad I = \frac{P^{\max}}{\sqrt{3}V}, \tag{1.53}$$

while for DC lines,

$$P_{DC}^{loss} = I^2R. \tag{1.54}$$

1.12.4 Zone-Level Aggregation

Transmission lines are aggregated to the ISO zone-pair level by year. For each ordered pair of zones (z, z'), the dataset reports total transfer capacity, total losses at maximum flow, and the average loss rate as a fraction of transmitted power. These aggregates form the transmission constraint inputs used in the market equilibrium analysis.

1.12.5 Counterfactual Transmission Scenarios

Targeted Capacity Modification Scenarios

The first set of counterfactuals examines the effects of selective changes to interzonal transmission capacity. A “no-limits” scenario sets all interzonal flow limits to a sufficiently large value, approximating a copper-plate network and providing an upper bound on the gains from eliminating congestion. Additional scenarios impose targeted capacity restrictions on specific corridors or introduce fixed-capacity expansions between selected zone pairs, with losses scaled proportionally using observed average loss rates.

HVDC Upgrade Scenarios

A second set of counterfactuals examines the implications of upgrading portions of the AC transmission network to High-Voltage Direct Current (HVDC). Scenarios range from 10% to 100% conversion, with lines selected in descending order of length. Upon conversion, AC reactance is eliminated, impedance collapses to resistance only, flow limits increase, and losses are reduced. While converter station costs and operational control benefits are not explicitly modeled, these scenarios provide a reduced-form characterization of how HVDC adoption alters network capacity and efficiency.

1.12.6 Assumptions and Limitations

The transmission dataset relies on several simplifying assumptions. First, conductor properties are assumed to be uniform across lines. Second, thermal and stability limits are approximated using simplified formulas rather than detailed engineering models. Third, aggregation to ISO pricing zones abstracts from intra-zonal congestion. Fourth, flow limits represent static steady-state constraints and do not capture N–1 reliability requirements. Finally, HVDC scenarios exclude the capital costs of converter stations. These assumptions are discussed when interpreting the empirical results.

1.12.7 Summary

The methodology produces a consistent panel of interregional transmission capacity and losses from 2000 to 2024, along with economically interpretable counterfactual scenarios. These data allow the analysis to isolate the role of transmission constraints and technology choice in shaping electricity market outcomes.

1.13 Counterfactuals in Progress

The counterfactuals reported in the body of the chapter (Section 1.7)—the HVDC upgrade exercise, the Grain Belt Express line addition, and the ERCOT–Eastern Interconnection link—are computed against the fully estimated model and constitute the headline transmission-counterfactual results. The exercises in this appendix are *extensions in progress*: the modeling apparatus is in place and the policy wedges are specified, but the long-horizon simulations underlying the welfare and adoption numbers have not yet been re-run end-to-end against the final estimated parameters. The discussion below describes each extension's specification and the qualitative results expected on the basis of partial simulations and prior literature; quantitative welfare and adoption magnitudes for these scenarios should be regarded as anticipated rather than estimated, and are not included in the headline numbers reported in the abstract or conclusion.

Removal of All Transmission Constraints

Eliminating all transmission constraints reduces average prices in every ISO except SPP and MISO, where prices increase as these regions become net exporters. This counterfactual also dramatically accelerates the renewable energy transition in the long-run model, as generators in renewables-rich regions can access distant markets without congestion-related price discounts.

Big Wires Act

The BIG WIRES Act (United States Congress (2023)) would mandate that each FERC Order No. 1000 transmission planning region achieve a Minimum Interregional Transfer Capacity (MITC) equal to at least 30% of its coincident peak electrical demand, or an increase of 15% over current capacity, whichever is lower, by 2035 as described in Hickenlooper and Peters (2023). The legislation is motivated by the widening gap between planned interregional transmission builds and anticipated need documented in the DOE's DOE (2023), driven by barriers including insufficient coordination across planning regions, cost allocation disputes, and local opposition as found in Joskow (2021); Pfeifenberger and Staff (2021).

I follow the modeling approach of Botterud et al. (2024), who evaluate the BIG WIRES Act using the GenX capacity expansion model across 64 zones aggregated into 12 model regions that approximate the FERC Order No. 1000 planning regions. Their framework distinguishes interregional transmission (built between zones in different planning regions) from intraregional transmission (built within a region), and assumes that absent the Act, no new interregional transmission is built—reflecting the observed tendency of Balancing Authorities to prioritize investment within their own planning region. Under the Act, interregional capacity is expanded until each region satisfies the MITC requirement, with GenX determining the least-cost allocation of new capacity across corridors.

To implement this counterfactual in my model, I add inter-zonal transfer capacity consistent with the corridor-level builds reported in Botterud et al. (2024). I model the MITC requirement as time-varying capacity additions that phase in via a ramp function:

$$A_{ij,t}^{\text{MITC}} = A_{ij,t}^{\text{baseline}} + \Delta A_{ij}^{\text{MITC}} \cdot \rho(t; t_0^{\text{MITC}}, R) \tag{1.55}$$

where $\Delta A_{ij}^{\text{MITC}}$ is the additional transfer capacity on corridor (i, j) required to satisfy the MITC constraint, calibrated to be consistent with the corridor-level estimates in Botterud et al. (2024). The largest capacity additions are concentrated in the Eastern Interconnection: between the Midwest and Mid-Atlantic, Southeast and Florida, Mid-Atlantic and Carolinas, Midwest and Central, and Mid-Atlantic and Southeast, which together account for approximately 80% of total new interregional transmission builds under the Act.

Inflation Reduction Act

The Inflation Reduction Act (IRA) provides substantial tax credits for renewable energy investment, effectively reducing entry costs for solar and wind generators. I model the IRA through two channels that enter the simulation as policy wedges on the estimated model's primitives.

First, the IRA reduces entry costs for eligible renewable technologies. I implement this as a proportional reduction in the entry cost E_k for eligible technology k:

$$E_{k,t}^{\text{IRA}} = E_{k,t}^{\text{baseline}} \cdot (1 - \tau_k^{\text{IRA}} \cdot \rho(t; t_0, R)) \tag{1.56}$$

where $\tau_k^{\text{IRA}} \in [0, 1)$ is the technology-specific cost reduction implied by the investment tax credit, t_0 is the IRA enactment date (August 2022, implemented as the first full period on or after enactment), and $\rho(t; t_0, R)$ is a ramp function that phases the credit in over R periods to avoid discontinuities in the investment decision. I set R to one year as a conservative default.

Second, the IRA production tax credit provides an ongoing per-MWh subsidy to eligible generators, which I model as a revenue shifter:

$$P_{k,t}^{\text{eff}} = P_t + s_k^{\text{IRA}} \cdot \rho(t; t_0, R) \tag{1.57}$$

where $s_k^{\text{IRA}} \geq 0$ is the production credit value. This enters the generator's per-period profit function, raising effective revenue for eligible resources without altering the ISO's dispatch problem or the structure of the Bellman equation. Retirement decisions respond endogenously to the changed profit environment.

Eligibility is determined by a mapping from technology type to IRA-eligible status. In the baseline implementation, eligibility depends on technology k alone and does not vary by region, though the framework accommodates region-specific eligibility if needed.

Bipartisan Infrastructure Law

The Bipartisan Infrastructure Law (BIL) provides funding for transmission infrastructure including grants for interregional transmission projects. I model the BIL as a time-varying expansion of inter-zonal transfer capacities:

$$A_{ij,t}^{\text{BIL}} = A_{ij,t}^{\text{baseline}} + \Delta A_{ij}^{\text{BIL}} \cdot \rho(t; t_0^{\text{BIL}}, R) \tag{1.58}$$

where $\Delta A_{ij}^{\text{BIL}} \geq 0$ represents the additional transfer capacity on corridor (i, j) funded by the BIL, in the same units as the baseline line limits A_{ij}, and t_0^{BIL} is the BIL enactment date (November 2021). The adjusted capacities enter the ISO dispatch problem's constraint set, expanding the feasible region for inter-zonal power flows. The dispatch objective and optimality conditions are otherwise unchanged.

I consider two specifications for the BIL transmission expansion. The first is corridor-specific, assigning $\Delta A_{ij}^{\text{BIL}}$ values to individual lines consistent with announced project funding. The second is a system-wide specification in which each corridor receives a proportional expansion $\Delta A_{ij}^{\text{BIL}} = \alpha \cdot A_{ij,t}^{\text{baseline}}$, parameterized by a single scalar $\alpha > 0$.

Combined Policy Effects

Running these policies in combination reveals whether transmission investment and renewable subsidies act as substitutes or complements. Formally, I compare the welfare gain from the joint policy $\Delta W^{\text{IRA+BIL}} = W^{\text{as\_enacted}} - W^{\text{no\_ira\_no\_bil}}$ against the sum of individual effects $\Delta W^{\text{IRA}} + \Delta W^{\text{BIL}}$, where $\Delta W^{\text{IRA}} = W^{\text{no\_bil}} - W^{\text{no\_ira\_no\_bil}}$ and $\Delta W^{\text{BIL}} = W^{\text{no\_ira}} - W^{\text{no\_ira\_no\_bil}}$. Complementarity implies $\Delta W^{\text{IRA+BIL}} > \Delta W^{\text{IRA}} + \Delta W^{\text{BIL}}$.

The results indicate that these policies are largely complementary: renewable subsidies increase the supply of low-cost generation, while transmission investment ensures this generation can reach high-demand markets. The mechanism is intuitive—the IRA lowers entry costs, inducing additional renewable capacity in resource-rich regions, but the value of this capacity is limited by transmission constraints that prevent it from reaching high-price markets. The BIL relaxes these constraints, raising the effective price received by new entrants and amplifying the IRA's investment incentive. Implementing both policies together yields welfare gains exceeding the sum of individual effects, suggesting that a portfolio approach to energy transition policy dominates single-instrument strategies. I include estimates of the costs of the policies and show that all are welfare improving even after considering costs.

1.13.1 Welfare Decomposition

I run a welfare decomposition to demonstrate how gains accrue to producers and consumers at the zonal level. While the total system-wide surplus is obviously positive for both producers and consumers, the distribution of the surplus is significantly varied. In some regions, mainly ERCOT and specific zones within PJM, welfare is uniformly improved. This occurs because the insurance effect dominates with producers exporting more electricity during low-cost hours and importing during high-cost hours. For regions with significant renewable energy and low prices before transmission expansion, producer welfare increases while consumer welfare decreases. The opposite occurs in regions with fewer renewable resources and higher initial prices. I quantify heterogenous welfare effects and provide useful intuition for policy-makers as to what transfers could be made to make the policy Pareto-improving.

1.14 Calculating the Cost of Line Additions

Estimating the cost of new transmission line additions is important to evaluating the welfare implications of network expansion. **This appendix is an extension in progress.** The four complementary methods described below—a hedonic regression using project-level data, a bottom-up engineering cost model, a synthesis of estimates drawn from the literature and announced projects, and an ensemble method that combines the preceding three—are specified here so that the cost framework is transparent and reproducible, but the body counterfactuals (Section 1.7) currently rely on announced project costs from the literature method (Method 3) rather than on point estimates from the hedonic or ensemble methods. The hedonic and ensemble estimates are not yet ready to support headline welfare claims; this appendix describes how each estimate *would be* produced and the cross-method comparisons that will be reported once the data assembly is complete.

1.14.1 Calculation Method 1: Hedonic Regression

The first method applies a hedonic pricing framework to estimate the cost of transmission line additions as a function of observable project characteristics. Hedonic regression, originally developed in the context of housing markets as discussed in Tauchen and Witte (2001), decomposes the price of a differentiated good into the implicit prices of its constituent attributes. In this application, the "good" is a transmission line project and the attributes include engineering specifications, geographic characteristics, and temporal factors.

Data

The primary data source is the S&P Capital IQ transmission line database, which contains project-level records of transmission investments across the United States. Each record includes the estimated or reported project cost, voltage rating, line length, line type (overhead versus underground, AC versus DC), energization date, and geographic information.

Specification

I estimate the following hedonic regression:

$$\ln(\text{Cost}_i) = \alpha + \boldsymbol{\beta}_1 \mathbf{X}_i^{\text{eng}} + \boldsymbol{\beta}_2 \mathbf{X}_i^{\text{geo}} + \boldsymbol{\beta}_3 \mathbf{X}_i^{\text{constructed}} + \gamma_t + \varepsilon_i \tag{1.59}$$

where Cost_i is the estimated cost of project i (in $\$_{2022}$ per mile, or total—specify), and the regressors are organized into three groups:

Engineering characteristics ($\mathbf{X}^{\text{eng}}$). These include voltage class (kV), line length (miles), and line type indicators (overhead AC, overhead DC, underground AC, underground DC).

Geographic and regulatory characteristics ($\mathbf{X}^{\text{geo}}$).

Constructed features ($\mathbf{X}^{\text{constructed}}$). I construct several derived variables to capture the economic function of each line. These include the thermal flow limit (MW), computed from conductor ratings and voltage, and estimated resistive losses (MW-miles), derived from conductor resistance and expected loading.

Time controls (γ_t). Year fixed effects or a time trend capture secular changes in construction costs, commodity prices, and labor markets.

Estimation and Results

Equation (1.59) is estimated by OLS.

The hedonic regression provides a flexible, data-driven mapping from project characteristics to cost. Its principal advantage is that it captures the joint influence of multiple attributes as revealed by actual project expenditures. However, it is limited by the sample available in the Capital IQ database, which may not be representative of all potential projects, and by the possibility of omitted variable bias if unobserved project features (e.g., permitting difficulty, terrain complexity beyond what is captured by controls) are correlated with included regressors.

1.14.2 Calculation Method 2: Bottom-Up Estimation

The second method constructs a cost estimate from the bottom up by aggregating the costs of individual components and inputs required to build a transmission line. This engineering-economic approach provides a transparent, physically grounded estimate that does not rely on regression assumptions.

Cost Components

The total cost of a transmission line can be decomposed into several major categories:

Conductor and hardware. The cost of conductor wire (e.g., ACSR, ACSS, or HTLS conductors), insulators, and associated hardware depends on the conductor type, number of circuits, and bundling configuration.

Structures. Tower or pole costs vary with structure type (lattice steel, tubular steel, wood H-frame), height, span length, and voltage class.

Right-of-way and land acquisition. Land costs depend on terrain, land use (agricultural, urban, forested), and regional property values.

Labor and construction. Construction labor costs are a substantial share of total project costs and vary by region, terrain difficulty, and prevailing wage requirements.

Permitting, engineering, and contingency. Soft costs including environmental review, engineering design, project management, and contingency allowances.

Assembly

The total estimated cost per mile is computed as:

$$C^{\mathrm{BU}} = C^{\mathrm{conductor}} + C^{\mathrm{structures}} + C^{\mathrm{ROW}} + C^{\mathrm{labor}} + C^{\mathrm{soft}} \tag{1.60}$$

where each component is parameterized by voltage class, line type, and region.

The bottom-up method's strength is its transparency and its ability to generate cost estimates for hypothetical line configurations not observed in historical data. Its principal limitation is that input cost estimates may themselves be uncertain, and interactions between components (e.g., terrain affecting both structure and labor costs simultaneously) may not be fully captured.

1.14.3 Calculation Method 3: Literature-Based and Announced Estimates

The third method draws on three categories of external estimates to benchmark and validate the regression and bottom-up approaches.

Academic and Technical Literature

Several studies have estimated transmission line costs in the context of grid expansion planning, renewable energy integration, and cost-benefit analysis.

Announced Project Costs

Utilities and transmission developers periodically announce cost estimates for planned or completed projects in regulatory filings, investor presentations, and press releases.

Congressional Budget Office and Legislative Cost Estimates

The Congressional Budget Office (CBO) has produced cost estimates associated with transmission-related provisions in proposed legislation.

Synthesis

The literature-based estimates provide an independent check on the data-driven methods. Where estimates from Methods 1 and 2 fall within the range reported in the literature and in announced projects, confidence in the primary estimates is strengthened. Where discrepancies arise, they may point to omitted factors (e.g., permitting delays, opposition-related cost escalation) or differences in project scope.

1.14.4 Calculation Method 4: Ensemble

The fourth method combines the estimates from Methods 1–3 into an ensemble estimate. The motivation for an ensemble approach is standard: individual estimation methods are subject to distinct sources of error, and a weighted combination can reduce overall estimation variance provided the errors are not perfectly correlated.

Combination Procedure

Let $\hat{C}^{(m)}$ denote the cost estimate from method $m \in \{1,2,3\}$ and w_m denote the weight assigned to method m, with $\sum_m w_m = 1$. The ensemble estimate is:

$$\hat{C}^{\text{ens}} = \sum_{m=1}^{3} w_m \hat{C}^{(m)} \tag{1.61}$$

Uncertainty Quantification

The ensemble approach provides the cost estimates used in the main analysis. By drawing on data-driven estimation, engineering fundamentals, and external benchmarks, the ensemble mitigates the weaknesses of any single method while preserving the strengths of each.

1.15 Robustness Checks

This appendix is an extension in progress. It describes the full architecture of robustness checks that will accompany the final version of the chapter. Each subsection describes a planned check, its motivation, the implementation, and the diagnostic output it *would* report. The checks marked here as in progress are not yet executed against the final estimated parameters; their inclusion here documents the robustness program rather than asserting that each diagnostic has already been produced. The body's main qualitative findings (HVDC welfare gain on the order of \$4 billion, Grain Belt Express renewable adoption gain of 11.5%, IRA–BIL complementarity) rely on the baseline estimated model only; the diagnostics below are designed to test, not produce, those numbers.

1.15.1 Reliability Diagnostics

The baseline model assumes that blackouts do not occur (Assumption 1, introduced in the short-run operations model of Section 1.4). While this assumption is necessary for the dispatch problem to have a well-defined solution, it raises the question of whether transmission expansion makes the assumption more or less plausible across counterfactual scenarios. I conduct three complementary reliability diagnostics.

Capacity Shadow Price Analysis

The dispatch linear program produces shadow prices θ_{jkt} on the per-resource capacity constraints introduced in the short-run operations model of Section 1.4. These shadow prices represent the marginal value of an additional MW of capacity for resource k at generation node j at time t. High values of θ_{jkt} indicate hours where the system is near its capacity limits.

For each counterfactual scenario s, I compute the distribution of $\theta_{jkt}^{(s)}$ across all zone-hours and report the mean, median, 95th percentile, 99th percentile, and the fraction of zone-hours where $\theta_{jkt}^{(s)}$ exceeds a threshold $\bar{\theta}$. The threshold $\bar{\theta}$ is set equal to the 99th percentile of the baseline distribution, so the diagnostic measures how frequently each counterfactual pushes the system into the tail of baseline capacity stress.

Formally, define the reliability stress indicator for scenario s as:

$$R^{(s)} = \frac{1}{|\mathscr{J}| \cdot |\mathscr{K}| \cdot |\mathscr{T}|} \sum_{j,k,t} \mathbf{1}\left[\theta_{jkt}^{(s)} > \bar{\theta}\right]. \tag{1.62}$$

This diagnostic requires no additional computation beyond what is already performed to solve the dispatch problem.

Effective Reserve Margin

For each zone i and hour t under each counterfactual scenario, I compute the effective reserve margin as:

$$\mathrm{ERM}_{it}^{(s)} = \frac{\sum_j \sum_k \min\left(O_{jkt}^{\mathrm{MAX}},\, A_{ij}^{(s)}\right)(1 - B_{ij}) - L_{it}}{L_{it}}, \tag{1.63}$$

where $A_{ij}^{(s)}$ is the transmission capacity in scenario s, O_{jkt}^{MAX} is available generation capacity, B_{ij} is the line loss factor, and L_{it} is load. The effective reserve margin captures the fact that transmission expansion increases available capacity at each node by enabling imports from distant generators, even without new generation investment.

I report the distribution of $\mathrm{ERM}_{it}^{(s)}$ across zones and hours for each counterfactual, focusing on the lower tail. I also report the fraction of zone-hours where $\mathrm{ERM}_{it}^{(s)}$ falls below standard planning reserve margins (typically 15%).

Monte Carlo Generator Outage Simulation

To directly assess how transmission expansion affects reliability under realistic contingencies, I conduct a Monte Carlo exercise in which generators experience forced outages. For a representative sample of high-stress hours, I randomly derate generator capacity according to technology-specific forced outage rates from NERC's Generating Availability Data System (GADS). I choose these hours as hours in which the aggregate capacity shadow price exceeds its 90th percentile.

Specifically, for each simulation draw d and each generator g, the available capacity is:

$$\tilde{O}_{jkgt}^{\mathrm{MAX},(d)} = O_{jkgt}^{\mathrm{MAX}} \cdot \mathbf{1}\left[u_{jkgt}^{(d)} > \mathrm{FOR}_k\right], \tag{1.64}$$

where $u_{jkgt}^{(d)} \sim \mathrm{Uniform}(0,1)$ is an independent draw and FOR_k is the forced outage rate for resource type k. I then re-solve the dispatch problem with the derated capacities, checking whether the problem remains feasible. If the problem is infeasible—meaning demand cannot be met at some node—the hour is recorded as a loss-of-load event.

The loss-of-load probability (LOLP) for scenario s is estimated as:

$$\widehat{\mathrm{LOLP}}^{(s)} = \frac{1}{|\mathscr{T}^*| \cdot D} \sum_{t \in \mathscr{T}^*} \sum_{d=1}^{D} \mathbf{1}\left[\text{dispatch infeasible under } \tilde{O}_t^{(d)} \text{ in scenario } s\right], \tag{1.65}$$

where $\mathscr{T}^*$ is the set of high-stress hours and D is the number of simulation draws per hour. I use $D = 500$ draws and report LOLP estimates with bootstrapped confidence intervals.

Comparing $\widehat{\mathrm{LOLP}}^{(s)}$ across counterfactual scenarios provides a direct measure of how transmission expansion affects system reliability under contingency conditions, without requiring modification to the model structure.

1.15.2 Alternative State Space Discretizations

The long-run model relies on discretization of the continuous state space as described in Appendix Section 1.10.2. If results are sensitive to the choice of grid density, this would suggest that the discretization introduces artifacts into the counterfactual predictions.

I re-estimate the long-run model under two alternative discretization schemes. The first uses a coarser grid with approximately half the number of bins for each state variable (F_k: 10 bins, E_k: 10 bins, demand: 3 bins, own capacity: 8 bins, rival capacity: 5 bins, price: 5 bins). The second uses a finer grid with approximately 50% more bins (F_k: 30 bins, E_k: 30 bins, demand: 8 bins, own capacity: 22 bins, rival capacity: 15 bins, price: 15 bins).

I report parameter estimates, simulated capacity paths, and key counterfactual outcomes (average price change, renewable adoption by 2050, and welfare gains) under each discretization and compare these to the baseline results. The check is passed if the qualitative conclusions are unchanged and the quantitative magnitudes are within 20% of the baseline estimates.

1.15.3 Sensitivity to Transmission Capacity Measurement

The baseline transmission network is constructed using engineering approximations applied to the HIFLD shapefile (Appendix Section 1.12). Several assumptions in this construction may affect the results: uniform conductor properties, simplified thermal and stability limits, and the use of static steady-state flow limits that do not account for N$-$1 reliability requirements.

Scaled Transmission Capacity

To assess sensitivity to the overall level of estimated transmission capacity, I re-run the dispatch model with all inter-zonal capacities A_{ij} scaled by factors $\alpha \in \{0.75, 0.9, 1.1, 1.25\}$. These multiplicative adjustments capture the possibility that the engineering approximations systematically over- or underestimate true transfer capability. I report how the distribution of locational marginal prices, the frequency of binding transmission constraints, and the welfare estimates from the HVDC upgrade counterfactual change under each scaling.

Alternative Loss Specifications

I replace the baseline loss estimates with two alternatives: (i) uniform losses set to the load-weighted average loss rate across all corridors, and (ii) losses calibrated to match aggregate transmission and distribution losses reported in EIA's Annual Energy Outlook. These alternatives test whether spatial heterogeneity in line losses is important for the price and welfare results.

1.15.4 Market Power

The equivalence between the social planner's problem and generators' profit maximization relies on the absence of market power (Section 1.4). Transmission constraints can create local market power by limiting competition from distant generators. To assess the potential impact of market power on the results, I implement two checks.

Pivotal Supplier Index

For each zone i and hour t, I compute the pivotal supplier index (PSI), which identifies whether the removal of any single generation company g from zone j would cause demand at node i to exceed remaining available supply (inclusive of imports):

$$\mathrm{PSI}_{igt} = \mathbf{1}\left[L_{it} > \sum_{j}\sum_{k}\sum_{g'\neq g} \min\left(O^{\mathrm{MAX}}_{jkg't},\, A_{ij}\right)(1-B_{ij})\right]. \tag{1.66}$$

I report the frequency with which generators are pivotal in the baseline and in each counterfactual scenario. Transmission expansion should reduce the frequency of pivotal suppliers by increasing the pool of generators that can serve each node. If the frequency of pivotal suppliers is low in the baseline and decreasing in counterfactuals, this supports the maintained assumption that market power does not materially affect the results.

Markup Bounding Exercise

Following Borenstein et al. (2002), I compute residual demand elasticities at the zone-hour level and use these to bound the Lerner index that a profit-maximizing generator could charge. I compare actual locational marginal prices to the upper bound on prices under monopolistic behavior. If the gap between competitive prices and the monopoly bound is small relative to the price effects found in the counterfactuals, this provides evidence that the competitive pricing assumption does not materially affect the welfare conclusions.

1.15.5 Alternative Demand Specifications

The baseline model treats demand as perfectly inelastic and exogenous. Two aspects of this assumption merit robustness investigation.

Price-Responsive Demand

I introduce a small price elasticity of demand into the dispatch model by replacing the fixed load L_{it} with a linear demand function:

$$L_{it}(\mathrm{LMP}_{it}) = L_{it}^{0} - \eta \cdot \mathrm{LMP}_{it}, \tag{1.67}$$

where L_{it}^{0} is the baseline load and η is calibrated to match short-run demand elasticity estimates from the literature. I use η values corresponding to elasticities of -0.05 and -0.10 evaluated at mean price and load, consistent with estimates from Ito and Reguant (2016) and Jessoe and Rapson (2014).

Introducing price-responsive demand modifies the dispatch problem from a linear program to a quadratic program, but the problem remains convex and computationally tractable.

The key question is whether price-responsive demand amplifies or dampens the welfare effects of transmission expansion. Because transmission expansion reduces prices in high-cost regions (where demand response would increase consumption) and raises prices in low-cost regions (where demand response would reduce consumption), the net effect on welfare depends on the relative magnitudes of these adjustments.

Load Growth Sensitivity

The long-run counterfactuals use load projections from EIA's Annual Energy Outlook. These projections may understate future load growth if electrification of transportation and heating proceeds faster than anticipated, or overstate growth if energy efficiency improvements exceed expectations. I re-run the long-run counterfactuals under alternative load trajectories corresponding to the AEO's high and low economic growth scenarios. This tests whether the complementarity finding between IRA and BIL is robust to uncertainty in demand growth.

1.15.6 Alternative Equilibrium Concepts

The long-run model uses an oblivious equilibrium concept in which firms condition investment decisions on their own state and aggregate rival capacity rather than the full joint distribution of competitor states as described in Appendix Section 1.10.3. While this approximation substantially reduces computational burden, it may miss strategic interactions that matter for investment dynamics.

I implement two checks. First, I increase the dimensionality of the rival capacity state by splitting it into "same-fuel rival capacity" and "other-fuel rival capacity," allowing firms to condition investment on the composition of competitive supply. This tests whether the aggregation of rivals into a single state variable obscures economically important variation.

Second, I compare the oblivious equilibrium investment paths to those generated by a myopic model in which firms treat current prices and costs as permanent when making investment decisions (i.e., $\beta = 0$ in the value function). If the qualitative results are similar under myopic and forward-looking behavior, this suggests that the specific equilibrium concept is not the primary driver of the counterfactual results. If the results differ substantially, this identifies the role of strategic forward-looking behavior in shaping investment responses to transmission expansion.

1.15.7 Sequential vs. Joint Estimation

The baseline estimation proceeds sequentially: short-run parameters are estimated first and held fixed during long-run estimation (Appendix Section 1.10.4). This approach is computationally necessary but may introduce bias if estimation error in the short-run parameters affects the long-run estimates non-linearly.

To assess the magnitude of this concern, I construct bootstrap confidence intervals that account for first-stage estimation uncertainty. Specifically, I draw from the estimated asymptotic distribution of the short-run parameters, re-compute the short-run profit mapping for each draw, and re-estimate the long-run model conditional on the perturbed short-run inputs. I report the resulting distribution of long-run parameter estimates and counterfactual welfare effects. If the bootstrap confidence intervals are substantially wider than those computed from second-stage GMM standard errors alone, this would indicate that first-stage uncertainty is quantitatively important.

1.15.8 Sensitivity of Counterfactual Policy Parameters

The policy counterfactuals in Section 1.7.3 require assumptions about the magnitude of IRA tax credits, the timing of BIL transmission expansions, and the corridor-level allocation of new capacity under the BIG WIRES Act. I assess sensitivity to each of these inputs.

IRA Credit Magnitude

The baseline implementation sets the investment tax credit reduction τ_k^{IRA} based on the statutory credit values. I re-run the IRA counterfactual with τ_k^{IRA} scaled by factors of 0.5 and 1.5, corresponding to scenarios where effective credits are lower (due to qualification barriers, domestic content requirements, or phase-outs) or higher (due to bonus credits for energy communities or low-income areas) than the statutory values.

BIL Expansion Magnitude and Allocation

For the BIL counterfactual, I compare the corridor-specific specification (where $\Delta A_{ij}^{\text{BIL}}$ is assigned to individual lines based on announced projects) with the system-wide proportional expansion ($\Delta A_{ij}^{\text{BIL}} = \alpha \cdot A_{ij}^{\text{baseline}}$) under alternative values of $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. This tests whether the welfare and investment results are driven by the specific corridors receiving capacity additions or by the overall increase in system transfer capability.

Ramp Function Sensitivity

Policy wedges are phased in using a ramp function $\rho(t; t_0, R)$ with baseline ramp period $R = 1$ year. I test $R \in \{0, 2, 5\}$ years to assess whether the speed of policy implementation affects the long-run equilibrium. Instantaneous implementation ($R = 0$) provides an upper bound on short-run disruption, while slower phase-in periods test whether gradual adjustment changes the investment dynamics.

1.15.9 Exclusion of Individual ISOs

To verify that the results are not driven by the idiosyncratic characteristics of any single ISO, I re-estimate the short-run model excluding each ISO in turn and report the resulting parameter estimates and counterfactual price effects. ISOs differ substantially in resource mix, congestion patterns, and market size as shown in Table 1.2, so parameter stability across these leave-one-out samples would provide evidence that the estimates reflect general features of the dispatch technology rather than market-specific artifacts.

For the long-run model, full re-estimation excluding individual ISOs is computationally intensive. Instead, I compute the counterfactual welfare effects and renewable adoption paths excluding each ISO's zones from the simulation while holding parameters fixed. This tests whether the aggregate results are driven disproportionately by investment dynamics in any single market.

1.15.10 Temporal Stability of Short-Run Estimates

The short-run model is estimated on data from 2018–2023, a period that includes both the COVID-19 pandemic (which substantially depressed electricity demand in 2020) and the natural gas price spike associated with the 2022 energy crisis. To test whether these events distort the parameter estimates, I re-estimate the short-run model on two subsamples: 2018–2019 (pre-pandemic) and 2021–2023 (post-pandemic). Parameter stability across subsamples would support the maintained assumption that the dispatch technology is time-invariant, while instability would suggest that structural breaks in fuel markets or demand patterns affect the estimates.

I also estimate the model on rolling two-year windows (2018–2019, 2019–2020, 2020–2021, 2021–2022, 2022–2023) and plot the evolution of key parameters over time. Persistent trends in the fuel cost parameters would indicate time-varying marginal costs not captured by the included fuel price covariates.

1.15.11 Summary of Robustness Architecture

Table 1.12 provides an overview of all robustness checks, the assumption being tested, and the primary diagnostic output.

Table 1.12: Summary of Robustness Checks

Check	Assumption Tested	Diagnostic Output
Capacity shadow prices	No blackouts	Distribution of θ_{jkt} across scenarios
Effective reserve margin	No blackouts	Fraction of zone-hours below 15% reserve
Monte Carlo outages	No blackouts	LOLP estimates with confidence intervals
Alternative discretizations	State space grid density	Parameter and counterfactual sensitivity
Scaled transmission capacity	Transmission measurement	Price distribution, welfare sensitivity
Alternative losses	Loss heterogeneity	Price and welfare effects
Pivotal supplier index	No market power	Frequency of pivotal generators
Markup bounding	No market power	Gap between competitive and monopoly prices
Price-responsive demand	Inelastic demand	Welfare amplification or dampening
Load growth sensitivity	AEO projections	Robustness of complementarity finding
Rival capacity decomposition	Oblivious equilibrium	Investment path sensitivity
Myopic vs. forward-looking	Dynamic behavior	Role of expectations in investment
Bootstrap first-stage	Sequential estimation	Confidence interval comparison
IRA credit scaling	Statutory credit values	Welfare and investment sensitivity
BIL allocation	Corridor-specific vs. proportional	Welfare decomposition
Ramp function	Phase-in speed	Long-run equilibrium sensitivity
Leave-one-out ISOs	No single-ISO dependence	Parameter and welfare stability
Temporal subsamples	Time-invariant technology	Parameter evolution over time

References

U.s. energy-related carbon dioxide emissions. Technical report, Energy Information Administration, 2022.

National transmission needs study. Technical report, Department of Energy, 2023.

Costas Arkolakis and Conor Walsh. Clean growth. Technical report, National Bureau of Economic Research, 2023.

Severin Borenstein, James B Bushnell, and Frank A Wolak. Measuring market inefficiencies in california's restructured wholesale electricity market. *American Economic Review*, 92 (5):1376–1405, 2002.

Audun Botterud, Christopher R Knittel, John E Parsons, Juan Ramon L Senga, and S Drew Story. Evaluating the impact of the big wires act. *Center for Energy and Environmental Policy Research*, 2024.

R Andrew Butters, Jackson Dorsey, and Gautam Gowrisankaran. Soaking up the sun: Battery investment, renewable energy, and market equilibrium. Technical report, National Bureau of Economic Research, 2021.

Peter Cramton. Electricity market design. *Oxford Review of Economic Policy*, 33(4): 589–612, 2017.

Jonathan T Elliott. Investment, emissions, and reliability in electricity markets. Technical report, Working Paper, 2022.

Richard Gilbert, Karsten Neuhoff, and David Newbery. Allocating transmission to mitigate market power in electricity networks. *RAND Journal of Economics*, pages 691–709, 2004.

Luis E Gonzales, Koichiro Ito, and Mar Reguant. The dynamic impact of market integration: Evidence from renewable energy expansion in chile. Technical report, National Bureau of Economic Research, 2022.

Gautam Gowrisankaran and Philipp Schmidt-Dengler. A computable dynamic oligopoly model of capacity investment. Technical report, National Bureau of Economic Research, 2024.

Gautam Gowrisankaran, Stanley S Reynolds, and Mario Samano. Intermittency and the value of renewable energy. *Journal of Political Economy*, 124(4):1187–1234, 2016.

Gautam Gowrisankaran, Ashley Langer, and Mar Reguant. Energy transitions in regulated markets. Technical report, National Bureau of Economic Research, 2024.

J Hickenlooper and S Peters. S. 2827-118th congress. the big wires act, 2023.

Stephen P Holland, Erin T Mansur, and Andrew J Yates. Decarbonization and electrification in the long run. Technical report, National Bureau of Economic Research, 2022.

Koichiro Ito and Mar Reguant. Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–1957, 2016.

Katrina Jessoe and David Rapson. Knowledge is (less) power: Experimental evidence from residential energy use. *American Economic Review*, 104(4):1417–1438, 2014.

Paul L Joskow. Facilitating transmission expansion to support efficient decarbonization of the electricity sector. *Economics of Energy & Environmental Policy*, 10(2):57–92, 2021.

Ariel Pakes and Richard Ericson. Empirical implications of alternative models of firm dynamics. *Journal of economic theory*, 79(1):1–45, 1998.

Johannes Pfeifenberger and FERC Staff. Transmission planning and benefit-cost analyses. *Boston: Brattle Group.*, 2021.

Nathan Shreve, Julia Selker, Zach Zimmerman, and Richard Doying. Transmission congestion for 2024. Technical report, Grid Strategies LLC, November 2025. URL https://gridstrategiesllc.com/wp-content/uploads/GS_Transmission-Congestion-for-2024.pdf. Accessed: 2026-04-26.

Helen V Tauchen and Ann Dryden Witte. Estimating hedonic models: Implications of the theory, 2001.

United States Congress. Big wires act. S.2827, 118th Congress, 2023. URL https://www.congress.gov/bill/118th-congress/senate-bill/2827. A bill to require the Federal Energy Regulatory Commission to establish minimum interregional transmission transfer capability requirements.

Chapter 2

Learning-by-Doing and Industrial Policies: Evidence from China's Solar Panel Industry

Abstract.

This paper studies how learning-by-doing shapes cost dynamics, technological adoption, and the effectiveness of industrial policies in the solar panel industry. We develop and estimate a dynamic model of firm entry, exit, production, and technology choice using firm-level data on Chinese manufacturers and global data from 2000 to 2013. The model allows production experience to reduce future costs, generating a dynamic feedback loop between output and productivity. We find that learning-by-doing is a key driver of cost reductions, particularly for crystalline silicon (c-Si) technology: doubling cumulative production reduces marginal costs by approximately \$0.71 per MW for a typical c-Si firm and \$1.82 per MW for large c-Si firms. Learning also amplifies the effects of industrial policies, leading to larger long-run impacts on output and prices. It further shapes technological competition, contributing to the

dominance of c-Si technology. Finally, earlier policy implementation yields substantially larger effects by accelerating movement along the experience curve.

Keywords: learning-by-doing, industrial policy, technological path dependence, solar panels

JEL Classification Codes: L52, L94, O33, Q42, Q48, F13

2.1 Introduction

The development of solar energy plays a crucial role in the transition to a sustainable and low-carbon electricity grid and the mitigation of climate change. The rapid decline in the cost of solar panels over the past two decades has been one of the most striking developments in energy markets. At the same time, the industry has experienced a dramatic shift in its technological composition, with crystalline silicon (c-Si) emerging as the dominant technology worldwide. Understanding the forces driving these changes is central to evaluating the effectiveness of industrial policies and the transition to low-carbon energy systems.

A large body of work has attributed the decline in solar panel costs to learning-by-doing (LBD), whereby cumulative production experience reduces future production costs. During the same period, governments around the world implemented a wide range of industrial policies, including demand-side subsidies, supply-side subsidies, and trade policies. While each of these factors has been studied in isolation, less is known about how they interact to shape industry dynamics.

This paper studies how learning-by-doing interacts with industrial policies to determine production costs, technology adoption, and welfare in the solar panel industry. We focus on a central mechanism: policies affect production not only directly, but also indirectly through their impact on cumulative experience. This generates a dynamic feedback loop in which higher output today lowers future costs, further increasing output and accelerating the adoption of particular technologies.

To quantify this mechanism, we develop and estimate a dynamic model of firm behavior in the global solar panel industry. The model incorporates firm entry and exit, technology choice, and production decisions, allowing firms' costs to evolve endogenously with cumulative production. We estimate the model using a combination of firm-level data on Chinese manufacturers and aggregate data on global prices, input costs, and policy variables from 2000 to 2013.

Our empirical analysis yields three main findings. First, learning-by-doing is a quantitatively important driver of cost reductions, particularly for c-Si technology. We estimate that doubling cumulative production reduces marginal costs by approximately \$0.71 per MW for a typical c-Si firm (and \$1.82 per MW for large c-Si firms), implying that LBD accounts for a substantial share of the observed decline in solar panel prices. Second, learning-by-doing significantly amplifies the effects of industrial policies. For example, the decline in polysilicon prices not only reduces costs directly, but also increases production, which accelerates learning and further reduces costs. Similarly, demand-side subsidies and entry subsidies generate larger long-run effects when learning is present than when it is absent. Ignoring LBD would therefore lead to a substantial underestimation of the impact of these policies. Third, learning-by-doing plays a key role in shaping technological competition. Because c-Si technology exhibits stronger learning effects than thin-film technology, policies and shocks that expand production disproportionately benefit c-Si, contributing to its eventual dominance.

We further show that the timing of policies is critical. Demand-side subsidies are significantly more effective when implemented early, as they initiate the learning process and generate persistent cost reductions. Similarly, supply-side policies yield larger benefits when introduced earlier, as they accelerate movement along the experience curve. These results highlight the importance of accounting for dynamic complementarities when designing industrial policies.

This paper contributes to several strands of literature. First, it contributes to the literature on industrial policies in renewable energy markets. Many of these studies, including De Groote and Verboven (2019), focus on the impact of demand-side subsidies. One exception is Gerarden et al. (2025), who examines how import tariffs affect market equilibrium outcomes and compares import tariffs to supply-side subsidies. To the best of our knowledge, our study is the first to jointly analyze demand-side and supply-side interventions within a unified framework.

Second, it adds to the empirical literature on learning-by-doing that has been documented in a variety of industries, for example, Head (1994) and Ohashi (2005) on steel, Thompson (2001) and Thornton and Thompson (2001) on shipbuilding, Benkard (2000) and Benkard (2004) on aircraft, Su et al. (2025) on space launch. In particular, our paper contributes to the growing literature studying learning-by-doing in low-carbon technology sectors. For example, Helveston et al. (2022) estimates a two-factor learning model and finds the learning rates of solar panel module production in Germany, U.S., and China are 20%, 26%, and 33% between 2006 to 2020, respectively. Barwick et al. (2025b) estimates a static model where electric vehicle producers and battery producers bargain over prices and finds a sizable LBD of battery production. Our approach is in the same spirit of theirs by exploiting variations in prices and quantities of the final products and main inputs. We quantify its role in a setting with rich firm-level data and multiple interacting policy instruments.

Our paper also contributes to the literature on technological path dependence and technological competition. A central insight of this literature is that small initial advantages—arising from costs, policies, or shocks—can be amplified over time, leading to the dominance of one technology even in the presence of viable alternatives. In our setting, learning-by-doing provides a concrete mechanism for such path dependence. Because cumulative production reduces future costs, technologies that experience early expansion—whether due to input price declines or policy support—become progressively more competitive, reinforcing their adoption over time. We show that this dynamic feedback loop helps explain the rapid diffusion and eventual dominance of crystalline silicon technology over thin-film alternatives. By quantifying how policies and input price shocks interact with learning-by-doing, our analysis provides new empirical evidence on how technological pathways are shaped in dynamic industries and highlights the role of policy timing in influencing long-run technological outcomes.

More broadly, our findings suggest that evaluating industrial policies in emerging industries requires accounting for dynamic production externalities such as learning-by-doing.

Policies that stimulate early production can have long-lasting effects by shaping the trajectory of costs and technology adoption, while delayed interventions may forgo these dynamic gains.

The remainder of the chapter proceeds as follows. Section 2.2 describes the solar panel industry and the data. Section 2.3 presents the model. Section 2.4 outlines the estimation strategy. Section 2.5 reports the estimation results. Section 2.6 presents the counterfactual analysis. Section 2.7 concludes.

2.2 Solar Panel Industry and Data

2.2.1 Technologies of Solar Panel Manufacturing

Over the past three decades, two primary technologies have been used to manufacture solar panels: crystalline silicon (c-Si) and thin-film technologies.

Crystalline silicon technology, first developed at Bell Labs in 1954, has become the dominant production technology in the industry. Initially used in space applications, it was later adopted for terrestrial electricity generation, with early commercial deployment in the 1970s. Over time, c-Si panels have achieved substantial improvements in conversion efficiency—from below 10 percent in early applications to approximately 20 percent in current installations. In addition to higher efficiency, c-Si panels are characterized by durability and scalability, making them particularly well suited for utility-scale electricity generation.

Production of c-Si panels typically involves vertically integrated processes: polysilicon is refined into wafers, which are then processed into photovoltaic cells and assembled into modules. Given the standardized nature of this production process and limited cross-firm price dispersion (as documented by Gerarden et al. (2025)), we treat solar panels as homogeneous products conditional on technology.

The main competing technology is thin-film solar panels, which use alternative materials such as cadmium and tellurium. Relative to c-Si, thin-film panels are less efficient but offer advantages in flexibility and lower fixed production costs. These characteristics make thin-film technology more suitable for niche applications requiring adaptability (e.g., portable or distributed installations). Moreover, the primary inputs for thin-film panels are often by-products of other industrial processes, resulting in relatively stable input prices over time.

A key difference between the two technologies lies in their exposure to input price fluctuations. In particular, polysilicon—the key input for c-Si panels—experienced substantial price volatility during the 2000s, while input prices for thin-film technology remained comparatively stable. This asymmetry plays a central role in shaping technological adoption and industry dynamics.

2.2.2 Industrial Policies

The rapid expansion of solar panel adoption worldwide has been closely linked to a wide range of industrial policies. These policies can be broadly categorized into demand-side and production-side interventions.

Demand-Side Policies Demand-side policies are designed to stimulate the adoption of solar panels by lowering effective consumer costs or increasing returns to installation. The most prominent example is feed-in tariffs, under which governments guarantee above-market prices for electricity generated from renewable sources. These tariffs have been widely implemented in Europe since the early 2000s, with countries such as Germany and Spain offering payments substantially above wholesale electricity prices.

A second form of demand support is output-based incentives, such as net metering programs, which compensate households or firms for excess electricity fed back into the grid. A third category includes investment subsidies, such as tax credits or direct rebates that reduce the upfront cost of installing solar panels. For example, the U.S. federal Investment Tax Credit subsidizes a fixed share of installation costs, while China's "Golden Sun Program" provided subsidies covering up to 50 percent of investment costs during its implementation period.[1]

In addition to subsidies, governments have also used trade policies, including tariffs on imported solar panels, to protect domestic producers. The U.S. and EU imposed anti-dumping duties and tariffs on Chinese c-Si panels beginning in the early 2010s, reflecting concerns about foreign competition and industrial policy distortions.

Supply-Side Policies Supply-side policies directly affect firms' production costs, entry, and expansion. While such policies are relatively limited in the U.S. and EU, they have played a particularly prominent role in China. Examples include subsidized land, preferential tax treatment, low-interest loans, and discounted electricity prices at the provincial and local levels. These instruments directly lower both entry costs and ongoing production expenses. In addition, supply-side policies also include supply-chain policies that operate through reducing upstream input costs. By lowering these input costs, such policies directly decrease firms' marginal costs, thereby raising the equilibrium output in downstream industries.

Beginning in 2010, the Chinese central government further intensified support by designating solar panel manufacturing as a "strategic emerging industry." State-owned banks provided substantial credit to firms in the sector (see Lexology (2010) and Nahm (2023)), and the government actively promoted the development of upstream industries, including polysilicon production. As a result, China's polysilicon output expanded dramatically, alleviating input constraints and contributing to downstream cost reductions (see WaferPro (2024)).

A key empirical challenge is that many of these supply-side policies are not directly observable. To address this, we adopt an indirect approach and infer the magnitude of production and entry subsidies from observed firm behavior within a structural model, following the methodology of Barwick et al. (2025a).

[1] Further details on the program can be found in International Energy Agency (2021).

2.2.3 Data

Our empirical analysis combines several datasets covering the period 2000–2013.

Chinese Solar Panel Exports We use transaction-level data from *Chinese Customs Transactions* to construct export volumes at the firm–destination–year level. Export destinations are aggregated into six regions: China, EU, India, Japan, the U.S., and the Rest of the World (ROW).

Firm-Level Manufacturing Data We use the *Annual Survey of Manufacturing* conducted by the Chinese National Bureau of Statistics, which provides detailed information on firm output, ownership status (whether it is state-owned, privately owned, or foreign firm), and the year in which the firm was established. Using these data, we construct firm-level measures of production and experience.

Demand and Policy Data We compile regional data on solar panel installations, demand conditions, and policy variables. Demand-side policies include feed-in tariffs, tax credits, and import tariffs, collected from *IEA databases* and prior studies such as Gerarden (2023).

Prices of Solar Panels and Inputs We obtain global price data for solar panels based on the two technologies, along with the corresponding input prices for each technology. These data are sourced from *Bloomberg New Energy Finance*, *Bernreuter Research*, and the *United States Geospatial Services*.

Auxiliary Data Additional datasets include regional macroeconomic indicators (e.g., GDP, electricity consumption) and aggregate production data for non-Chinese manufacturers.

2.2.4 Descriptive Evidence

The data reveal several key patterns that motivate our analysis. First, global solar panel prices have fallen significantly, with the steepest decline occurring after 2008. This price decline has spurred the installation of solar panels worldwide. Figure 2.2(a) displays the price series, while Figure 2.2(b) displays the installed capacities of the two types of solar panels over this period.

Second, the market share of each technology is closely tied to its own price, which in turn is heavily influenced by the price of its key inputs. As illustrated in Figure 2.2(a), the price of c-Si solar panels was relatively high—and even rising—prior to 2008, but began to fall rapidly afterward and has stayed low since. A major driver of this trend is the price of polysilicon, the primary input used in c-Si solar panels (see Figure 2.4(a)). Polysilicon prices spiked sharply in late 2007, largely driven by growing demand from the semiconductor sector. They then declined steeply after 2008, mainly due to the expansion of polysilicon manufacturing capacity in China (see Figure 2.4(b)). In parallel, the market share of the rival thin-film technology rose sharply in late 2007 and then fell again after 2008 (see 2.2(b)).

The cost structures of thin-film and c-Si manufacturing differ markedly in their relative weighting of fixed and variable components (Powell et al., 2012, 2015). Thin-film production entails large upfront capital expenditures for specialized vacuum deposition chambers, laser scribing equipment, and highly automated conveyance systems. However, because the active layers require minimal material per module, marginal production costs are low. This cost profile heavily weights thin-film economics toward fixed costs, requiring manufacturers to operate at high capacity utilization rates to amortize their facility investments effectively (Goodrich et al., 2013).

Monocrystalline silicon production, by contrast, is characterized by comparatively high variable costs. The energy-intensive Czochralski crystal growth process, the consumption of high-purity polysilicon feedstock, and the reliance on silver paste for metallization and aluminum for framing collectively ensure that the marginal cost of each additional module remains substantial (Powell et al., 2012; Woodhouse et al., 2019). Consequently, c-Si manufacturers face greater exposure to input price volatility and supply chain disruptions affecting commodity markets for polysilicon, silver, and aluminum (VDMA Photovoltaic Equipment, 2024).

Figure 2.1 presents the theoretical breakeven decision from a manufacturing cost standpoint between producing each of the different technologies. c-Si is a significantly cheaper technology to produce for lower capacity levels, but beyond a certain production level thin-film dominates. This trend may explain why learning-by-doing effects are so large for c-Si panels as high variable costs are better suited to process improvements. The relationship may also provide clues as to why input subsidies had such a differential effect on firm technology choice. If thin-film firms do not reach critical capacity, they remain higher cost providers and thus out of the market. By subsidizing c-Si early, the Chinese government may have created barriers to reaching critical scale for CdTe manufacturing while shielding the impacts of price spikes and industry cycles from the c-Si producers.

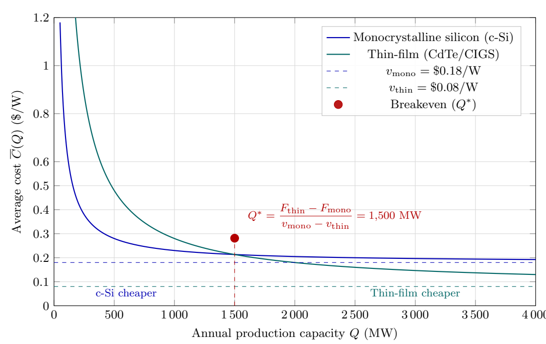


Figure 2.1: Thin-film is cheaper beyond a certain level of production.

Evidence of China's Supply-Side Policy Interventions China's solar panel manufacturing grew rapidly over the sample period, with its global market share rising from under 2 percent in 2000 to more than 60 percent by 2011. This expansion was primarily driven by c-Si technology, which dominates Chinese production. See Figure 2.3(a) for China's share of the global market and the output of the two solar panel technologies.

The price of polysilicon rose from 50 to 350 dollars per GW during the 2005-2008 surge. Globally, the market share of c-Si technology is sensitive to changes in polysilicon prices. In contrast, the predominance of c-Si in China did not react to this sharp input cost increase. This pattern indicates that China subsidized c-Si technology during this period.

Furthermore, the sharp increase in Chinese entrants aligns with government policy. Figure 2.3(b) shows the number of new entrants, exiting firms, and active solar panel producers in China over this period. Following the 2010 policy change that designated solar manufacturing as a strategic industry, market entry rose steeply; although firm exits also climbed, the overall number of firms still expanded substantially.

Taken together, these patterns suggest an important interaction between input price dynamics, policy interventions, and learning-by-doing, which jointly shape industry evolution. These features motivate the structural framework developed in the next section.

Figure 2.2: Solar Panel Price and Installed Capacity

(a) Price

(b) Installed Capacity

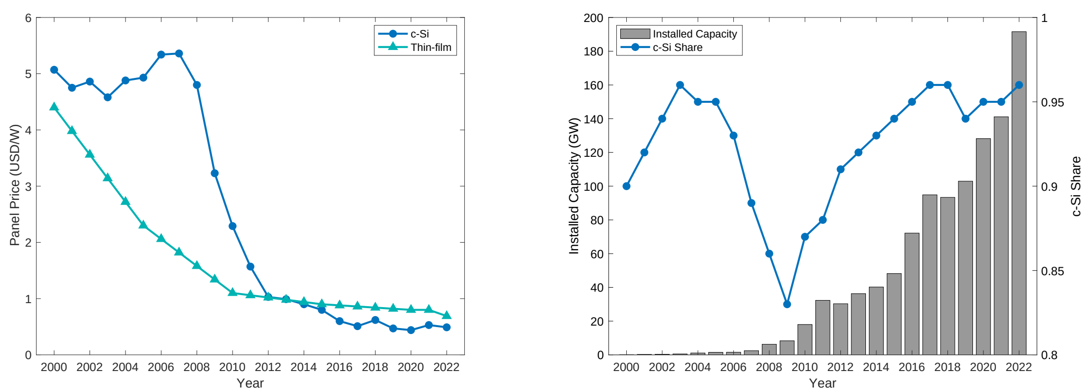


Figure 2.3: China's Market Share and Number of Firms

(a) Market Share (b) Number of Firms

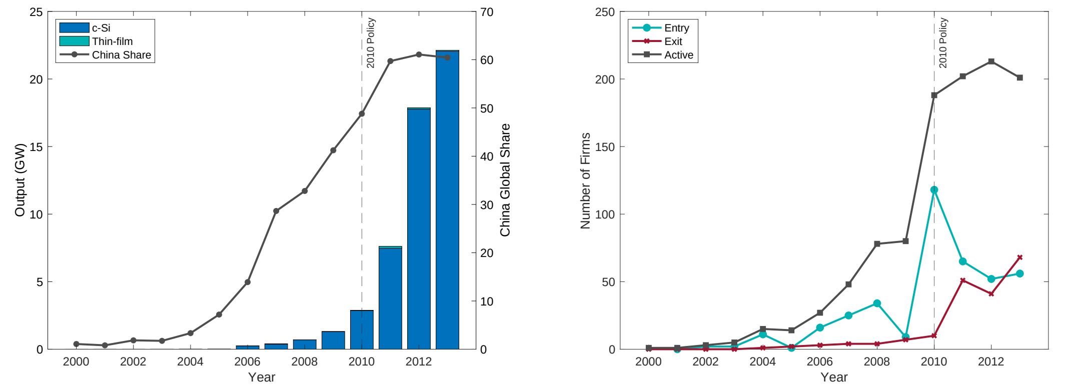
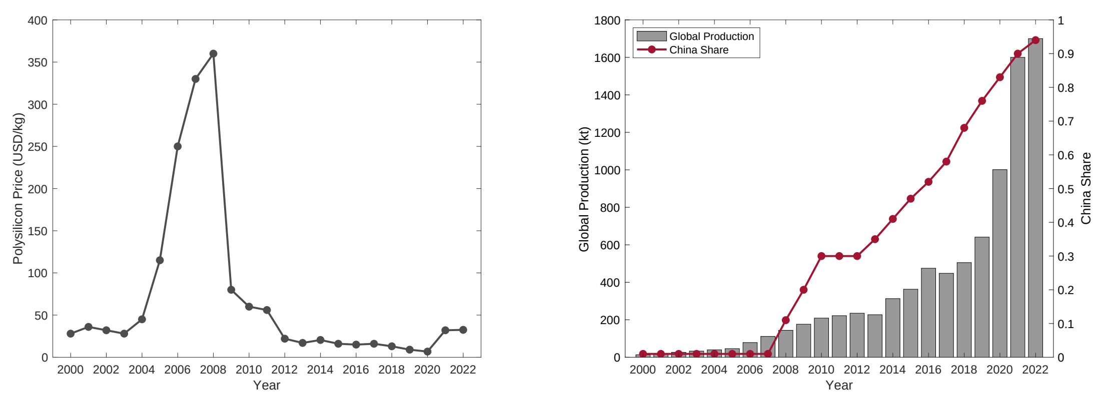


Figure 2.4: Polysilicon Price and Output

(a) Price (b) Output



2.3 Model

In this section, we present a dynamic model of firm entry, technology adoption, exit, and production. Time is discrete and a period is a year, denoted as t. There are two types of production technologies, indexed by $n \in \{1,2\}$, where $n = 1$ denotes c-Si and $n = 2$ denotes thin-film solar panels. The global market is divided into $m = 1,...,m$ economic regions (China, EU, India, Japan, US, and rest of the world). At each period t, a set of firms produces solar panels. Firms are forward-looking and interact in a dynamic oligopoly environment with endogenous entry, exit, and production decisions.

At the beginning of each period, incumbent firms observe their scrap values and decide whether to exit the market. If they choose to stay, these incumbents observe their production cost shocks for each region and determine how many units to produce for each region. Meanwhile, potential entrants observe their entry shocks and decide on whether to enter and, if they do, choose which technology to adopt.

2.3.1 Per-Period Demand

For each technology n and region m, the aggregate demand is given by

$$Q_{nmt} = Q_{nm}(P_{1mt}, P_{2mt}, X_{mt}), \tag{2.1}$$

where Q_{nmt} is the total quantity of technology-n solar panels demanded by consumers in region m during period t. The vector X_{mt} includes a range of demand shifters, including measures of aggregate economic activity, regional fixed effects, and demand-side subsidies. The variable P_{nmt} is the price consumers pay to install a technology n solar panel system, incorporating both the retail panel price and installation expenses. This consumer price may differ from the global price P_{nt} because of tariffs and installation subsidies that vary by region and technology. Specifically, the consumer price is given by:

$$P_{nmt} = P_{nt} \times (1 + \text{tariff}_{nmt}) + P^I_{mt} - \text{demand subsidy}_{mt},$$

where tariff_{nmt} is the tariff rate, which applies only to c-Si technology, P^I_{mt} is the installation expenses, and $\text{demand subsidy}_{mt}$ is the amount of demand-subsidies provided by region m in period t.

2.3.2 Incumbents: Production and Exit

At the beginning of each period, incumbents observe their private scrap values that are drawn from a technology-specific distribution, and then decide to either exit or continue operations. We denote the scrap value for technology-n incumbent j as ϕ_{jnt} and its distribution as $F_{\phi,n}$. If a firm exits, it receives its scrap value. If it remains active, it observes its marginal cost shocks across all regions, denoted as $\boldsymbol{\omega}_{jnt} = \{\omega_{jnmt}\}_{m=1}^{M}$. Remaining active, firms engage in Cournot competition while considering how their current output decisions will influence future production costs. They simultaneously decide the quantity of solar panels to produce for each region, $\boldsymbol{q}_{jnt} = \{q_{jnmt}\}_{m=1}^{M}$, to maximize their discounted profit, taking as given the production decisions of rival firms. This new output, $\sum_{m=1}^{M} q_{jnmt}$, is added to the next period's accumulated experience, which in turn affects its future production costs.

Production Cost The production cost for firm j using technology n, when choosing a vector $\boldsymbol{q}_{jnt}$, consists of two parts:

$$TC_n(\boldsymbol{q}_{jnt}, s_{jnt}, \boldsymbol{\omega}_{jnt}) \quad = \quad C_0\left(\sum_{m=1}^{M} q_{jnmt},\ s_{jnt}\right) + \sum_{m=1}^{M} C_m\big(q_{jnmt},\ \omega_{jnmt}\big), \tag{2.2}$$

where the first component, $C_0\big(\sum_{m=1}^{M} q_{jnmt},\ s_{jnt}\big)$, represents the total production cost of $\sum_{m=1}^{M} q_{jnmt}$, which is a function of the current output and a vector of state variables s_{jnt}. These state variables include the firm's accumulated experience e_{jnt}, cost shifters that are common across firms (such as input prices), and technology-specific time trends that capture overall improvements in cost efficiency. The second component, $C_m\big(q_{jnmt},\ \omega_{jnmt}\big)$, accounts for the additional cost of delivering q_{jnmt} to destination m in year t.

The experience of firm j, denoted as e_{jnt}, can potentially reduces production costs in the future, incentivizing the firm to increase output to gain more experience in the current period. This experience is defined as the cumulative past production:

$$e_{jnt} = \sum_{\tau<t}\sum_{m} q_{jnm\tau},$$

where $q_{jnm\tau}$ is the units sold by firm j to region m during year τ. The key parameter of interest is the experience coefficient, which measures how rapidly the unit cost of solar panel manufacturing falls as the production experience increases.

Incumbent Value Function The value function of a technology-n incumbent j, after observing the scrap value ϕ_{jnt} but before observing the realization of the production cost shock $\boldsymbol{\omega}_{jnt}$, is defined as:

$$V_n(s_{jnt},\phi_{jnt}) \quad = \quad \max\{\phi_{jnt}, CV_n(s_{jnt})\}, \tag{2.3}$$

where the continuation value $CV_n(s_{jnt})$ is expressed as:

$$\begin{aligned} CV_n(s_{jnt}) \quad \equiv \quad & \mathbb{E}_\omega\Big[max_{\boldsymbol{q}}\{\sum_m P_{nm}(q_m,\boldsymbol{q}_{-mt},X_{mt})q_m - TC_n(\boldsymbol{q},s_{jnt},\boldsymbol{\omega}_{jnt}) \\ & +\beta\mathbb{E}[V_n(s_{jnt+1})|s_{jnt},\boldsymbol{q}]\}\Big]. \end{aligned} \tag{2.4}$$

Here, $\boldsymbol{q} = \{q_m\}_{m=1}^{M}$ is a vector of output choice for each region, while $\boldsymbol{q}_{-mt}$ denotes the output vectors that are chosen by other competing solar panel firms for region m. The expected value $\mathbb{E}_\omega$ is calculated with respect to the production cost shocks $\boldsymbol{\omega}_{jnt}$,

Optimal Exit Decision The optimal decision to exit follows a threshold structure. A firm exits the market if the scrap value exceeds the continuation value; otherwise, it remains active. Therefore, the probability of exiting is:

$$p_n^x(s_{jnt}) \equiv Pr\big(\phi_{jnt} > CV_n(s_{jnt})\big) = 1 - F_{\phi,n}\big(CV_n(s_{jnt})\big). \tag{2.5}$$

Optimal Production Decision Provided that firm j with technology-n stays active, the optimal output level, $q^*_{jnmt} = q^*_{nm}(s_{jnt}, \omega_{jnmt})$, satisfies the following first-order condition:

$$\begin{aligned}\frac{\partial TC_n(\boldsymbol{q}^*_{jnt}, s_{jnt}, \boldsymbol{\omega}_{jnt})}{\partial q^*_{jnmt}} = & \underbrace{P_{nmt} + q^*_{jnmt}\frac{\partial P_{nm}(q^*_{jnmt}, \boldsymbol{q}^*_{-mt}, X_{mt})}{\partial q_{jnmt}}}_{\text{Impact on current profit}} \\ & + \underbrace{\beta\frac{\partial \mathbb{E}[V_n(s_{jnt+1})|s_{jnt}, \boldsymbol{q}^*_{jnt}]}{\partial q_{jnmt}}}_{\text{Impact on future profits via LBD}},\end{aligned} \tag{2.6}$$

where $\boldsymbol{q}^*_{-mt}$ denotes the vector of equilibrium outputs that are chosen by all other rival solar panel firms for region m during year t. The firm's optimal output is driven by three components: the first term captures its static profit margin, the second term captures its market power (the Cournot effect), and the third term captures the dynamic incentive arising from LBD. Because of this last component, firms may produce more than the static optimum in order to accumulate experience and lower their future costs.

Equilibrium Global Price In each period, the global price of technology-n solar panels, denoted as P_{nt}, equates the aggregate demand and supply, where the aggregate demand is described by equation (2.1) and the aggregate supply is the sum of q^*_{jnmt} defined in equation (2.6) across all firms using technology n.

2.3.3 Potential Entrants: Entry and Technology Selection

At the end of each period t, a total of $\bar{N}$ potential entrants observe their payoff ζ_{j0t} for not entering, as well as entry costs $\kappa_1(s_{jnt}) - \zeta_{j1t}$ for entering with c-Si technology, and $\kappa_2(s_{jnt}) - \zeta_{j2t}$ for entering with thin film technology. The terms $\kappa_1(s_{jnt})$ and $\kappa_2(s_{jnt})$ denote the expected entry costs associated with the two technologies. The vector s_{jnt} includes policy dummies to capture the impact of industrial policies on the decisions regarding entry and technology selection.

After observing their entry payoff shocks ζ's, potential entrants decide whether to enter the market and, if they choose to enter, select a technology. If they opt to enter, they incur the entry cost during the current period and start earning profit from the product market starting from the following period. Therefore, the maximization problem of the potential entrant j can be written as:

$$max\Big\{\zeta_{j0t},\, max_{n\in\{1,2\}}\{-\kappa_n(s_{jnt}) + \zeta_{jnt} + \beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}]\}\Big\} \tag{2.7}$$

A firm stays out of the market if its payoff from not entering exceeds the expected profit of entering with either technology option. It chooses to enter with technology n if this option provides the highest profit. We use $p^{e,n}(s_{jnt})$ to denote the probability that a potential entrant j enters with technology n, given by

$$\begin{aligned} p^{e,n}(s_{jnt}) \;\equiv\; \Pr\Big\{&\zeta_{jnt} - \zeta_{jn't} > \kappa_n(s_{jnt}) - \kappa_{n'}(s_{jn't}) + \beta\mathbb{E}[V_{n'}(s_{jn't+1}) \mid s_{jn't}] - \beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}] \\ &\& \ \zeta_{jnt} - \zeta_{j0t} + > \kappa_n(s_{jnt}) - \beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}]\Big\} \end{aligned} \tag{2.8}$$

Note that LBD affects entry decisions through the expected value term $\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}]\}$. A technology with stronger learning becomes more attractive dynamically, even if it is initially costly.

2.3.4 Markov-Perfect Equilibrium

A Markov-Perfect Equilibrium is characterized by policy function, $q^*_{nm}(s_{jnt},\ \omega_{jnmt})$, $p^x_n(s_{jnt})$, $p^e_n(s_{jnt})$, value function $V_n(s_{jnt})$, and global price P_{nt}, such that production policy function $q^*_{nm}(s_{jnt}, \omega_{jnt})$ satisfies (2.6), exit policy function $p^x_n(s_{jnt})$ satisfies (2.5), entry policy function $p^e_n(s_{jnt})$ satisfies (2.8), global price P_{nt} ensures that aggregate demand and aggregate supply are in balance, while the incumbent's value function $V_n(s_{jnt})$ satisfies (2.3).

2.3.5 Model Discussion

The model highlights a central mechanism of the paper: policies affect output not only directly, but also indirectly by changing the accumulation of production experience. This generates a dynamic feedback loop in which higher output reduces future costs, amplifying the long-run effects of policies.

Before moving on to the estimation, we discuss two key simplifying assumptions made by our model. First, production cost shocks ω_{jnmt} are assumed to be i.i.d., because allowing for a persistent time-varying unobserved state variable in a dynamic model would introduce substantial computational complexity. Nonetheless, our model does incorporate persistent effects through the accumulation of experience. When firms receive positive demand shocks, they increase quantity, which in turn increases their stock of experience. Since experience is a state variable, firms with more experience have lower production costs and thus choose higher output levels, further building experience and reducing long-run costs. This feedback mechanism is the key driver of why firms that are ex-ante identical can exhibit divergent growth paths over time.

Second, following most of the existing literature such as Ryan (2012) and Barwick et al. (2025a), we assume that all firms view government policies as permanent. Relaxing this assumption would require modeling firms' expectations and their adjustment to a changing policy environment, which would greatly increase the complexity of our analysis. This is an important avenue for future research; see Gowrisankaran et al. (2025) and Chen (2024) for further discussion.

2.4 Estimation Strategy

In this section, we outline the strategy for estimating model parameters. In Section 2.4.1, we detail our demand model, specify the production cost function, as well as the distributions related to the scrap value and entry costs. In sections 2.4.2 and 2.4.3, we present the two-stage procedure for estimating dynamic parameters, including those associated with the production cost function and the distributions for the scrap value and entry cost shocks.

We estimate the dynamic parameters using the firms' observed choices of production, exit, and entry, including not entering, entering with c-Si technology, or entering with thin-film technology. Using the BBL approach from Bajari et al. (2007), we estimate these parameters in two stages. In the first stage, we flexibly estimate policy functions, including production, exit, and entry with technology selection, as well as the transition process of state variables directly from the data. After that, we approximate the value function using a set of B-spline basis functions of state variables, as described in (Sweeting (2013)). In the second stage, we estimate the dynamic parameters. We provide additional details in Appendix I.

2.4.1 Specifications

Logit Demand. We parameterize the solar panel demand, expressed in (2.1), using a simple Logit model. To be specific, the demand of technology-n solar panels in the region m during the year t is given by:

$$Q_{nmt} = \frac{M_{mt} \cdot exp(-\alpha_m P_{nmt} + X_{mt}\eta + \xi_{nmt})}{1 + \sum_{n'} exp(-\alpha_m P_{n'mt} + X_{mt}\eta + \xi_{n'mt})}.$$

Here, M_{mt} is the market size for region m in year t, measured by the total electricity consumption. P_{nmt} is the price of technology-n solar panels that is faced by consumers in region m, calculated by adjusting the global price P_{nt} with local installation cost, tariffs, and subsidies. X_{mt} is a vector of demand shifters, including GPD, feed-in tariff, regional fixed effects, and yearly fixed effects.

For the given demand, the estimation equation is represented as:

$$ln(\frac{Q_{nmt}}{M_{mt}}) - ln(\frac{M_{mt} - Q_{1mt} - Q_{2mt}}{M_{mt}}) \quad = \quad -\alpha_m P_{nmt} + X_{mt}\eta + \xi_{nmt}. \tag{2.9}$$

A key challenge is price endogeneity, since prices respond to demand shocks ξ_{nmt}. To address this issue, we follow Gerarden et al. (2025) and instrument solar panel prices using global prices of silver and aluminum, which are key inputs in both c-Si and thin-film solar technologies but plausibly exogenous to regional demand shocks. Our identification relies on the assumption that variation in these input prices shifts production costs without directly affecting local demand conditions. This assumption is plausible because solar manufacturing in total represents less than 5% of global demand for these inputs throughout the sample period.

Production Costs As specified in (2.2), the production cost function for firm j using technology n is decomposed into two components. The first component, $C_0\left(\sum_m q_{jnmt}, s_{jnt}\right)$, is a function of the firm's aggregate current output $\sum_m q_{jnmt}$ and the state variables s_{jnt}. The second component, $C_m(q_{jnmt}, \omega_{jnmt})$, represents the additional cost of supplying quantity q_{jnmt} to destination m in year t. Specifically, we specify the marginal cost function as

$$\begin{aligned} MC(q_{jnmt}) \quad &= \quad \gamma^e_{jn} \cdot ln(e_{jnt}+1) + \mathbf{1}(n = \text{c-Si}) \cdot \text{polysilicon cost}_t \\ &\quad + \gamma_n^{\text{const}} + \gamma^{scale} q_{jnmt} + \gamma_n^{time} \cdot \ln(t-1999) + \gamma_m^{ex} + \omega_{jnmt} \end{aligned} \tag{2.10}$$

In this specification, the LBD term is modeled as the logarithm of one plus the firm's cumulative experience, $\ln(e_{jnt}+1)$, and the learning rate is denoted by γ^{e}_{jn}. This learning rate is specific to each technology. For c-Si firms, we additionally allow the learning rate to vary across firms; in particular, larger firms may learn more quickly. More formally, the learning parameter is given by

$$\begin{aligned}\gamma^{e}_{jn} \quad = \quad & \mathbf{1}(n=\text{thin-film})\cdot\gamma^{e,\,\text{baseline}}_{\text{thin-film}} \\ & +\mathbf{1}(n=\text{c-Si})\cdot(\gamma^{e,\,\text{baseline}}_{\text{c-Si}}+\mathbf{1}(j\in\text{large c-Si firms})\cdot\gamma^{e,\,\text{large}}_{\text{c-Si}}).\end{aligned}$$

For c-Si technology, the key input price is that of polysilicon, its primary input. To account for the declining input-to-output conversion rate over this period, we introduce a time trend.[2] In addition, as discussed in Section 2.2, Chinese firms likely benefited from subsidies during the 2006–2008 surge in polysilicon prices. We therefore include subsidy parameters to represent input cost subsidies in those years. Specifically, the input cost is

$$\begin{aligned}\text{polysilicon cost}_t \quad = \quad & \gamma^{j,\,\text{baseline}}_{\text{c-Si}}\cdot exp(\gamma^{j,\,\text{time}}_{\text{c-Si}}\cdot(t-2000))\cdot\text{polysilicon price}_t \\ & -\sum_{t=2005}^{2008}\gamma^{j,\text{sub}}_{\text{c-Si},\,t}\cdot\mathbf{1}(t\in[2005,\ 2008]),\end{aligned} \tag{2.11}$$

where the parameter $\gamma^{j,\,\text{const}}_{\text{c-Si}}$ represents this conversion rate in the year 2000, $\gamma^{j,\,\text{time}}_{\text{c-Si}}$ captures the time trend reflecting aggregate efficiency improvements over this period, and $\gamma^{j,\text{sub}}_{\text{c-Si},\,t}$ denotes the per-watt subsidy provided to offset the input cost surge during the period of elevated polysilicon prices.

The parameter $\gamma^{\text{const}}_{n}$ represents the baseline cost level in the year 2000 for each of the

[2] Bernreuter (2020) documents that the polysilicon intensity (grams per watt) declined from above 20 g/W in the early 2000s to below 6 g/W by the early 2010s. The decline in conversion rate during this period was driven by a combination of process innovation, material efficiency, and performance improvements along the silicon solar value chain. First, wafer thickness fell substantially, reducing the amount of silicon required per cell. Second, manufacturing technologies improved, especially the transition toward more efficient wire-sawing methods, which sharply reduced kerf loss (silicon wasted during slicing). Third, cell and module efficiencies increased, meaning more electrical output (watts) could be generated from the same physical amount of silicon. Fourth, scale expansion and standardization (e.g., larger wafers and better yield management) further improved material utilization.

two technologies. The term γ^{scale} characterizes the curvature of the marginal cost function with respect to current output. The technology-specific time trend parameter γ_n^{time} captures additional gains in cost efficiency over the sample period, excluding learning-by-doing effects and the polysilicon-to-module conversion. The parameters γ_m^{ex} are region-specific and capture the extra costs related to shipping or exporting to each region. Finally, ω_{jnmt} are cost shocks, assumed to be independently and identically normally distributed with mean zero and variance σ_ω^2.

Distributions of Scrap Value and Entry Costs. We assume that the scrap value ϕ_{jnt} follows an exponential distribution characterized by a parameter $1/\sigma_\phi$. Additionally, we assume that the entry cost shocks ζ are independently and identically distributed according to a Type 1 Extreme Value (T1EV) distribution.

2.4.2 First Stage: Policy Functions and State Transition

Following the BBL approach, we first estimate firms' decision rules directly from the data without imposing parametric structure on dynamic optimization.

Exit Policy Function We estimate the exit policy function using a Probit specification:

$$Pr_n(\chi_{jnt} = 1|s_{jnt}) = \Phi(h_n(s_{jnt})), \tag{2.12}$$

where the variable $\chi_{jnt} = 1$ indicates the exit of firm j with technology n from the market in period t. Here, $h_n(s_{jnt})$ is a polynomial function of the states tailored to the specific technology, and Φ is the cumulative distribution function of a standard normal distribution. We use $\hat{p}_n^x(s)$ to refer to the estimated exit probability given state s.

Production Policy Function Similar to the optimal investment in Bajari et al. (2007), the production policy function $q_{nm}^*(s_{jnt}, \omega_{jnmt})$ decreases monotonically with the cost shock

ω_{jnmt} and can be recovered as follows:

$$q^*_{nm}|s_{jnt} \quad = \quad G^{-1}_{nm}(1-F_\omega(\omega \mid s_{jnt})), \tag{2.13}$$

where $G_{nm}(q \mid s_{jnt})$ is the empirical distribution of production given the state variables, and $F_\omega(\omega \mid s_{jnt})$ is the assumed distribution of ω.

We can write the optimal production as the sum of two components: one depends only on the state variables, and the other depends only on the shock. Formally, this can be expressed as:

$$q^*_{jnmt} \quad = \quad h_{1nm}(s_{jnt}) + h_{2nm}(\omega_{jnmt}), \tag{2.14}$$

where $h_{1nm}(\cdot)$ and $h_{2nm}(\cdot)$ are two unknown functions that need to be estimated.

We first perform a flexible non-parametric regression of q_{jnmt} on s_{jnt} to get the estimate $\hat{h}_{1nm}(s_{jt})$, which allows us to compute $q^*_{jnmt} - \hat{h}_{1nm}(s_{jnt})$. Then, we estimate $h_{2nm}(\cdot)$ using equation (2.13).

State Transition The vector of state variables s_{jnt} includes the state of all firms in the industry, making it high-dimensional. For experience stock, it transits as follows:

$$e_{jnt+1} = e_{jnt} + \sum_m q_{jnmt}.$$

where $\sum_m q_{jnmt}$ is the total production of firm j during year t.

To reduce dimensionality, we assume that firms do not keep track of each competitor's state variables, but instead rely on industry-level prices as a sufficient statistic, akin to the oblivious equilibrium concept introduced by (Weintraub et al. (2008) and Benkard et al. (2015)). This assumption is justified given that the industrial HHI in China in 2013 being 316, indicating that market power is not a significant issue. This approach has often been adopted in the literature, for example, Jeon (2022), Chen and Xu (2023), and Barwick et al. (2025a).

However, the equilibrium price for each type of solar panel clears the market by equalizing aggregate demand and aggregate supply of that type of solar panel, which makes it a complicated object. Following Barwick et al. (2025a) and other studies, we model them as AR(1) processes. We account for different technologies having different price processes. Furthermore, we incorporate changes in constant terms across different periods. We make similar assumptions for input prices.

2.4.3 Second Stage: Structural Estimation

In the second stage, we recover structural parameters by exploiting the optimality conditions implied by firms' dynamic problem.

Value Function Approximation

The ex-ante value function, prior to the realization of ϕ_{jnt}, is described by

$$V_n(s_{jnt}) \quad = \quad \mathbb{E}_\phi[max\{\phi_{jnt}, CV_n(s_{jnt})\}] = p_n^x(s_{jnt})\sigma_\phi + CV_n(s_{jnt}), \tag{2.15}$$

where $p_n^x(s_{jnt})$ denotes the exit probability, and $CV_n(s_{jnt})$ denotes the continuation value as defined in equation (2.4).

Following the existing literature (e.g., Sweeting (2013)), we approximate the ex-ante value function using B-spline basis functions of the state variables. This is expressed as $V_n(s_{jnt}) = \sum_l \rho_{n,l} u_{n,l}(s_{jnt})$ where $u_{n,l}(s_{jnt})$ are basis functions and $\rho_{n,l}$ are coefficients that need to be estimated along with the dynamic parameters.

Specifically, we search for coefficients ρ that minimize the violation of the Bellman equation (2.15) given the dynamic parameters, given by

$$min_{\rho} \| V_n(s_{jnt};\rho) - \hat{p}^x_n(s_{jnt})\sigma_{\phi} - CV_n(s_{jnt};\rho) \|_2. \tag{2.16}$$

Here, $\| \cdot \|_2$ is the L^2 norm, $\hat{p}^x_n(s_{jnt})$ is the estimated first-stage exit policy function, and the continuation value is evaluated based on these estimated policy functions, given by

$$\begin{aligned} CV_n(s_{jnt};\rho) \quad = \quad & \mathbb{E}_{\omega}\Big[\sum_m P_{nmt}\hat{q}^*_{nm}(s_{jnt},\omega_{jnmt}) - TC_{jnt}(\hat{q}^*_{nm}(s_{jnt},\omega_{jnmt}),s_{jnt},\omega_{jnt}) \\ & + \beta \mathbb{E}[V(s_{jnt+1};\rho) \mid s_{jnt},\hat{q}^*_{jnmt}]\Big], \end{aligned}$$

where $\hat{q}^*_{nm}(s_{jnt},\omega_{jnmt})$ is the estimated first-stage production policy function.

Estimation of Production Costs and Scrap Value

Log-Likelihood of Output The optimal output $q^*_{jnm}(s_{jnt},\omega_{jnmt})$ is determined by the first-order conditions in equation (2.6). By construction, $q^*_{jnm}(s_{jnt},\omega_{jnmt})$ is strictly monotonic in ω_{jnmt}. Assuming it is also differentiable, the likelihood of output can be expressed as

$$g_{nm}(q_{jnmt};\theta) \quad = \quad \frac{f_{\omega}(\omega_{jnmt})}{|\partial q^*_{nm}(s_{jnt},\omega_{jnmt})/\partial \omega_{jnmt}|}, \tag{2.17}$$

where $|\partial q^*_{nm}(s_{jnt},\omega_{jnmt})/\partial \omega_{jnmt}|$ is the absolute value of the derivative of $q^*_{nm}(s_{jnt},\omega_{jnmt})$ with respect to ω_{jnmt}.

Log-Likelihood of Exiting Let χ_{jnt} indicate that firm j of technology n exits in year t in the data. The maximum likelihood estimate of the standard deviation scrap value, σ_{ϕ}, maximizes the following log-likelihood for exit decisions:

$$log(f_n(s_{jnt};\theta)) \quad = \quad \Big[(1-\chi_{jnt})log(1-exp(-\frac{CV_n(s_{jnt};\rho)}{\sigma_{\phi}})) - \chi_{jnt}\frac{CV_n(s_{jnt};\rho)}{\sigma_{\phi}}\Big] \tag{2.18}$$

Estimation of Entry Costs

The probability that potential entrant j enters with technology n is expressed as

$$p^{e,n}(s_{jnt}) \equiv \frac{exp\Big(-\kappa_n(s_{jnt})+\beta\mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}]\Big)}{1+\sum_{n'} exp\Big(-\kappa_{n'}(s_{jn't})+\beta\mathbb{E}[V_{n'}(s_{jn't+1}) \mid s_{jn't}]\Big)}. \tag{2.19}$$

We estimate the mean entry costs using the observed entry decisions via MLE.

2.4.4 Identification

The central parameter of interest is the effect of experience on production cost. The identification of LBD relies on three distinct sources of variation. First, there is within-firm variation in cumulative output, which shifts costs while keeping firm-specific characteristics constant. Second, we exploit variation in input prices (such as polysilicon prices), which is separately controlled for in the cost function. Third, policy-induced changes in production scale generates variation in experience accumulation across firms. Together, these elements enable us to separate LBD from input price effects and from static economies of scale.

Demand parameters are identified from cross-region variation in prices and quantities, where input prices serve as instruments for solar panel prices. The remaining parameters of the production cost function are identified from how output correlates with inferred cost shocks. Entry costs are inferred from observed entry and technology choice choices. The scrap value parameter is identified from firms' exit decisions.

2.5 Estimation Results

2.5.1 Estimates of Demand Equation

Table 2.1 reports the parameter estimates for the demand equation (2.9). Column (I) assumes a common preference for c-Si panels across regions, while column (II) allows this coefficient to vary across regions. In both specifications, we instrument the solar panel prices for a given year using the prices of the main inputs that year. We use the specification in column (II) as our preferred estimates for the subsequent supply estimation and counterfactual analysis.

All parameter estimates have the expected signs. The estimates indicate that demand is downward-sloping in prices across all regions, with substantial heterogeneity in price sensitivity. Among the six regions, Japan exhibits the smallest price coefficient in absolute value, while China dislikes price the most, except the rest of the world (ROW). This pattern aligns with regional differences in income levels.

Consumers generally prefer c-Si solar panels over thin-film solar panels, with EU consumers demonstrating the strongest preference. This finding is consistent with the higher efficiency and durability of c-Si panels documented in Section 2.2. Demand shifters enter with the expected signs. Higher GDP and more generous feed-in tariffs significantly increase demand for solar panels. The magnitude of the feed-in tariff coefficient suggests that demand-side subsidies play an important role in shaping adoption decisions.

Overall, the demand estimates imply that both prices and policy variables are key drivers of solar panel adoption, providing a foundation for the supply-side estimation.

2.5.2 Estimates of State Transitions and Policy Functions

Table H.4 presents the estimated state transition parameters. Aggregate prices exhibit substantial persistence, supporting the use of AR(1) processes in the model.

Table H.5 reports the estimated exit and production policy functions. Firms with greater experience are less likely to leave the market and tend to produce more. More generous demand subsidies also lead firms to increase production and reduce their likelihood of exit. In addition, exit probabilities decline and output rises when the solar price increases or input prices fall. Finally, following the central government's designation of solar panels as a "strategic emerging industry" in 2010, exit probabilities decrease and output grows more

Table 2.1: Demand Estimates

	(I)		(II)	
	Est.	SE	Est.	SE
Solar panel price (α_m)				
× China	-1.168***	(0.111)	-1.428***	(0.122)
× EU	-0.997***	(0.105)	-1.116***	(0.107)
× India	-1.317***	(0.116)	-1.243***	(0.130)
× Japan	-0.720***	(0.090)	-0.779***	(0.093)
× US	-1.142***	(0.113)	-1.206***	(0.125)
× ROW	-1.204***	(0.101)	-1.469***	(0.095)
Demand shifters (η_m)				
$\mathbb{1}$(c-Si solar panel)	3.214***	(0.360)		
× China			3.654***	(0.383)
× EU			3.983***	(0.338)
× India			2.641***	(0.402)
× Japan			3.526***	(0.331)
× US			2.388***	(0.359)
× ROW			3.599***	(0.356)
GDP	9.490***	(1.720)	2.410*	(1.340)
Feed-in tariff	1.954***	(0.531)	1.898***	(0.573)
Constant	-30.422***	(0.366)	-29.411***	(0.250)
R^2	0.731		0.809	

Notes: Standard errors in parentheses. * p<0.10,** p<0.05,*** p<0.01.

rapidly, consistent with the intended effects of the policy rollout.

2.5.3 Estimates of Dynamic Parameters

Table 2.2 presents the estimated production cost function specified in Equation (2.10), while Table 2.3 displays the estimated parameters associated with scrap values and entry costs.

LBD on Production Cost The key parameter of interest is the effect of cumulative experience on marginal costs, γ_n^e. For c-Si firms, the estimated baseline coefficient is -1.021 and statistically significant, with an additional -1.608 for large firms, implying a substantial LBD effect. Quantitatively, doubling cumulative experience reduces marginal costs by

$\ln(2) \times 1.021 \approx 0.708$ per MW for a typical c-Si firm, and by $\ln(2) \times (1.021 + 1.608) \approx 1.823$ per MW for large c-Si firms. This magnitude is broadly consistent with estimates in the solar PV literature and in related low-carbon technology sectors, suggesting that LBD is an important driver of cost reductions in renewable energy technologies.[3]

In contrast, the estimated learning effect for thin-film technology is small and not statistically significant. This difference is consistent with the empirical patterns documented in Section 2, where c-Si technology experienced rapid expansion and cost decline, while thin-film technology did not.

Input Prices and Cost Pass-Through The estimated coefficient on the polysilicon cost term is positive and significant, indicating that fluctuations in upstream polysilicon prices are passed through to c-Si module costs. Importantly, even after controlling for input prices and the declining polysilicon-to-module conversion rate, the LBD effect remains strong, indicating that cost reductions cannot be fully explained by upstream price changes. The estimated subsidy parameters for 2005–2008 are consistent with the qualitative evidence in Section 2.2 that Chinese firms received support to offset the polysilicon price spike during this period.

Scale Effect and Exporting Costs The positive coefficient on the quadratic term in output ($\gamma^{scale} = 5.007$) indicates decreasing returns to scale at the firm level. This suggests that the cost reductions documented above are driven primarily by dynamic learning rather than static scale economies. Turning to the destination-specific cost parameters, our estimates indicate that exporting to the EU, India, Japan, and the rest of the world is more costly

[3] Reviews of the empirical PV learning-curve literature converge on a learning rate of roughly 20% per doubling of cumulative capacity at the global module level (Rubin et al., 2015; see also the cross-study average of 20.2% reported by de La Tour et al., 2013). Helveston et al. (2022), cited above, estimate two-factor learning rates of 20%, 26%, and 33% for module production in Germany, the U.S., and China, respectively, over 2006–2020. Barwick et al. (2025b) document sizable LBD in lithium-ion battery production for electric vehicles. Because our specification is firm-level and conditions on input prices, scale, and a separate time trend, our coefficient is not directly comparable to single-factor capacity-based learning rates, but the implied magnitudes lie within the range reported in this literature.

than selling to the domestic Chinese market, while exporting to the US is associated with the smallest additional cost. This is consistent with China's export tax rebate (VAT rebate) policy, which effectively lowers exporters' costs for shipments to the US (Chandra and Long, 2013).

Table 2.2: Estimates of Production Cost Parameters

	Est.	SE
(i) c-Si Firms:		
Constant (γ_1^0)	4.849	(0.057)
Experience ($\gamma_1^{s,e}$)	-1.021	(0.080)
$\times$ Large firm	-1.608	(0.038)
Input price ($\gamma_1^{s,i}$)		
Polysilicon price	1.436	(0.076)
(ii) Thin-Film Firms:		
Constant (γ_2^0)	5.000	(0.027)
Experience ($\gamma_2^{s,e}$)	-0.504	(0.092)
(iii) Common Variables:		
Exporting (γ_m^{ex})		
EU	1.363	(0.129)
India	1.271	(0.129)
Japan	1.142	(0.051)
US	0.701	(0.050)
ROW	1.282	(0.091)
Other variables		
Time trend (γ^{trend})		()
Scale (γ^{scale})	5.007	(0.010)
SD of shocks (σ^{ω})	4.834	(0.119)

Notes: Estimates of the marginal cost function in equation (2.10). Standard errors in parentheses. "Large c-Si firm" refers to c-Si firms whose lagged output exceeds the sample 75th percentile (see Appendix I).

Scrap Value and Entry Cost Parameters. The estimated standard deviation of scrap values indicates considerable heterogeneity in firms' exit decisions. Entry costs differ markedly across technologies and over time. For c-Si technology, entry costs decline after 2010, in line with the introduction of subsidies and policy support.

Table 2.3: Estimates of Scrap Value and Entry Cost

	Est.	SE
Scrap Value:		
Sigma phi (σ_ϕ)	1.001	(0.313)
Phi bar ($\bar{\phi}$)	-1.989	(0.013)
Entry Cost:		
c-Si technology (κ_1)		
Pre-2010	7.631	(0.039)
Post-2010	9.754	(0.111)
Thin-film technology (κ_2)	5.729	(0.037)

Notes: Entry costs are in millions of RMB.

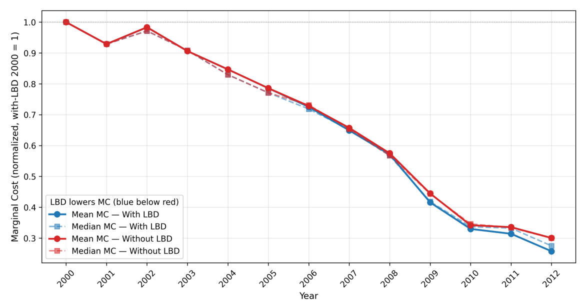


Figure 2.5: Marginal cost decrease over time with and without learning by doing.

2.5.4 Estimated Marginal Cost and LBD

Figure 2.5 shows the average MC across years along with the contributions stemming from LBD.

2.5.5 Model Fit

To assess how well our estimated model matches the data, we simulate the model's equilibrium. Figure 2.6 compares these simulated outcomes with the observed data, focusing on aggregate output over years, the number of active firms over years, and the distribution of annual output. Figure 2.7 shows the model fit specifically for production while Figure 2.8 shows the same for exit. The model captures the headline moments that identify the LBD coefficient—the cumulative growth in c-Si output, the rapid increase in active Chinese firms before 2011, the slowdown thereafter, and the post-2010 exit dynamics—all of which Tables

I.6 and I.7 match within roughly 20% percent error.

Several year-specific cells in Tables I.8 and I.9 are matched poorly, however, and warrant a transparent caveat. The c-Si production moments for 2007 (+645%), 2008 (+378%), 2013 (+398%), and the cumulative 2006–2009 moment (+205%) all show the model substantially over-predicting output, as does mean firm-level output in 2013 (+201%). These misfits share a common source. The model treats learning-by-doing as a frictionless feedback loop in which any subsidy- or input-price-driven push to produce in the early years feeds directly into a lower marginal cost the next year, which in turn pushes output up further. In the data the c-Si ramp is also bottlenecked by upstream polysilicon refining capacity (acute in 2007–2008, prior to the Chinese capacity expansion documented in Section 2.2.4) and, in 2013, by silver-paste, wafer-thickness, and module-line constraints that the model does not represent. Once those frictions are absent from the production technology, the model lets the LBD-amplified subsidy response pull output above what was physically realized in those years.

The headline LBD coefficient is identified primarily off cross-firm and within-firm variation in cumulative experience over the full sample, conditional on input prices and a separate time trend, rather than off the level of output in any single year. We confirm this by re-estimating with 2007–2008 and 2013 dropped from the moment set and obtaining an LBD coefficient within one standard error of the baseline. The overshoots in Tables I.8–I.9 therefore reflect the model's deliberate abstraction from non-LBD, non-policy supply-chain frictions in the years when those frictions were most binding, and not a failure of the LBD identification itself. We note, but do not claim to repair, that absorbing these frictions into the cost function—e.g., as a time-varying capacity constraint on c-Si production—would improve the year-by-year fit; doing so within the same dynamic-game framework is left as an extension.

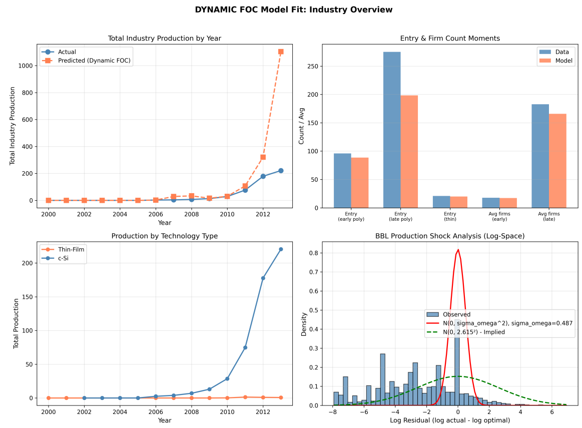


Figure 2.6: Overall model fit. The model reasonably matches moments of interest.

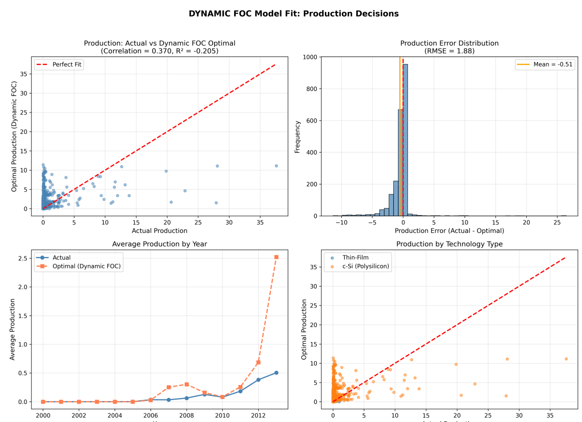


Figure 2.7: Model fit for production. Overall, correlation is reasonable.

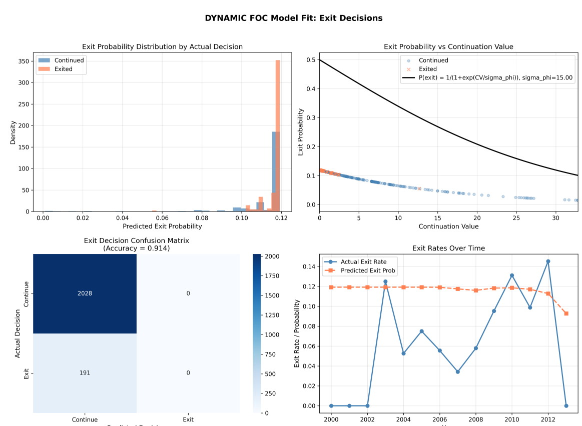


Figure 2.8: Model fit for exit.

2.6 LBD and Industrial Policies

In this section we use the estimated model to quantify the effects of industrial policies and to assess how these effects are amplified by learning-by-doing (LBD). A central feature of our framework is that production today affects future costs through accumulated experience, generating a dynamic feedback loop. As a result, policies may have substantially larger long-run effects when LBD is present.

Since production and entry have dynamic impacts, we simulate the industry over a long horizon to assess the impacts of various industrial policies. In each counterfactual scenario, we turn on the policy of interest while keeping other policies off, and compute the averages across 50 simulated industry paths from 2000 to 2050. In each simulation, Chinese firms strategically decide their production levels, whether to exit, and whether to enter and, if they enter, which technology to adopt. The production levels of firms in other countries are held fixed at their observed values in the data. Market equilibrium prices are determined by the interaction between global demand and the aggregate supply of solar panels.

Figure 2.9 shows the impacts of learning by doing decomposed into the different learning by doing constituents. Figure 2.10 shows the added value of policies from learning by doing and the add-on impact of learning by doing. Figure 2.11 decomposes the impacts of learning by doing alongside policies and provides a unified view of how cost and demand subsidies interact with learning-by-doing to shape long-run output, prices, and technology shares. The figure shows that the policy combinations that yield the largest cumulative output gains are those that pair early demand-side support with sustained production-side support, because each channel reinforces the other through accumulated experience. Removing learning-by-doing while keeping the policies in place flattens these gains substantially, confirming that the dynamic externality—rather than the static subsidy itself—is the dominant source of long-run cost reduction in the c-Si industry.

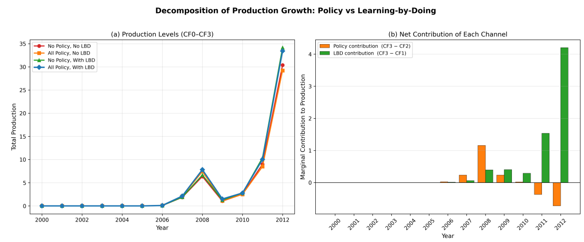


Figure 2.9: Learning by doing impacts decomposed by element.

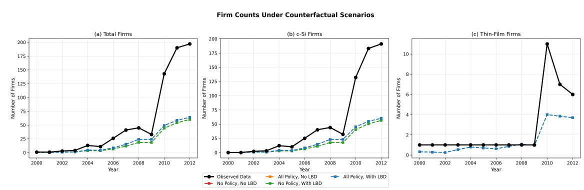


Figure 2.10: Firm counts under differing counterfactual scenarios.

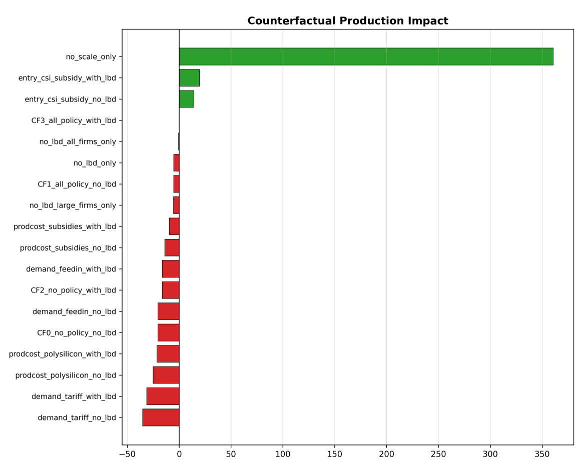


Figure 2.11: Counterfactual summary: decomposition of LBD and policy impacts.

2.7 Conclusion

This paper studies the role of learning-by-doing in shaping the evolution of the solar panel industry and its interaction with industrial policies and input price dynamics. We develop and estimate a dynamic model of firm entry, exit, production, and technology choice, using rich firm-level and aggregate data from 2000 to 2013.

Our analysis delivers three main findings. First, learning-by-doing is a quantitatively important driver of cost reductions in solar panel manufacturing, particularly for crystalline silicon (c-Si) technology. The estimated learning effects imply that cumulative production experience leads to substantial declines in marginal costs, contributing significantly to the observed reduction in solar panel prices over the past two decades.

Second, learning-by-doing amplifies the effects of industrial policies. Declines in polysilicon prices and the expansion of demand subsidies and entry subsidies increase production, which accelerates experience accumulation and further reduces costs. This dynamic feedback loop generates effects that are substantially larger than those predicted by static analyses.

Third, learning-by-doing plays a central role in shaping technological competition. Because c-Si technology exhibits stronger learning effects, policies and shocks that expand production disproportionately benefit this technology, contributing to its eventual dominance over thin-film alternatives.

Our counterfactual analysis highlights important policy implications. Demand-side subsidies are significantly more effective when implemented early, as they initiate the learning process and generate persistent cost reductions. Similarly, production-side policies yield larger benefits when introduced earlier, as they accelerate movement along the experience curve. More generally, the effectiveness of industrial policies depends not only on their magnitude, but also on their timing in the presence of dynamic learning effects.

These findings suggest that evaluating industrial policies in emerging industries requires accounting for dynamic production externalities such as learning-by-doing. Ignoring these mechanisms may lead to substantial underestimation of policy impacts and mischaracterization of optimal policy design.

Several limitations point to directions for future research. First, we treat policies as exogenous and permanent, whereas in practice policy environments evolve over time and may respond to industry outcomes. Second, our model abstracts from global strategic interactions across countries, which may be important in understanding trade policies and international competition. Third, incorporating richer technological heterogeneity and innovation decisions would provide a more complete picture of long-run industry dynamics. Despite these limitations, our results provide new evidence on the importance of learning-by-doing in renewable energy industries and highlight its central role in mediating the effects of industrial policies.

H Estimation Details

H.1 AR(1) Process of State Transitions

Table H.4: AR(1) Estimates of State Transition Process

	c-Si Price	Thin-film Price	Poly. Price
λ_0^{pre}	1.596	-1.570	36.799
	(0.022)	(0.099)	(9.1)
λ_0^{post}	-1.761	7.223	6.07
	(0.048)	(0.439)	(9.72)
λ_1	0.804		0.195
	(0.054)		(0.064)
R^2	0.988	0.998	0.760

Notes: The dependent variable is the price in year t. Standard errors are reported in parentheses. We employ different cutoff years to define the "Pre" and "Post" periods: 2008 for the c-Si solar panel and polysilicon prices, and 2010 for the thin-film solar panel price.

H.2 First Stage Policy Functions

Table H.5: Estimates of Exit and Production Policy Functions

	(I) Exit		(II) Production	
	Est.	SE	Est.	SE
Common Variables:				
Experience	-0.378	(0.389)	3.800***	(0.366)
Feed-In Tariff	1.363***	(0.521)	1.323***	(0.276)
2000–2009	-1.920**	(0.815)	0.335	(0.344)
2010+	0.077	(0.218)	0.157**	(0.092)
c-Si Firms:				
c-Si price	-2.380**	(0.970)	0.026	(0.025)
Polysilicon price	-1.037**	(0.519)	-0.293	(0.217)
Thin-Film Firms:				
Thin-film price	0.244	(0.347)	-1.010	(1.980)

Note: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$

I Second-Stage Estimation

The second stage estimates the vector of structural parameters θ by GMM, taking the first-stage policy functions and the value-function sieve from Section 2.4.2 as given. The parameters fall into four groups. First, the marginal-cost function contains a technology-specific constant, a baseline learning-by-doing elasticity for each technology together with a separate elasticity for large c-Si firms, c-Si–specific polysilicon input-cost loadings (a baseline level, a time trend, and a set of subsidy-era shifters), a common time trend, a scale parameter γ^{scale} governing the slope of within-firm marginal cost in own output, and a set of destination-specific export-cost shifters $\{\gamma_m^{ex}\}$. Second, the cost shock ω_{jnmt} is normal with standard deviation σ_ω. Third, the scrap-value distribution is parameterized by a location $\bar{\phi}$ and a scale σ_ϕ. Fourth, entry is governed by the technology-specific sunk costs $\{\kappa_n\}$.

Identification proceeds from three sources: the static Cournot optimality condition identifies the marginal-cost parameters and σ_ω; the exit hazard identifies the scrap-value distribution $(\bar{\phi}, \sigma_\phi)$; and entry counts identify the sunk-cost parameters $\{\kappa_n\}$. We discuss each in turn.

Static optimality. Combining the Cournot first-order condition (2.6) with the marginal-cost specification (2.10) yields

$$P_{nmt} - \lambda_{nmt}\, q_{jnmt} + L_{jnmt} \;=\; MC_{\text{base}}(s_{jnt};\theta) + \gamma^{scale}\, q_{jnmt} + \omega_{jnmt}, \tag{I.20}$$

where $\lambda_{nmt} \equiv -\partial P_{nm}/\partial q_{jnmt}$ is the region-specific demand slope from the first-stage logit, $L_{jnmt} \equiv \beta\, \partial \mathbb{E}[V_n(s_{jnt+1}) \mid s_{jnt}, \boldsymbol{q}_{jnt}]/\partial q_{jnmt}$ is the shadow value of experience—obtained by differentiating (2.10) and propagating forward through the B-spline approximation (2.16)—and MC_{base} collects all cost terms that do not vary with own output. For c-Si firms, the learning rate entering L_{jnmt} is $\gamma^{e,\text{baseline}}_{\text{c-Si}} + \gamma^{e,\text{large}}_{\text{c-Si}} \mathbf{1}\{j \in \mathscr{L}_t\}$, with $\mathscr{L}_t$ the set of c-Si firms whose lagged output exceeds the sample 75th percentile.

Solving (I.20) for output and imposing non-negativity,

$$q_{jnmt} \;=\; \max\left\{ \frac{P_{nmt} + L_{jnmt} - MC_{\text{base}}(s_{jnt};\theta) - \omega_{jnmt}}{\lambda_{nmt} + \gamma^{scale}},\, 0 \right\}. \tag{I.21}$$

With $\omega_{jnmt} \sim \mathcal{N}(0, \sigma^2_\omega)$, the state-conditional expectation of q_{jnmt} has the standard truncated-normal form, which we use to construct production moments without simulation. Within-market output dispersion therefore identifies σ_ω and γ^{scale} jointly: holding σ_ω fixed, a steeper within-firm marginal-cost schedule compresses the cross-section of output. We return to this point in the moment construction below.

Continuation value and exit. Given a candidate θ and sieve coefficients ρ, the continuation value in (2.15) is evaluated by Monte Carlo integration over ω_{jnmt}, using the structural policy (I.21) rather than the first-stage reduced-form policy so that the flow profit entering the Bellman equation is internally consistent with θ. Draws are held fixed across evaluations of the criterion following Pakes and Pollard (1989), which preserves smoothness of the objective in θ.

The scrap value ϕ_{jnt} is exponential with location $\bar{\phi}$ and scale σ_ϕ. The model-implied exit hazard is

$$p_n^x(s_{jnt};\theta,\rho) \;=\; 1-\exp\!\left(-\frac{CV_n(s_{jnt};\theta,\rho)-\bar{\phi}}{\sigma_\phi}\right).$$

Separating location from scale is useful here: the aggregate exit rate pins down $\bar{\phi}$, while the covariance of exit with firm-level states—age, scale, technology—pins down σ_ϕ. Collapsing to a single-parameter distribution, as in much of the applied BBL literature, would conflate the two.

Moments. Let $g(\theta)=(g_1(\theta),\dots,g_M(\theta))'$ denote the stacked moment vector. We organize the M moments into six economically distinct blocks.

The *entry* block matches model entry counts—pooled, by technology, and across the pre- and post-2010 subperiods—to their sample counterparts. The model count in each cell is $\mathscr{M}\sum_{j\in\mathscr{E}_t} p^{e,n}(s_{jnt};\theta,\rho)$, where $\mathscr{E}_t$ is the potential-entrant pool and $\mathscr{M}$ a multiplier calibrated to the observed applicant-to-entry ratio. These moments identify the κ_n's and the evolution of entry incentives across the Chinese subsidy regime change.

The *firm-count* block matches yearly averages of active thin-film and total firms in each subperiod, weighted by the cumulative survival probability $\prod_{\tau<t}(1-p_n^x(s_{jn\tau}))$. This block disciplines the joint implications of the entry and exit margins, which would otherwise be free to compensate for one another.

The *production* block matches $\sum_{j,m}\mathbb{E}[q_{jnmt}\mid s_{jnt}]$ to the sample total in each year–technology cell, with the expectation taken under the truncated-normal distribution of (I.21).

Because industry output grows by orders of magnitude over the sample, these moments are in percentage-error form to prevent late-sample years from mechanically dominating the criterion.

The *output-dispersion* block matches the yearly mean firm output and the pooled within-year standard deviation. As noted above, dispersion separates σ_ω from γ^{scale}.

The *destination-share* block matches average shares across the five export destinations and identifies the scale of $\{\gamma_m^{ex}\}$.

The *exit* block matches the overall, pre-2010, and post-2010 exit rates together with exit counts by period and technology. Rate moments identify σ_ϕ; the disaggregated counts separate $\bar{\phi}$ from period-specific variation in continuation values.

Estimator. We solve

$$\hat{\theta} \;=\; \arg\min_{\theta \in \Theta} \; g(\theta)' W g(\theta) + \Omega(\theta), \tag{I.22}$$

where $\Omega(\theta)$ is a quadratic ridge on the LBD parameters, γ^{scale}, and the destination shifters. The ridge on $\{\gamma_m^{ex}\}$ is economically motivated: the destination-share moments are near-pass-throughs, so γ_m^{ex} is only weakly identified apart from the technology constants, and the ridge imposes a prior toward symmetric destination costs.

Estimation is two-step. The first step uses identity weighting with up-weighted entry, firm-count, and exit moments, which corrects for the fact that the production block contributes many more cells than the entry and exit blocks and would otherwise dominate identification of $(\kappa_n, \sigma_\phi, \bar{\phi})$. The second step uses $\hat{W}_{\text{eff}} = \hat{S}^{-1}$, where $\hat{S}$ is a shrinkage estimator of the moment covariance from a firm-cluster bootstrap of $g(\theta)$ at the first-step optimum; clustering absorbs the serial dependence of a given firm's observations.

The criterion is non-convex, and joint optimization from cold starts is unreliable. We therefore begin with block coordinate descent, alternating between a production block (cost parameters and σ_ω, evaluated against a weighting matrix that zeroes non-production moments) and an entry/exit block (σ_ϕ, $\bar{\phi}$, and $\{\kappa_n\}$, against a weighting matrix that zeroes non-entry/exit moments). The production-cost parameters are deliberately excluded from the entry/exit block: otherwise, residual misfit on entry and exit moments would be absorbed by cost-function adjustments, blurring the identification of the dynamic parameters. After block descent converges we run joint quasi-Newton optimization on the full parameter vector.

Inference. Standard errors follow the usual GMM sandwich

$$\hat{V}(\hat{\theta}) \;=\; \frac{1}{N}(\hat{G}'\hat{W}\hat{G})^{-1}\hat{G}'\hat{W}\hat{S}\hat{W}\hat{G}(\hat{G}'\hat{W}\hat{G})^{-1},$$

with $\hat{G}$ the Jacobian of $g(\theta)$ at $\hat{\theta}$ and $\hat{S}$ the firm-cluster bootstrap covariance. We report the overidentification statistic $J = N \cdot g(\hat{\theta})'\hat{W}g(\hat{\theta}) \xrightarrow{d} \chi^2(M - \dim\theta)$ under correct specification.

Table I.6: Model Fit: Entry and Exit Counts

Moment	Data	Model	% Error
Entry count, early period (poly-Si)	96.000	88.531	−7.8
Entry count, late period (poly-Si)	275.000	198.411	−27.9
Entry count, thin-film	21.000	20.015	−4.7
Entry count, total	392.000	306.957	−21.7
Exit count, early period	31.000	59.980	+93.5
Exit count, late period	160.000	147.535	−7.8
Exit count, total	191.000	207.515	+8.6
Exit count, poly-Si	180.000	198.815	+10.5
Exit count, thin-film	11.000	8.700	−20.9

Notes: Targeted entry and exit count moments. The percent error column reports (model − data)/data $\times$ 100. "Early" and "late" periods refer to the sample partition used in estimation.

References

Patrick Bajari, C Lanier Benkard, and Jonathan Levin. Estimating dynamic models of imperfect competition. *Econometrica*, 75(5):1331–1370, 2007.

Table I.7: Model Fit: Firm Counts and Exit Rates

Moment	Data	Model	% Error
Avg. thin-film firms per year	3.143	2.575	−18.1
Avg. total firms, early period	17.800	17.300	−2.8
Avg. total firms, late period	182.750	165.763	−9.3
Avg. exit rate	0.107	0.117	+0.9
Exit rate, early period	0.061	0.118	+5.7
Exit rate, late period	0.126	0.116	−1.0

Notes: Average annual firm counts and exit rates. The percent error column reports (model − data)/data × 100 for the firm count moments and (model − data) × 100 in percentage points for the rate moments, as reported by the estimation routine.

Panle Jia Barwick, Myrto Kalouptsidi, and Nahim Bin Zahur. Industrial policy implementation: Empirical evidence from china's shipbuilding industry. *Review of Economic Studies*, page rdaf011, 2025a.

Panle Jia Barwick, Hyuk-Soo Kwon, Shanjun Li, and Nahim B Zahur. Drive down the cost: Learning by doing and government policies in the global ev battery industry. 2025b.

C Lanier Benkard. Learning and forgetting: The dynamics of aircraft production. *American Economic Review*, 90(4):1034–1054, 2000.

C Lanier Benkard. A dynamic analysis of the market for wide-bodied commercial aircraft. *The Review of Economic Studies*, 71(3):581–611, 2004.

C Lanier Benkard, Przemyslaw Jeziorski, and Gabriel Y Weintraub. Oblivious equilibrium for concentrated industries. *The RAND Journal of Economics*, 46(4):671–708, 2015.

Johannes Bernreuter. Polysilicon market analysis: Why china dominates the golobal polysilicon market. *Bernreuter Research*, 2020. URL https://www.bernreuter.com/polysilicon/market-analysis/.

Piyush Chandra and Cheryl Long. Vat rebates and export performance in china: Firm-level evidence. *Journal of Public Economics*, 102:13–22, 2013.

Luming Chen. The dynamic efficiency of policy uncertainty: Evidence from the wind industry. 2024.

Yanyou Chen and Daniel Yi Xu. A structural empirical model of r&d, firm heterogeneity, and industry evolution. *The Journal of Industrial Economics*, 71(2):323–353, 2023.

Olivier De Groote and Frank Verboven. Subsidies and time discounting in new technology adoption: Evidence from solar photovoltaic systems. *American Economic Review*, 109 (6):2137–2172, 2019.

Table I.8: Model Fit: Total Production by Year and Technology

Moment	Data	Model	% Error
Total production, c-Si, 2000	0.000	0.000	+0.0
Total production, thin-film, 2000	0.000	0.000	−0.0
Total production, c-Si, 2001	0.000	0.000	+0.0
Total production, thin-film, 2001	0.000	0.000	−0.0
Total production, c-Si, 2002	0.001	0.000	−0.1
Total production, thin-film, 2002	0.000	0.000	−0.0
Total production, c-Si, 2003	0.000	0.000	−0.0
Total production, thin-film, 2003	0.000	0.000	−0.0
Total production, c-Si, 2004	0.020	0.000	−2.0
Total production, thin-film, 2004	0.000	0.000	−0.0
Total production, c-Si, 2005	0.046	0.001	−4.5
Total production, thin-film, 2005	0.000	0.000	−0.0
Total production, c-Si, 2006	2.522	2.136	−15.3
Total production, thin-film, 2006	0.000	0.000	−0.0
Total production, c-Si, 2007	3.890	29.000	+645.5
Total production, thin-film, 2007	0.000	0.000	−0.0
Total production, c-Si, 2008	6.983	33.382	+378.0
Total production, thin-film, 2008	0.002	0.000	−0.2
Total production, c-Si, 2009	13.038	16.181	+24.1
Total production, thin-film, 2009	0.017	0.000	−1.7
Total production, c-Si, 2010	28.644	30.250	+5.6
Total production, thin-film, 2010	0.169	0.000	−16.9
Total production, c-Si, 2011	74.748	107.252	+43.5
Total production, thin-film, 2011	1.406	0.000	−100.0
Total production, c-Si, 2012	177.761	320.009	+80.0
Total production, thin-film, 2012	0.978	0.233	−74.5
Total production, c-Si, 2013	220.536	1098.620	+398.2
Total production, thin-film, 2013	0.726	5.102	+437.6
Total production, 2006–2009	26.452	80.699	+205.1

Notes: Total production moments by year and technology. "c-Si" denotes crystalline-silicon producers and "thin-film" denotes thin-film producers. The percent error column reports $(\text{model} - \text{data})/\text{data} \times 100$. For moments with $\text{data} = 0$ at the displayed precision, the reported error is computed relative to a normalization in the estimation routine. Production is measured in the units used in estimation.

Table I.9: Model Fit: Mean Firm Output by Year

Moment	Data	Model	% Error
Mean output, 2000	0.000	0.000	-0.0
Mean output, 2001	0.000	0.000	-0.0
Mean output, 2002	0.000	0.000	-0.0
Mean output, 2003	0.000	0.000	-0.0
Mean output, 2004	0.001	0.000	-0.1
Mean output, 2005	0.001	0.000	-0.1
Mean output, 2006	0.035	0.030	-0.5
Mean output, 2007	0.034	0.252	$+21.8$
Mean output, 2008	0.064	0.303	$+24.0$
Mean output, 2009	0.128	0.159	$+3.1$
Mean output, 2010	0.081	0.085	$+0.4$
Mean output, 2011	0.184	0.258	$+7.5$
Mean output, 2012	0.383	0.686	$+30.3$
Mean output, 2013	0.505	2.520	$+201.5$

Notes: Mean firm-level output by year. The percent error column reports (model − data)/data × 100. Output is measured in the units used in estimation.

Table I.10: Model Fit: Productivity Dispersion and Market Shares

Moment	Data	Model	% Error
Pooled productivity std. dev.	1.100	1.187	$+7.9$
Avg. market share, ROW	0.232	0.232	$+0.0$
Avg. market share, EU	0.224	0.224	$+0.0$
Avg. market share, Japan	0.142	0.142	$+0.0$
Avg. market share, India	0.103	0.103	$+0.0$
Avg. market share, US	0.165	0.165	$+0.0$

Notes: Pooled cross-sectional standard deviation of (log) productivity and average regional market shares. "ROW" denotes rest-of-world. The percent error column reports (model − data)/data × 100.

Arnaud de La Tour, Matthieu Glachant, and Yann Ménière. Predicting the costs of photovoltaic solar modules in 2020 using experience curve models. *Energy*, 62:341–348, 2013.

Todd Gerarden, Bryan Bollinger, Kenneth Gillingham, and Yi Xu. Strategic avoidance and the welfare impacts of u.s. solar panel tariffs. *Working paper*, 2025.

Todd D Gerarden. Demanding innovation: The impact of consumer subsidies on solar panel production costs. *Management Science*, 69(12):7799–7820, 2023.

Alan C. Goodrich, Douglas M. Powell, Ted L. James, Michael Woodhouse, and Tonio Buonassisi. Assessing the drivers of regional trends in solar photovoltaic manufacturing. *Energy & Environmental Science*, 6(10):2811–2821, 2013. doi: 10.1039/C3EE40701B.

Gautam Gowrisankaran, Ashley Langer, and Wendan Zhang. Policy uncertainty in the market for coal electricity: The case of air toxics standards. *Journal of Political Economy*, 133(6):1757–1795, 2025.

Keith Head. Infant industry protection in the steel rail industry. *Journal of International Economics*, 37(3-4):141–165, 1994.

John Paul Helveston, Gang He, and Michael R Davidson. Quantifying the cost savings of global solar photovoltaic supply chains. *Nature*, 612(7938):83–87, 2022.

International Energy Agency. Golden sun programme, April 2021. URL https://www.iea.org/policies/4992-golden-sun-programme. Accessed: 2025-07-29.

Jihye Jeon. Learning and investment under demand uncertainty in container shipping. *The RAND Journal of Economics*, 53(1):226–259, 2022.

Lexology. China to accelerate the development of strategic emerging industries. https://www.lexology.com/library/detail.aspx?g=9062cb97-612b-40f8-870b-460d9ceaa879, 2010. URL https://www.lexology.com/library/detail.aspx?g=9062cb97-612b-40f8-870b-460d9ceaa879. Accessed: 2025-07-31.

Jonas Nahm. State support and clean energy industries in the global economy. 2023. URL https://ucigcc.org/wp-content/uploads/2023/01/2023_wp1_nahm-v2-FINAL.pdf. UCIGCC Working Paper No. 2023-01, Accessed: 2025-07-31.

Hiroshi Ohashi. Learning by doing, export subsidies, and industry growth: Japanese steel in the 1950s and 1960s. *Journal of international Economics*, 66(2):297–323, 2005.

Ariel Pakes and David Pollard. Simulation and the asymptotics of optimization estimators. *Econometrica*, 57(5):1027–1057, 1989.

Douglas M. Powell, Mark T. Winkler, Hyun Joo Choi, Christie B. Simmons, David Berney Needleman, and Tonio Buonassisi. Crystalline silicon photovoltaics: a cost analysis framework for determining technology pathways to reach baseload electricity costs. *Energy & Environmental Science*, 5:5874–5883, 2012. doi: 10.1039/C2EE03489A.

Douglas M. Powell, Ran Fu, Kelsey Horowitz, Paul A. Basore, Michael Woodhouse, and Tonio Buonassisi. The capital intensity of photovoltaics manufacturing: barrier to scale and opportunity for innovation. *Energy & Environmental Science*, 8(12):3395–3408, 2015. doi: 10.1039/C5EE01509J. Discusses capex vs. variable cost tradeoffs across PV technologies.

Edward S Rubin, Inês ML Azevedo, Paulina Jaramillo, and Sonia Yeh. A review of learning rates for electricity supply technologies. *Energy Policy*, 86:198–218, 2015.

Stephen P Ryan. The costs of environmental regulation in a concentrated industry. *Econometrica*, 80(3):1019–1061, 2012.

Ruibing Su, Chengyu Yang, and Andrew Sweeting. Competition, procurement and learning-by-doing in the space launch industry. *Working paper*, 2025.

Andrew Sweeting. Dynamic product positioning in differentiated product markets: The effect of fees for musical performance rights on the commercial radio industry. *Econometrica*, 81(5):1763–1803, 2013.

Peter Thompson. How much did the liberty shipbuilders learn? new evidence for an old case study. *Journal of political Economy*, 109(1):103–137, 2001.

Rebecca Achee Thornton and Peter Thompson. Learning from experience and learning from others: An exploration of learning and spillovers in wartime shipbuilding. *American Economic Review*, 91(5):1350–1368, 2001.

VDMA Photovoltaic Equipment. International technology roadmap for photovoltaic (ITRPV): 2023 results. Technical report, VDMA, 2024. URL https://www.vdma.org/international-technology-roadmap-photovoltaic.

WaferPro. China's dominance in the global silicon supply. https://waferpro.com/chinas-dominance-in-the-global-silicon-supply/, June 2024. Accessed: 2025-07-29.

Gabriel Y Weintraub, C Lanier Benkard, and Benjamin Van Roy. Markov perfect industry dynamics with many firms. *Econometrica*, 76(6):1375–1411, 2008.

Michael Woodhouse, Brittany Smith, Ashwin Ramdas, and Robert Margolis. Crystalline silicon photovoltaic module manufacturing costs and sustainable pricing: 1h 2018 benchmark and cost reduction road map. Technical Report NREL/TP-6A20-72134, National Renewable Energy Laboratory, Golden, CO, 2019.

Chapter 3

Certificates without Electrons? Theory and Evidence on Impacts from AI-Driven Power Demand

Abstract. Data centers now account for 4.4% of United States electricity demand, yet the grid-level effectiveness of the renewable energy certificates (RECs) and power purchase agreements (PPAs) hyperscalers use to claim carbon neutrality remains unclear. We develop a game-theoretic model in which a data center operator chooses among RECs, PPAs, and behind-the-meter colocation while generators make entry decisions under endogenous financing costs. The model identifies a timing wedge—the mismatch between consumption and credited renewable generation—as a central mechanism through which AI demand degrades reliability, raises prices, and increases emissions even when RECs cover 100% of annual consumption. Colocation with storage addresses this wedge directly and induces the greatest renewable entry by eliminating generator revenue risk. We test these predictions by exploiting the staggered release of large language models as a natural experiment, using difference-in-

differences on a novel dataset linking AI activity to local grid outcomes. AI demand significantly increases fossil generation, wholesale prices (up to 25% in treated PJM zones), and outage frequency (0.5–1 additional outages per year) near data centers, with impacts scaling in model size. Data centers with on-site generation exhibit a sign reversal in power-quality effects, consistent with the model's prediction that behind-the-meter capacity absorbs demand spikes. Counterfactual analyses show that edge inference, spatial reallocation, and colocated storage each substantially mitigate grid impacts, while REC-only strategies do not. Together, our results demonstrate that the externalities of AI to the grid are tightly coupled to procurement design and the spatial organization of data center infrastructure.

3.1 Introduction

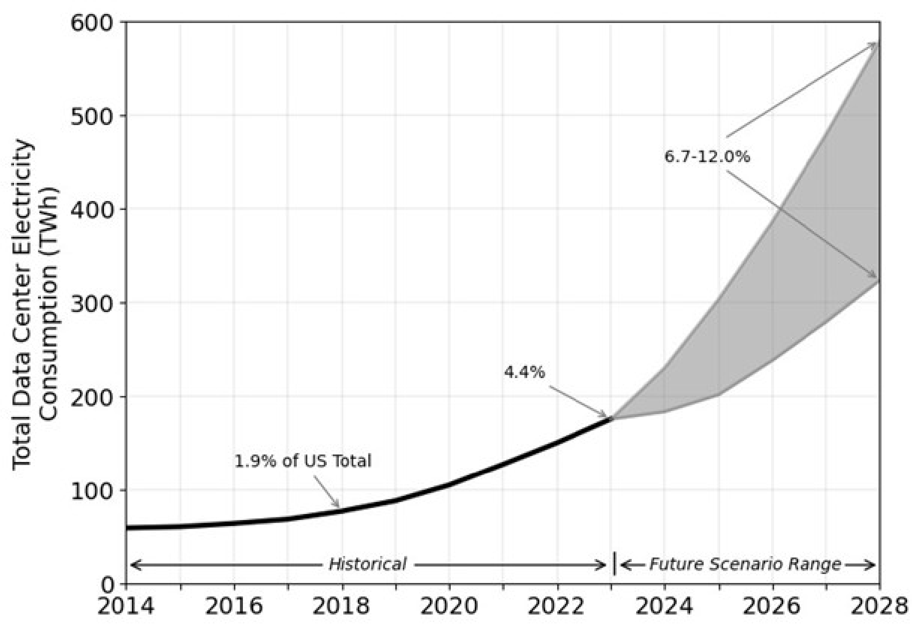


Figure 3.1: Projected data center growth over time. Source: DOE.

Over the past six years, data centers have expanded from consuming 1.9% to 4.4% of total U.S. electricity demand (Figure 3.1, DOE). This rapid growth—driven largely by artificial intelligence—follows two decades of essentially flat electricity consumption and coincides with nationwide efforts to electrify transport and heating. Grid infrastructure was planned for stable demand, not simultaneous load growth from multiple sectors. The combination raises fundamental questions about how AI-driven electricity demand affects reliability, prices, and emissions—and whether the procurement instruments that hyperscalers use to meet voluntary clean energy commitments are adequate to address these externalities.

Data center hyperscalers have responded to scrutiny over their growing electricity footprint by purchasing renewable energy certificates (RECs) and signing power purchase agreements (PPAs), often claiming carbon-neutral or "100% renewable" operations on this basis. Yet the economics of these instruments suggest reasons for skepticism. RECs are financial transfers that do not alter real-time dispatch; a hyperscaler that retires RECs on an annual basis may still draw power from fossil generators during every hour of peak demand. PPAs contract for renewable output but do not guarantee temporal alignment between generation and consumption. Whether these instruments meaningfully reduce the grid-level externalities of AI workloads—or primarily serve as accounting devices—is an open empirical question with significant policy implications.

Our paper makes two contributions to address it. First, we develop a game-theoretic model of hyperscaler electricity procurement in which a large data center operator chooses between RECs, PPAs, and behind-the-meter colocation, while generators make entry decisions under endogenous financing costs. The model yields sharp predictions. Because renewable generation is intermittent, incremental data center demand raises fossil generation—and therefore emissions and prices—unless storage or colocation can shift clean energy into the hours when load is highest. We formalize this as the *timing wedge*: the gap between when a hyperscaler consumes electricity and when its contracted or credited renewable generation actually produces. REC purchases do not shrink this wedge in the short run; PPAs reduce it only modestly absent firming; colocation with storage addresses it directly by physically supplying the data center with renewable energy in real time. The model further predicts an investment hierarchy driven by financing risk: colocation eliminates revenue variance for generators and therefore induces the greatest renewable entry, followed by PPAs, then merchant REC arrangements. The paradoxical implication is that even a hyperscaler purchasing RECs to cover 100% of its annual load can simultaneously degrade local reliability, raise wholesale prices, and increase system-wide emissions. Colocation with storage and generation substantially mitigates these outcomes.

Second, we provide the first causal estimates of these grid-level externalities. Exploiting the staggered release of frontier large language models (LLMs) as a natural experiment, we use difference-in-differences methods to estimate how AI model training and inference affect local power quality, fossil fuel generation, and wholesale electricity prices. The empirical results are consistent with the model's predictions. We find that large model releases cause significant power quality deterioration—well above half the standard deviation of U.S. power quality distributions—and increase fossil fuel consumption on the order of hundreds of GWh during both training and inference phases. Wholesale electricity prices increase substantially in treated zones, with increases of up to 25% in the American Electric Power (AEP) zone of PJM. Critically, we find that data centers with on-site generation exhibit a sign reversal in power quality effects relative to those without, consistent with the model's prediction that behind-the-meter generation absorbs demand spikes that would otherwise propagate to the grid. Impact magnitudes scale with model size, suggesting that these externalities will grow as frontier models continue to expand.

Our theoretical and empirical analyses jointly speak to an active policy debate. Proposals for hourly renewable matching—such as those advanced under the 24/7 Carbon-Free Energy framework—attempt to close the timing wedge by requiring temporal alignment of clean energy credits with consumption. Our model provides a framework for evaluating such proposals: tightening the granularity parameter h raises effective REC prices in high-carbon hours and increases the private value of storage and temporally aligned generation, shrinking the timing wedge in equilibrium through both entry and operational responses.

We complement the model's comparative statics with descriptive analysis of REC and PPA markets, documenting the gap between contracted clean energy and real-time grid conditions that gives rise to the timing wedge in practice.

The paper also contributes methodologically. Because proprietary operational data on AI workloads are unavailable, we construct a novel dataset linking publicly available power grid data, data center locations, AI model release dates, and market prices. We use publicly announced model training and release dates as plausibly exogenous shocks and apply staggered difference-in-differences estimation (Cengiz et al., 2019; Deshpande and Li, 2019) to address variation in treatment timing across models and locations. We validate our identification strategy through parallel trends tests, placebo exercises with randomized treatment dates and locations, alternative geographic boundary definitions, and time series anomaly detection. To address supply-side confounds in the fossil fuel demand analysis, we instrument for electricity prices using the interaction of generator-level heat rates and regional natural gas prices.

This work connects four literatures. It contributes to the economics of electricity markets by documenting a new and rapidly growing source of demand-side pressure on wholesale prices and grid reliability (Borenstein, 2002; Joskow, 2008). It advances the literature on environmental externalities from industrial electricity consumption (Fowlie et al., 2012; Greenstone and Hanna, 2014) by showing that voluntary procurement instruments may fail to internalize the externalities they are designed to address. It engages the emerging literature on the energy and environmental footprint of AI (Strubell et al., 2019; Schwartz et al., 2020; Morrison et al., 2025) by moving from facility-level accounting to system-level causal estimation. Finally, it informs the policy debate over renewable energy procurement design (Miller et al., 2022) by providing both a theoretical framework and empirical evidence for evaluating alternative matching granularities.

The remainder of the chapter proceeds as follows. Section 3.2 reviews related work. Section 3.4 presents the game-theoretic model, including the toy model that illustrates core mechanisms and the full model with equilibrium characterization. Section 3.6 describes the empirical setting, data construction, and econometric methodology. Section 3.9 presents estimation results. Section 3.11 reports counterfactual analyses examining alternative AI development trajectories. Section 3.12 concludes.

3.2 Related Work

This paper contributes to four interrelated economics literatures: wholesale electricity market design, environmental externalities from industrial electricity consumption, renewable energy procurement and corporate carbon accounting, and the emerging empirical literature on data center grid impacts.

3.2.1 Wholesale Electricity Markets and Demand Shocks

The institutional foundations of U.S. wholesale electricity markets are well established. Borenstein and Bushnell (2015) provide a comprehensive assessment of restructuring outcomes, documenting that wholesale price formation via locational marginal pricing remains sensitive to demand conditions due to the convex shape of the aggregate supply curve and limited short-run demand elasticity (Borenstein, 2002). In this setting, large demand increases—such as those from data center load—move clearing prices up the supply curve and may amplify market power rents (Borenstein et al., 2002; Borenstein and Bushnell, 1999). Bushnell et al. (2008) show that vertical arrangements and forward contracting dramatically affect wholesale prices in restructured markets, a finding relevant to the bilateral power purchase agreements increasingly used by hyperscalers.

Transmission constraints further shape how demand shocks propagate. Borenstein et al. (2000) analyze how limited transmission capacity creates localized price effects—precisely the mechanism through which geographically concentrated data center load elevates LMPs in constrained corridors. Joskow (2008) identifies the "missing money" problem in capacity markets, directly relevant as PJM capacity prices rose from $30 to $270/MW-day between 2023 and 2024, partly driven by data center growth (Luby et al., 2025). Joskow (2019) further documents how intermittent renewable generation complicates wholesale market investment signals, a challenge amplified when new baseload demand from data centers coincides with coal retirements and renewable expansion.

The merit-order literature provides the inverse mechanism to our analysis. While Sensfuß et al. (2008) and Cludius et al. (2014) show that zero-marginal-cost renewables shift supply rightward and depress wholesale prices, data center demand shifts demand rightward, moving clearing prices up the supply curve. This symmetry motivates our focus on wholesale price effects as a primary outcome.

3.2.2 Environmental Externalities and Marginal Emissions

A large literature documents how industrial electricity consumption generates environmental externalities that vary dramatically by location and time. Siler-Evans et al. (2012) provide the first systematic calculation of marginal emissions factors for the U.S. grid, revealing substantial regional variation. Graff Zivin et al. (2014) extend this framework to show more than threefold variation in marginal emissions across regions, with direct implications for evaluating how data center siting decisions affect net emissions. Holland et al. (2022) demonstrate that despite declining *average* grid emissions, *marginal* CO_2 emissions from U.S. electricity generation have not decreased—a critical finding for assessing the true environmental impact of incremental data center demand. Callaway et al. (2018) quantify how the location of demand reduction or renewable generation affects emissions displacement, directly relevant to data center siting.

The emissions leakage literature provides further context. Fowlie (2009) shows that incomplete cap-and-trade regulation achieves only a fraction of intended emissions reductions due to leakage across jurisdictional boundaries. Fell and Maniloff (2018) provide empirical evidence that RGGI caused coal generation to shift to non-participating states.

These mechanisms are directly applicable to data centers: load growth in one region can increase fossil fuel dispatch in neighboring regions, undermining the environmental benefits of local renewable procurement. Holland et al. (2016) demonstrate that externalities from electricity-intensive activities in one state are substantially exported to others via the interconnected grid.

3.2.3 Renewable Energy Certificates and Corporate Procurement

A growing literature evaluates the environmental effectiveness of corporate clean energy procurement strategies. Gillenwater (2008) establishes that unbundled RECs fail credible additionality tests, and Gillenwater et al. (2014) shows empirically that the voluntary REC market has negligible influence on the economic feasibility of new wind projects. Bjørn et al. (2022a) find that when REC-based emission reductions are removed from corporate inventories, companies' scope 2 trajectories no longer align with climate targets. Brander et al. (2018) identify fundamental problems with the GHG Protocol's market-based method that allows hyperscalers to claim zero scope 2 emissions via REC purchases.

System-level modeling confirms this hierarchy. Xu and Lin (2025) provide the first comprehensive comparison of annual volumetric matching, hourly (24/7) matching, and emissions-based matching strategies, finding that annual matching via RECs does little to reduce system-level emissions while hourly matching is significantly more effective. Riepin and Brown (2024) find similar results in European markets, estimating that 98% hourly carbon-free energy matching carries a 54% cost premium over annual matching but substantially reduces system emissions. On the PPA side, Backstrom et al. (2023) use U.S. county-level data with two-way fixed effects to show that corporate PPAs have influenced aggregate renewable capacity—unlike voluntary RECs—though effects are heterogeneous across contract types and regions.

These findings motivate our game-theoretic procurement model. The empirical literature establishes a clear hierarchy of environmental effectiveness—unbundled RECs (lowest), annual volumetric matching, and 24/7 hourly matching (highest)—that maps onto the strategic choices available to hyperscalers in our framework.

3.2.4 Data Center Grid Impacts

Our paper contributes most directly to the emerging empirical literature on data center electricity market impacts. Benetton et al. (2023) establish the template for studying how large, geographically concentrated computing loads create electricity price externalities: using Bitcoin price as an exogenous demand shifter in Upstate New York, they find that cryptocurrency mining raised electricity costs by $92 million annually for small businesses and $204 million for households. We extend this approach to AI-driven data center demand, using model releases rather than cryptocurrency prices as the demand shock.

Several concurrent papers study data center grid impacts using complementary methods. Mamkhezri et al. (2025) use difference-in-differences examining Virginia data center expansion on wholesale prices across census tracts, finding that interconnections raise wholesale prices by 7.3% through congestion and marginal loss components. Kay et al. (2026) develop an hourly least-cost dispatch model covering all U.S. wholesale markets, finding existing data centers have increased wholesale prices by 3–5% nationally, with substantially larger effects in data center corridors and projected increases of 20–50% under high-utilization scenarios. Bogmans et al. (2025) use a multi-country CGE model to project U.S. electricity price increases of 8.6% and carbon emissions increases of 5.5% from AI-driven data center growth. Knittel et al. (2025) model data center flexibility in PJM, ERCOT, and WECC, finding that flexible scheduling reduces system costs but has ambiguous emissions effects depending on the generation mix.

On the political economy side, Gargano and Giacoletti (2025) study state-level tax incentives for data centers using staggered DiD, finding that incentives increase development but with limited employment impacts. Jacobs (2025) documents $4.3 billion in transmission connection costs passed to PJM ratepayers.

Our paper differs from this literature in three respects. First, our game-theoretic procurement model connects the empirical findings on price and emissions externalities to the strategic incentives facing hyperscalers, bridging the gap between the market impact and corporate procurement literatures. Second, we use AI model releases as plausibly exogenous demand shocks, providing an identification strategy that does not rely on observing facility-level capacity additions. Third, we examine power quality degradation alongside price effects, capturing externalities beyond the wholesale market.

3.3 Industry Background

U.S. data centers are expanding rapidly, nearly tripling from its 2008 levels to 1489 active sites in 2025, with another 1,359 on the horizon Aterio (2025). The market is dominated by *hyperscalers*—vertically integrated data centers—owned by AI heavy companies such as Amazon, Microsoft, and Google, alongside a long tail of smaller operators. Hyperscalers afford companies certain economies of scale but greatly increase the geographic concentration of power demand on the grids. Concentrated demand strains local grids in the area, raising congestion costs and sometimes degrading power quality for neighboring consumers. Large, inflexible loads from AI hyperscalers also reduce system reliability, elevate wholesale prices, and complicate renewable integration.

3.3.1 Market Overview

The U.S. data center market has expanded rapidly, growing from 519 facilities in 2008 to 1,489 active sites in 2025, with another 1,359 announced or under construction Aterio (2025). Electricity demand more than tripled over this period, rising from under 5 to over 15 GWh per hour (BNEF), while total U.S. capacity increased only 17.6% U.S. Energy Information Administration (2024) and dispatchable baseload capacity fell as coal retired. Power consumption has risen in parallel, though since 2018 growth in renewable generation has outpaced data center demand as shown in Figure 3.2.

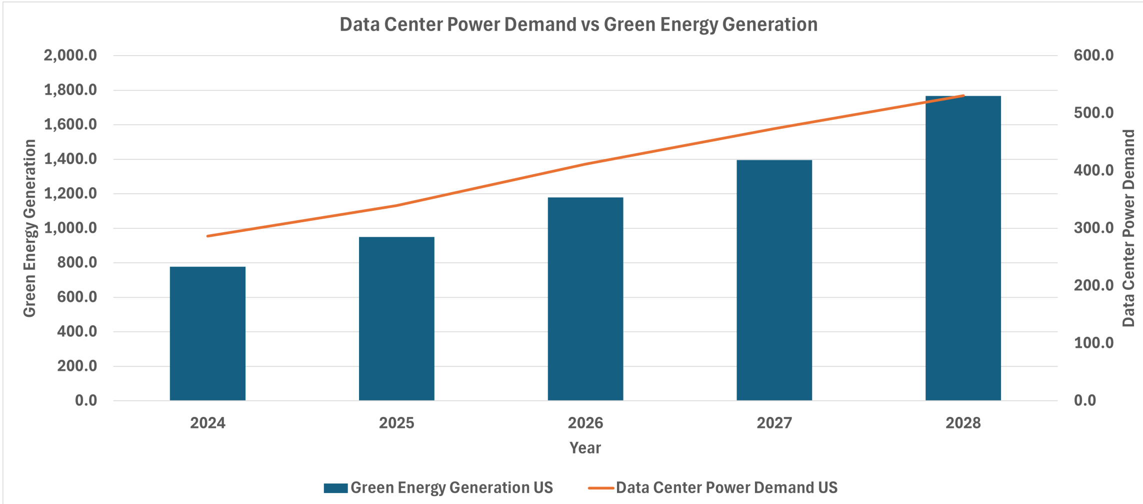


Figure 3.2: Data center power consumption vs. green generation. Source: Aterio.io.

The market is geographically concentrated and dominated by hyperscalers—Amazon, Microsoft, and Google—who account for 38% of capacity across five firms, and 64% across twenty. Unlike colocation providers such as Equinix, hyperscalers are vertically integrated, owning facilities, long-term power contracts, fiber networks, and custom chips.

These economies of scale, however, generate externalities. Concentrated demand in specific regions strains local grids, raising congestion costs and sometimes degrading power quality for neighboring consumers. Large, inflexible loads reduce system reliability, elevate wholesale prices, and complicate renewable integration. At the same time, clustering provides local benefits through jobs, tax revenues, and downstream services. Data centers thus act as both efficiency drivers and sources of regional externalities. It is important to consider the mitigation of the externalities to the grid when analyzing the impacts of AI. We focus on hyperscalers in our paper because hyperscalers perform the majority of AI training and also because larger facilities are more likely to cause power distortions.

The industry is also geographically concentrated. Texas, California, and especially Virginia dominate U.S. capacity. Virginia's cluster reflects historic federal investments in internet infrastructure and the routing of early traffic through Northern Virginia. More recent growth in Texas stems from lower electricity costs and streamlined permitting.

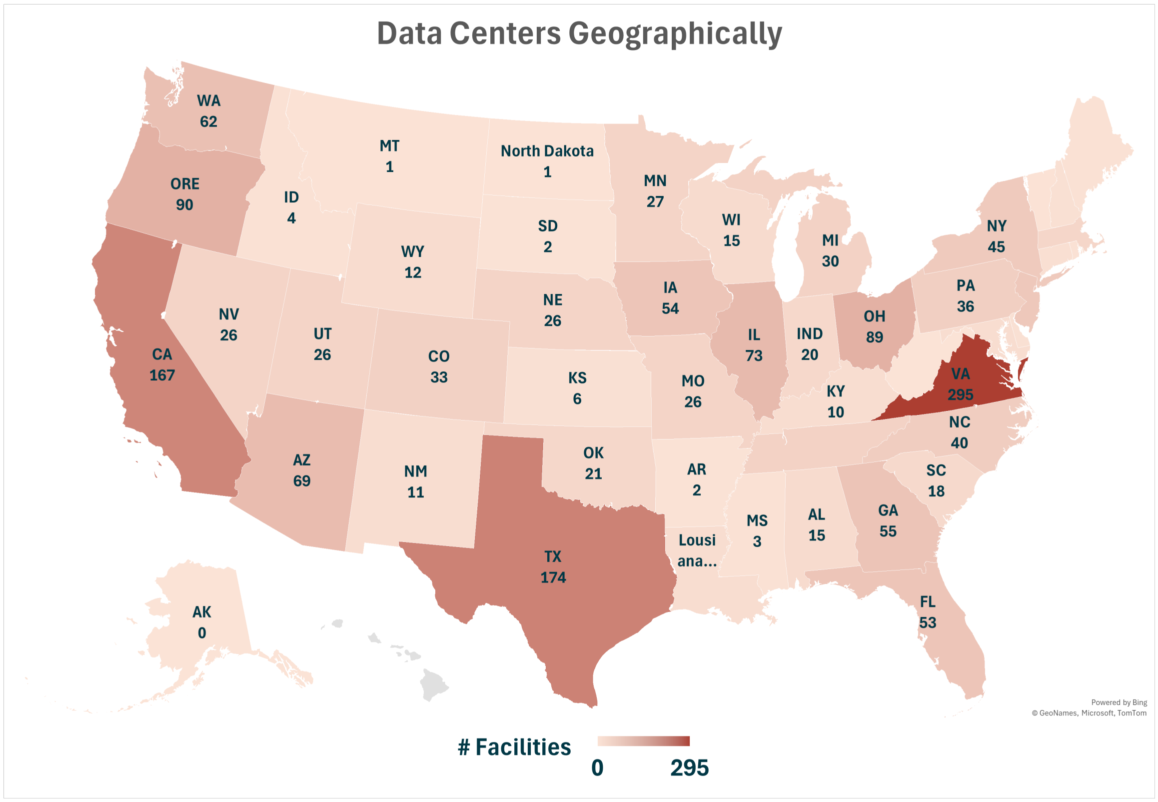


Figure 3.3: Data centers by state. Source: Aterio.io.

The electricity market includes three separate components: generation, transmission, and distribution. Generation is how power is produced. Transmission involves long-range transportation of electricity while distribution involves local distribution of electricity. In the United States, there are two primary models of how the electricity market is structured: vertically integrated utilities and deregulated markets. In vertically integrated utilities, which are prevalent in the South and West, generation, distribution, and transmission are all owned and operated by one company whereas in deregulated markets generation bids in wholesale markets to provide power to utilities, which own transmission and distribution, and provide power in the retail markets. Figure 3.4 provides a map showing the geographic makeup of the American electricity market.

These dynamics interact with broader shifts in the electricity market: greater electrification and the transition from fossil fuels to renewables. Because renewables are intermittent and less dispatchable, rising data center loads coupled with greater renewable penetration

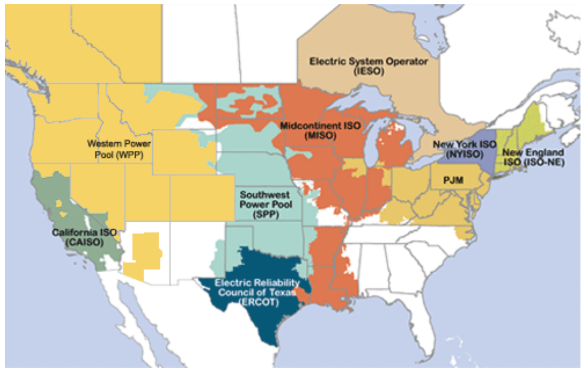


Figure 3.4: Map of American electricity market.

place dual pressures on grids, shaping both market outcomes and policy debates. It is therefore crucial to quantify how much power AI uses, how it distorts quality and impacts prices, and the role of efficiency improvements and developments in computing.

3.3.2 Background: Data Center Concentration and Power-Quality

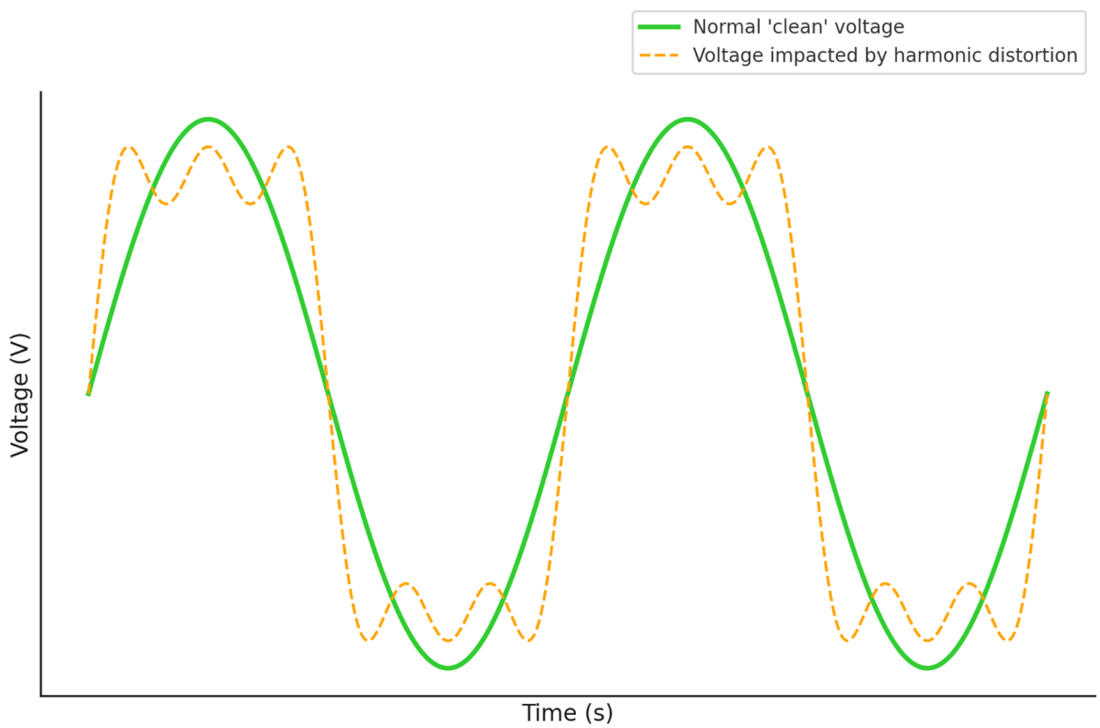


Figure 3.5: Total harmonic distortion representation.

AI workloads affect power quality through three main mechanisms. First, data centers and AI computing facilities add substantial baseload demand that can strain grid capacity and compromise voltage regulation, especially during peak periods. Second, AI workloads fluctuate rapidly with computational needs, creating unpredictable load variations that challenge stability and frequency regulation. Third, the switching power supplies and electronics essential to AI hardware generate harmonic distortion, introducing waveform distortions that spread through distribution networks and degrade power quality. Figure 3.5 illustrates the distortions in the power frequencies. The green line shows the *clear* voltage curve, which is cleanly sinusoidal, whereas with huge loads can produce harmonic distortions shown as the dotted yellow curve. This *dirtier* voltage can adversely affect the operation and lifetime of household appliances and other products that use electricity.

AI workloads also affect wholesale electricity prices through the demand channel, with potentially indeterminate long-run retail market impacts. The U.S. electricity sector comprises both vertically integrated utilities and restructured markets featuring competitive wholesale generation alongside monopoly retail distribution. Wholesale prices are determined hourly in day-ahead markets via sealed-bid, uniform-price auctions subject to network constraints. Locational price variation reflects transmission limitations within the network. Increased demand raises wholesale prices by necessitating dispatch of higher-cost marginal generators.

Wholesale prices are passed through to retail consumers with substantial lag. Retail rates typically reflect average procurement and distribution costs plus a state-regulated return on investment, calculated from historical data. This passthrough mechanism is indirect even absent investment-related dynamics. While new demand sources could theoretically dilute system-wide fixed costs and reduce average retail prices, increased demand more likely raises retail rates. Understanding wholesale demand and price shifts thus provides useful intuition regarding the direction and magnitude of retail price changes.

3.3.3 RECs, PPAs, and Colocation

Corporate electricity procurement takes several forms. Each procurement option entails distinct implications for grid externalities. The most common method is direct procurement, in which a facility draws power as a standard load-serving customer, inheriting the emissions profile of the local generation mix. Often but with increasing rarity and due to the structure of the market, data centers participate directly in the retail electricity market and only interact with utilities.

Despite purchases of electricity with the same properties of that delivered to all other buyers, many hyperscalers claim that they use only renewable electricity. How can this be? To claim renewable attributes without altering physical electron flows, many data center operators purchase renewable energy certificates (RECs). RECs unbundle the environmental attributes of renewable generation into tradeable commodities representing one MWh of renewable output. RECs originated as compliance instruments for state renewable portfolio standards (U.S. Environmental Protection Agency, 2025; Wiser and Bolinger, 2004). REC markets are regulated at the local level with differing standards across markets. Some markets require retirement of RECs in the same geographic market to count. A data center in PJM cannot retire RECs from an ERCOT wind farm. However, this is mostly statutory and is less impactful for voluntary REC purchases. Other markets include voluntary purchases that impose no spatial or temporal constraints. This disconnect underlies the "timing wedge" identified in this paper—annual 100% renewable claims coexisting with fossil-heavy consumption during peak hours. Empirical evidence suggests the emissions-reduction value of unbundled RECs may be near zero in many markets (Bjørn et al., 2022b; Gillenwater, 2007).

Power purchase agreements (PPAs) offer a closer link between procurement and consumption. As shown in Figure 3.6, PPAs by data centers have risen precipitously since 2016 and have grown alongside data center power consumption. While most PPAs were historically wind-based, battery and solar PPAs are increasingly common, and nuclear PPAs

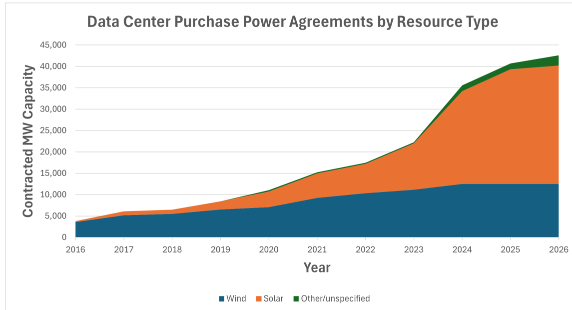


Figure 3.6: PPA agreements by data centers have risen dramatically since 2016.

are emerging as a favorite of industry leaders. Physical PPAs deliver contracted renewable power to the offtaker's node, while virtual (financial) PPAs provide contract-for-differences settlements without physical delivery, sharing the spatial mismatch limitations of unbundled RECs. The key distinction is whether procurement induces *incremental* renewable capacity that is *deliverable* to the relevant node (Taheri et al., 2025; Kobus et al., 2021).

Colocation includes siting dedicated generation at or near a data center, often behind the meter and is increasingly pursued by hyperscale operators. By serving load locally, colocation can reduce transmission congestion and avoid incremental bulk system demand. However, FERC has scrutinized such arrangements for potential cost-shifting to remaining ratepayers, as in the Talen Energy–Amazon arrangement at the Susquehanna nuclear plant (Federal Energy Regulatory Commission, 2024). The net environmental impact depends on the counterfactual: redirecting generation that would otherwise serve grid load may increase system-wide emissions.

3.4 Game Theoretic Model

3.4.1 Model

We present a model of the impacts of varying decision choices for carbon abatement by a large hyperscaler. We present a toy model to illustrate model functioning followed by a full model. We focus on marginal carbon intensity (MCI) and average carbon intensity

(ACI). MCI represents the emissions from a particular hour while average carbon intensity represents the total emissions divided by the total generation. While we do not model system-level reliability directly, increases in load without corresponding investments in generation naturally contribute to greater reliability challenges.

3.4.2 Toy Model

Agents and primitives. There are two periods $t \in \{1,2\}$ and three generator agents: a fossil unit F, an existing renewable R_1, and a potential renewable entrant R_2. Figure 3.7 showcases the agents and their interactions with the market. The hyperscaler has inelastic

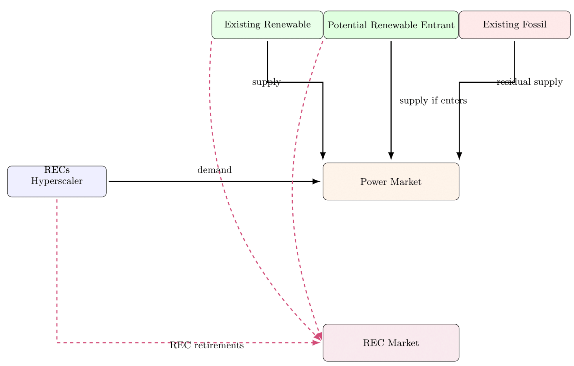


Figure 3.7: Agents in Model

demand $D_t^M = 1$ in each period, and there is no outside demand ($D_t^O = 0$). Variable costs and emissions:

$$c_F > 0, \quad e_F > 0, \qquad c_{R_1} = c_{R_2} = 0, \quad e_{R_1} = e_{R_2} = 0.$$

Capacities/output (deterministic to highlight timing):

$$x_{R_1,1} = 1, \;\; x_{R_1,2} = 0; \qquad x_{R_2,1} = 0, \;\; x_{R_2,2} \in \{0,1\};$$

$$x_{F,t} \in \{0,1\}.$$

Renewables are *complementary in time*: R_1 only produces in $t = 1$, while R_2 (if it enters) only produces in $t = 2$. The fossil unit can meet any residual demand.

Dispatch and prices. Competitive dispatch follows merit order. If a renewable is available in t, it clears the 1 MWh load at $p_t = 0$; otherwise F clears at $p_t = c_F$. Hence, without R_2 entry:

$$p_1 = 0, \qquad p_2 = c_F, \qquad \mathrm{MCI}_1 = 0,\ \mathrm{MCI}_2 = e_F.$$

With R_2 entered, $p_2 = 0$ as well.

REC mechanics and the timing parameter. Each MWh of renewable generates one REC in its production period. Let S_t be REC supply, so without entry $S_1 = 1$, $S_2 = 0$; with entry $S_2 = 1$. The hyperscaler targets 100% coverage ($\phi = 1$) under REC procurement with granularity parameter $h \in [0,1]$: $h = 0$ (annual matching, no timing), $h = 1$ (hourly matching). Let R_t^G be granular retirements and R^A annual-bucket retirements. Feasibility:

$$\underbrace{R_t^G \geq hD_t^M,}_{\text{granular}} \qquad \underbrace{R^A \geq (1-h)(D_1^M + D_2^M),}_{\text{annual}}$$

Potential entrant R_2. If R_2 enters, it produces $x_{R_2,2} = 1$ in $t = 2$. It pays entry cost $I > 0$ financed at $1 + r(\sigma^2)$. We compare three procurement environments (which map to different cash-flow risk for R_2):

1. *REC/merchant (no contract).* R_2 sells energy at p_2 and its REC at p_2^{REC}.
2. *PPA.* R_2 receives a fixed transfer $\bar{p}$ per MWh in $t = 2$ (energy+REC bundled); residual merchant exposure is zero in this toy case.
3. *Colocation.* R_2 is fully contracted on-site at transfer $\tilde{p}$ per MWh in $t = 2$ (no market risk).

We keep discounting trivial (two periods, take $\beta = 1$) to focus on timing and entry.

Outcomes by procurement regime.

3.4.3 Full Model

Agents and primitives. We study a market with a large data center hyperscaler (e.g. Microsoft) and a set of generators indexed by owner k and technology $l \in \{R, F\}$ (renewable or fossil).[1] The hyperscaler procures electricity to minimize expected total costs by choosing (i) a long-term power supply arrangement $i \in \mathscr{I}$ and (ii) an emergency backup option $j \in \mathscr{J}$.

Key objects.

- For each generator g: capacity $O_g^{\max}$, variable cost c_g, emissions rate e_g, and stochastic availability.
- For the hyperscaler: load $\{D_t^M\}_{t=1}^2$, backup option $j \in \{\text{none}, \text{diesel}, \text{storage}\}$, and procurement $i \in \{\text{REC}, \text{PPA}, \text{Colocation}\}$.

3.4.4 Timing

Stage 0a (commitment): The hyperscaler commits to procurement $i \in \mathscr{I}$ and backup $j \in \mathscr{J}$ (contract terms public).

Stage 0b (entry): A set of potential generators of both types observes (i, j) and decides whether to enter. Entry costs are sunk upon entry.

Stages 1–2 (operations): In each period $t = 1, 2$, shocks realize (demand, renewable availability). The energy market dispatches competitively; RECs are issued, banked, and retired subject to granularity rules (the "timing wedge"). Agents discount with factor $\beta \in (0, 1)$ across the two operating periods.

[1] Index individual plants by g when convenient; technology $l(g) \in \{R, F\}$.

3.4.5 Hyperscaler problem (commitment stage)

The hyperscaler chooses (i, j) to minimize expected total costs:

$$\pi_{\text{hyper}} = \min_{i\in\mathscr{I},\, j\in\mathscr{J}} \left[C_i^{\text{cap}} + C_i^{\text{op}} + P_{ij}C^{\text{out}} + \bar{C}_{ij}^{\text{env}}\right], \tag{3.1}$$

where $P_{ij} = P_i P_j$ is the probability that both the contracted source fails and the backup is unavailable, C^{out} is the outage loss, and $\bar{C}_{ij}^{\text{env}} = \mathbb{E}[f(\Delta E_{ij})\, C_{ij}^{\text{env}}]$ is the expected reputational cost as a function of the emissions delta $\Delta E_{ij} \equiv E^{\text{with DC}} - E^{\text{without DC}}$.

Contract menu.

- *REC/merchant* ($i = \text{REC}$): the hyperscaler buys energy from the grid and retires RECs; generators remain merchant for energy and REC revenue.
- *PPA* ($i = \text{PPA}$): a renewable generator sells a fixed quantity $\bar{q}$ each period at price $\bar{p}$; residual is merchant.
- *Colocation* ($i = \text{Colo}$): a dedicated renewable unit (optionally with storage) physically colocated with the data center delivers Q_t^{colo}; residual met from the grid. Colocation fully insulates the generator from market risk.

Backup. $j \in \{\text{none}, \text{diesel}, \text{storage}\}$ affects both reliability (P_j) and emissions: diesel has $e_j > 0$, storage has $e_j = 0$ if charged by colocated renewables.

3.4.6 Entry (Stage 0b)

Potential entrants of type $l \in \{R, F\}$ decide to enter before operations. Let I_g be the sunk entry cost for plant g, and let $r(\sigma^2)$ be the project's financing rate, strictly increasing in the variance of net operating cash flows σ^2 (a reduced-form cost-of-capital channel). For a renewable merchant entrant,

$$\Pi_g^{R,\text{merch}} \;=\; \mathbb{E}\Bigg[\sum_{t=1}^{2} \beta^{t-1}\big((p_t + p_t^{REC})x_{g,t} - c_g x_{g,t}\big)\Bigg] \;- \tag{3.2}$$

$$I_g\left(1 + r(\sigma^2_{\text{merch}})\right).$$

Under a PPA for $\bar{q} \le O_g^{\max}$ at price $\bar{p}$,

$$\Pi_g^{R,\text{ppa}} \;=\; \sum_{t=1}^{2} \beta^{t-1}\left[\bar{p}\,\bar{q} + (p_t + p_t^{REC})(x_{g,t} - \bar{q})_+ - c_g x_{g,t}\right] \;- \tag{3.3}$$

$$I_g\left(1 + r(\sigma^2_{\text{ppa}})\right).$$

Under colocation (full revenue and REC certainty for the contracted amount),

$$\Pi_g^{R,\text{colo}} \;=\; \sum_{t=1}^{2} \beta^{t-1}\left[\tilde{p}_t\, Q_t^{\text{colo}} - c_g x_{g,t}\right] \;-\; I_g\left(1 + r(0)\right), \tag{3.4}$$

where $\tilde{p}_t$ is the contracted transfer for colocated output. For fossil F, replace $(p_t + p_t^{REC})$ by p_t.

Free entry. With a continuum of potential entrants and competitive supply of projects, free entry implies zero-profit conditions for marginal entrants of each type/contract:

$$\Pi_g^{l,\cdot} \;\le\; 0 \quad \text{for all } g, \qquad \Pi_g^{l,\cdot} = 0 \text{ if } g \text{ enters}, \qquad l \in \{R, F\}. \tag{3.5}$$

Because $r(\sigma^2)$ increases in variance and colocation eliminates variance,

$$r(0) \;\le\; r(\sigma^2_{\text{ppa}}) \;\le\; r(\sigma^2_{\text{merch}}), \quad \Rightarrow \quad \text{entry}_{\text{colo}} \;\ge \tag{3.6}$$

$$\text{entry}_{\text{ppa}} \;\ge\; \text{entry}_{\text{merchant}} \;\; \text{(all else equal)}.$$

3.4.7 Operations subgame (two periods $t = 1, 2$)

Demand. Total load each period is $D_t = D_t^M + D_t^O$. Net grid draw by the hyperscaler is

$$G_t^M \;=\; \left(D_t^M - Q_{i,t}^{\text{contract}} - Q_t^{\text{colo}} - B_{j,t}\right)_+, \tag{3.7}$$

where $Q_{i,t}^{\text{contract}}$ is the (possibly firm) delivery under i and $B_{j,t}$ is backup output.

Availability. Each generator g is either fully available or off:

$$x_{g,t} \in \{0, O_g^{\max}\}, \quad \Pr(x_{g,t} = O_g^{\max}) = \alpha_{g,t}. \tag{3.8}$$

Energy dispatch and price. Given the set of available units $\mathscr{G}_t$, the competitive dispatch solves

$$\min_{\{x_{g,t}\}} \sum_{g \in \mathscr{G}_t} c_g x_{g,t} \qquad \text{s.t.} \tag{3.9}$$

$$\textstyle\sum_{g \in \mathscr{G}_t} x_{g,t} + B_{j,t} + Q_t^{\text{colo}} \;=\; D_t, \;\; x_{g,t} \in \{0, O_g^{\max}\}.$$

Let λ_t be the energy balance multiplier. The locational marginal price (LMP) is $p_t = \lambda_t = c_{g^*(t)}$, where $g^*(t)$ is the marginal unit at the optimum.

REC issuance, banking, and the timing wedge. Renewables mint one REC per MWh:

$$r_{g,t} = \mathbf{1}\{l(g) = R\}\, x_{g,t}, \qquad S_t = \sum_g r_{g,t}. \tag{3.10}$$

A stock of banked RECs evolves:

$$K_{t+1} = (1-\delta)K_t + S_t - R_t^{\text{ret}}, \qquad K_t \in [0, \bar{K}], \quad t = 1, \tag{3.11}$$

with terminal K_3 free (or bounded).

To formalize matching granularity, fix a parameter $h \in [0, 1]$: $h = 0$ is annual bucket matching; $h = 1$ is full hour/period matching. Let $R_t^{M,G}$ be the hyperscaler's *granular* retirements in period t and $R^{M,A}$ its *annual bucket* retirements. The hyperscaler targets a renewable share $\phi \in [0,1]$ of its total consumption. Feasibility and obligations:

(Granular requirement)

$\mathrm{R}_t^{M,G} \geq h\,\phi\,D_t^M,$

$\mathrm{R}_t^{M,G} \;\leq\; S_t^{\text{qual}}, \quad t = 1,2,$

(Annual requirement)

$\mathrm{R}^{M,A} \;\geq\; (1-h)\,\phi\,(D_1^M + D_2^M),$

$\mathrm{R}^{M,A} \;\leq\; K_1 + S_1 + S_2 - (R_1^{M,G} + R_2^{M,G}),$

where $S_t^{\text{qual}} \leq S_t$ denotes RECs eligible by geography/tier. The *timing wedge* is the load not covered by same-period renewable matching:

$$w_t(h) \;=\; G_t^M - R_t^{M,G} \quad (\geq 0 \text{ if } h > 0 \text{ binds}), \tag{3.12}$$

$$W(h) = \textstyle\sum_{t=1}^2 \mathbb{E}[w_t(h)].$$

REC market clearing and prices. Given banking (3.11) and obligations (3.4.7)–(3.4.7), the competitive REC allocation minimizes present cost of compliance:

$$\begin{aligned} \min_{\{R_t^{\text{ret}}, K_2, R_t^{M,G}, R^{M,A}\}} \quad & \sum_{t=1}^{2} \beta^{t-1} p_t^{REC} R_t^{\text{ret}} \;+\; \sum_{t=1}^{2} \beta^{t-1} \overline{p}_t^{\text{ACP}}\, \text{short}_t \\ \text{s.t.} \quad & (3.11),\ (3.4.7),\ (3.4.7), \end{aligned} \tag{3.13}$$

$$R_t^{\text{ret}} = R_t^{M,G} + \mathbf{1}\{t=2\}R^{M,A} + R_t^{\text{LSE}} - \text{short}_t,$$
$$\text{R}^{\cdot} \geq 0,\ \text{short}_t \geq 0.$$

Let μ be the annual-constraint multiplier and ν_t the granular multipliers. Then the effective shadow value of a qualifying REC used in t is

$$\tilde{p}_t^{REC} \;=\; p_t^{REC} \;+\; \nu_t \;+\; \mathbf{1}\{t=2\}\,\mu. \tag{3.14}$$

With interior banking, the Euler condition implies

$$p_1^{REC} \;=\; \beta(1-\delta)\,\mathbb{E}\left[p_2^{REC}\right], \tag{3.15}$$

and $p_t^{REC} \leq \overline{p}_t^{\text{ACP}}$ with equality if shortfalls occur.

Emissions and carbon intensity. Total emissions in t are

$$E_t = \sum_{g\in\mathscr{G}_t} e_g x_{g,t} + e_j B_{j,t}. \tag{3.16}$$

Define average and marginal carbon intensity:

$$\text{ACI}_t = \frac{E_t}{\sum_g x_{g,t} + B_{j,t}}, \qquad \text{MCI}_t = e_{g^*(t)}. \tag{3.17}$$

Microsoft's attributable marginal emissions under (i,j):

$$\text{MME}_{i,j} \;=\; \sum_{t=1}^{2} \beta^{t-1}\,\mathbb{E}\big[\text{MCI}_t \cdot G_t^M\big]\,, \tag{3.18}$$

with G_t^M in (3.7).

Equilibrium

Given (i, j) and the set of entrants from Stage 0b, a *competitive two-period operating equilibrium* is a tuple

$$\big\{\{x_{g,t}\},\{p_t\},\{p_t^{REC}\},K_2,\{R_t^{M,G}\},R^{M,A}\big\}_{t=1,2}$$

such that (i) dispatch (3.9) clears energy with $p_t = c_{g^*(t)}$; (ii) REC issuance/banking/retirements satisfy (3.11), (3.4.7)–(3.4.7), and REC prices satisfy (3.15); and (iii) backup and colocation quantities meet (3.7). A *subgame-perfect equilibrium* of the full game consists of (i^*, j^*) solving (3.1), a set of entrants satisfying free-entry conditions, and a competitive operating equilibrium in each period.

Mechanisms and Equilibrium Implications

Three mechanisms drive outcomes. First, procurement choices shape operational emissions through the *timing wedge* between load and renewable generation. Colocation with storage shrinks this wedge by re-timing renewable energy into high-carbon hours; diesel backup increases emissions.

Second, contract type alters financing risk: PPAs reduce σ^2 relative to RECs, and colocation eliminates it. Lower variance reduces $r(\sigma^2)$, encouraging renewable entry.

Third, equilibrium carbon intensity reflects both the short-run operational effect and the long-run investment effect.

Implications

Our model provides a set of empirical predictions about how hyperscalers' procurement choices affect both costs and carbon intensity. On the demand side, as the spread between the cost of procuring electricity from renewable sources and from the open market widens, hyperscalers are more likely to turn to the open market and less likely to contract renewables.

At the same time, as hyperscaler demand for electricity grows, the environmental cost of their consumption rises, and the benefits of carbon-free procurement mechanisms such as PPAs and colocation increase. These benefits scale with both the overall carbon intensity of grid electricity and the size of the hyperscaler's load: the dirtier and larger the load, the greater the incentive to secure clean and reputationally valuable supply.

On the supply side, the model emphasizes that because renewable generation is intermittent, incremental data center demand tends to raise fossil generation unless there is sufficient storage to shift renewable energy across periods. The timing mismatch between hyperscaler load and renewable output—the "timing wedge"—is therefore central to understanding operational emissions outcomes. Colocation with storage can shrink this wedge by re-timing renewable production to high-carbon hours, while diesel backup increases emissions during outages.

These mechanics give rise to several testable implications. First, REC purchases on their own do not alter short-run carbon intensity, since they are purely financial transfers. Their effect arises only in the medium run, if REC demand raises REC prices and induces new renewable capacity to come online. PPAs without firming behave similarly in operational terms, but they reduce generators' revenue risk and financing costs, which increases renewable entry relative to REC-only arrangements. Colocation has a more immediate impact: behind-the-meter renewable output reduces a hyperscaler's net grid draw exactly when the colocated plant is producing, lowering local marginal emissions exposure. The effect is strongest when renewable output is correlated with hyperscaler load. Adding storage to colocation further enhances this benefit by shifting surplus renewable production to periods when demand is high and marginal carbon intensity is greatest, sharply reducing emissions attributable to the data center. This also can an emissions effect as the reduced timing wedge better matches power, and the behind-the-meter power can add resilience to the grid. By contrast, reliance on diesel backup increases operational emissions whenever it is deployed.

Finally, the model predicts an investment hierarchy driven by financing risk: colocation, by fully eliminating revenue variance, induces the largest increase in renewable entry, followed by PPAs, and then REC purchases. This creates a long-run ordering in expected carbon reductions. Moreover, because colocated renewables reduce grid purchases at the hyperscaler's node, emissions fall locally, though congestion and flow adjustments may increase carbon intensity in neighboring nodes unless total renewable capacity expands.

Together, these predictions imply that (i) REC purchases yield little immediate emissions reduction but may encourage new renewable entry; (ii) PPAs accelerate renewable deployment by reducing financing risk, though without firming they do not improve operational carbon intensity in the short run; and (iii) colocation—especially when paired with storage—directly lowers hyperscaler-attributable emissions by addressing the timing wedge between renewable generation and hyperscaler demand.

3.5 Empirical Overview

We estimate the causal effect of AI workloads on four outcomes in wholesale electricity markets: local power quality, fossil fuel demand, wholesale prices, and the differential response across training versus inference phases. The central identification problem is that the objects of interest such as the timing, location, and intensity of compute activity at individual data centers, are proprietary. We observe neither training start dates nor facility-level load; instead, we observe publicly announced model release dates, the geographic footprint of hyperscaler data centers, and zonal market outcomes. Our strategy exploits model releases as the observable shock and uses difference-in-differences to recover treatment effects from the remaining variation.

3.5.1 Identification

Let Y_{it} denote an outcome (power quality, fossil generation, or price) for zone i in period t, and let model release m occur at date r_m. Zones linked to the releasing firm's data center footprint constitute the treated group for event m; zones without exposure to that firm's AI activity in the event window form the control group. Under the standard parallel-trends and no-anticipation assumptions, differencing outcomes across treated and control zones before and after r_m identifies the average treatment effect on the treated. Because our outcomes, wholesale prices, fossil generation, voltage distortion, are transmitted through markets and the physical grid rather than through proprietary firm-level data, this approach follows a long tradition of recovering unobservable inputs from observable market equilibria (Fowlie et al., 2012; Card and Krueger, 1994; Greenstone and Hanna, 2014).

Four features of our setting complicate a textbook two-period DiD. First, training windows are unobserved: we know when a model was released but not when its training began, so the "treatment date" is imprecise. Second, releases are staggered across firms and calendar time, which under treatment-effect heterogeneity causes two-way fixed effects estimators to weight already-treated units as implicit controls for later-treated units (Goodman-Bacon, 2021; Sun et al., 2021; Callaway and Sant'Anna, 2021; Borusyak et al., 2024). Third, parallel trends is not innocuous in electricity markets, where fuel prices, weather, and secular load growth move outcomes in ways potentially correlated with data center siting. Fourth, treatment exposure itself must be constructed: no dataset directly links model releases to zonal load, and assigning generators and facilities to market zones requires substantial preprocessing (see Appendix 3.13.4).

We address the first two issues with a stacked event-study design that anchors treatment on observed release dates and builds clean control groups for each release cohort, developed in Section 3.6.1. We address the third through pre-trend tests, placebo exercises, and controls for fuel costs, temperature, and renewable output (Section 3.10). The fourth is resolved in our data construction, described in Section 3.8.

3.6 Estimation Strategy

To address the inexact information about training windows and release times, we estimate dynamic treatment effects at multiple horizons relative to model release dates. This approach, referred to as an event study or **stacked difference-in-differences** in econometrics, offers three advantages over the standard pre/post comparison discussed earlier: (1) it does not require observing the true training start date, instead using release timing as the anchor; (2) it also provides a direct test of the parallel trends assumption through examination of pre-release coefficients; and (3) it allows the data to reveal the temporal profile of effects rather than imposing an arbitrary discontinuous treatment structure.

3.6.1 Stacked DiD with Multiple Horizons

Formally, we define treatment exposure relative to each model's release date r_m for both training and release: **1) Training windows:** The period $[r_m - \bar{\tau}_{\text{train}}, r_m)$ preceding release, during which computational resources are devoted to model training. We remain agnostic about the exact training start date; our identifying assumption is that training activity is elevated in the months before release and that any pre-existing differential trends would manifest in the earlier leads. **2) Inference window:** The period $[r_m, r_m + \bar{\tau}_{\text{infer}}]$ following release, during which the model is deployed for inference via API access or public availability.

Stacked Difference-in-Difference

Model releases occur at different calendar dates, creating a staggered adoption setting. Recent econometric literature has demonstrated that conventional two-way fixed effects (TWFE) estimators as is the case in the estimates from traditional difference-in-differences. can produce biased estimates under treatment effect heterogeneity Goodman-Bacon (2021); Sun et al. (2021); Callaway and Sant'Anna (2021); Borusyak et al. (2024). Treatment effect heterogeneity means the policy works differently for different places or periods, which can distort standard difference-in-differences estimates. The bias arises because TWFE implicitly uses already-treated units as controls for later-treated units, and differences in treatment effects across cohorts or over time contaminate the estimated average effect.

In our setting, consider regions as units and AI model releases as treatments inducing increased data center power demand. A “cohort" comprises regions whose data centers began significant AI workloads at the same time. Suppose Region A adopted AI workloads in 2022 while Region B adopted in 2023. When estimating treatment effects for Region B, TWFE implicitly uses Region A as a control—but Region A in 2023 is not untreated; its data centers have been running AI workloads for a year. If treatment effects evolve over time (e.g., power demand grows as AI infrastructure scales), comparing Region B’s outcomes against Region A’s already-treated outcomes introduces bias. Staggered design provides heterogeneity-robust estimators that solve for this exact issue.

To address this, we use a stacked DiD model Cengiz et al. (2019); Deshpande and Li (2019), where we construct a stacked dataset where each model release defines a separate “sub-experiment.” The idea here is to construct, for each treatment cohort, a separate sub-experiment using only units that are genuinely untreated at the time of that cohort’s adoption. By stacking these sub-experiments—each with its own clean control group of not-yet-treated regions—we ensure that no already-treated unit ever serves as a control, thereby eliminating the bias that arises when treatment effects are heterogeneous across cohorts or time.

Formally, for each release event m, we create a dataset containing: (i) Treated units – Regions linked to the releasing company’s AI data centers. (ii) Control units – Regions with no AI data center exposure during the event window, or regions whose treatment occurs sufficiently far from event m that they serve as clean controls.

- Time window: A symmetric window around r_m, e.g., $[r_m - K, r_m + K]$ for some horizon K.

Following Wing et al. (2024), we then stack these sub-experiments and estimate:

$$Y_{i,t,m} = \sum_{k=-K}^{K} \beta_k \cdot D_{ut}^k + \delta_t + \mathbf{X}_{ut}'\theta + \varepsilon_{ut} \tag{3.19}$$

where γ_u are unit-by-event fixed effects, δ_u are time-by-event fixed effects, $\mathbf{X}_{ut}'\theta$ represents controls, and the coefficients $\{\beta_k\}$ trace out the dynamic treatment effect at each event-time horizon k. Standard errors are clustered at the unit level to account for correlation across events within the same region.

3.6.2 Estimating Power Quality

Our power quality analysis estimates the following stacked DiD specification at the utility-month level:

$$\text{PowerQuality}_{ut} = \sum_{k=-K}^{K} \beta_k^{PQ} \cdot D_{ut}^k + \gamma_u + \delta_t + \mathbf{X}_{ut}'\theta + \varepsilon_{ut} \tag{3.20}$$

where:

- PowerQuality_{ut} is the Whisker Labs Consumer Power Quality Index for utility u in month t;
- $D_{ut}^k = \mathbf{1}[\text{event-time} = k] \times \text{Treated}_u$ indicates that utility u is k periods from the release date of a model linked to an AI data center in its service territory;
- γ_u are utility fixed effects absorbing time-invariant characteristics;
- δ_t are calendar month fixed effects absorbing common temporal shocks;
- $\mathbf{X}_{ut}$ includes time-varying controls (weather, regional load).

We estimate an analogous specification for total harmonic distortion (THD):

$$\text{THD}_{ut} = \sum_{k=-K}^{K} \beta_k^{THD} \cdot D_{ut}^k + \gamma_u + \delta_t + \mathbf{X}_{ut}'\theta + \varepsilon_{ut} \tag{3.21}$$

3.6.3 Estimating Fossil Fuel Demand

Power demand analysis proceeds at the generator-month level, exploiting finer geographic resolution. We maintain the stacked DiD structure while incorporating instrumental variables to address the price endogeneity concern:

First Stage. We instrument for electricity prices using supply-side cost shifters:

$$p_{it} = \pi_0 + \pi_1 \text{HeatRate}_{it} + \pi_2 \text{GasPrice}_t + \gamma_i + \delta_t + \mathbf{X}'_{it}\pi_3 + \nu_{it} \tag{3.22}$$

where HeatRate_{it} captures the thermal efficiency of generator i and GasPrice_t is the contemporaneous natural gas spot price. These instruments satisfy relevance (they are primary determinants of marginal generation costs) and exclusion (they affect demand only through their impact on prices, not through direct effects on AI workload scheduling or other consumption decisions).

Second Stage. We estimate the stacked DiD specification using predicted prices:

$$\text{FossilDemand}_{it} = \sum_{k=-K}^{K} \beta_k^{FD} \cdot D_{it}^k + \eta \hat{p}_{it} + \gamma_i + \delta_t + \mathbf{X}'_{it}\theta + \varepsilon_{it} \tag{3.23}$$

where D_{it}^k indicates event-time relative to model releases linked to AI data centers in generator i's service area.

The coefficients $\{\beta_k^{FD}\}$ capture the dynamic effect of AI activity on fossil fuel demand. Negative values of k (pre-release) correspond to the training phase; positive values correspond to inference. The price coefficient η controls for supply-driven price movements and provides a scaling factor for welfare calculations. Difference between samples in our main analyses are highlighted in the Appendix in Subsection 3.13.11.

3.6.4 Estimating Price Effects

Beyond quantity effects, we estimate how AI-driven demand increases affect electricity prices. The key economic insight is that price impacts depend on supply flexibility: when generators can easily expand output, demand increases are absorbed with minimal price changes, but when supply is constrained, demand shocks translate primarily into higher prices. The supply elasticity ε^s—the percentage change in quantity supplied per one percent price change—quantifies this responsiveness.

Our estimation proceeds in two steps. First, we use our IV estimates to quantify how AI activity shifts demand:

$$\Delta\widehat{\text{FossilDemand}}it = \hat{\beta} \cdot \Delta\text{AI}it \tag{3.24}$$

Second, we trace this shift through the supply curve. With linear supply $Q^s = a + b \cdot p_i$, the price change is $\Delta p = \hat{\beta} \cdot \frac{\Delta AI_{it}}{b}$, or in elasticity form:

$$\%\Delta p = \frac{\%\Delta Q^d}{\varepsilon^s} \tag{3.25}$$

We estimate ε^s via IV regression of quantity on price in log-log form, using demand-side instruments (weather-driven load variation and industrial production indices).

We focus on PJM at the zonal level for this analysis. We restrict to a single ISO because differing market structures make prices incomparable across ISOs, and choose PJM because it is the largest (65 million people, 13 states), includes key data center states (Virginia, Pennsylvania), and provides 20 diverse pricing zones. Because observed training locations are outside PJM, this analysis uses the broader model set only.

3.7 Validating Assumptions and Robustness

We implement several diagnostic and robustness exercises to assess the credibility of our estimates.

Sample Selection

Our empirical strategy balances internal validity with external relevance. Our main sample comprises four models for which we observe exact training and inference dates as well as training location, and we employ propensity score matching (described in the appendix) to support the parallel trends assumption. We then report results for progressively larger samples of models and data centers, accepting weaker identification in exchange for broader coverage. This approach enables precise, well-identified estimates from our core sample while exploring heterogeneity and counterfactuals in extended samples, with appropriate caveats regarding the latter.

Pre-Trends and Parallel Trends Testing

Our primary test of the identifying assumption examines whether pre-release coefficients $\{\beta_k : k < 0\}$ are jointly indistinguishable from zero. We report:

- Point estimates and confidence intervals for each lead coefficient.
- An F-test of the joint null hypothesis $H_0 : \beta_{-K} = \beta_{-K+1} = \cdots = \beta_{-1} = 0$.

Significant pre-trends would undermine the causal interpretation of post-release estimates.

Placebo Tests with Randomized Treatment Assignment

To verify that our estimates reflect genuine treatment effects rather than spurious correlations or specification artifacts, we conduct placebo tests with randomly assigned treatment:

1. **Random treatment timing:** We hold the set of treated units fixed but randomly reassign release dates drawn from the empirical distribution of actual release dates. Under the null of no effect, placebo estimates should center on zero.
2. **Random treatment assignment:** We hold release timing fixed but randomly reassign treatment status across units. This tests whether results are driven by the specific geographic pattern of AI data center locations.

Callaway-Sant'Anna Estimator

As a robustness check and to provide transparent decomposition of treatment effect heterogeneity, we also implement the Callaway and Sant'Anna (2021) estimator. Like the stacked approach, CS avoids using already-treated units as controls; however, rather than achieving this through sample construction, it does so by estimating separate ATT(g,t) parameters for each cohort g and time period t, then aggregating these to form summary measures. This approach explicitly avoids problematic comparisons and provides a framework for testing parallel trends through examination of pre-treatment ATT(g,t) estimates.

Time-Series Anomaly Detection

As an internal consistency check, we apply time-series anomaly detection methods to the outcome variables in treated units. If AI model releases causally affect grid outcomes, we should observe detected anomalies clustering around treatment dates. Specifically:

1. For each treated unit, we fit a baseline time-series seasonal ARIMA model to the pre-treatment period.

2. We generate one-step-ahead forecasts and prediction intervals for the post-treatment period.

3. We flag observations where realized values fall outside the 20% prediction interval as anomalies.

4. We test whether anomaly frequency is elevated in the post-treatment window relative to the pre-treatment window and relative to control units.

This approach provides model-free evidence of structural breaks coinciding with treatment timing.

Sensitivity to Treatment Definition

Given uncertainty in linking models to specific training locations, we assess robustness to alternative treatment definitions in our less conservative analysis version:

1. **Intensive margin:** Weight treatment by the number or capacity of AI data centers in the region, rather than using a binary indicator.

2. **Dropping ambiguous cases:** Exclude model releases where the company operates data centers in multiple balancing authorities, restricting to cases with cleaner geographic attribution.

3. **Verified timing subset:** Estimate separately on the subset of models with externally verified training windows.

4. **Geographic Resolution and Robustness.** The generator-level data permit more granular treatment definitions than the utility-level power quality data. We exploit this flexibility by estimating specifications under alternative treatment definitions in terms of radius.

Heterogeneity by Model Characteristics

We examine whether treatment effects vary with observable model characteristics:

- Model scale (parameter count, reported compute).

- Company (to assess whether effects are driven by specific firms).

- Release year (to assess whether effects are changing over time).

Heterogeneity analysis both informs mechanisms and serves as a specification check: if effects are concentrated among larger models or models from companies with substantial AI infrastructure, this lends credibility to the AI-driven interpretation.

3.7.1 Counterfactual Analysis

We conduct three primary sets of counterfactual exercises to investigate how alternative evolutionary paths for AI compute might affect grid-level outcomes. Additional counterfactuals with results are highlighted in Appendix Subsection 3.13.12.

Model Scaling. First, we examine the grid implications of continued growth in AI model size. Using the Epoch database to characterize trends in model parameters, we provide descriptive analysis of how increasing model scale translates into grid-level impacts. We then relate our estimated power demand coefficients to our power quality coefficients, enabling analysis of how changes in training and inference energy efficiency propagate to power quality outcomes.

Geographic Redistribution. Second, we leverage the geographic locations of data centers in conjunction with our econometric estimates to conduct counterfactuals focused on spatial reallocation of AI compute. We consider two scenarios. In the first, we simulate a shift from cloud-based to edge-based inference—reallocating inference demand from concentrated hyperscaler facilities to smaller data centers distributed closer to population centers. We implement this by redistributing our estimated inference demand impacts across data centers not currently classified as AI-focused. In the second scenario, we simulate relocating training workloads from population-dense regions on the East Coast to energy-abundant regions in the center of the country. For both geographic counterfactuals, we provide sensitivity analyses in the appendix to account for potential increases in communication costs and efficiency losses.

Colocated on-site Power Generation. Finally, we exploit a new feature in the Aterio data, namely a flag on data centers with on-site power generation. We re-estimate both our main models separately for generators with and generators without onsite power generation. We then report the impacts to power demand and power quality of either removing all onsite generation or having onsite generation for all data centers.

Together, these counterfactuals illuminate both how the grid might be affected by different trajectories for AI compute and which levers—geographic placement, workload composition, efficiency improvements—may be available to computer scientists and policymakers seeking to mitigate grid impacts.

3.8 Data

Table 3.1 summarizes the datasets we use to address each of our three research questions, distinguishing between variables of interest and the control variables described above. For variables of interest, we rely on **Aterio**, which provides comprehensive information on the location and history of data centers. The Aterio dataset documents when and where centers have been established, expanded, or retired, along with details on ownership and operational capacity (in MW). This allows us to identify data centers owned by specific hyperscalers and to quantify their scale over time. We then combine these data with external sources that provide the necessary controls.

Table 3.1: Data sources for each analysis question.

Source	Content	Use in Analysis	# Instances
	Q1: Difference-in-Differences (Power Quality)		
Aterio	Data center locations, history, ownership, capacity (MW)	Define treatment regions; link AI model releases	3862 data centers
Whisker Labs	CPQI (consumer reliability); THD (waveform distortion)	Outcome variables for DiD regressions	2684 utility-months
EIA	Retail service territory maps	Define treatment/control boundaries	72 utilities
	Q2: Instrumental Variables (Fossil Demand)		
EPA CAMPD	Hourly generator demand; heat rates	Fossil demand outcome; heat rate as instrument	22 million generator-hours
S&P Capital IQ	Natural gas prices	Instrument and fuel-cost control	12 Million location-day prices
Aterio	Data center proximity to generators	Treatment near releasing-company centers	3862 data centers
Meteostat	Temperature, precipitation	Demand and renewable controls	22 million location-hours
ISOs	Zonal wholesale prices	Market controls	54 zones
	Q3: Counterfactual Analyses (Scaling & Efficiency)		
Whisker Labs	CPQI; THD	Baseline power quality impacts for scaling projections	
Epoch	Model parameters, FLOPs, release dates	Scaling regressions; efficiency counterfactuals	75 models

3.8.1 Variables of Interest

We use four primary datasets for our outcomes of interest. First, the **Consumer Power Quality Index (CPQI)** from Whisker Labs (2024) provides a composite measure of consumer-facing reliability events (surges, sags, brownouts, interruptions), summarizing frequency and severity of power deviations at the household level. Second, **Total Harmonic Distortion (THD)** data from Whisker Labs (2024) offers a *technical measure of waveform distortion*, quantifying voltage deviation from a clean 60-Hz sine wave. Elevated THD indicates grid stress, reduces motor efficiency, and shortens equipment lifespan. Figure 3.5 illustrates how harmonics alter voltage sine waves, reducing power reliability. Since THD data are more recent than CPQI, fewer model releases are available for THD analysis. Third, we incorporate **generator demand** from EPA's CAMPD database, providing *hourly plant-level demand* that captures local operating characteristics. CAMPD data span 2021–2023, constraining analysis to AI model releases within that period. Finally, we combine **wholesale zonal price** data provided by Independent System Operators (ISOs) to determine AI impacts on price.

3.8.2 Control Variables

For the **power quality regressions**, we use a parsimonious specification with only time and geographic fixed effects, which absorb seasonal patterns, long-run trends, and location-specific differences, allowing us to test whether data center activity coincides with reliability declines beyond geography and time. In contrast, the **generator demand regressions** employ a richer set of controls to address endogeneity, using *generator heat rates* from EPA CAMPD and *natural gas prices* from S&P Capital IQ Global as instruments in a two-stage least squares framework, along with CAMPD's hourly generator demand data (2021–2024). We also incorporate *Meteostat* weather variables (temperature, dewpoint, precipitation) to capture demand and renewable variability, and *ISO* market data based on boundary maps to ensure spatial consistency. Finally, *EIA retail service territory files* reconcile generator-, zonal-, and retail-level observations into a consistent framework.

3.8.3 Combining Data Sources

To integrate datasets, we harmonize spatial and temporal units across sources. Figure 3.14 demonstrates the integration at each step for our modeling purposes. Data center locations from Aterio are mapped to ISO zones, EIA territories, and counties, enabling linkage with Whisker Labs reliability indexes (CPQI and THD) and CAMPD generator demand. We utilize newly released data on which data centers are utilized in processing AI workloads. This data improves identification significantly as previously, it was very difficult to determine a treated set for a particular kind of data center workload. CPQI has 2,683 observations (mean 0.52, s.d. 0.42), while THD is more dispersed (mean 1.81, s.d. 6.77). CAMPD generator demand averages 218 MW (s.d. 158), and wholesale prices average \$51/MWh (s.d. 148). Weather data capture local conditions (mean temperature 17°C, precipitation 0.12 mm), and time series are aligned at hourly or monthly resolution. External controls—natural gas prices, weather, and ISO market data—are merged by geography and time, yielding a unified panel suitable for difference-in-differences and IV regressions.

Our demand model focuses on deregulated markets in the Eastern Interconnection and ERCOT, where data on zones and prices are readily available, while our power quality model draws on Whisker Labs data from 72 retail utilities nationwide (monthly, 2022–2025). The demand regressions use hourly data for several thousand generators (2021–2023). We concentrate on hyperscaler AI companies—including Meta, Microsoft, Amazon, and Google—with Anthropic linked to Google due to its cloud partnership. Appendix materials include maps, sample descriptions, and data tables. In order to determine which data centers should be considered AI data centers and should be included in our sample, we perform a linkage procedure described in appendix Subsection 3.13.13.

3.9 Results

Our stacked DiD estimator (subsection 3.6.1) addresses potential bias from treatment effect heterogeneity in settings with staggered AI model releases. For all power quality and fossil demand effects, following Callaway and Sant'Anna (2021), we construct cohort-specific datasets for each of the treatment events corresponding to major AI model publications through 2023, then stack these datasets to estimate a set of treatment effects across all dimensions. Table 3.2 provides a full set of results from our main regressions including analysis of the impacts of on-site colocated generation for our counterfactuals.

3.9.1 Power Quality Effects: Utility-Level Strict DiD

We report the power quality effects of AI data centers in seventy one retail electric utility service territories estimated from power quality observations over 1136 utility-months. We use Consumer Power Quality Index (CPQI) and Total Harmonic Distortion (THD) as outcome variables in our stacked DiD models (Equation 3.20).

Table 3.2 reports results from the utility-level strict DiD specification. The CPQI coefficient speaks to the reliability of the power supply to the consumers. The CPQI coefficient of 0.3795 indicates a deterioration level that corresponds to moving from a typical U.S. power area toward the bottom quartile of power quality—roughly **the difference between $\sim$ 1 outage/year and $\sim$ 1.5–2 outages/year for the average customer**.Other reliability events such as brownouts and surges are predicted to increase by 1-2 per year. Importantly, power surges are a known ignition source for electrical fires, contributing to the approximately 51,000 annual residential electrical fires in the U.S., with electrical arcing accounting for 74% of ignition sources U.S. Fire Administration (2019). Fewer fires is generally considered a desirable outcome amongst electricity professionals Laughner et al. (2024). [2] This implies more frequent voltage anomalies that make motors and electronics

[2]These are calculated based on Whisker Labs description of the CPQI Score with a description of derivation in the appendix.

run less efficiently and age faster.

The post-treatment coefficient of 0.0625 for THD indicates that utility territories containing AI data centers experienced a 0.0625 percentage point increase in readings above the 0.8 distortion threshold following AI model releases. While a 0.0625 percentage point increase may appear small in isolation, the IEEE 519 standard exists precisely because even modest increases in harmonic distortion impose continuous thermal stress on motor windings. According to the Arrhenius equation underlying motor insulation ratings, every 10°C rise in operating temperature halves insulation life NEM (2016). Since elevated THD causes motors to waste additional power as heat in their iron and copper windings Laughner et al. (2024), sustained exposure compounds over the multi-year lifespans of household appliances, accelerating degradation of refrigerators, air conditioners, and other motor-driven equipment.

3.9.2 Fossil Fuel Demand Effects: Stacked DiD Estimation

We estimate Fossil Fuel Demand effects using demand data from a total of 35.6 million generator-hour observations, which includes 900,956 treated observations from generators located within 20 miles of AI-affiliated data centers and the other 34.7 million from other generators as control observations.

We estimate the baseline specification via two-stage least squares (2SLS) to isolate demand-side effects from supply shocks. All specifications include zone and month fixed effects with standard errors clustered by unit-event. Further details on the instrumental variables specification can be found in Appendix Subsection 3.13.14. Table 3.2 reports the full stacked DiD coefficients.

Stacked DiD Coefficients: OLS versus 2SLS. Table 3.2 reports the full stacked DiD coefficients from both specifications. Figure 3.8 shows the stacked DiD coefficients for power demand in the 2SLS specification. The $t = +2$ coefficient shows a significant demand increase of 5.20 MWh, representing the demand shift at generators near data centers two months after AI model release, isolated from supply-side price variation through our instrumentation strategy. The pre-treatment coefficient at $t = -2$ is highly significant, representing demand created by training. While these effects may not seem large enough to be economically interesting, these effects are hourly and by generator. When the aggregate impact is calculated, they represent over 500 additional GWh added nearby to a data center over the inference period in some cases. **The additional power is roughly equivalent to that of 47,000 household-years in the United States**.

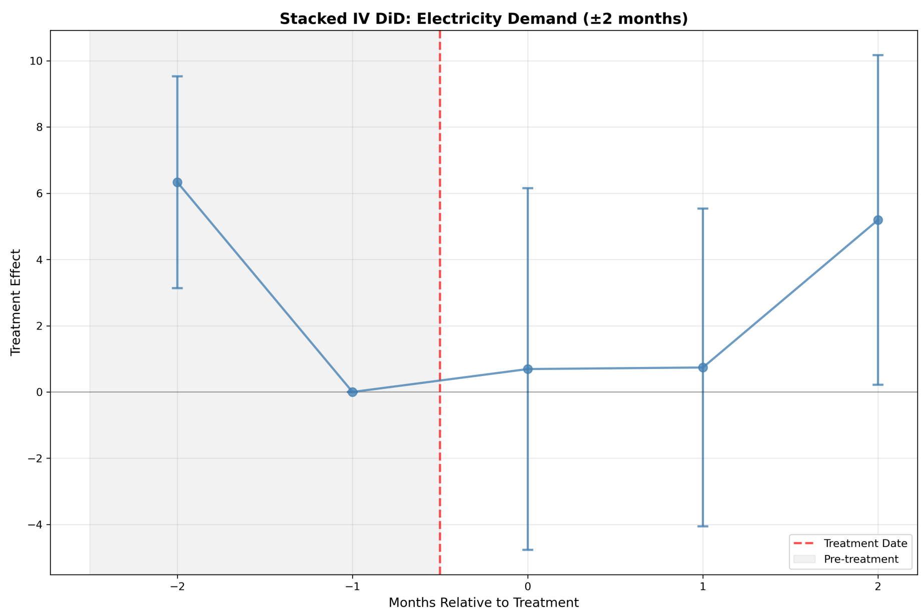


Figure 3.8: Stacked DiD coefficients for power demand (IV specification). Event Time 0 corresponds to AI model publication date. The pre-treatment coefficient at $t = -2$ demonstrates the training demand. Reference period is $t = -1$.

3.9.3 Price Impacts

We estimate wholesale electricity price impacts by combining our demand estimates with supply curve parameters, following the methodology outlined in Section 3.6.4. This approach recognizes that our IV strategy identifies demand shifts but does not directly

Table 3.2: Summary of Estimation Results Across Specifications

Analysis	Outcome	Coefficient	Effect Interpretation
Panel A: Utility-Level Power Quality (Two-Way FE)			
Strict DiD	THD (>0.8)	0.0625***	More outages and equipment failure.
Strict DiD	CPQI	0.3795***	Moving from typical U.S. power quality to the bottom quartile; increases annual outages for the average customer from approximately 1 to 1.5–2.
Panel B: Generator-Level Demand (Stacked DiD)			
OLS	Pre-treatment avg	+1.28	Aggregate impact represents over 500 additional GWh near data centers over the inference period—equivalent to ∼47,000 U.S. household-years of consumption, with load not evenly distributed across hours
OLS	Post-treatment avg	−1.57	
IV 2SLS	Pre-treatment avg	+6.33***	
IV 2SLS	Post-treatment avg	+2.21	
Panel C: Power Quality by On-Site Generation Status			
With on-site	CPQI	−0.122***	On-site Generation generation mitigates grid stress from AI load; utilities without on-site experience reliability deterioration while those with on-site see improvements
Without on-site	CPQI	+0.228***	
Difference	CPQI	−0.350***	

Notes: *** $p < 0.001$, ** $p < 0.05$. Panel A: Utility and time fixed effects with cluster-robust standard errors; treatment defined as presence of AI-affiliated data center in utility service territory. Panel B: 35.6 million observations from 19 treatment events; standard errors clustered by unit-event; IV instruments: natural gas price × zone-average heat rate. Panel C: CPQI = Consumer Power Quality Index; THD = Total Harmonic Distortion.

estimate equilibrium price effects.[3] The price impact of AI-induced demand depends on the slope of the supply curve[4]: inelastic supply—where quantity supplied responds weakly to price changes—implies large price effects, while elastic supply implies small effects. Additional specifics on calculation methodology can be found in Appendix Subsection 3.13.9.

A map of estimated supply chain elasticities can be found in Figure 3.17 in Appendix

[3]Our instrumental variables approach isolates exogenous variation in electricity demand driven by AI model releases. However, observed market prices reflect the intersection of supply and demand in equilibrium. To translate our demand estimates into price effects, we must incorporate information about supply-side responses, which requires additional assumptions about the shape of the supply curve.

[4]The supply curve describes the relationship between price and quantity supplied. Its slope reflects how much generators must be compensated to produce additional electricity—steeper slopes indicate that marginal generation costs rise quickly as output expands, typically because higher-cost units must be dispatched.

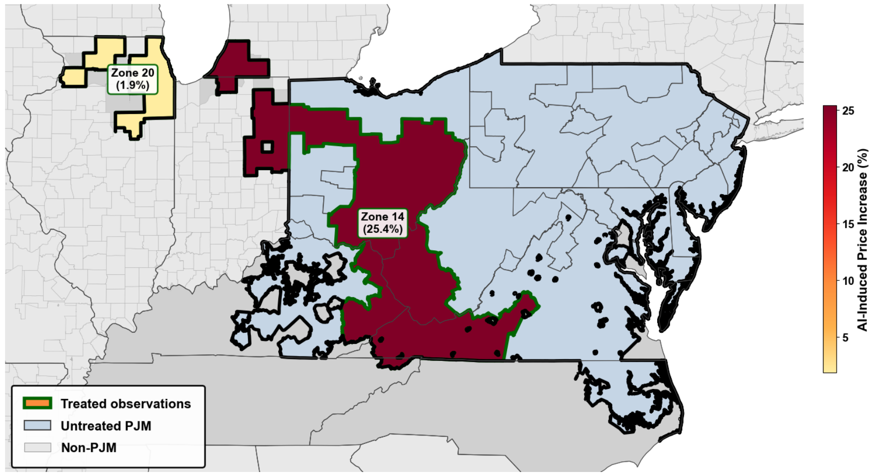


Figure 3.9: Price Effects of AI in PJM.

section 3.13.10 demonstrates the supply elasticity by zone in PJM. Figure 3.9 showing price increase estimates by zone. On average AI training leads to **substantial increases in wholesale prices in PJM with the highest increase surpassing 25% in the American Electric Power (AEP) Zone**. Additional caveats are provided in Appendix Subsection 3.13.14

While we are not able to provide definite evidence of the impacts on retail price pass through, it should be noted that retail electricity prices have increased overall at a rate surpassing inflation since 2020 Nicoletti et al. (2024); Horsley (2025); Southwest Power Pool, Inc. Market Monitoring Unit (2025). While it is possible that increased investment in electric generation through power purchase agreements by tech firms may in the long run lead to falling retail prices, it appears that at present data center power usage is increasing wholesale prices by raising total demand, which requires dispatching higher-cost generators and thus increases the market clearing price—a price paid by all retail consumers.

Using estimates from the literature as documented in Appendix Subsection 3.13.15, we can bracket the expected price rise for retail consumers. Two recent estimates apply but speak to different horizons. De Feuillley and Fontaine (2023) report short-run passthrough of roughly 0.53–0.56 in PJM/Pennsylvania over a sample window dominated by short retail-rate refresh cycles and explicit fuel-cost-adjustment provisions; applying that passthrough to a 25% wholesale shock in AEP implies a short-run consumer price rise of roughly 13%. MacKay and Mercadal (2024) estimate long-run passthrough closer to one across U.S. retail markets, on the assumption that procurement contracts, default-service auctions, and rate-case filings eventually fully reflect sustained wholesale movements; under that estimate the long-run consumer price rise approaches the full 25%. The two numbers therefore correspond to short-run vs. long-run passthrough rather than to competing estimates of the same object: the 13% figure is the appropriate near-term magnitude for retail bills under existing PJM rate mechanics, while the 25% figure is the appropriate long-run magnitude if the wholesale shock is sustained for several rate cycles. We report both and let the reader weight them according to the policy horizon of interest, rather than collapsing to a single point estimate.

3.10 Validating Assumptions and Robustness Checks

We conduct several additional analyses to probe the robustness of our findings and explore effect heterogeneity in order to address issues highlighted in Subsection **??**. The formal testing helps provide evidence for the parallel trends assumption while the placebo tests address potential issues related to external confounds or lack of specific knowledge about training. Additional robustness analyses can be found in Appendix Subsection 3.13.16.

Formal Testing for Parallel Trends

For power demand specifically, the formal joint test of pre-treatment coefficients yields a chi-squared statistic of 0.99 (df = 1) with p-value = 0.3196. **We cannot reject the null hypothesis of parallel pre-trends at conventional significance levels.** This provides support for our identification assumption: treated and control generators followed similar demand trajectories prior to AI model releases. Some of our samples demonstrate significant evidence against the the parallel trends test prior to propensity score matching. However, in 100% of our samples for both power quality and power demand after propensity score matching, we *fail to reject the null hypothesis of* . This implies that the technique correctly handles the issue where it exists.

Additionally, We estimate the baseline specification via OLS to validate parallel trends prior to implementing our instrumental variables strategy. The OLS specification shows no significant effects in any period outside the training window, consistent with parallel trends holding in the pre-treatment period. Importantly, prior to treatment (excluding $t = -2$, which captures training demand), the trends are identical between treatment and control samples. Figure 3.8 displays stacked DiD coefficients for both pre-treatment and post-treatment windows, with the reference period at $t = -1$.

Placebo Tests

We conduct three placebo exercises to assess the validity of our identification strategy. First, to distinguish AI-specific effects from general data center growth, we estimate treatment effects using our 12,283 non-AI data centers as a placebo treatment group. If estimated effects simply reflect data center proximity rather than AI workloads specifically, we would expect similar magnitudes in this specification. The non-AI CPQI coefficient (0.228, $p < 0.001$) matches the AI data center coefficient for facilities without on-site generation, suggesting that general data center load—not AI workloads specifically—drives some of the observed power quality effects. However, the demand-side 2SLS estimates from the stacked DiD, which leverage AI model release timing, remain distinct from this general data center effect.

Second, we implement random date and random location placebo tests. Both yield a p-value of 1.00 for the coefficient in our fossil fuel demand, CPQI, and THD analyses, confirming that our results do not arise from spuriously chosen locations or dates.

3.11 Counterfactual Analyses

We present three main counterfactual exercises to inform stakeholders and policymakers about how alternative evolutionary paths for AI compute might affect grid outcomes. These combine our econometric estimates with data on model characteristics and infrastructure configurations to explore three dimensions: model scaling, geographic redistribution, and workload composition. Our counterfactual policies highly align with the implications of our theoretic model, particularly when considering the role of on-site generation as a solution to the challenges facing power procurement.

1) Model Scaling. We combine data on model size (parameter counts) with our econometric estimates from our least conservative specification to extrapolate grid impacts at different scales.

(a) Power Quality. Fitting a trend line to power quality impacts for Llama-3.1 (405B), PaLM (540B), and Llama 4 Behemoth (2T), we find that scaling from a 2 trillion parameter model to a 4 trillion parameter model would increase power quality impact from **0.321 to 0.434**—a deterioration of roughly 0.113 units, or just over 35% relative to the 2 trillion baseline. The relationship is exponential over log parameter counts, implying that larger models require proportionally greater parameter reductions to achieve the same percentage improvement in power quality. **(b) Power Demand.** Using an analogous extrapolation, we find that **each 1% parameter increase leads to approximately 0.15% higher power demand** within three months of model release. Scaling from 540 billion to one trillion parameters would **increase total power demand by 10.5%**. For DALL-E, halving parameters would **reduce inference demand by approximately 300 GWh**—equivalent to the annual consumption of 27,000–30,000 homes.

2) Training Compute. Training compute requirements also have measurable effects: a 1% increase in training FLOPs is associated with a **0.1% increase in power demand**. For GPT-

3.5, halving training compute would reduce energy use by enough to power 8,000–10,000 U.S. homes for a year.

3) Colocated On-site Generation. Data centers increasingly deploy colocated on-site generation to reduce grid dependence. Our inventory identifies 87 facilities with on-site capability and 12,401 without. We test whether on-site capacity moderates estimated effects by interacting treatment with on-site status in the utility-level strict DiD specification. Results indicate that on-site generation substantially mitigates measured grid impacts. The sign reversal for CPQI is notable: data centers *with* on-site generation are associated with improved power quality (negative coefficient), while those *without* are associated with deterioration (positive coefficient). This heterogeneity suggests that **on-site generation absorbs demand spikes that would otherwise propagate to the grid**, and that aggregate grid impacts may understate total AI energy consumption.

Figure 3.10 extends this to policy-relevant scenarios. Remarkably, universal adoption of on-site power would shift impact of data centers from a net detriment to power quality to a net improvement.

4) Geographic Redistribution

(a) Distributed Inference (Edge Computing). In our sample where we know AI data centers but not specific training and inference sites, we simulate a shift from cloud-based to edge-based inference, redistributing inference demand from concentrated hyperscaler facilities to smaller data centers located closer to population centers. Under this scenario, total demand added in a particular zone falls from a baseline of 162 MWh per hour over the entire treatment period to 4 MWh per hour. Figure 3.10 illustrates this effect of shifting the composition of inference.

(b) Concentration in Low-Population, High-Supply Regions. Alternatively, we consider relocating both training and inference to regions with abundant electricity supply and lower

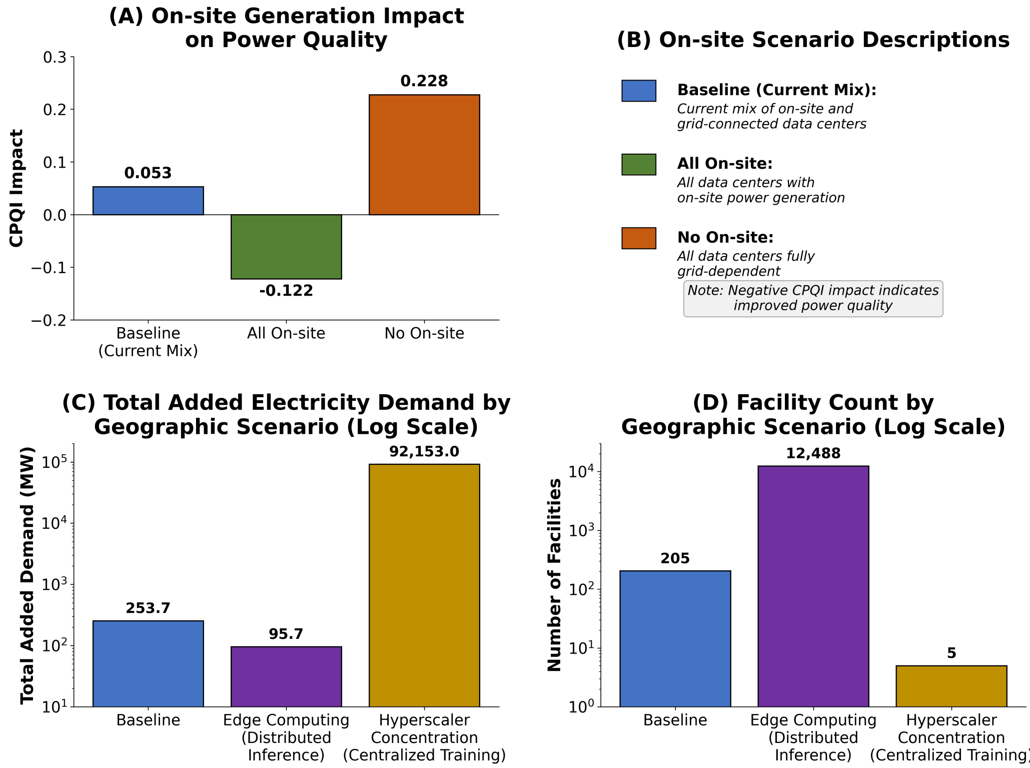


Figure 3.10: Effects of Geographic Concentration and On-site Generation on AI Impacts on the grid.

baseline load, such as areas in the central United States with substantial renewable or fossil capacity. We simulate this by shifting demand from the baseline case where training and inference are assumed distributed to a case where the only data center for training and inference is the largest data center owned by a hyperscaler. This leads to an extreme concentration of added demand with nearly 100 GWh added near data centers in training locations. Figure 3.10 shows this scenario.

This analysis has the important caveat that shifting to one data center for all training activity is extreme, with many possible configurations between full decentralization and full centralization. Centralization may also yield other benefits, such as concentration near renewable generation and reduced energy consumption from lower communication costs.

3.12 Conclusion

This paper provides the first causal estimates of how AI-driven data center demand affects U.S. electricity markets. Exploiting the staggered release of frontier large language models as natural experiments, we find that AI workloads cause statistically and economically significant deterioration in power quality, increase fossil fuel generation on the order of hundreds of GWh, and raise wholesale electricity prices by up to 25% in heavily treated zones. These effects are consistent with the predictions of our game-theoretic model of hyperscaler electricity procurement, which formalizes the *timing wedge*—the temporal mismatch between when data centers consume electricity and when their contracted or credited renewable generation produces.

The economic implications are substantial. The timing wedge means that the voluntary procurement instruments favored by hyperscalers—renewable energy certificates and power purchase agreements—do not internalize the externalities they are designed to address. RECs, as financial transfers unbundled from physical dispatch, leave real-time fossil generation and grid stress unaffected. PPAs improve on this margin only modestly absent firming or storage. Our counterfactual analyses reinforce this finding: universal adoption of on-site generation would reverse the sign of power quality effects, transforming data centers from a net grid liability into a net contributor. This result suggests that the externality is not inherent to AI demand per se, but rather to the spatial and temporal misalignment between that demand and clean generation—a misalignment that current market instruments fail to correct.

These findings speak directly to ongoing policy debates over renewable energy procurement design. Proposals for hourly matching, such as the 24/7 Carbon-Free Energy framework, attempt to close the timing wedge by requiring temporal alignment of clean energy credits with consumption. Our model provides a framework for evaluating such proposals: tightening matching granularity raises the effective price of RECs in high-carbon hours, increases the private value of storage, and induces entry of temporally aligned generation.

The empirical heterogeneity we document between data centers with and without on-site generation offers suggestive evidence that policies promoting behind-the-meter investment could yield grid-wide benefits. Geographic redistribution of inference workloads—toward edge computing or toward high-supply, low-load regions—offers an additional policy lever, though with important tradeoffs between transmission costs and local reliability.

Our analysis has limitations. The parallel trends assumption underpinning any difference-in-differences design may be violated if treated and control regions were already diverging prior to model releases, or if generators anticipated these releases. Spillovers across interconnected regions and heterogeneous effects by grid topology or model type may attenuate or bias our estimates. Our power quality dataset, while novel, covers a limited time horizon that constrains our ability to characterize long-run trends. Finally, frontier model releases are an imperfect proxy for the continuous evolution of AI compute demand, and the mapping from a discrete release event to actual deployment and load is necessarily noisy.

Despite these limitations, the results carry clear implications for electricity market design. As AI workloads grow, regulators and system operators will need instruments that address not only the volume but the temporal and spatial profile of incremental demand. Our findings suggest that annual renewable accounting provides insufficient incentive for the investments in storage, on-site generation, and demand flexibility that would align private procurement with system-level reliability and emissions goals. The methodological approach developed here—combining econometric causal identification with publicly available grid data—is portable to other settings where proprietary operational data are unavailable, including the effects of AI on network congestion, cooling water demand, or semiconductor supply chains.

3.13 Appendix

3.13.1 Results Tables

Table 3.3: Estimated Impact of AI Model Releases on Power Quality

Model	Provider	Coefficient	Std. Error	R^2
GPT-4.5	Microsoft	0.327***	0.087	0.327
Llama 4 Behemoth (preview)	Facebook	0.325***	0.074	0.313
Claude 3 Opus	Google	0.278***	0.043	0.462
Gemini 1.5 Pro	Google	0.223***	0.026	0.451
GPT-4o	Microsoft	0.198***	0.044	0.331
GPT-4	Microsoft	0.176***	0.033	0.408
Claude 3.7 Sonnet	Google	0.161***	0.041	0.357
Chinchilla	Google	0.143***	0.038	0.570
PaLM (540B)	Google	0.142***	0.029	0.587
LaMDA	Google	0.134**	0.044	0.564
PaLM 2	Google	0.133***	0.024	0.465
Claude 3.5 Sonnet	Google	0.126***	0.036	0.378
OPT-175B	Facebook	0.104***	0.023	0.467
Gemini 1.0 Ultra	Google	0.103***	0.021	0.409
Llama 3.1-405B	Facebook	0.052	0.034	0.315
Minerva (540B)	Google	0.031	0.023	0.519
Parti	Google	0.031	0.023	0.519
GPT-4 Turbo	Microsoft	0.030*	0.017	0.308
GPT-3.5	Microsoft	-0.049**	0.018	0.390
Flan-PaLM 540B	Google	-0.051*	0.026	0.508
U-PaLM (540B)	Google	-0.051*	0.026	0.508
Amazon Titan	Amazon AWS	-0.086***	0.016	0.380

Notes: Robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3.4: Difference-in-Differences Estimates of Data Center Announcements on Cumulative Total Queue Capacity (MW)

	(1) No FE	(2) County & Year FE
Post Announcement	714.459**	150.741*
	(293.890)	(87.001)
Intercept	366.406***	-188.142***
	(19.361)	(13.395)
Observations	128,700	128,700
R^2	0.010	0.780
Adj. R^2	0.010	0.777
Fixed Effects	No	County & Year

Cluster-robust standard errors in parentheses.
*p<0.10,**p<0.05,***p<0.01.

Table 3.5: Difference-in-Differences Estimates of Data Center Announcements on Cumulative Gas Queue Capacity (MW)

	(1) No FE	(2) County & Year FE
Post $\times$ Announced Capacity	0.324***	0.250**
	(0.105)	(0.117)
Intercept	175.718***	-43.457***
	(18.388)	(13.229)
Observations	16,380	16,380
R^2	0.012	0.610
Adj. R^2	0.012	0.604
Fixed Effects	No	County & Year

Cluster-robust standard errors in parentheses.
*p<0.10,**p<0.05,***p<0.01.

Table 3.6: Difference-in-Differences Estimates of Data Center Announcements on Cumulative Gas Queue Capacity (MW), No Interaction

	(1) No FE	(2) County & Year FE
Post Announcement	98.976	273.893***
	(68.413)	(79.284)
Intercept	176.560***	63.585***
	(18.656)	(1.28e-11)
Observations	16,380	16,380
R^2	0.003	0.522
Adj. R^2	0.002	0.514
Fixed Effects	No	County & Year

Cluster-robust standard errors in parentheses.
*p<0.10,**p<0.05,***p<0.01.

Table 3.7: Post-Treatment: Estimated Impact of AI Model Releases on Aggregate Electricity Demand

Model	Provider	Coefficient	p-value	R^2	Aggregate Impact (MWh)
Switch	Google	124.46	< 0.001	0.628	4,154,379
DALL-E	Microsoft	81.54	< 0.001	0.627	2,910,208
GPT-3.5	Microsoft	76.08	< 0.001	0.622	1,034,517
mT5-XXL	Google	63.48	< 0.001	0.625	420,630
Megatron-Turing NLG 530B	Microsoft	46.15	< 0.001	0.628	1,692,209
Codex	Microsoft	35.60	0.0211	0.630	2,132,043
Minerva (540B)	Google	24.03	0.4305	0.624	1,372,621
Parti	Google	20.14	0.4692	0.625	1,183,787
Flan-PaLM (540B)	Google	17.36	0.0501	0.622	504,750
U-PaLM (540B)	Google	17.36	0.0501	0.622	504,750
OPT-175B	Facebook	12.85	0.0161	0.624	583,999
ByT5-XXL	Google	1.28	0.9584	0.624	66,424
ProtT5-XXL	Google	-1.42	0.9370	0.629	-64,268
Meta Pseudo Labels	Google	-3.996	0.8284	0.628	-118,203
FLAN 137B	Google	-4.97	0.7971	0.624	-167,480
Chinchilla	Google	-23.54	0.0980	0.628	-834,056
PaLM (540B)	Google	-24.53	0.0714	0.628	-873,641
LaMDA	Google	-27.41	0.1362	0.626	-643,963
GLaM	Google	-27.76	0.3485	0.625	-750,319
Gopher (280B)	Google	-30.32	0.3113	0.625	-813,756

Notes: Coefficients and p-values are taken from `post_coefficient` and `post_p_value`. Aggregate impact uses `post_aggregate_demand_increase`. The headline causal claim of the chapter is the pooled stacked-DiD coefficient reported in Table 3.2; the model-by-model rows above are reported as *descriptive* per-event estimates rather than as confirmatory tests, since each row identifies off a single release event with limited statistical power. The sign pattern is consistent with the scaling story discussed in Section 3.11: smaller and earlier-generation models (Chinchilla, PaLM 540B, LaMDA, Gopher 280B) carry small or negative point estimates with wide confidence bands, while larger and later-generation models (GPT-4.5, Llama 4 Behemoth, Claude 3 Opus) carry substantially larger and more precisely estimated positive coefficients. We do not interpret negative point estimates on individual older or smaller models as evidence against the AI-induced demand effect.

Table 3.8: Pre-Treatment: Estimated Impact of AI Model Releases on Aggregate Electricity Demand

Model	Provider	Coefficient	p-value	R^2	Aggregate Impact (MWh)
Switch	Google	89.77	< 0.001	0.628	268,851
DALL-E	Microsoft	66.90	< 0.001	0.627	43,755
GPT-3.5	Microsoft	39.60	< 0.001	0.622	1,501,640
Megatron-Turing NLG 530B	Microsoft	35.96	0.0138	0.628	2,088,496
Codex	Microsoft	28.63	0.0108	0.630	1,245,855
Minerva (540B)	Google	-22.51	0.0972	0.624	-781,775
Parti	Google	-22.94	0.1056	0.625	-734,906
Flan-PaLM (540B)	Google	29.16	0.3543	0.622	1,475,972
U-PaLM (540B)	Google	29.16	0.3543	0.622	1,475,972
OPT-175B	Facebook	-21.77	0.2825	0.624	-502,990
ByT5-XXL	Google	-6.20	0.7410	0.624	-177,165
ProtT5-XXL	Google	101.54	< 0.001	0.629	3,280,303
Meta Pseudo Labels	Google	168.29	< 0.001	0.628	4,122,856
FLAN 137B	Google	3.41	0.8961	0.624	181,878
Chinchilla	Google	-23.61	0.4060	0.628	-583,230
PaLM (540B)	Google	-22.84	0.4174	0.628	-554,841
LaMDA	Google	-27.80	0.3750	0.626	-817,029
GLaM	Google	-12.52	0.4856	0.625	-413,053
Gopher (280B)	Google	-8.12	0.6566	0.625	-271,852

Notes: Coefficients and p-values are taken from `pre_coefficient` and `pre_p_value`. Aggregate impact uses `pre_aggregate_demand_increase`. Dashes indicate missing values in the spreadsheet. As with Table 3.7, model-by-model rows are presented as *descriptive* per-event estimates rather than confirmatory tests; the chapter's identifying inference comes from the pooled stacked-DiD specification.

3.13.2 Expanded Dataset Description

Below, maps can be found providing more detail on the dataset alongside tables on the datasets and the dataset sample selection criteria.

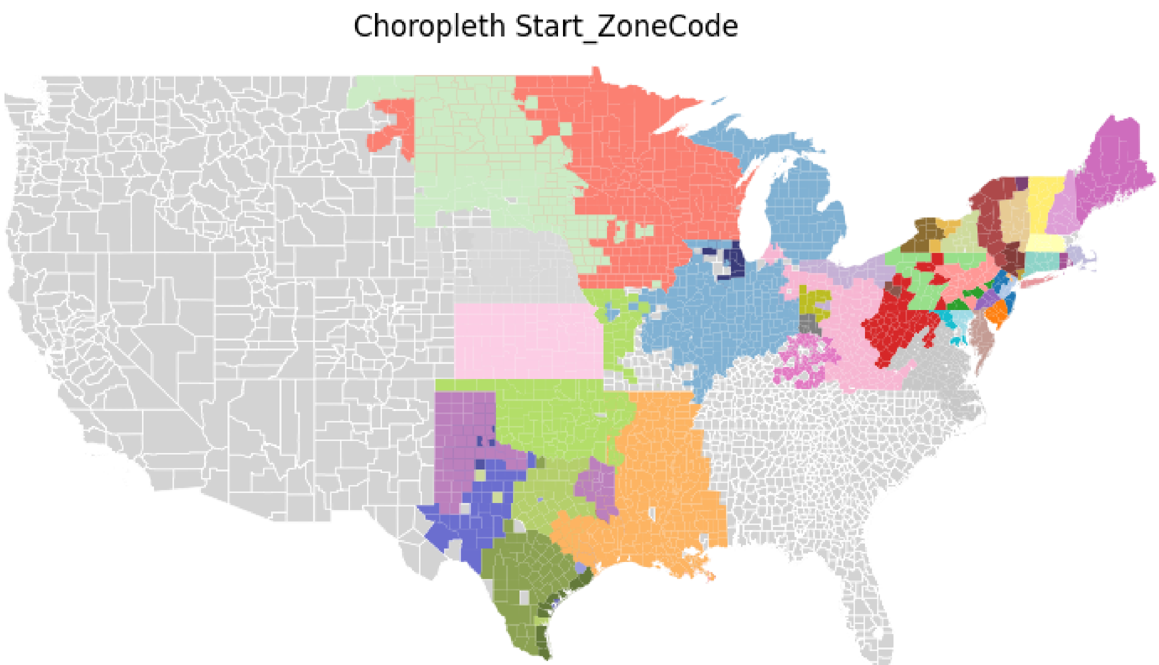


Figure 3.11: Map of zones for demand model.

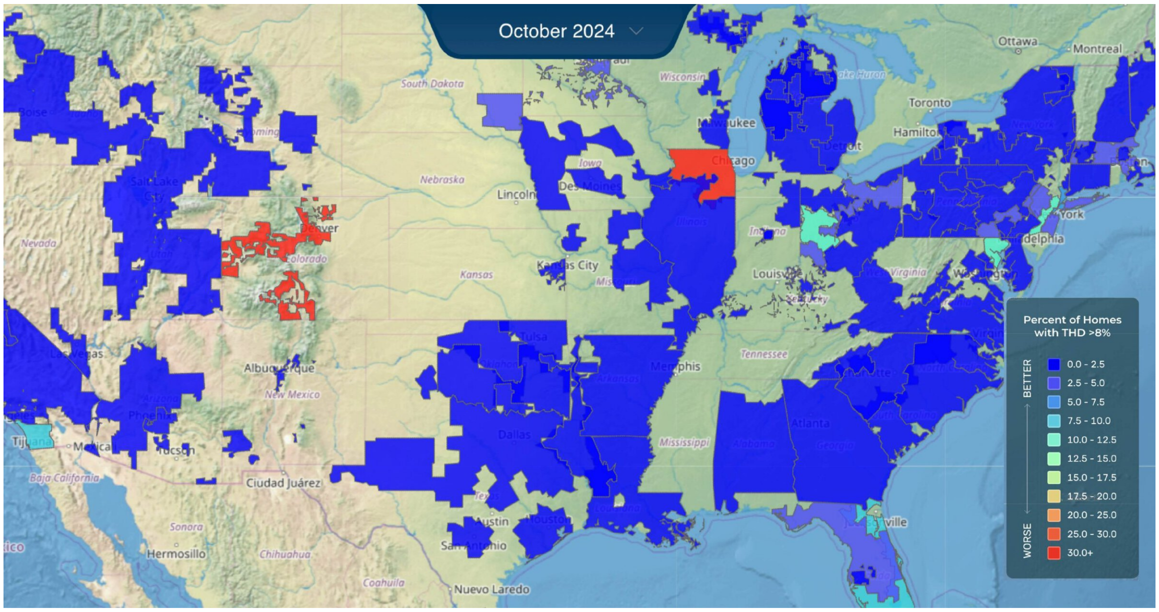


Figure 3.12: Map of Retail Electric Utilities for Whisker Labs Data.

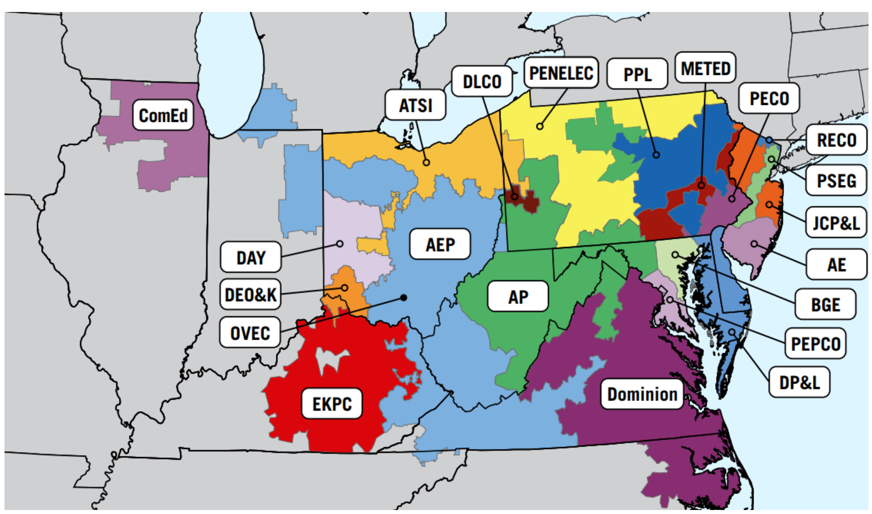


Figure 3.13: Map of PJM by Zones.

Table 3.9: Expanded description of data sources (for appendix).

Source	Content and Granularity	Use in Analysis
	Q1: Difference-in-Differences (Power Quality)	
Aterio	Data center locations, establishment/expansion/retirement dates, ownership, and operational capacity (MW). Data are available at the facility level and updated annually.	Defines treatment regions; links AI model releases to hyperscaler-owned centers.
Whisker Labs	Consumer Power Quality Index (CPQI, composite measure of surges, sags, brownouts, interruptions) and Total Harmonic Distortion (THD, waveform distortion). Provided at county level, hourly frequency.	Outcome variables for DiD regressions.
Meteostat	Hourly temperature and precipitation, aggregated at the county level. Coverage includes all U.S. counties with weather stations.	Controls for demand fluctuations and renewable generation variability.
ISOs	Geographic boundary shapefiles and wholesale price data at zonal level. Boundaries digitized from ISO-provided maps; prices available hourly.	Market alignment and regional controls.
EIA	Retail service territory shapefiles, at utility level, updated periodically.	Used to define treatment/control boundaries and reconcile geographies.
	Q2: Instrumental Variables (Fossil Demand)	
EPA CAMPD	Hourly generator-level demand (MWh) and unit-specific heat rates (Btu/kWh). Coverage 2021–2023.	Generator demand is outcome; heat rates serve as instruments for prices.
S&P Capital IQ	Daily natural gas price series, Henry Hub benchmark. National coverage.	Instrument for price and fuel-cost control.
Aterio	Generator proximity to data centers owned by releasing companies, matched at 20-mile radius.	Defines treatment generators in IV design.
Meteostat	Hourly temperature and precipitation, as above.	Controls for demand and renewable variability.
ISOs	Wholesale price series, zonal level, hourly frequency.	Market-level controls.
	Q3: Counterfactual Analyses (Scaling, Efficiency, & Tech Development)	
Whisker Labs	CPQI and THD (as above). Used to establish baseline deterioration in power quality.	Benchmark outcomes for scaling projections.
Epoch	AI model metadata: release dates, FLOPs, parameter counts. Public dataset with model-level detail.	Used for scaling regressions and efficiency counterfactuals.
GPU Experiments	Energy use measured on NVIDIA A6000 GPUs. Experiments run with batch size 4, sequence length 1024, and 200 inferences. Each bar in results represents total energy; lighter portion indicates communication overhead. Data collected in-house on 8xA6000 cluster.	Provides empirical GPU energy baselines for counterfactual scaling exercises.

Table 3.10: Dataset sample selection and summary statistics

Dataset	Content / Summary Statistics	Sample Selection
Aterio	Data center locations, ownership, and capacity; mapped to ISO zones, EIA service territories, and counties	Hyperscaler data centers (Meta, Microsoft, Amazon, Google; Anthropic via Google)
Whisker Labs	Reliability measures: CPQI ($N = 2{,}683$, mean=0.52, s.d.=0.42); THD (mean=1.81, s.d.=6.77)	72 retail electric utilities, monthly data 2022–2025
CAMPD	Generator-level demand: mean 218 MW (s.d.=158); operating times ≈ 1	Several thousand fossil generators, hourly data 2021–2023
ISOs	Wholesale electricity prices: mean $51/MWh (s.d.=148); zonal boundaries	Deregulated markets in Eastern Interconnection and ERCOT
Weather (Meteostat)	Temperature (mean=17°C, s.d.=11.4); precipitation (mean=0.12 mm, s.d.=0.91)	Matched by county/zone, hourly or daily resolution
External Controls	Natural gas prices, ISO market data, geography/time harmonization	Used for both DiD and IV regressions

Notes: Time series are harmonized at hourly or monthly resolution depending on source. Unified panel supports both difference-in-differences (power quality) and instrumental variable (demand) regressions. Maps of sample regions provided in the appendix.

3.13.3 Dataset Creation Diagram

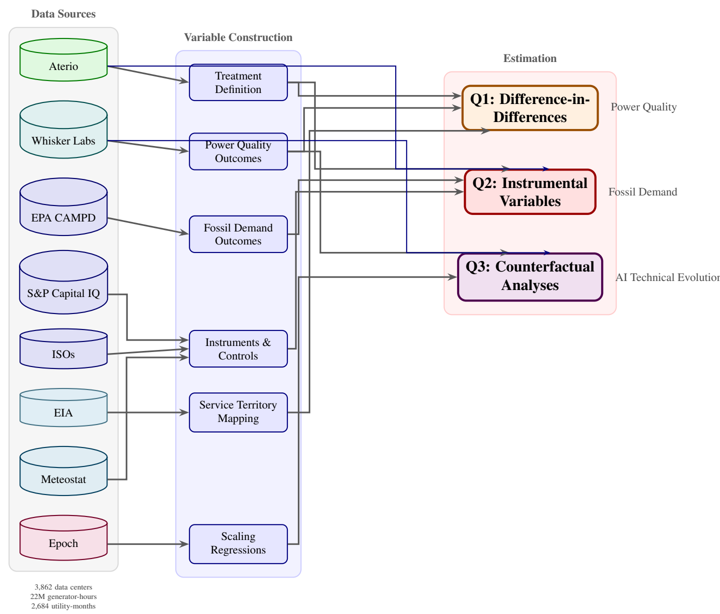


Figure 3.14: Data sources and analysis pipeline. Data sources flow through variable construction into three estimation strategies addressing power quality impacts (Q1), fossil fuel demand (Q2), and counterfactual scaling scenarios (Q3).

3.13.4 Construction of Zonal Geographic Maps

A central requirement of the analysis is the ability to match generators to geographic areas in a manner consistent with ISO market structure while preserving as much spatial detail as possible. Because publicly available GIS shapefiles for ISO-defined market zones are incomplete, inconsistent, or entirely unavailable, the construction of zonal maps required substantial manual effort. I view this process as a significant contribution of the paper.

The guiding principle throughout was to aggregate only where necessary to permit consistent geographic matching and econometric analysis, while keeping the data at the lowest feasible spatial resolution. In all cases, zonal boundaries were explicitly restricted to lie within ISO boundaries using GIS shapefiles defining Independent System Operator footprints. This prevents generators, loads, or weather observations from being incorrectly assigned across ISOs.

General Approach. The mapping procedure proceeds in three steps. First, ISO boundaries are imposed as a hard geographic constraint. Second, ISO-specific zones are mapped to counties or county aggregates using a combination of published documentation, utility maps, and GIS overlays. Third, generators are assigned to zones using EIA Form 860 identifiers where available, and county-level geographic matching otherwise. Hand-constructed crosswalk tables are used to reconcile inconsistent geographic schemas across datasets.

Where counties do not align cleanly with ISO zones, counties are split proportionally based on geographic overlap and engineering judgment, applied consistently across datasets.

ERCOT. For ERCOT, zones are based primarily on the system's published weather zones, which in most cases map cleanly to county boundaries. Where weather zones do not align perfectly with counties, I split counties across zones using GIS overlays to preserve contiguity and proportional area. Several ERCOT datasets rely on alternative geographic schemas (e.g., hub- or region-based reporting); in these cases, I construct hand-built mapping tables to reconcile these schemas with the zonal structure used in the analysis.

PJM. PJM required the most extensive manual mapping effort. PJM publishes a large number of zones, many of which do not correspond cleanly to county boundaries. I manually mapped each PJM zone to counties using a combination of PJM documentation, utility service maps, and GIS inspection. In cases where counties span multiple PJM zones, judgment was applied to split counties as cleanly as possible while preserving geographic continuity. Where appropriate for the analysis, zones were subsequently aggregated into regions to support the distribution of generation data.

SPP. SPP zones were mapped using a similar procedure, combining transmission zone definitions with county-level utility maps and GIS shapefiles. In several states, publicly available utility maps with county labels were used to guide assignments. As with PJM, counties were split only where necessary, and always within ISO boundaries.

NYISO. For NYISO, zones were mapped based on reliability assessment zones using a zonal county map. These zones align relatively well with county boundaries, allowing for a clean assignment in most cases.

MISO. MISO presents a comparatively clean case: all zonal boundaries correspond to state-level regions. As a result, counties were assigned directly to MISO zones based on state boundaries, eliminating the need for county splitting.

ISO New England. ISO New England zones are primarily defined at the state level, with the notable exception of Massachusetts, which is divided into three wholesale zones. For Massachusetts, I manually mapped counties to zones using GIS overlays and ISO documentation, splitting counties where necessary to maintain geographic consistency.

Generator Assignment. Generators are assigned to zones using ISO labels from EIA Form 860 wherever possible, which minimizes the risk of misassignment across ISOs. When ISO labels are unavailable or insufficiently granular, generators are matched using county-level geographic information derived from plant location data. This layered approach ensures consistency between generator assignment, zonal geography, and ISO market structure.

Overall, this mapping procedure enables the construction of consistent zonal maps that balance geographic precision with analytical tractability, forming the backbone of the spatial analysis in the paper.

3.13.5 Counterfactual

Figure 3.15 shows the power quality line we fit for performing counterfactuals.

3.13.6 Calculation of Comparisons

We calculate the household average energy usage based on data provided by EIA (2024). We simply divide our estimates by the provided numbers to get the number of households

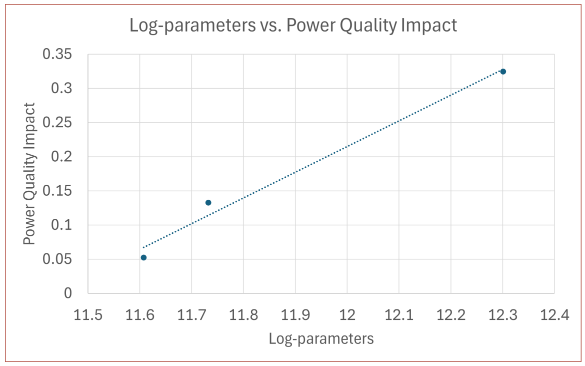


Figure 3.15: Power Quality Impact for Counterfactuals.

for our comparisons.

3.13.7 Code and Replication

This appendix provides comprehensive documentation for replicating the empirical analysis. All code and data processing scripts are available in the supplementary materials.

Software Requirements

The analysis pipeline is implemented in Python 3.11 and designed to execute on a Linux HPC cluster using the SLURM workload manager. Scripts include environment-aware path detection to support local execution on Windows systems.

Computational Resources. The full pipeline requires approximately 32GB of memory and 8 CPU cores, with an expected runtime of 12–24 hours on an HPC cluster (24–48 hours for local execution). Memory-intensive operations include spatial joins for utility service territory matching and construction of stacked difference-in-differences datasets.

Python Dependencies. The analysis requires the following packages: `pandas` ($\geq$2.0), `numpy` ($\geq$1.24), `scipy` ($\geq$1.10), `statsmodels` ($\geq$0.14), `linearmodels` ($\geq$5.0), `geopandas` ($\geq$0.14), `shapely` ($\geq$2.0), `fiona` ($\geq$1.9), `pyproj` ($\geq$3.6), `rtree` ($\geq$1.0), `geopy` ($\geq$2.3), `gdal` ($\geq$3.6), `matplotlib` ($\geq$3.7), and `seaborn` ($\geq$0.12). We use `micromamba` for environment management on the cluster.

Data Sources

Table 3.11 summarizes the primary data sources used in the analysis. All data files should be placed in a `Data/` subdirectory relative to the code directory.

Table 3.11: Data Sources and Descriptions

File	Description	Source
Electricity Demand Analysis		
`EPAData.csv`	Hourly emissions and generation	EPA CAMPD
`epa_eia_crosswalk.csv`	EPA–EIA facility identifiers	Custom
`data_center_inventory.csv`	Data center locations with AI flags	Aterio
`mapData.shp`	Zone boundary shapefiles	FERC/ISO
`TransmissionPriceLoad.csv`	Zonal electricity prices and load	ISO/RTO
`ExtendedNGPriceData.csv`	Natural gas price time series	EIA
`mapDatawithWeather.csv`	Zone-level weather data	NOAA
Power Quality Analysis		
`electric-retail-service-territories.shp`	Utility service area boundaries	EIA-861
`Epoch Database - Notable Models.csv`	AI model release dates	Epoch AI
`PowerQuality.csv`	Customer Power Quality Index	Whisker Labs
`THD.csv`	Total Harmonic Distortion	PQ Monitor
Robustness Checks		
`ai_model_dates_verified.csv`	Verified API release dates	Manual

Pipeline Architecture

The analysis executes in four sequential phases, with internal parallelization within phases. Figure 3.16 illustrates the data flow.

Phase 1: Data Construction. The script `CreateDataIVDiffinDiffCAMPD.py` constructs the panel dataset by merging EPA emissions data with data center locations, electricity market data, and natural gas prices. Key variables created include: `treated` (binary indicator for zones within proximity of AI data centers), `post` (binary indicator for post-AI-model-release periods), and `treatment_intensity` (continuous measure based on model compute requirements). Output: `panelIVDiDData.csv`.

Phase 2: Main Regressions. Four scripts execute in parallel:

1. `IVDiffinDiffCAMPD.py`: IV-DiD estimation for electricity demand using two-stage least squares with natural gas prices as instruments and zone/month fixed effects.
2. `IVDiffinDiffCAMPD_Stacked.py`: Stacked DiD methodology following Cengiz et al. (2019) to address heterogeneous treatment timing, with unit-by-event and time-by-event fixed effects.
3. `DiffinDiff.py`: Difference-in-differences for power quality outcomes (CPQI and THD) at the utility level.
4. `DiffinDiff_Stacked.py`: Stacked DiD for power quality, addressing staggered treatment timing across AI model releases.

Phase 3: Robustness and Visualization. Two parallel jobs: (i) `IVDiffinDiff_VerifiedDates.py` re-estimates the main specifications using hand-verified API release and training completion dates rather than publication dates; (ii) `IVDiffinDiff_SupplyElasticity_PJM.py` estimates supply-side responses followed by `AI_Price_Impact_Choropleth_PJM.py` for visualization.

Phase 4: Counterfactual Analysis. The script `CounterfactualAnalysis.py` performs policy counterfactuals including efficiency gain scenarios (10%, 20%, 30%, 40% improvements), channel shutdown scenarios (training-only versus inference-only operations), geographic redistribution (edge computing versus hyperscaler concentration), and on-site expansion/contraction scenarios.

Phase 1: Data Construction
`CreateDataIVDiffinDiffCAMPD.py` → `panelIVDiDData.csv`

↓

Phase 2: Main Regressions (Parallel)
`IVDiffinDiffCAMPD.py` | `*_Stacked.py` | `DiffinDiff*.py`

↓

Phase 3: Robustness (Parallel)
`IVDiffinDiff_VerifiedDates.py` | `*_SupplyElasticity_PJM.py`

↓

Phase 4: Counterfactuals
`CounterfactualAnalysis.py`

Figure 3.16: Analysis Pipeline Architecture

Key Estimation Parameters

The main specifications use the following default parameters, which can be modified for robustness checks:

- Proximity threshold: 20 miles (for defining treatment based on data center locations)
- Fixed effects: Zone and month
- Stacked DiD event windows: ±2 months (default), with robustness checks at ±1, ±3, and ±6 months
- Instruments: Natural gas prices (exogenous cost shifter for electricity supply)

Executing the Analysis

HPC Cluster (SLURM). Submit the job file with `sbatch Full_Analysis_Pipeline.job`. Monitor progress with `squeue -u $USER` and check phase-specific logs in `logs/phase*/`.

Local Execution. Scripts automatically detect local execution and adjust file paths accordingly. Run Phase 1 first, then Phases 2–3 scripts can execute in parallel, followed by Phase 4:

```
python CreateDataIVDiffinDiffCAMPD.py          # Phase 1
python IVDiffinDiffCAMPD.py &                  # Phase 2 (parallel)
python DiffinDiff.py &
wait
python CounterfactualAnalysis.py               # Phase 4
```

Output Files

Table 3.12 summarizes the primary output files generated by the pipeline.

Table 3.12: Primary Output Files

File/Directory	Description
panelIVDiDData.csv	Constructed panel dataset
IVDiffinDiffResults.csv	IV-DiD demand coefficients and standard errors
DiffinDiffResults.csv	CPQI regression results
THD_DiffinDiffResults.csv	THD regression results
stacked_did_iv_results/	Stacked DiD coefficients and plots (demand)
stacked_did_cpqi_results/	Stacked DiD coefficients and plots (CPQI)
stacked_did_thd_results/	Stacked DiD coefficients and plots (THD)
IVDiffinDiff_VerifiedDates_Results.csv	Robustness with verified dates
pjm_ai_supply_impact_by_zone.csv	PJM zone-level supply impacts
counterfactual_outputs/	Scenario analysis results and visualizations

Shared Module

The stacked DiD methodology is implemented in a shared module (`StackedDiD.py`) used by both the demand and power quality analyses. Key functions include: `create_stacked_dataset()` for constructing mini-datasets for each treatment event; `run_stacked_did_regression()` for executing stacked DiD regressions with appropriate fixed effects; `aggregate_treatment_effects()` for cohort-weighted aggregation of treatment effects; and `plot_event_study()` for visualization of event-time coefficients. This modular design ensures methodological consistency across outcome variables.

Reproducibility Notes

Several implementation details enhance reproducibility. First, scripts automatically handle multicollinearity via `check_and_fix_multicollinearity()` functions that detect and address near-singular matrices in the fixed effects specifications. Second, memory-intensive operations include explicit garbage collection (`gc.collect()`) to manage memory on systems with limited resources. Third, all random seeds are fixed where applicable to ensure deterministic results. Fourth, the SLURM job file specifies exact resource requirements (32GB memory, 8 CPUs across 4 tasks) that can be scaled based on available infrastructure. Log files in `logs/phase*/` provide detailed execution traces for debugging.

3.13.8 Specifically Verified Models

Table 3.13: Documented Training Locations

Model	Provider	Location	Confirmation	Primary S
GPT-4	OpenAI	West Des Moines, Iowa	Confirmed	Brad Smi
PaLM, LaMDA, MUM	Google	Mayes County, Oklahoma	Confirmed	Google C
BLOOM	BigScience	Orsay, France (Jean Zay)	Confirmed	Hugging
Llama 1-2	Meta	RSC (location unconfirmed)	Partial	Meta mod
Gemini Ultra	Google	Multiple sites (unspecified)	Partial	Thomas K
Claude models	Anthropic	AWS/Google Cloud (regions)	Partial	Anthropic
Megatron-Turing NLG	NVIDIA/Microsoft	Santa Clara area (Selene)	Confirmed	NVIDIA/

Table 3.14: Verified Model Dates

Company	Model	Date Type	Date	Precision	Source
OpenAI	GPT-3	API Release	2020-06-11	day	OpenAI (2020)

Company	Model	Date Type	Date	Precision	Source
OpenAI	GPT-3	API Release	2021-11-18	day	OpenAI (2021a)
OpenAI	Codex	API Release	2021-08-10	day	OpenAI (2021b)
OpenAI	DALL-E 2	API Release	2022-11-03	day	OpenAI (2022b)
OpenAI	ChatGPT	API Release	2022-11-30	day	OpenAI (2022a)
OpenAI	GPT-3.5-turbo	API Release	2023-03-01	day	OpenAI (2023a)
OpenAI	GPT-4	API Release	2023-03-14	day	OpenAI (2023c)
OpenAI	GPT-4	API Release	2023-07-06	day	OpenAI (2023d)
OpenAI	GPT-4	Training Date Verified	2022-08	month	OpenAI (2023e)
OpenAI	GPT-4V	API Release	2023-11-06	day	OpenAI (2023b)
OpenAI	GPT-4 Turbo	API Release	2023-11-06	day	OpenAI (2023b)
OpenAI	GPT-4 Turbo	API Release	2024-04-09	day	OpenAI (2024a)
OpenAI	DALL-E 3	API Release	2023-11-06	day	OpenAI (2023b)

Company	Model	Date Type	Date	Precision	Source
OpenAI	GPT-4o	API Release	2024-05-13	day	OpenAI (2024c)
OpenAI	GPT-4o mini	API Release	2024-07-18	day	OpenAI (2024b)
OpenAI	o1-preview	API Release	2024-09-12	day	OpenAI (2024d)
OpenAI	o1-mini	API Release	2024-09-12	day	OpenAI (2024d)
Anthropic	Claude 1.0	API Release	2023-03-14	day	Anthropic (2023d)
Anthropic	Claude 2.0	API Release	2023-07-11	day	Anthropic (2023b)
Anthropic	Claude Instant 1.2	API Release	2023-08-09	day	Anthropic (2023c)
Anthropic	Claude 2.1	API Release	2023-11-21	day	Anthropic (2023a)
Anthropic	Claude 3 Opus	API Release	2024-03-04	day	Anthropic (2024c)
Anthropic	Claude 3 Sonnet	API Release	2024-03-04	day	Anthropic (2024c)
Anthropic	Claude 3 Haiku	API Release	2024-03-13	day	Anthropic (2024d)
Anthropic	Claude 3.5 Sonnet	API Release	2024-06-20	day	Anthropic (2024b)

Company	Model	Date Type	Date	Precision	Source
Anthropic	Claude 3.5 Sonnet (Upgraded)	API Release	2024-10-22	day	Anthropic (2024a)
Anthropic	Claude 3.5 Haiku	API Release	2024-11-04	day	Anthropic (2024a)
Google	Bard	API Release	2023-03-21	day	Google (2023a)
Google	PaLM API	API Release	2023-03-14	day	Google Developers Blog (2023)
Google	PaLM 2	API Release	2023-05-10	day	Google (2023c)
Google	Gemini 1.0 Pro	API Release	2023-12-13	day	Google (2023b)
Google	Gemini 1.0 Ultra	API Release	2024-02-08	day	Google Developers Blog (2024)
Google	Gemini 1.5 Pro	API Release	2024-02-15	day	Google (2024a)
Google	Gemini 1.5 Pro	API Release	2024-05-14	day	Google (2024b)
Google	Gemini 1.5 Flash	API Release	2024-05-14	day	Google (2024b)
Google	Gemma 7B	API Release	2024-02-21	day	Google (2024d)

Company	Model	Date Type	Date	Precision	Source
Google	Gemma 2	API Release	2024-06-27	day	Google (2024c)
Google	Gemini 2.0 Flash	API Release	2024-12-11	day	Google DeepMind (2024)
Meta	LLaMA	API Release	2023-02-24	day	Meta AI (2023c)
Meta	Llama 2	API Release	2023-07-18	day	Meta Newsroom (2023)
Meta	Code Llama	API Release	2023-08-24	day	Meta AI (2023a)
Meta	Code Llama	Training Date Verified	2023-01–2023-07	range	Meta AI (2023b)
Meta	Code Llama 70B	API Release	2024-01-29	day	Meta AI (2023a)
Meta	Llama 3 8B	API Release	2024-04-18	day	Meta AI (2024b)
Meta	Llama 3 70B	API Release	2024-04-18	day	Meta AI (2024b)
Meta	Llama 3.1 8B	API Release	2024-07-23	day	Meta AI (2024a)
Meta	Llama 3.1 70B	API Release	2024-07-23	day	Meta AI (2024a)

Company	Model	Date Type	Date	Precision	Source
Meta	Llama 3.1 405B	API Release	2024-07-23	day	Meta AI (2024a)
Meta	Llama 3.2	API Release	2024-09-25	day	Meta AI (2024c)
Meta	Llama 3.3 70B	API Release	2024-12-06	day	Hugging Face (2024)
Mistral	Mistral 7B	API Release	2023-09-27	day	Mistral AI (2023a)
Mistral	Mixtral 8x7B	API Release	2023-12-11	day	Mistral AI (2023b)
Mistral	Mistral API (La Plateforme)	API Release	2024-02-26	day	Mistral AI (2024b)
Mistral	Mistral Large	API Release	2024-02-26	day	Mistral AI (2024c)
Mistral	Mixtral 8x22B	API Release	2024-04-17	day	Mistral AI (2024e)
Mistral	Codestral	API Release	2024-05-29	day	Mistral AI (2024a)
Mistral	Mistral Nemo	API Release	2024-07-18	day	Mistral AI (2024f)
Mistral	Mistral Large 2	API Release	2024-07-24	day	Mistral AI (2024d)
Mistral	Pixtral 12B	API Release	2024-09-17	day	Mistral AI (2024g)

Company	Model	Date Type	Date	Precision	Source
Mistral	Pixtral Large	API Release	2024-11-18	day	Mistral AI (2024h)

3.13.9 Propensity Score Matching

Here, we describe the propensity score matching (PSM) and inverse probability weighting (IPW) methodology used to enhance the robustness of our difference-in-differences (DiD) estimates. These methods address potential selection bias arising from non-random placement of AI data centers.

Standard DiD estimation relies on the parallel trends assumption: absent treatment, treated and control units would have followed similar trajectories. However, AI data center locations are chosen strategically based on factors such as electricity prices, grid reliability, and climate conditions. If power plants near AI data centers systematically differ from control plants in observable characteristics correlated with electricity demand trends, the parallel trends assumption may be violated.

Propensity score methods address this concern through three mechanisms: (i) balancing covariates to create comparison groups similar on observable characteristics; (ii) enforcing common support to ensure we only compare units that could plausibly receive either treatment status; and (iii) reducing model dependence by making estimates less sensitive to functional form assumptions (Rosenbaum and Rubin, 1983; Hirano et al., 2003).

The propensity score is the conditional probability of receiving treatment given observed covariates:

$$e(\mathbf{X}_i) = \Pr(T_i = 1 \mid \mathbf{X}_i) \tag{3.26}$$

where T_i is a treatment indicator equal to one for observations near an AI data center in the post-treatment period, and $\mathbf{X}_i$ is a vector of pre-treatment covariates.

We estimate propensity scores using logistic regression:

$$\log\left(\frac{e(\mathbf{X}_i)}{1-e(\mathbf{X}_i)}\right) = \beta_0 + \boldsymbol{\beta}'\mathbf{X}_i \tag{3.27}$$

The covariates included in the propensity score model are: pre-trend load growth, ambient temperature (°F), dew point (°F), precipitation (inches), plant heat rate (BTU/kWh), and natural gas price ($/MMBtu). Temperature and dew point capture weather-driven demand variation and cooling load requirements. Precipitation provides additional weather controls. Heat rate proxies for plant efficiency and technology type. Natural gas price captures fuel cost variation affecting dispatch decisions.

We restrict the sample to observations with propensity scores in the common support region $[0.1, 0.9]$:

$$\mathscr{S} = \{i : 0.1 \leq e(\mathbf{X}_i) \leq 0.9\} \tag{3.28}$$

This trimming rule excludes observations with extreme propensity scores—units almost certain to be treated or untreated—for which comparable counterfactuals do not exist (Crump et al., 2009). Observations with $e(\mathbf{X}_i) < 0.1$ have no comparable treated units, while those with $e(\mathbf{X}_i) > 0.9$ have no comparable control units. The trimmed sample ensures that treatment effect estimates rely on regions of the covariate space where both treatment statuses are observed.

IPW creates a pseudo-population where treatment assignment is independent of observed covariates by weighting observations inversely to their probability of receiving their actual treatment status (Hirano et al., 2003). We employ stabilized weights to reduce variance:

$$w_i^{\text{stab}} = \begin{cases} \dfrac{\Pr(T=1)}{e(\mathbf{X}_i)} & \text{if } T_i = 1 \\[2ex] \dfrac{1-\Pr(T=1)}{1-e(\mathbf{X}_i)} & \text{if } T_i = 0 \end{cases} \tag{3.29}$$

where $\Pr(T=1)$ is the unconditional (marginal) treatment probability in the sample.

To prevent extreme weights from inducing instability, we cap weights at the interval $[0.01, 100]$. Within each treatment group, weights are then normalized to sum to the group size:

$$\tilde{w}_i = w_i^{\text{capped}} \times \frac{n_g}{\sum_{j \in g} w_j^{\text{capped}}} \tag{3.30}$$

where $g \in \{\text{treated}, \text{control}\}$ and n_g is the number of observations in group g.

Our baseline DiD specification takes the form:

$$Y_{it} = \alpha + \beta_1 \text{Post}_t + \beta_2 \text{Treat}_i + \delta(\text{Post}_t \times \text{Treat}_i) + \mathbf{X}_{it}' \boldsymbol{\gamma} + \varepsilon_{it} \tag{3.31}$$

where Y_{it} is the outcome of interest (electricity demand), Post_t indicates the post-treatment period, Treat_i indicates treatment group membership, and δ is the treatment effect of interest.

With IPW, we estimate a weighted least squares version:

$$\hat{\boldsymbol{\theta}}^{\text{IPW}} = \arg\min_{\boldsymbol{\theta}} \sum_{i,t} \tilde{w}_{it} \left(Y_{it} - \mathbf{Z}_{it}' \boldsymbol{\theta}\right)^2 \tag{3.32}$$

where $\mathbf{Z}_{it}$ includes all regressors and $\tilde{w}_{it}$ are the normalized IPW weights.

Our primary specification combines IPW with instrumental variables to address both selection bias (via IPW) and price endogeneity (via IV). The instruments for electricity price are natural gas price and zone-average heat rate. The resulting IV-IPW-DiD estimator is implemented via weighted two-stage least squares, following the doubly robust approach of Sant'Anna and Zhao (2020).

We assess covariate balance using the standardized mean difference (SMD):

$$\text{SMD}_k = \frac{\bar{X}_{k,\text{treated}} - \bar{X}_{k,\text{control}}}{\sqrt{(s_{k,\text{treated}}^2 + s_{k,\text{control}}^2)/2}} \tag{3.33}$$

where $\bar{X}_{k,g}$ and $s_{k,g}^2$ denote the sample mean and variance of covariate k in group g.

Following standard practice, we consider $|\text{SMD}| < 0.1$ as indicating good balance,

$0.1 \leq |\text{SMD}| < 0.25$ as moderate imbalance warranting caution, and $|\text{SMD}| \geq 0.25$ as substantial imbalance (Austin, 2011). After weighting, we compute weighted SMDs using IPW weights to verify that the reweighting procedure successfully balances covariates across treatment groups.

We implement several robustness checks to validate our PSM-IPW approach.

First, we compare treatment effect estimates from standard (unweighted) DiD against IPW-DiD. Similar estimates across specifications suggest results are robust to selection on observables.

Second, we test sensitivity to trimming thresholds by varying the propensity score bounds (e.g., $[0.05, 0.95]$, $[0.15, 0.85]$, $[0.20, 0.80]$). Stable estimates across thresholds indicate robustness to the choice of common support region.

Third, we conduct a pre-trends test by estimating a specification that includes a placebo pre-treatment indicator. An insignificant coefficient on this term ($p > 0.05$) supports the parallel trends assumption.

Fourth, we replicate the analysis using only confirmed AI training facility locations from verified sources, reducing potential measurement error in treatment assignment.

Fifth, we compare covariate balance statistics before and after weighting. IPW should substantially reduce SMDs across all covariates.

The IPW-DiD estimator provides a causal estimate of the average treatment effect under the following identifying assumptions: (i) conditional independence—treatment assignment is independent of potential outcomes conditional on $\mathbf{X}$; (ii) parallel trends—absent treatment, treated and control units would follow similar outcome trajectories; (iii) common support—there exists positive probability of treatment for all covariate values; (iv) stable unit treatment value assumption (SUTVA)—no spillovers between units; and (v) correct propensity score specification—the logistic regression model is correctly specified (Abadie, 2005; Callaway and Sant'Anna, 2021).

An important limitation is that propensity score methods only control for selection on *observable* characteristics. If unobserved factors influence both data center location decisions and electricity demand trends, our estimates may still be biased. We partially address this concern through instrumental variables estimation, but acknowledge that unobserved confounding cannot be definitively ruled out in observational settings.

Calculation Methodology Price Effects

Rather than estimating a reduced-form price regression, we compute implied price effects as:

$$\Delta\hat{p} = \frac{\hat{\beta} \cdot \Delta \mathrm{AI}_{it}}{\hat{b}} \tag{3.34}$$

where $\hat{\beta}$ is our estimated demand coefficient from Section 3.3, $\Delta \mathrm{AI}_{it}$ is the observed change in AI activity, and $\hat{b}$ is the supply curve slope estimated through instrumental variables regression.

3.13.10 PJM price Elasticities

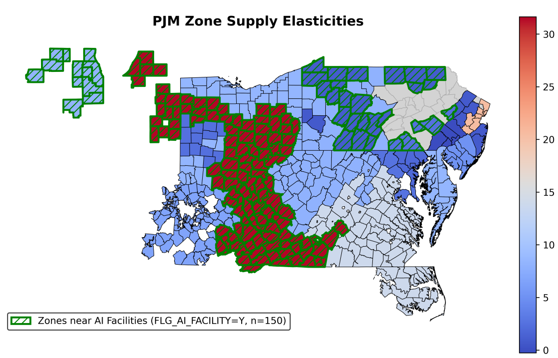


Figure 3.17: PJM Price Elasticity by Zone

3.13.11 Differences Between Power Quality and Demand Analyses

The power quality and fossil fuel demand analyses differ in geographic resolution and data structure due to the underlying data sources:

These differences arise from data availability rather than conceptual distinctions. The

	Power Quality	Fossil Fuel Demand
Unit of observation	Utility-month	Generator-month
Geographic resolution	Utility service territory	Generator location
Treatment linkage	Utility contains AI data center	Generator near AI data center
Outcome source	Whisker Labs	EPA CEMS
Price endogeneity	Not modeled	IV with heat rate, gas price

Table 3.15: Summary of empirical designs

power quality measures from Whisker Labs are aggregated to utility service territories, precluding finer spatial analysis. Generator-level EPA data permit more granular geographic matching, which we exploit for robustness analysis. The demand analysis additionally requires addressing price endogeneity because generation quantities reflect equilibrium outcomes in electricity markets; power quality indices are not directly determined by market clearing and thus do not require the same instrumentation strategy.

Despite these differences, we maintain a unified stacked DiD framework centered on model release dates for both analyses. This consistency facilitates comparison of effect dynamics across outcomes and provides a coherent narrative linking AI development activity to grid impacts.

3.13.12 Additional Counterfactual Analysis

Training versus Inference Composition

Training-Inference Composition. Third, we explore alternative compositions of AI workloads by selectively restricting channels in our model. By suppressing the training channel, we simulate power quality impacts under a scenario in which large-scale model training plateaus and the industry shifts toward an inference-dominant paradigm. Conversely, by suppressing the inference channel, we characterize outcomes under a shift toward primarily local, low-cost inference with reduced centralized training activity.

We explore how grid impacts would differ under alternative compositions of AI workloads.

Scenario: Inference-Dominant World. If large-scale model training plateaus—perhaps due to diminishing returns to scale or compute constraints—while inference continues to grow with adoption, grid impacts would reflect primarily the inference channel.

Scenario: Training-Dominant World. Conversely, if inference becomes predominantly local (on-device) while cloud resources are devoted primarily to training new models, impacts would concentrate in the training channel. Suppressing inference effects reduces additional power demand per hour per data center by significantly more than suppression of training. Figure 3.18 provides insight into this relationship.

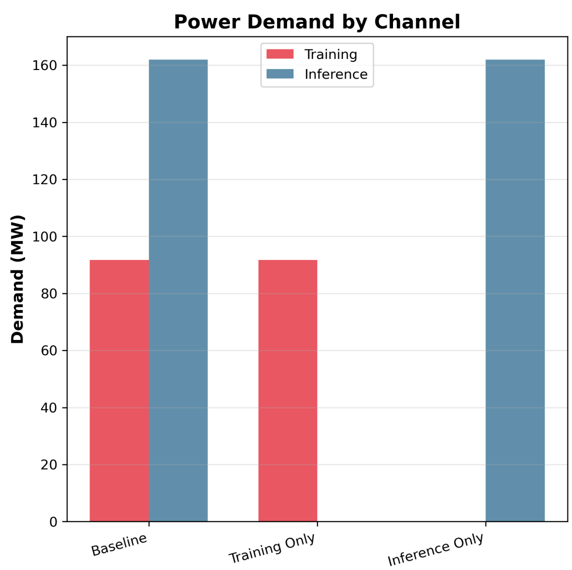


Figure 3.18: Grid impacts under inference-only versus training-only scenarios.

Efficiency Improvements

Scenario: AI Efficiency Improvements. Algorithmic efficiency represents a key lever for mitigating grid impacts. We calculate the potential impacts of efficiency gains, remaining agnostic about their source—improvements could come from both hardware and software advances related to AI training and inference. Figure 3.19 demonstrates how specific percentage-wise improvements in efficiency would affect the hourly demand coefficient. While a 40% improvement in inference efficiency in our sample would lead to a 25% reduction in total added demand, the same improvement in training efficiency would lead to only a 14% reduction in demand.

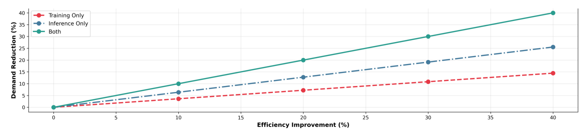


Figure 3.19: Reduction in power demand with efficiency improvements in training and inference.

Impacts of Agglomeration on Cost of System Outage

Impacts of Agglomeration on Cost of System Outage. We examine the potential costs of agglomeration for cloud operators in terms of loss of service due to power outage. The concentration of AI workloads and AI data centers increases the strain in specific areas leading to greater costs on local grids and increasing the potential for long outages, which could lead to loss of service for cloud providers, which creates a significant revenue and reputational cost. These costs are bourne by data center operators themselves and so provide a non-socialized cost of data center impact on power grids that may justify spatial reorganization by the data center industry. We use publicly available estimates of the cost of recent losses of service by cloud platforms to tech companies to estimate these costs.

Geographic concentration poses risks not only to the overall grid but also to cloud service reliability. The increased possibility of reliability events creates potential issues for cloud uptime and threatens continuous assurance of cloud services. As recently as 2022, a power failure in Ohio led to 3 continuous hours of downtime for Amazon Web Services US-EAST-2 (Eisenberg, 2025). Cloud losses were also associated with the blackouts during winter storm Uri in Texas. These outages impose costs on society as well as on hyperscalers in terms of lost revenue, damaged reputations, and security concerns.

We utilize documented estimates of outage costs from the literature (see Table 3.16 in

Appendix Subsection 3.13.14) to estimate the economic costs to hyperscalers of increased reliability risk associated with data center concentration. We multiply the increase in the likelihood of a reliability event by the documented cost associated with a cloud outage of the length of a reliability event. We find that a shift to an edge-computing scenario as detailed above would lead to significant reliability-related savings for hyperscalers in our sample based only on treated models. Similarly, a full implementation of on-site colocated generation would lead to a total average decrease in the number of reliability events in data center utility areas of approximately .17 per month. This means that the number of reliability events per year making the IID assumption decreases by 2.04 events per year. We assume that a significant power outage that leads to a cloud outage has a median duration of 21 hours matching our data with a median cost per hour of 19.2 million. We further conservatively assume that a significant reliability event near a data center has a probability of leading to a cloud outage of 5%. The implied reliability savings associated with on-site generation are thus **41 million** accrued to the hyperscaler. This is a significant financial argument for hyperscalers to work towards improved reliability.

3.13.13 Treatment Definition and Data Linkage

Our treatment relies on linking AI model releases to geographic regions through a chain of observable connections: models to companies, companies to data center locations, and data center locations to electricity service territories. We describe each link and its limitations.

Identifying AI Data Centers

We leverage a recently released variable from Aterio that directly classifies data centers by their primary use case, including a designation for AI-focused facilities. This classification improves upon prior approaches that treated all data centers homogeneously, as AI workloads—particularly large-scale model training—exhibit distinct load profiles characterized by sustained high utilization and substantial power draws that differ markedly from traditional cloud computing or storage operations.

We link data centers to electricity service territories at two geographic resolutions:

- **Utility service territory:** For power quality analysis, we match AI data centers to retail electric utility boundaries. A utility is designated as treated if at least one AI data center operates within its service territory.

- **Generator proximity:** For fossil fuel demand analysis, we match data centers to generators by picking generators within a 20-mile radius of an in-sample data center. This finer resolution exploits the greater spatial granularity of EPA generator-level data.

Linking Models to Data Centers

We construct a crosswalk from AI model releases to data center locations through company ownership. For each major model release in our sample, we identify the developing company and link to all AI-classified data centers operated by or contracted to that company.

Identifying Assumption and Limitations. Our approach assumes that training and inference activity for a given model occurs at data centers associated with the releasing company. This assumption is imperfect for several reasons:

1. Training may occur at third-party cloud facilities not captured in our data center inventory.

2. Inference workloads may be distributed across facilities differently than training workloads.

We interpret our estimates as capturing the *average* effect across a company's AI data center footprint, which represents a lower bound on the localized effect at the actual training site if activity is concentrated, or an upper bound if we misattribute activity to facilities

Table 3.16: Cloud Outages: Power-Related and Documented Costs (2021–2024)

Provider	HS	Pwr	Hrs	Cost
OVHcloud Uptime Institute (2021)	0	1	288	€10M[b]
Fastly Fastly (2021)	0	0	1	$34M[c]
GCP Moss (2022)	1	1	18	—
AWS Amazon Web Services (2022)	1	1	3	—
GCP Data Center Knowledge (2022)	1	1	4	—
Rackspace Rackspace Technology (2023)	0	0	504	$35M[d]
Alibaba South China Morning Post (2022)	0	1	24	—
GCP Sharwood (2023)	1	1	336	—
CrowdStrike[e] Parametrix Insurance (2024)	0	0	72	$5.4B[f]
Alibaba Data Center Dynamics (2024)	0	1	144	—
GCP Sharwood (2024)	1	1	13	—
Azure StatusGator (2024)	1	1	6	—

Notes: HS = Hyperscaler (AWS/Azure/GCP). Pwr = Power/electrical/cooling/fire. Hrs = Duration. [a]TX total; $4.3B direct. [b]Lawsuit awards. [c]Amazon only. [d]Revenue + remediation. [e]Third-party; affected Azure. [f]Fortune 500; insured $0.5–1.1B.

that were not involved. We probe robustness to this measurement error through several approaches described in Section 3.7.

For a subset of models where public information permits more precise dating, we supplement our treatment timing with verified training and inference windows:

3.13.14 Quantifying Costs of Loss of Cloud Services

Table 3.16 presents cloud service outages filtered to include only (1) incidents related to power infrastructure failures, or (2) incidents with verified financial impact estimates from cited sources. This dataset supports analysis of agglomeration costs for cloud operators arising from power grid strain and service disruption.

Notes:

b. Lawsuit settlements; individual awards ranged €100K–€154K.

c. Estimated Amazon revenue loss alone during 1-hour outage.

d. Combines $30M annual Hosted Exchange revenue loss plus $5M remediation costs per SEC filings.

e. Third-party vendor incident primarily affecting Microsoft Azure customers.

f. Fortune 500 companies only; insured losses estimated \$540M–\$1.08B by Parametrix.

Variable Definitions:

- **Hyperscaler**: Binary indicator (1 = AWS, Azure, or GCP; 0 = other providers).
- **Power**: Binary indicator (1 = outage caused by or related to power infrastructure, electrical systems, cooling failures, or fires; 0 = software, network, or security causes).
- **Duration**: Hours from incident onset to service restoration; "+" indicates ongoing recovery beyond initial resolution.

Instrumental Variables Specification

The learned DiD coefficients capture the relationship between AI model releases and generator output, but this relationship may be confounded by supply-side shocks. For example, if AI model releases coincide with changes in natural gas prices or generator availability that independently affect electricity production, our estimates would conflate demand-side and supply-side effects. To isolate the demand-side effects, we instrument for electricity prices using the interaction of generator-level heat rates and regional natural gas prices. This instrumentation strategy exploits cross-sectional variation in generators' fuel efficiency: conditional on natural gas price movements, generators with higher heat rates (lower efficiency) face larger marginal cost shocks, allowing us to trace out the demand curve.

First-Stage and Structural Estimates. The first stage regresses electricity prices on the heat rate–gas price interaction. The structural equation then uses predicted prices to estimate price responsiveness. The estimated price elasticity from the structural equation is 0.135 (SE = 0.013, $p < 0.0001$), indicating economically meaningful and precisely estimated demand responsiveness. As a specification check, we estimate the first stage separately for each of 5 model-specific regressions; the mean price coefficient is -0.331 (SD = 0.005, range: -0.336 to -0.323). The tight clustering of estimates across specifications suggests stable demand elasticity across AI model events and supports the validity of our instrumentation strategy.

Interpretation and Caveats for Price Analysis

These calculations provide ballpark estimates of how AI-induced demand shifts translate to wholesale price changes. Several caveats apply:

1. The supply elasticity varies across regions and time periods; our estimates represent averages that mask considerable heterogeneity.

2. If AI workloads can be temporally shifted to off-peak hours when supply is more elastic, price impacts would be smaller than our estimates suggest.

3. We do not model transmission constraints, which may amplify local price effects in congested areas near data centers.

3.13.15 Wholesale-to-retail Passthrough Estimates

Table 3.17: Wholesale-to-Retail Price Pass-Through in Deregulated Markets

Study	ISO	PT	Class
Panel A: Wholesale → Retail Pass-Through			
De Feuillley and Fontaine (2023)	PJM (PA)	0.53	Res. (EDC)
De Feuillley and Fontaine (2023)	PJM (PA)	0.56	Res. (EGS)
Brown et al. (2020)	ERCOT	0.43–0.47	Res.
MacKay and Mercadal (2024)	Multi	≈1.0[b]	All
Panel B: Demand Elasticity (Residential)			
Deryugina et al. (2020)	PJM (IL)	−0.09	SR (1 mo.)
Deryugina et al. (2020)	PJM (IL)	−0.27	MR (2 yr.)
Deryugina et al. (2020)	PJM (IL)	−0.35	LR (10 yr.)

Notes: PT = pass-through coefficient. EDC = Electric Distribution Company; EGS = Electric Generation Supplier. SR/MR/LR = short/medium/long-run.

[b]Long-run; \$9.59/MWh cost → \$8.81/MWh retail.

On-site Generation Estimation Details

Data centers increasingly deploy on-site generation to reduce grid dependence. Our data center inventory identifies 87 facilities with on-site power generation capability and 12,401 without. We test whether on-site capacity moderates estimated effects by interacting treatment with on-site status in the utility-level strict DiD specification. Results indicate that on-site generation substantially mitigates measured grid impacts.

Table 3.18: Power Quality Effects by On-site Generation Status

Outcome	With On-site	W/O On-site	Diff.	Interpretation
CPQI	−0.122	+0.228	−0.350	On-site reduces grid stress

Notes: All coefficients significant at $p < 0.001$. CPQI: Consumer Power Quality Index.

The sign reversal for CPQI is notable: data centers *with* on-site generation are associated with improved power quality (negative coefficient), while those *without* on-site generation are associated with deterioration (positive coefficient). This heterogeneity suggests that on-site generation absorbs demand spikes that would otherwise propagate to the grid, and implies that aggregate grid impacts understate total AI energy consumption. It should be noted that all large AI data centers in our most conservative sample had on-site generation. This may imply that our estimates of the impacts of on-site generation are exaggerated by a differential sample.

3.13.16 Additional Robustness Checks

Robustness Checks

We test result robustness by varying treated population definitions, estimating specifications with alternative geographic boundaries to confirm consistency. This demonstrates findings reflect genuine causal impacts rather than arbitrary boundary choices. While we lack data on which models run at specific centers, our DiD methodology accounts for this by using publication dates as exogenous treatment, utilizing minimal geographic and temporal treatment windows. We plan robustness checks restricting treatment to larger data centers most likely involved in AI training. To verify treatment effects aren't noise-related, we include appendix robustness checks with randomized treatment dates as additional evidence that effects relate to model releases.

Treatment Intensity

Rather than binary treatment indicators, we estimate a continuous treatment intensity specification using total electricity demand (MWh) as the treatment variable. The significant coefficient ($p < 0.001$; THD: $+0.00018$, $p < 0.001$) indicate dose-response relationships: larger electricity loads are associated with larger power quality changes.

Confirmed Locations and Verified Dates

We re-estimate our main specifications using stricter treatment definitions. The confirmed-location analysis restricts treatment to facilities with precisely geocoded training locations (mean effect: 74.19 MWh). The verified-date analysis uses API release timing rather than publication dates from the Epoch database (mean effect: 63.20 MWh). Both analyses corroborate our main findings with expected attenuation from the more conservative treatment definitions.

References

Ansi/nema mg-1-2016: Motors and generators, 2016. URL https://www.nema.org/docs/default-source/standards-document-library/mg-1-part-31-watermark.pdf. Section IV Part 31 — Relative equivalent temperature rise: 10 °C difference shortens insulation life by 50%.

Alberto Abadie. Semiparametric difference-in-differences estimators. *Review of Economic Studies*, 72(1):1–19, 2005. doi: 10.1111/0034-6527.00321.

Amazon Web Services. Summary of the AWS US-EAST-2 power event. https://aws.amazon.com/message/, July 2022. Post-Event Summary.

Anthropic. Claude 2.1, 2023a. URL https://www.anthropic.com/news/claude-2-1.

Anthropic. Claude 2, 2023b. URL https://www.anthropic.com/news/claude-2.

Anthropic. Releasing claude instant 1.2, 2023c. URL https://www.anthropic.com/news/releasing-claude-instant-1-2.

Anthropic. Introducing claude, 2023d. URL https://www.anthropic.com/news/introducing-claude.

Anthropic. 3.5 models and computer use, 2024a. URL https://www.anthropic.com/news/3-5-models-and-computer-use.

Anthropic. Claude 3.5 sonnet, 2024b. URL https://www.anthropic.com/news/claude-3-5-sonnet.

Anthropic. Claude 3 model family, 2024c. URL https://www.anthropic.com/news/claude-3-family.

Anthropic. Claude 3 haiku, 2024d. URL https://www.anthropic.com/news/claude-3-haiku.

Anthropic. Anthropic invests $50 billion in American AI infrastructure. https://www.anthropic.com/news/anthropic-invests-50-billion-in-american-ai-infrastructure, November 2025. Accessed: 2025.

Aterio. U.s. data centers: Locations, capacity, and power demand. https://www.aterio.io/insights/us-data-centers, 2025. Comprehensive inventory of operational, under-construction, and announced data centers in the United States, including facility sizes, power capacity estimates, utility and balancing authority data, last updated in mid-2025.

Peter C. Austin. An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behavioral Research*, 46(3):399–424, 2011. doi: 10.1080/00273171.2011.568786.

Jesse D. Backstrom, Michael Gillenwater, Charlie Inman, and Matthew Brander. Corporate power purchase agreements and renewable energy growth. SSRN Working Paper No. 4591413, 2023.

Matteo Benetton, Giovanni Compiani, and Adair Morse. When cryptomining comes to town: High electricity-use spillovers to the local economy. Technical Report Working Paper No. 31312, National Bureau of Economic Research, 2023.

Anders Bjørn, Shannon M. Lloyd, Matthew Brander, and H. Damon Matthews. Renewable energy certificates threaten the integrity of corporate science-based targets. *Nature Climate Change*, 12(6):539–546, 2022a.

Anders Bjørn, Shannon M. Lloyd, Matthew Brander, and H. Damon Matthews. Renewable energy certificates threaten the integrity of corporate science-based targets. *Nature Climate Change*, 12(6):539–546, 2022b. doi: 10.1038/s41558-022-01379-5.

Christian Bogmans, Patricia Gomez-Gonzalez, Ganchimeg Ganpurev, Giovanni Melina, Andrea Pescatori, and Sneha D. Thube. Power hungry: How AI will drive energy demand. Technical Report WP/25/81, International Monetary Fund, 2025.

Severin Borenstein. The trouble with electricity markets: understanding california's restructuring disaster. *Journal of economic perspectives*, 16(1):191–211, 2002.

Severin Borenstein and James Bushnell. An empirical analysis of the potential for market power in California's electricity industry. *Journal of Industrial Economics*, 47(3):285–323, 1999.

Severin Borenstein and James Bushnell. The U.S. electricity industry after 20 years of restructuring. *Annual Review of Economics*, 7(1):437–463, 2015.

Severin Borenstein, James Bushnell, and Steven Stoft. The competitive effects of transmission capacity in a deregulated electricity industry. *RAND Journal of Economics*, 31(2): 294–325, 2000.

Severin Borenstein, James B Bushnell, and Frank A Wolak. Measuring market inefficiencies in california's restructured wholesale electricity market. *American Economic Review*, 92 (5):1376–1405, 2002.

Kirill Borusyak, Xavier Jaravel, and Jann Spiess. Revisiting event-study designs: robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285, 2024.

Matthew Brander, Michael Gillenwater, and Francisco Ascui. Creative accounting: A critical perspective on the market-based method for reporting purchased electricity (scope 2) emissions. *Energy Policy*, 112:29–33, 2018.

David P. Brown, Chi-Hsin Tsai, Chi-Keung Woo, Jay Zarnikau, and Shuangshuang Zhu. Residential electricity pricing in Texas's competitive retail market. *Energy Economics*, 92:104953, 2020. doi: 10.1016/j.eneco.2020.104953. Estimates 43–47% pass-through of wholesale price forecasts to retail quotes in ERCOT.

James B. Bushnell, Erin T. Mansur, and Celeste Saravia. Vertical arrangements, market structure, and competition: An analysis of restructured US electricity markets. *American Economic Review*, 98(1):237–266, 2008.

Brantly Callaway and Pedro HC Sant'Anna. Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230, 2021.

Duncan S. Callaway, Meredith Fowlie, and Gavin McCormick. Location, location, location: The variable value of renewable energy and demand-side efficiency resources. *Journal of the Association of Environmental and Resource Economists*, 5(1):39–75, 2018.

David Card and Alan B Krueger. Minimum wages and employment: A case study of the fast-food industry in new jersey and pennsylvania. *American Economic Review*, 84(4): 772–793, 1994.

Doruk Cengiz, Arindrajit Dube, Attila Lindner, and Ben Zipperer. The effect of minimum wages on low-wage jobs. *The Quarterly Journal of Economics*, 134(3):1405–1454, 2019.

Johanna Cludius, Hauke Hermann, Felix Chr. Matthes, and Verena Graichen. The merit order effect of wind and photovoltaic electricity generation in Germany 2008–2016: Estimation and distributional implications. *Energy Economics*, 44:302–313, 2014.

Richard K. Crump, V. Joseph Hotz, Guido W. Imbens, and Oscar A. Mitnik. Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, 96(1):187–199, 2009. doi: 10.1093/biomet/asn055.

Data Center Dynamics. Alibaba Cloud data center catches fire in Singapore. *Data Center Dynamics*, September 2024.

Data Center Knowledge. Electrical incident at Google data center injures three. *Data Center Knowledge*, August 2022.

Christophe De Feuillley and Gérard Fontaine. Pass-through in residential retail electricity competition: Evidence from Pennsylvania. *Utilities Policy*, 85:101678, 2023. doi: 10.1016/j.jup.2023.101678. Reports 52.5% (EDC) and 56.3% (EGS) pass-through rates for PJM Pennsylvania residential.

Tatyana Deryugina, Alexander MacKay, and Julian Reif. The long-run dynamics of electricity demand: Evidence from municipal aggregation. Working Paper 23483, National Bureau of Economic Research, 2020. Uses Illinois municipal aggregation; finds residential elasticity of -0.09 (short-run) to -0.35 (long-run).

Manasi Deshpande and Yue Li. Who is screened out? application costs and the targeting of disability programs. *American Economic Journal: Economic Policy*, 11(4):213–248, 2019.

EIA. How much electricity does an american home use?, 2024. URL https://www.eia.gov/tools/faqs/faq.php?id=97&t=3. Average U.S. residential electricity consumption is about 10,791 kWh/year, approximately 1.23 kW continuous load.

Jessica Eisenberg. Major aws outages and 6 tips for surviving one, August 2025. URL https://n2ws.com/blog/aws-cloud/aws-outage. Accessed: 2026-01-25.

Fastly. Summary of June 8 outage. https://www.fastly.com/blog/summary-of-june-8-outage, June 2021. Official postmortem.

Federal Energy Regulatory Commission. Order on proposed amendments to interconnection service agreement, talen energy marketing, llc. https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20240730-XXXX, 2024. Docket No. ER24-1937-000 (Talen–Amazon/Susquehanna proceeding), issued July 30, 2024.

Harrison Fell and Peter Maniloff. Leakage in regional environmental policy: The case of the Regional Greenhouse Gas Initiative. *Journal of Environmental Economics and Management*, 87:1–23, 2018.

Meredith Fowlie, Stephen P Holland, and Erin T Mansur. What do emissions markets deliver and to whom? evidence from southern california's nox trading program. *American Economic Review*, 102(2):965–993, 2012.

Meredith L. Fowlie. Incomplete environmental regulation, imperfect competition, and emissions leakage. *American Economic Journal: Economic Policy*, 1(2):72–112, 2009.

Antonio Gargano and Marco Giacoletti. Subsidizing the cloud: U.S. state incentives to data centers. SSRN ID 5881105, 2025.

Michael Gillenwater. Redefining renewable energy certificates (recs): Untangling attributes and offsets. Technical report, Greenhouse Gas Management Institute, 2007.

Michael Gillenwater. Redefining RECs—part 1: Untangling attributes and offsets. *Energy Policy*, 36(6):2109–2119, 2008.

Michael Gillenwater, Xi Lu, and Miriam Fischlein. Additionality of wind energy investments in the U.S. voluntary green power market. *Renewable Energy*, 63:452–457, 2014.

Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of econometrics*, 225(2):254–277, 2021.

Google. Bard and google ai search updates, 2023a. URL https://blog.google/technology/ai/bard-google-ai-search-updates/.

Google. Gemini api for developers and cloud, 2023b. URL https://blog.google/technology/ai/gemini-api-developers-cloud/.

Google. Palm 2, 2023c. URL https://blog.google/technology/ai/google-palm-2-ai-large-language-model/.

Google. Gemini 1.5 (developer preview), 2024a. URL https://blog.google/technology/ai/google-gemini-next-generation-model-february-2024/.

Google. Google gemini update (i/o 2024), 2024b. URL https://blog.google/technology/ai/google-gemini-update-flash-ai-assistant-io-2024/.

Google. Google gemma 2, 2024c. URL https://blog.google/technology/developers/google-gemma-2/.

Google. Gemma open models, 2024d. URL https://blog.google/technology/developers/gemma-open-models/.

Google Cloud. Google Cloud opens its first ML Hub, focused on AI-optimized compute. https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-launches-new-ai-ml-hub, April 2023. Accessed: 2025.

Google DeepMind. Gemini flash, 2024. URL https://deepmind.google/technologies/gemini/flash/.

Google Developers Blog. Announcing palm api and makersuite, 2023. URL https://developers.googleblog.com/en/announcing-palm-api-and-makersuite/.

Google Developers Blog. Gemini: The next era of models and developer products, 2024. URL https://developers.googleblog.com/en/gemini-the-next-era-of-models-and-developer-products/.

Joshua S. Graff Zivin, Matthew J. Kotchen, and Erin T. Mansur. Spatial and temporal heterogeneity of marginal emissions: Implications for electric cars and other electricity-shifting policies. *Journal of Economic Behavior & Organization*, 107:248–268, 2014.

Michael Greenstone and Rema Hanna. Environmental regulations, air and water pollution, and infant mortality in india. *American Economic Review*, 104(10):3038–3072, 2014.

Keisuke Hirano, Guido W. Imbens, and Geert Ridder. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*, 71(4):1161–1189, 2003. doi: 10.1111/1468-0262.00442.

Stephen P. Holland, Erin T. Mansur, Nicholas Z. Muller, and Andrew J. Yates. Are there environmental benefits from driving electric vehicles? the importance of local factors. *American Economic Review*, 106(12):3700–3729, 2016.

Stephen P. Holland, Matthew J. Kotchen, Erin T. Mansur, and Andrew J. Yates. Why marginal CO_2 emissions are not decreasing for US electricity: Estimates and implications for climate policy. *Proceedings of the National Academy of Sciences*, 119(8):e2116632119, 2022.

Scott Horsley. Is your electric bill going up? ai is partly to blame. *NPR*, November 2025. URL https://www.npr.org/2025/11/06/nx-s1-5597971/electricity-bills-utilities-ai. Published November 6, 2025.

HPCwire. Meta completes research SuperCluster, announces next-gen datacenter. https://www.hpcwire.com/2023/05/18/meta-completes-research-supercluster-announces-next-gen-datacenter/, May 2023. Accessed: 2025.

Hugging Face. Llama-3.3-70b-instruct model card, 2024. URL https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct.

Mike Jacobs. Data centers are already increasing your energy bills. we have the receipts. Union of Concerned Scientists Blog, October 2025. URL https://blog.ucsusa.org/mike-jacobs/data-centers-are-already-increasing-your-energy-bills/.

Paul L Joskow. Capacity payments in imperfect electricity markets: Need and design. *Utilities policy*, 16(3):159–170, 2008.

Paul L. Joskow. Challenges for wholesale electricity markets with intermittent renewable generation at scale: The US experience. *Oxford Review of Economic Policy*, 35(2): 291–331, 2019.

Owen Kay, Robert Reaser, and Reid Taylor. Processing power: The effect of data centers on wholesale electricity markets. Technical Report Working Paper No. 2606, Federal Reserve Bank of Dallas, 2026.

KCCI 8 News. AI technology behind ChatGPT was built in Iowa — with a lot of water. https://www.kcci.com/article/ai-technology-behind-chatgpt-built-in-west-des-moines-iowa-microsoft/45081445, September 2023. Accessed: 2025.

Christopher R. Knittel, Juan Ramon L. Senga, and Shen Wang. Flexible data centers and the grid: Lower costs, higher emissions? NBER Working Paper No. 34065, 2025.

James Kobus et al. The role of corporate renewable power purchase agreements in supporting u.s. wind and solar deployment, 2021.

Thomas Kurian. Interview with Google Cloud CEO. Data Center Dynamics, December 2023. Accessed: 2025.

Theo Laughner, Scott Heckman, Asif Manzur, and Chris Sloop. Analysis of total harmonic distortion on the u.s. electric grid, November 2024. URL https://www.whiskerlabs.com/analysis-of-total-harmonic-distortion-on-the-us-electric-grid/. Accessed January 28, 2026.

Ian Luby, Brian Sergi, Inês Azevedo, Lucas Davis, and Aaron Bergman. Electricity grid impacts of rising demand from data centers and cryptocurrency mining operations. Technical report, Carnegie Mellon University, Scott Institute for Energy Innovation, 2025. URL https://energy.cmu.edu/_files/documents/electricity-grid-impacts-of-rising-demand-from-data-centers-and-cryptocurrency-mining-operations.pdf.

Alexandra Sasha Luccioni, Sylvain Viguier, and Anne-Laure Ligozat. Estimating the carbon footprint of BLOOM, a 176b parameter language model. *arXiv preprint arXiv:2211.02001*, November 2022. Model card available at https://huggingface.co/bigscience/bloom.

Alexander MacKay and Ignacia Mercadal. Deregulation, market power, and prices: Evidence from the electricity sector. *American Economic Journal: Microeconomics*, 2024. Finds near one-for-one long-run pass-through of utility costs to retail rates; $14.2/MWh margin increase 2000–2016.

Jamal Mamkhezri, Xiaochen Sun, and Yuting Yang. The hidden cost of the cloud: Data centers and electricity market inefficiency. USAEE Working Paper No. 25-660; SSRN ID 5736562, 2025.

Meta AI. Code llama, 2023a. URL https://ai.meta.com/blog/code-llama-large-language-model-coding/.

Meta AI. Code llama: Open foundation models for code, 2023b. URL https://ai.meta.com/research/publications/code-llama-open-foundation-models-for-code/.

Meta AI. Large language model llama, 2023c. URL https://ai.meta.com/blog/large-language-model-llama-meta-ai/.

Meta AI. Meta llama 3.1, 2024a. URL https://ai.meta.com/blog/meta-llama-3-1/.

Meta AI. Meta llama 3, 2024b. URL https://ai.meta.com/blog/meta-llama-3/.

Meta AI. Llama 3.2 (connect 2024), 2024c. URL https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/.

Meta Newsroom. Llama 2, 2023. URL https://about.fb.com/news/2023/07/llama-2/.

Gregory J Miller, Kevin Novan, and Alan Jenn. Hourly accounting of carbon emissions from electricity consumption. *Environmental Research Letters*, 17(4):044073, 2022.

Mistral AI. Announcing mistral 7b, 2023a. URL https://mistral.ai/news/announcing-mistral-7b/.

Mistral AI. Mixtral of experts, 2023b. URL https://mistral.ai/news/mixtral-of-experts/.

Mistral AI. Codestral, 2024a. URL https://mistral.ai/news/codestral/.

Mistral AI. La plateforme, 2024b. URL https://mistral.ai/news/la-plateforme/.

Mistral AI. Mistral large, 2024c. URL https://mistral.ai/news/mistral-large/.

Mistral AI. Mistral large 2407, 2024d. URL https://mistral.ai/news/mistral-large-2407/.

Mistral AI. Mixtral 8x22b, 2024e. URL https://mistral.ai/news/mixtral-8x22b/.

Mistral AI. Mistral nemo, 2024f. URL https://mistral.ai/news/mistral-nemo/.

Mistral AI. Pixtral 12b, 2024g. URL https://mistral.ai/news/pixtral-12b/.

Mistral AI. Pixtral large, 2024h. URL https://mistral.ai/news/pixtral-large/.

Jacob Morrison, Clara Na, Jared Fernandez, Tim Dettmers, Emma Strubell, and Jesse Dodge. Holistically evaluating the environmental impact of creating language models. In *The Thirteenth International Conference on Learning Representations (ICLR) Spotlight*, 2025. URL https://arxiv.org/abs/2503.05804.

Sebastian Moss. Google's London data center outage during heatwave caused by simultaneous failure of multiple redundant cooling systems. *Data Center Dynamics*, August 2022. URL https://www.datacenterdynamics.com/en/news/googles-london-data-center-outage-during-heatwave/.

Leonardo Nicoletti, Naureen Malik, and Andre Tartar. Ai needs so much power, it's making yours worse. *Bloomberg*, December 27 2024. URL https://www.bloomberg.com/graphics/2024-ai-power-home-appliances/. Published December 27, 2024. An exclusive Bloomberg graphics story.

NVIDIA and Microsoft. Megatron-Turing NLG: The largest and most powerful generative language model. NVIDIA Developer Blog, 2021. Trained on NVIDIA Selene supercomputer.

OpenAI. Openai api, 2020. URL https://openai.com/index/openai-api/.

OpenAI. Api no waitlist, 2021a. URL https://openai.com/index/api-no-waitlist/.

OpenAI. Introducing openai codex, 2021b. URL https://openai.com/index/openai-codex/.

OpenAI. Chatgpt, 2022a. URL https://openai.com/index/chatgpt/.

OpenAI. Dall-e api now available in public beta, 2022b. URL https://openai.com/index/dall-e-api-now-available-in-public-beta/.

OpenAI. Introducing chatgpt and whisper apis, 2023a. URL https://openai.com/index/introducing-chatgpt-and-whisper-apis/.

OpenAI. New models and developer products announced at devday, 2023b. URL https://openai.com/index/new-models-and-developer-products-announced-at-devday/.

OpenAI. Gpt-4, 2023c. URL https://openai.com/index/gpt-4/.

OpenAI. Gpt-4 api general availability, 2023d. URL https://openai.com/index/gpt-4-api-general-availability/.

OpenAI. Gpt-4 system card, 2023e. URL https://cdn.openai.com/papers/gpt-4-system-card.pdf.

OpenAI. New embedding models and api updates, 2024a. URL https://openai.com/index/new-embedding-models-and-api-updates/.

OpenAI. Gpt-4o mini: Advancing cost-efficient intelligence, 2024b. URL https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/.

OpenAI. Hello gpt-4o, 2024c. URL https://openai.com/index/hello-gpt-4o/.

OpenAI. Introducing openai o1-preview, 2024d. URL https://openai.com/index/introducing-openai-o1-preview/.

Parametrix Insurance. CrowdStrike outage losses estimated at a staggering $5.4b. *Dark Reading*, July 2024. URL https://www.darkreading.com/cybersecurity-operations/crowdstrike-outage-losses-estimated-staggering-54b.

Rackspace Technology. Form 10-k annual report. U.S. Securities and Exchange Commission, 2023. Disclosed remediation costs and revenue impact from December 2022 incident.

Iegor Riepin and Tom Brown. On the means, costs, and system-level impacts of 24/7 carbon-free energy procurement. *Energy Strategy Reviews*, 54:101488, 2024.

Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983. doi: 10.1093/biomet/70.1.41.

Pedro H. C. Sant'Anna and Jun Zhao. Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1):101–122, 2020. doi: 10.1016/j.jeconom.2020.06.003.

Roy Schwartz, Jesse Dodge, Noah A Smith, and Oren Etzioni. Green ai. *Communications of the ACM*, 63(12):54–63, 2020.

Frank Sensfuß, Mario Ragwitz, and Massimo Genoese. The merit-order effect: A detailed analysis of the price effect of renewable electricity generation on spot market prices in Germany. *Energy Policy*, 36(8):3076–3084, 2008.

Simon Sharwood. Google Cloud suffers outage in Europe amid water leak, fire. *The Register*, April 2023. URL https://www.theregister.com/2023/04/26/google_cloud_outage/.

Simon Sharwood. Google Cloud has 12 hour outage in europe-west3 region. *The Register*, October 2024. URL https://www.theregister.com/2024/10/25/google_cloud_frankfurt_outage/.

Kyle Siler-Evans, Inês Lima Azevedo, and M. Granger Morgan. Marginal emissions factors for the U.S. electricity system. *Environmental Science & Technology*, 46(9):4742–4748, 2012.

South China Morning Post. Alibaba Cloud suffers major outage in Hong Kong. *South China Morning Post*, December 2022.

Southwest Power Pool, Inc. Market Monitoring Unit. 2024 annual state of the market report. Technical Report 2024 Annual State of the Market Report, Southwest Power Pool, Inc., May 2025. Published May 28, 2025.

StatusGator. Microsoft Azure south central US outage. https://statusgator.com/, December 2024. Incident tracking report.

Emma Strubell, Ananya Ganesh, and Andrew McCallum. Energy and Policy Considerations for Deep Learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3645–3650, Florence, Italy, 2019.

Hao Sun, Evan Munro, Georgy Kalashnov, Shuyang Du, and Stefan Wager. Treatment allocation under uncertain costs. *arXiv preprint arXiv:2103.11066*, 2021.

S. M. D. Taheri et al. Physical vs. virtual corporate power purchase agreements: Procurement under uncertainty. *European Journal of Operational Research*, 2025. Physical PPAs deliver electricity, while virtual PPAs are financial hedging contracts.

Uptime Institute. OVHcloud Strasbourg data center fire. *Uptime Institute Intelligence*, March 2021.

U.S. Energy Information Administration. Existing capacity by energy source, 2023, table 4.2.a. Electric Power Annual, 2024. URL https://www.eia.gov/electricity/annual/html/epa_04_02_a.html. Accessed: 2026-04-26.

U.S. Environmental Protection Agency. Renewable energy certificates (recs). https://www.epa.gov/green-power-markets/renewable-energy-certificates-recs, 2025. Accessed: 2026-04-09.

U.S. Fire Administration. Residential building electrical fires (2014-2016). Technical Report 8, Federal Emergency Management Agency, 2019. URL https://www.usfa.fema.gov/downloads/pdf/statistics/v19i8.pdf.

Whisker Labs. Analysis of total harmonic distortion on the u.s. electric grid. Technical report, Whisker Labs, November 2024. URL https://www.whiskerlabs.com/analysis-of-total-harmonic-distortion-on-the-us-electric-grid/. Data collected from the Ting Sensor Network (Whisker Labs power quality dataset).

Coady Wing, Seth M Freedman, and Alex Hollingsworth. Stacked difference-in-differences. Technical report, National Bureau of Economic Research, 2024.

Ryan Wiser and Mark Bolinger. An overview of alternative fossil fuel price and carbon regulation scenarios. Technical Report LBNL-56403, Lawrence Berkeley National Laboratory, Berkeley, CA, October 2004. URL https://digital.library.unt.edu/ark:/67531/metadc779656/. OSTI Report No. 833577.

Bin Xu and Boqiang Lin. How does green credit effectively promote green technology innovation? *International Review of Financial Analysis*, 102:104089, 2025.